



AutoCAD Object Summary

This appendix contains a reference to the full AutoCAD object model. The first part of the appendix provides information about properties, methods, and events that are common to a number of the AutoCAD objects. The second and major part of the appendix explains the methods, properties, and events of each object and collection in alphabetical order.

AutoCAD Collections

The AutoCAD object model has the following collections:

AcadBlock	AcadModelSpace
AcadBlocks	AcadPaperSpace
AcadDictionaries	AcadPlotConfigurations
AcadDictionary	AcadPopupMenu
AcadDimStyles	AcadPopupMenu
AcadDocuments	AcadRegisteredApplications
AcadFileDependencies	AcadSelectionSet
AcadGroup	AcadSelectionSets
AcadGroups	AcadTextStyles
AcadHyperlinks	AcadToolbar
AcadLayers	AcadToolbars
AcadLayouts	AcadUCSs
AcadLineTypes	AcadViewports
AcadMenuBar	AcadViews
AcadMenuGroups	

Common Collection Methods

All the collections in the AutoCAD object model support the `Item` method, which is used to select an item from the collection.

Name	Returns	Description
Item	Object	Returns the object at the given index in the collection. Parameter: Index As Variant (Integer or String). If the Index value is a String, it must match an existing object's name.

Common Collection Property

All collections in the AutoCAD object model support the following property.

Name	Returns	Description
Count	Long	Gets the number of items in the collection.

The Application Property

Just one property is common to all the objects in the AutoCAD object model: the `Application` property.

Name	Returns	Description
Application	AcadApplication	Gets the <code>AcadApplication</code> object, which represents the application's frame controls and path settings and provides the means to navigate down the object hierarchy. This property's value is read-only.

The AcadObject Object

This object represents methods and properties common to the following objects:

Acad3DFace	AcadDictionaries
Acad3DPolyline	AcadDictionary
Acad3DSolid	AcadDim3PointAngular
AcadArc	AcadDimAligned
AcadAttribute	AcadDimAngular
AcadAttributeReference	AcadDimArcLength
AcadBlock	AcadDimDiametric
AcadBlockReference	AcadDimension
AcadBlocks	AcadDimOrdinate
AcadCircle	AcadDimRadial

AcadDimRadialLarge	AcadPolyfaceMesh
AcadDimRotated	AcadPolygonMesh
AcadDimStyle	AcadPolyline
AcadDimStyles	AcadPViewport
AcadDynamicBlockReferenceProperty	AcadRasterImage
AcadEllipse	AcadRay
AcadEntity	AcadRegion
AcadExternalReference	AcadRegisteredApplication
AcadGroup	AcadRegisteredApplications
AcadGroups	AcadShape
AcadHatch	AcadSolid
AcadLayer	AcadSortentsTable
AcadLayers	AcadSpline
AcadLayout	AcadTable
AcadLayouts	AcadTableStyle
AcadLeader	AcadText
AcadLine	AcadTextStyle
AcadLineType	AcadTextStyles
AcadLineTypes	AcadTolerance
AcadLWPolyline	AcadTrace
AcadMInsertBlock	AcadUCS
AcadMLine	AcadUCSs
AcadModelSpace	AcadView
AcadMText	AcadViewport
AcadOle	AcadViewports
AcadPaperSpace	AcadViews
AcadPlotConfiguration	AcadXLine
AcadPlotConfigurations	AcadXRecord
AcadPoint	

Common AcadObject Methods

The following table presents common AcadObject methods.

Name	Description
Delete	Deletes a specified object. If an object in a collection is deleted, all the remaining items are reassigned a new index based on the current count. Note that you can't delete collections, so you can't use this method for collections without creating an error.
GetExtensionDictionary	Returns AcadDictionary and gets the extension dictionary associated with an object. If an object doesn't have an extension dictionary, this method will create one for that object and return it.
GetXData	Gets the extended data (XData), that is, instance-specific data, associated with an object. A NULL string value for the AppName parameter will return all the data attached to the object, regardless of the application that created it. Supplying an application name will return only the data that was created by the specified application. Parameters: AppName As String, XDataType As Variant (array of Integers), and XDataValue As Variant (array of Variants).
SetXData	Sets the extended data (XData), that is, instance-specific data associated with an object. Parameters: XDataType As Variant (array of Integers) and XDataValue As Variant (array of Variants).

Common AcadObject Properties

The following table presents common AcadObject properties.

Name	Returns	Description
Application	Object	Gets the Application object. This property's value is read-only. The Application object represents the application's frame controls and path settings and provides the means to navigate down the object hierarchy.
Document	Object	Gets the document or drawing in which the object belongs. This property's value is read-only.
Handle	String	Gets the handle of an object. This property's value is read-only. An object ID and a handle are the two ways of referencing an object. A handle stays the same in a drawing for the lifetime of the object.
HasExtensionDictionary	Boolean	Determines whether the object has an extension dictionary associated with it. This property's value is read-only.
ObjectID	Integer	Gets the object ID of the object. This property's value is read-only. You can use an object ID or a handle to reference an object. It's better to use a handle unless you plan to work with certain ObjectARX functions that require an object ID.
ObjectName	String	Gets the AutoCAD class name of the object. This property's value is read-only.
OwnerID	Integer	Gets the object ID of the owner (i.e., the parent object). This property's value is read-only.

Common AcadObject Event

Apart from the AcadApplication and AcadDocument objects, which expose a variety of events, AutoCAD provides only one other event, the Modified event. This event is common to all the objects listed previously. When coding in VBA, you must provide an event handler for all objects enabled for the Modified event; otherwise, your application may crash. No events are fired while a modal dialog box is displayed.

Event	Description
Modified	Triggered whenever an object is modified. Note that setting the value of a property even if that value remains unchanged is considered to be a modification. Parameter: Entity As AcadEntity, the object in the drawing that is modified. The AcadEntity object is covered in the next section.

The AcadEntity Object

The AcadEntity object represents a generic object that exposes all the methods and properties common to a set of AutoCAD objects known as the *Drawing Objects*. These objects, which define all the drawing capabilities (including drawing planar and solid objects, inserting text and hyperlinks, and linking external documents) are as follows:

Acad3DFace	AcadDimRotated
Acad3DPolyline	AcadEllipse
Acad3DSolid	AcadEntity
AcadArc	AcadExternalReference
AcadAttribute	AcadHatch
AcadAttributeReference	AcadLeader
AcadBlockReference	AcadLine
AcadCircle	AcadLWPolyline
AcadDim3PointAngular	AcadMInsertBlock
AcadDimAligned	AcadMLine
AcadDimAngular	AcadMText
AcadDimArcLength	AcadOle
AcadDimDiametric	AcadPoint
AcadDimension	AcadPolyfaceMesh
AcadDimOrdinate	AcadPolygonMesh
AcadDimRadial	AcadPolyline
AcadDimRadialLarge	AcadPViewport

AcadRasterImage	AcadTable
AcadRay	AcadText
AcadRegion	AcadTolerance
AcadShape	AcadTrace
AcadSolid	AcadXline
AcadSpline	

Common AcadEntity Methods

In addition to the methods inherited from the AcadObject object, the AcadEntity object supports the following methods.

Name	Returns	Description
ArrayPolar	Variant	Creates an array of selected objects in a polar pattern. The distance is determined from the array's center point to a reference point on the last object selected, which in turn depends on the type of object previously selected. AutoCAD uses the center point of a circle or arc, the insertion point of a block or shape, the start point of text, and one end point of a line or trace. Note that you can't use this method for the AcadAttributeReference object. Note also that you can't use this method while iterating through a collection. Parameters: NumberOfObjects As Integer, AngleToFill As Double, and CenterPoint As Variant. The NumberOfObjects parameter specifies the number of objects to be created in the polar array. This must be a positive integer greater than 1. The AngleToFill parameter specifies the angle to fill in radians. A positive value indicates counterclockwise rotation. This parameter can't be equal to 0. The CenterPoint parameter specifies the 3-D WCS coordinates indicating the center point for the polar array.
ArrayRectangular	Variant	Creates an array of selected objects in a rectangular pattern. AutoCAD builds the rectangular array along a baseline defined by the current snap rotation angle, which is 0 by default. Note that you can't use this method for the AcadAttributeReference object. Note also that you can't use this method while iterating through a collection. Parameters: NumberOfRows As Integer, NumberOfColumns As Integer, NumberOfLevels As Integer, DistBetweenRows As Double, DistBetweenCols As Double, and DistBetweenLevels As Double. The NumberOfRows parameter specifies the number of rows in the rectangular array. This must be a positive number. If NumberOfColumns is 1, then NumberOfRows must be greater than 1. The NumberOfColumns parameter specifies the number of columns in the rectangular array. This must be a positive number. If NumberOfRows is 1, then NumberOfColumns must be greater than 1. The NumberOfLevels parameter specifies the number of levels in a 3-D array. The DistBetweenRows parameter specifies the distance between the rows. If positive, rows are added upward from the base entity; otherwise, they're added downward. The DistBetweenCols parameter specifies the distance between the columns. If positive, columns are added to the right of the base entity; otherwise, they're added to the left. The DistBetweenLevels parameter specifies the distance between the array levels.

Name	Returns	Description
CopyObject		Duplicates the given object to the same location. Note that you can't use this method for the <code>AcadAttributeReference</code> object. Note also that you can't use this method while iterating through a collection.
GetBoundingBox		Gets two points of a box enclosing the specified object. Parameters: <code>MinPoint As Variant</code> and <code>MaxPoint As Variant</code> (both of these are three-element arrays of Doubles).
Highlight		Sets the highlight status for the given object or for all objects in a given selection set. Note that you'll see the changes only once the drawing is updated. Parameter: <code>HighlightFlag As Boolean</code> .
IntersectWith	Variant	Returns the points at which one object intersects another object in the drawing as an array of Doubles. The object supplied as the parameter can be any drawing object. However, this method isn't supported by the <code>AcadPolygonMesh</code> and <code>AcadPViewport</code> objects. Parameters: <code>IntersectObject As Object</code> and <code>option As AcExtendOption</code> . For a list of the values of the <code>AcExtendOption</code> enumerated type, see Appendix B.
MirrorObject		Creates a mirror image copy of a planar object around an axis, defined by two points. This method places the reflected image into the drawing and retains the original object. To remove the original object, use the <code>Delete</code> method. To manage the reflection properties of text objects, use the <code>MIRRTEXT</code> system variable. Note that you can't use this method for the <code>AcadAttributeReference</code> object. Note also that you can't use this method while iterating through a collection. Parameters: <code>Point1 As Variant</code> and <code>Point2 As Variant</code> .
Mirror3D	Object	Creates a mirror image copy of a planar object reflected in a plane, defined by three points. AutoCAD checks to see whether the object to be copied owns any other object. If it does, it performs a copy on those objects as well. The process continues until all owned objects have been copied. Note that you can't use this method for the <code>AcadAttributeReference</code> object. Note also that you can't use this method while iterating through a collection. Parameters: <code>Point1 As Variant</code> , <code>Point2 As Variant</code> , and <code>Point3 As Variant</code> .
Move		Moves an object along a vector, defined by two points. Parameters: <code>FromPoint As Variant</code> and <code>ToPoint As Variant</code> .
Rotate		Rotates an object around a base point. Parameters: <code>BasePoint As Variant</code> and <code>RotationAngle As Double</code> . The <code>RotationAngle</code> parameter should be specified in radians.
Rotate3D		Rotates an object around an axis in 3-D space, defined by two points, which become the axis of rotation. Parameters: <code>Point1 As Variant</code> , <code>Point2 As Variant</code> , and <code>RotationAngle As Double</code> . The <code>RotationAngle</code> parameter should be in radians.
ScaleEntity		Scales an object equally in the X, Y, and Z directions. Parameters: <code>BasePoint As Variant</code> and <code>ScaleFactor As Double</code> . The <code>BasePoint</code> parameter is a set of 3-D WCS coordinates and is the only common point in the original and scaled system. The <code>ScaleFactor</code> parameter determines the level of scaling. A value greater than 1 enlarges the object. A value between 0 and 1 shrinks the object. The scale factor must be greater than 0.
TransformBy		Moves, scales, and rotates an object given a 4×4 transformation matrix. See the AutoCAD documentation for more information. Parameter: <code>TransformationMatrix As Variant</code> (a 4×4 array of Doubles).
Update		Updates the object to the drawing screen.

Common AcadEntity Properties

In addition to the properties inherited from the AcadObject, the AcadEntity object supports the following properties.

Name	Returns	Description
Color	AcColor	Gets or sets the color of an entity. For a list of possible values for the AcColor enumerated type, see Appendix B.
EntityName	String	Gets the class name of the object. Retained for backward compatibility only. Use the VBA TypeName function instead. This property's value is read-only.
EntityType	AcEntityName	Gets the entity type of the object. Retained for backward compatibility only. Use the VBA TypeOf keyword instead. This property's value is read-only. For a list of the values for the AcEntityName enumerated type, see Appendix B.
Hyperlinks	AcadHyperlinks	Gets the AcadHyperlinks collection for an entity. This property's value is read-only.
Layer	String	Gets or sets the layer for an entity.
Linetype	String	Gets or sets the linetype of an entity.
LinetypeScale	Double	Gets or sets the linetype scale of an entity. The default value is 1.0.
Lineweight	AcLineWeight	Gets or sets the lineweight of an individual entity or the default lineweight for the drawing. For a list of possible values for the AcLineWeight enumerated type, see Appendix B.
PlotStyleName	String	Gets or sets the object's plot-style name.
TrueColor	AcCmColor	Gets or sets the RGB color index, color name, or Pantone color book settings.
Visible	Boolean	Specifies whether the object (or application) is visible.

Note Because this object inherits from AcadObject, it supports the Modified event.

AcadDimension Object

The AcadDimension object represents a generic object with a set of properties that are common to the nine dimension objects, namely, the following:

AcadDim3PointAngular	AcadDimOrdinate
AcadDimAligned	AcadDimRadial
AcadDimAngular	AcadDimRadialLarge
AcadDimArcLength	AcadDimRotated
AcadDimDiametric	

In addition, it inherits all the methods and properties of the AcadEntity and AcadObject objects.

Common AcadDimension Properties

The AcadDimension object inherits all the properties of the AcadEntity and AcadObject objects, as well as the common Application property. It also supports the following properties.

Name	Returns	Description
DecimalSeparator	String	Gets or sets the character used as the decimal separator for dimension values. The initial value for this property is the period (.), though any character will be accepted as a valid value for this property. Use this property only when the UnitsFormat property is set to acDimDecimal. This property overrides the value of the DIMSEP system variable.
Normal	Variant	Gets or sets the 3-D (Z-axis) normal unit vector for the dimension object.
Rotation	Double	Gets or sets the rotation angle (in radians) for the dimension line, relative to the X-axis with positive values going counterclockwise when viewed along the Z-axis toward the origin.
ScaleFactor	Double	Gets or sets the object's relative X scale factor. The initial value for this property is 1.0000.
StyleName	String	Gets or sets the name of the object's style. Use the AcadDimStyle object to change the attributes of a given dimension style. The name given must already be defined in the drawing.
SuppressTrailingZeros	Boolean	Gets or sets whether leading/trailing zeros are suppressed in the object's values. It overrides the value of the DIMZIN system variable. The initial value for both these properties is False.
TextColor	AcColor	Gets or sets the color of the text. Use a color index number from 0 to 256 or one of the constants listed here: acByBlock (where AutoCAD draws objects in the default color) or acByLayer (where AutoCAD draws objects in the color specified for the layer). For a list of possible values for the AcColor enumerated type, see Appendix B.
TextGap	Double	Gets or sets the distance between the text and the dimension line when the dimension line is broken to make room for the text. A negative value creates basic dimensioning, that is, dimension text with a box around it. This property overrides the value of the DIMGAP system variable.
TextHeight	Double	Gets or sets the height of the text. The initial value for this property is 0.1800.
TextMovement	AcDimTextMovement	Gets or sets how text is drawn when it's moved. The initial value for this property is acMoveTextNoLeader. For a list of possible values for the AcDimTextMovement enumerated type, see Appendix B.

Continued

Name	Returns	Description
TextOverride	String	Gets or sets the text string, which has a maximum length of 256 characters. The user string replaces the calculated dimension value. You can revert to the calculated dimension value by setting the text to a NULL string (""). You can append or prefix text to the primary dimension value by using a closed set of brackets (<>) to represent the value. The primary dimension value will replace the brackets when the string is displayed.
TextPosition	Variant	Gets or sets the position of the text as a set of WCS coordinates.
TextPrefix	String	Gets or sets the prefix of the dimension value. The initial value for this property is NULL. To turn off an established prefix, set this property equal to a single period.
TextRotation	Double	Gets or sets the angle of the text in radians. The valid range is 0 to 6.28.
TextStyle	String	Gets or sets the style of the text. The initial value for this property is STANDARD.
TextSuffix	String	Gets or sets the suffix of the dimension value. The initial value for this property is NULL. To turn off an established suffix, set this property equal to a single period.
ToleranceDisplay	AcDimToleranceMethod	Gets or sets whether the tolerances are displayed with the text. It overrides the value of the DIMTOL system variable. The initial value for this property is acTolNone. For a list of possible values for the AcDimToleranceMethod enumerated type, see Appendix B.
ToleranceHeightScale	Double	Gets or sets the scale factor of the text's height for the tolerance values relative to the object's general text height. The initial value for this property is 1.0000. This property is available only when the ToleranceDisplay property is set to any value other than acTolNone.
ToleranceJustification	AcDimToleranceJustify	Gets or sets the vertical justification of tolerance values relative to that of the object's general text. The initial value for this property is acTolMiddle. This property is available only when the ToleranceDisplay property is set to any value other than acTolNone. For a list of possible values for the AcDimToleranceJustify enumerated type, see Appendix B.
ToleranceLowerLimit	Double	Gets or sets the minimum tolerance for text. The initial value for this property is 0.0000. This property is available only when the ToleranceDisplay property is set to acTolDeviation or acTolLimits.
TolerancePrecision	AcDimPrecision	Gets or sets the precision of tolerance values in the primary dimensions. This property isn't available when ToleranceDisplay is set to acTolNone. For a list of possible values for the AcDimPrecision enumerated type, see Appendix B.

Name	Returns	Description
ToleranceSuppressLeadingZeros	Boolean	Gets or sets whether leading zeros in tolerance values are suppressed. It overrides the value of the DIMTZIN system variable. The initial value for this property is False.
ToleranceSuppressTrailingZeros	Boolean	Gets or sets whether trailing zeros in tolerance values are suppressed. It overrides the value of the DIMTZIN system variable. The initial value for this property is False.
ToleranceUpperLimit	Double	Gets or sets the maximum tolerance for text. The initial value for this property is 0.0000. This property is available only when the ToleranceDisplay property is set to acTolSymmetrical, acTolDeviation, or acTolLimits.
VerticalTextPosition		Gets or sets the vertical position of the text relative to the dimension line. It overrides the value of the DIMTAD system variable. The initial value for this property is acVertCentered. For a list of possible values for the AcDimVerticalJustification enumerated type, see Appendix B.

Note Because this object inherits from `AcadObject`, it supports the Modified event.

AutoCAD Object Reference

The following sections detail the remaining AutoCAD object model objects derived from the base class objects detailed earlier in this appendix.

Acad3DFace Object

The `Acad3DFace` object represents a 3-D triangle or a quadrilateral plane section. You create it using the `Add3DFace` method of the `AcadBlock`, `AcadModelSpace`, or `AcadPaperSpace` object. You can specify different Z coordinates for each corner point of a `Acad3DFace` object, and you can build solid objects by joining 3-D faces together (e.g., a cube consists of six 3-D faces joined together). You control which edges of the face are visible through the `SetInvisibleEdge` method, which allows accurate modeling of objects with holes.

Acad3DFace Object Methods

The `Acad3DFace` object inherits all the methods of the `AcadEntity` and `AcadObject` objects. It also supports the following methods.

Name	Returns	Description
GetInvisibleEdge	Boolean	Gets the visibility setting for an edge of a <code>Acad3DFace</code> object at a given index. It's <code>True</code> if the edge is invisible and <code>False</code> otherwise. Parameter: <code>Index</code> As Integer.
SetInvisibleEdge		Sets the visibility state of an edge at a given index. To see any changes in visibility, the drawing must be regenerated. Parameters: <code>Index</code> As Integer and <code>State</code> As Boolean. The <code>Index</code> parameter must be in the range 0 to 3. Set the <code>State</code> parameter to <code>True</code> for an invisible edge and <code>False</code> otherwise.

Acad3DFace Object Properties

The `Acad3DFace` object inherits all the properties of the `AcadEntity` and `AcadObject` objects, as well as the common `Application` property. It also supports the following properties.

Name	Returns	Description
Coordinate	Variant	Gets or sets the coordinate of a single vertex in the object. This will replace the specified vertex of the object. Use standard array-handling techniques to process the values contained in this property. It returns a three-element array of Doubles containing 3-D coordinates in WCS. Note that the Z coordinate will default to 0 on the active UCS. Parameter: <code>Index</code> As Integer (the index in the zero-based array of vertices for the vertex you want to set or query).
Coordinates	Variant	Gets or sets the coordinates for each vertex in the object. This will replace any existing coordinates for the specified object. Use standard array-handling techniques to process the coordinates contained in this property. Note that you can't change the number of coordinates in the object by using this property; you can change only the location of existing coordinates. It returns a three-element array of Doubles containing 3-D coordinates in WCS. Note also that the Z coordinate will always default to 0 on the active UCS.
VisibilityEdge1	Boolean	Gets or sets whether edge 1 is visible. For this property and the three that follow, <code>True</code> means the edge is visible and <code>False</code> means the edge isn't visible.
VisibilityEdge2	Boolean	Gets or sets whether edge 2 is visible.
VisibilityEdge3	Boolean	Gets or sets whether edge 3 is visible.
VisibilityEdge4	Boolean	Gets or sets whether edge 4 is visible. In the case of a triangle, this property will still return a value, because a triangle is defined as a quadrilateral in which point 3 is equal to point 4.

Note Because this object inherits from `AcadObject`, it supports the `Modified` event.

Acad3DPolyline Object

The Acad3DPolyline object represents a 3-D polyline of straight-line segments. You create it using the Add3DPoly method of the AcadBlock, AcadModelSpace, or AcadPaperSpace object.

Acad3DPolyline Object Methods

The Acad3DPolyline object inherits all the methods of the AcadEntity and AcadObject objects. It also supports the following methods.

Name	Returns	Description
AppendVertex		Appends a vertex to the polyline, which is an array of 3-D coordinates. Parameter: vertex As Variant (a three-element array of Doubles).
Explode	Variant	Explodes the polyline into its constituent lines and returns them as an array of Objects.

Acad3DPolyline Object Properties

The Acad3DPolyline object inherits all the properties of the AcadEntity and AcadObject objects, as well as the common Application property. It also supports the following properties.

Name	Returns	Description
Closed	Boolean	Gets or sets whether the polyline is closed (True) or open (False).
Coordinate	Variant	Gets or sets the coordinate of a single vertex in the object. This will replace the specified existing vertex of the object. Use standard array-handling techniques to process the values contained in this property. It returns a Variant (three-element array of Doubles). These coordinates can be converted to and from other coordinate systems using the TranslateCoordinates method. Parameter: Index As Integer (the index in the zero-based array for the vertex you want to set or query).
Coordinates	Variant	Gets or sets the coordinates for each vertex in the object. This will replace any existing coordinates for the specified object. Use standard array-handling techniques to process the coordinates contained in this property. When you set the coordinates for a polyline, if you supply fewer coordinates than the object currently possesses, the polyline will be truncated. Any fit points applying to the truncated vertices will also be truncated. If you supply more coordinates than the object currently possesses, the extra vertices will be appended to the polyline. It returns the vertex points as a Variant (a three-element array of Doubles) containing the X, Y, and Z coordinates for the vertices in WCS. These coordinates can be converted to and from other coordinate systems using the TranslateCoordinates method.
Type	Ac3DPolylineType	Gets or sets the type of the 3-D polyline. You can use this property to convert a 3-D polyline into a spline. For a list of the values of the Ac3DPolylineType enumerated type, see Appendix B.

Note Because this object inherits from AcadObject, it supports the Modified event.

Acad3DSolid Object

The Acad3DSolid object represents a solid object with free-form surface support. You don't create general Acad3DSolid objects, as there's no Add3DSolid method supported by the AcadBlock, AcadModelSpace, and AcadPaperSpace objects. Instead, you use this object to represent a whole series of 3-D shapes for which no specific AutoCAD object exists: boxes, cones, cylinders, spheres, wedges, tori (donuts), and others. You can then combine these shapes to create more complex solids by joining or subtracting them or finding their intersecting (overlapping) volume. You can also create solids by sweeping a 2-D object along a path or revolving it around an axis.

To create an Acad3DSolid object, use one of the following methods, all of which are common to the AcadBlock, AcadModelSpace, and AcadPaperSpace objects:

AddBox	AddExtrudedSolidAlongPath
AddCone	AddRevolvedSolid
AddCylinder	AddSphere
AddEllipticalCone	AddTorus
AddEllipticalCylinder	AddWedge
AddExtrudedSolid	

See the section about the AcadBlock object for more details on each of these.

Acad3DSolid Object Methods

The Acad3DSolid object inherits all the methods of the AcadEntity and AcadObject objects. It also supports the following methods.

Name	Returns	Description
Boolean		Performs a destructive operation returning the union, intersection, or subtraction between the object and another Acad3DSolid or AcadRegion object. Parameters: Operation As AcBooleanType and SolidObject As Acad3DSolid. For a list of possible values for the AcBooleanType enumerated type, see Appendix B.
CheckInterference	Acad3DSolid	Checks for interference between the object and another Acad3DSolid object and creates an object representing the interference if required. Parameters: Object As Acad3DSolid and CreateInterferenceSolid As Boolean. The second parameter is True if an interference solid is required and False otherwise.
SectionSolid	AcadRegion	Creates an AcadRegion object representing the intersection of the object with the plane containing the specified points. Parameters: Point1 As Variant, Point2 As Variant, and Point3 As Variant.

Name	Returns	Description
SliceSolid	Acad3DSolid	Creates an Acad3DSolid object representing the portion of the object on one side of the plane containing the specified points. Parameters: Point1 As Variant, Point2 As Variant, Point3 As Variant, and Negative As Boolean. The fourth parameter determines whether the resulting solid should be returned on the negative side of the plane, which is True if this is the case and False otherwise. The original Acad3DSolid object retains the positive side of the slice.

Acad3DSolid Object Properties

The Acad3DSolid object inherits all the properties of the AcadEntity and AcadObject objects, as well as the common Application property. It also supports the following properties.

Name	Returns	Description
Centroid	Variant	Gets the center of area or mass for a solid as a 2-D coordinate. It returns a two-element array of Doubles. This property is read-only.
MomentOfInertia	Variant	Gets the moment of inertia for the solid as a 3-D coordinate. It returns a three-element array of Doubles. This property is read-only.
PrincipalDirections	Variant	Gets the principal directions of the solid as X, Y, and Z coordinates calculated on the current coordinate system. This property is read-only.
PrincipalMoments	Variant	Gets the principal moments property of the solid as X, Y, and Z coordinates calculated on the current coordinate system. This property's value is read-only.
ProductOfInertia	Variant	Gets the product of inertia of the solid as X, Y, and Z coordinates calculated on the current coordinate system. This property's value is read-only.
RadiiOfGyration	Variant	Gets the radius of gyration of the solid as X, Y, and Z coordinates calculated on the current coordinate system. This property's value is read-only.
Volume	Double	Gets the object's volume. This property's value is read-only.

Note Because this object inherits from AcadObject, it supports the Modified event.

AcadApplication Object

The AcadApplication object represents an instance of the AutoCAD application and is accessed in VBA by using ThisDrawing.Application. The AcadApplication object is also the global object for the ActiveX interface, and all methods and properties are available in the

global namespace. The object supports 19 methods, which control the loading or listing of the currently loaded external applications and interface objects, and 20 properties, which reflect the properties of the main application window. In addition, more than 20 events are associated with the AcadApplication object.

AcadApplication Object Methods

The AcadApplication object supports the following methods.

Name	Returns	Description
Eval		Executes a line of VBA code without the need to create a macro. Parameter: Expression As String.
GetAcadState	AcadState	Returns an AcadState object that will monitor the state of the AutoCAD application from other applications.
GetInterfaceObject	Object	Loads a program ID into AutoCAD as an in-process server. Although the object will be loaded into AutoCAD, it won't show up in its type library. The object will have its own type library. This method lets you connect to an ActiveX Automation server. Parameter: ProgID As String.
ListARX	Variant	Returns the currently loaded AutoCAD ARX applications as an array. It returns empty if none are loaded.
LoadARX		Loads an AutoCAD ARX application. Parameter: Name As String.
LoadDVB		Loads an AutoCAD VBA project file. Parameter: Name As String.
Quit		Closes the drawing file and the AutoCAD application. If AutoCAD is in the middle of a command, an error message will be generated. If the modified document hasn't been saved, a prompt message will be generated.
RunMacro		Runs a VBA macro. Parameter: MacroPath As String. If the full path name isn't specified, AutoCAD will use its search path to find the macro or search all currently loaded projects. Syntax: RunMacro "MyProject.Module1".
UnloadARX		Unloads an unlocked AutoCAD ARX application. Parameter: Name As String. Don't attempt to unload the file acadvba.arx.
UnloadDVB		Unloads an AutoCAD VBA project file. Parameter: Name As String.
Update		Updates the screen.
ZoomAll		Zooms the current active model space or paper space viewport to display the entire drawing. For a 2-D view, AutoCAD displays the whole drawing or the current extent, whichever is greater. For a 3-D view, this method is equivalent to ZoomExtents.
ZoomCenter		Zooms the current active model space or paper space viewport as specified by the parameters. Parameters: Center As Variant and Magnify As Double.
ZoomExtents		Zooms the current active model space or paper space viewport to the extent of the drawing.

Name	Returns	Description
ZoomPickWindow		Zooms the current active model space or paper space viewport to a window defined by points picked on the screen.
ZoomPrevious		Zooms the current active model space or paper space viewport to its previous state.
ZoomScaled		Zooms the current active model space or paper space viewport to the specified scale factor. Parameters: Scale As Double and ScaleType As AcZoomScaleType. For the values of the second parameter, see Appendix B.
ZoomWindow		Zooms the current active model space or paper space viewport as specified by the parameters. Parameters: LowerLeft As Variant and UpperRight As Variant. Parameters are three-element arrays of Doubles specifying WCS coordinates.

AcadApplication Object Properties

The AcadApplication object supports the following properties.

Name	Returns	Description
ActiveDocument	AcadDocument	Gets or sets a reference to the active document.
Application	AcadApplication	Gets the AcadApplication object, which represents the application's frame controls and path settings and provides the means to navigate down the object hierarchy.
Caption	String	Gets the text seen by the user in the window caption bar. This property is read-only. Note that if the caption is modified using the Win32 API function SetWindowText, the caption property will return the modified caption text.
Documents	AcadDocuments	Gets the Documents collection. This property is read-only.
FullName	String	Gets the full name, including the path, of the application. This property's value is read-only.
Height	Integer	Gets or sets the height of the application window in pixels.
LocaleID	Long	Gets the regional settings for the session as defined by the Windows 95 or Windows NT operating systems. This property's value is read-only.
MenuBar	AcadMenuBar	Gets the AcadMenuBar object for the session. This property's value is read-only.
MenuGroups	AcadMenuGroups	Gets the MenuGroups collection for the session.
Name	String	Gets the name of the application. This property's value is read-only.
Path	String	Gets the path of the application without the file name. This property's value is read-only.
Preferences	AcadPreferences	Gets the application's AcadPreferences object. This property's value is read-only.

Continued

Name	Returns	Description
StatusID	Boolean	Gets the current active status of the viewport. It's True if the specified viewport is active and False otherwise. This property's value is read-only. Parameter: VportObj As AcadViewport.
VBE	VBAIDE	Gets the VBAIDE extensibility object. This isn't available if the acadvba.arx application hasn't been loaded. This property's value is read-only.
Version	String	Gets the version of the AutoCAD application. This property's value is read-only. For AutoCAD 2000, the value is 15.0.
Visible	Boolean	Gets or sets whether the application is visible.
Width	Integer	Gets or sets the width of the application window in pixels.
WindowLeft	Integer	Gets or sets the distance between the left edge of the application window and the left edge of the screen in pixels. This distance will establish the X coordinate of the upper-left corner (0,0) of the application.
WindowState	AcWindowState	Gets or sets the state of the application window. Possible values are acMin (minimized), acMax (maximized), or acNorm (normal).
WindowTop	Integer	Gets or sets the distance between the top of the application window and the top of the screen in pixels. This distance will establish the Y coordinate of the upper-left corner (0,0) of the application.

AcadApplication Object Events

The following events are triggered during various stages in the lifetime of the AcadApplication object. Note that no events will be fired while a modal dialog box is displayed.

Event	Description
AppActivate	Occurs before the main application window is activated.
AppDeactivate	Occurs before the main application window is deactivated.
ARXLoaded	Occurs after an ObjectARX application is loaded. Parameter: FullPathName As String.
ARXUnloaded	Occurs after an ObjectARX application is unloaded. Parameter: FullPathName As String.
BeginCommand	Occurs after a command is issued but before it is completed. Parameter: CommandName As String.
BeginFileDrop	Occurs when a file is dropped on the main application window. Parameters: FileName As String and Cancel As Boolean. If Cancel is set to True, this aborts the loading of the drawing, LISP file, or ARX file; otherwise, loading is allowed to continue.
BeginLISP	Occurs after a request to evaluate a LISP expression is received. Parameter: FirstLine As String. FirstLine won't have any case conversion of the alpha characters.
BeginModal	Occurs before a modal dialog box is displayed. You should never issue a message box from within an event handler for the BeginModal event.

Event	Description
BeginOpen	Occurs after a request to open an existing drawing is received. Parameter: Filename As String.
BeginPlot	Occurs after a request to print a drawing is received. Parameter: DrawingName As String.
BeginQuit	Occurs just before the session ends. Parameter: Cancel As Boolean.
BeginSave	Occurs after a request to save a drawing is received. Parameter: Filename As String.
EndCommand	Occurs after a command is completed. Parameter: CommandName As String.
EndLISP	Occurs after a LISP expression is executed.
EndModal	Occurs after a modal dialog box is dismissed.
EndOpen	Occurs after an existing drawing is opened. Parameter: Filename As String.
EndPlot	Occurs after a document is sent to the printer. Parameter: DrawingName As String.
EndSave	Occurs after a drawing is saved. Parameter: Filename As String.
LISPCancelled	Occurs when the evaluation of a LISP expression is canceled.
NewDrawing	Occurs before a drawing is created.
SysVarChanged	Occurs when the value of a system variable is changed. Parameters: SysVarName As String and NewVal As Variant.
WindowChanged	Occurs when the application window is changed. Parameter: WindowState As Integer (acMin, acMax, or acNorm).
WindowMovedOrResized	Occurs after one document window is moved or resized. Parameters: HwndFrame As Long and Moved As Boolean. If Moved is set to True, the window was moved; otherwise, it was resized.

AcadArc Object

The AcadArc object represents a circular arc, which is always drawn counterclockwise from the start point to the end point. The StartPoint and EndPoint properties of an arc are calculated through the StartAngle, EndAngle, and Radius properties. The AcadArc object is created using the AddArc method of the AcadBlock, AcadModelSpace, or AcadPaperSpace object.

AcadArc Object Methods

The AcadArc object inherits all the methods of the AcadEntity and AcadObject objects. It also supports the following method.

Name	Returns	Description
Offset	Variant	Creates a new arc by offsetting the current arc by a specified distance, which must be a nonzero number. If the offset is positive, a concentric arc with a larger radius is created. If the offset is negative, a smaller arc is created. The return value isn't an arc but a Variant, which is an array of Objects. Parameter: Distance As Double.

AcadArc Object Properties

The AcadArc object inherits all the properties of the AcadEntity and AcadObject objects, as well as the common Application property. It also supports the following properties.

Name	Returns	Description
ArcLength	Double	Gets the length of the arc. This property's value is read-only.
Area	Double	Gets the enclosed area of the arc. This property's value is read-only.
Center	Variant	Gets or sets the center of the arc as an array of the Doubles.
EndAngle	Double	Gets or sets the end angle of the arc in radians.
EndPoint	Variant	Gets the end point of the arc as an array of three Doubles. This property's value is read-only.
Normal	Variant	Gets or sets the 3-D (Z-axis) normal unit vector for the arc.
Radius	Double	Gets or sets the radius of the arc.
StartAngle	Double	Gets or sets the start angle of the arc in radians.
StartPoint	Variant	Gets the start point of the arc as an array of three Doubles. This property's value is read-only.
Thickness	Double	Gets or sets the distance the AcadArc object is extruded above or below its elevation, in a Z-axis direction. The default is 0.0.
TotalAngle	Double	Gets the total angle of the arc in radians. This property's value is read-only.

Note Because this object inherits from AcadObject, it supports the Modified event.

AcadAttribute Object

The AcadAttribute object represents an object appearing as a text string that describes the characteristics of an attribute reference, or, in other words, an attribute definition. You create it using the AddAttribute method of the AcadBlock, AcadModelSpace, or AcadPaperSpace object.

AcadAttribute Object Methods

The AcadAttribute object inherits all the methods of the AcadEntity and AcadObject objects. It supports no other methods.

AcadAttribute Object Properties

The AcadAttribute object inherits all the properties of the AcadEntity and AcadObject objects, as well as the common Application property. It also supports the following properties.

Name	Returns	Description
Alignment	AcAlignment	Gets or sets the horizontal and vertical alignments of the attribute. For a list of the values for the AcAlignment enumeration, see Appendix B.
Backward	Boolean	Gets or sets whether the text is backward.
Constant	Boolean	Gets or sets whether the attribute has the same value in every occurrence. AutoCAD doesn't prompt for a value of constant attributes.
FieldLength	Integer	Gets or sets the field length of the attribute.
Height	Double	Gets or sets the height of the attribute.
HorizontalAlignment	AcHorizontalAlignment	Gets or sets the horizontal alignment of the attribute.
InsertionPoint	Variant	Gets the insertion point of the attribute.
Invisible	Boolean	Gets or sets whether the attribute is invisible. An invisible attribute will not be displayed or plotted. This property's value is read-only.
Mode	Integer	Gets or sets the mode of the attribute definition. Values: acAttributeModeNormal, acAttributeModeInvisible, acAttributeModeConstant, acAttributeModeVerify, or acAttributeModePreset.
Normal	Variant	Gets or sets the 3-D (Z-axis) normal unit vector for the attribute.
ObliqueAngle	Double	Gets or sets the oblique angle of the attribute and is measured from the vertical axis. The units are radians within the range of -85 to +85 degrees. A positive value denotes a lean toward the right.
Preset	Boolean	Gets or sets whether the attribute is preset. A preset attribute sets the attribute to its default value when the user inserts the block.
PromptString	String	Gets or sets the prompt string for the attribute that appears when a block containing this attribute is inserted. The default for this string is the TagString property.
Rotation	Double	Gets or sets the rotation angle (in radians) for the attribute, relative to the X-axis, with positive values going counterclockwise when viewed along the Z-axis toward the origin.
ScaleFactor	Double	Gets or sets the scale factor for the attribute. It must be positive. A scale factor greater than 1 enlarges the object. A scale factor from 0 to 1 shrinks the object. The initial value for this property is 1.0.
StyleName	String	Gets or sets the name of the style used with the object, the default being the current style.
TagString	String	Gets or sets the tag string, which identifies each occurrence of the attribute. The tag string identifies each occurrence of the attribute and may contain any characters other than spaces and exclamation marks. Lowercase letters are changed to uppercase.

Continued

Name	Returns	Description
TextAlignmentPoint	Variant	Gets or sets the alignment point for text, returning a three-element array of Doubles. This property's value is read-only when the Alignment property is set to acAlignmentLeft.
TextGenerationFlag	AcTextGenerationFlag	Gets or sets the attribute text generation flag. Values: acTextFlagBackward and acTextFlagUpsideDown. The values can be combined.
TextString	String	Gets or sets the text string for the attribute.
Thickness	Double	Gets or sets the distance the Attribute object is extruded above or below its elevation, in a Z-axis direction. The default is 0.0.
UpsideDown	Boolean	Specifies whether the text is upside down.
Verify	Boolean	Specifies whether the user is prompted for verification when inserting the block.
VerticalAlignment	AcVerticalAlignment	Gets or sets the vertical alignment of the attribute. For a list of possible values for the AcVerticalAlignment enumeration, see Appendix B.

Note Because this object inherits from AcadObject, it supports the Modified event.

AcadAttributeReference Object

The AcadAttributeReference object represents an object containing text that links to a block or to an instance of the AcadAttribute object. These objects can't be directly created but are added to a drawing when a block containing an attribute definition is inserted. You achieve this by calling the InsertBlock method, which is common to the AcadBlock, AcadModelSpace, and AcadPaperSpace objects. The GetAttributes method of the AcadBlockReference object returns an array of all attribute references attached to the inserted block.

AcadAttributeReference Object Methods

The AcadAttributeReference object inherits all the methods of the AcadEntity and AcadObject objects. It supports no other methods.

AcadAttributeReference Object Properties

The AcadAttributeReference object inherits all the properties of the AcadEntity and AcadObject objects, as well as the common Application property. It also supports all the other properties supported by the AcadAttribute object except for Mode, Preset, PromptString, and Verify. It supports no other properties.

Note Because this object inherits from AcadObject, it supports the Modified event.

AcadBlock Object

The AcadBlock object represents a block definition containing a name and a set of objects. There's no limit to the number of objects a block can contain. Three kinds of block exist: a simple block, an Xref block, and a layout block. Chapter 12 discusses blocks.

AcadBlock Object Methods

In addition to the methods inherited from the AcadObject object, the AcadBlock object supports the following methods.

Name	Returns	Description
Add3DFace	Acad3DFace	Creates an Acad3DFace object using the four vertices. If the last parameter is omitted, a three-sided face is created. Points must be entered in a clockwise or counterclockwise order to create the object. Parameters: Point1 As Variant, Point2 As Variant, Point3 As Variant, and Point4 As Variant.
Add3DMesh	AcadPolygonMesh	Creates a free-form AcadPolygonMesh object representing an M×N free-form mesh from the given array of coordinates. Parameters: M As Integer, N As Integer, and PointsMatrix As Variant. The size of the mesh in both the M and N directions is limited to between 2 and 256. The PointsMatrix parameter is a 3×M×N array of 3-D WCS coordinates. All the coordinates for row M must be supplied before any vertices for row M+1.
Add3DPoly	Acad3DPolyline	Creates an Acad3DPolyline object from the given array of 3-D WCS coordinates. The polyline will be created according to the order of the coordinates in the array. Parameter: PointsArray As Variant.
AddArc	AcadArc	Creates an AcadArc object as specified by the parameters. Parameters: Center As Variant (an array of three Doubles representing 3-D WCS coordinates), Radius As Double, StartAngle As Double, and EndAngle As Double. The start and end angles must be in radians. If StartAngle is greater than EndAngle, a clockwise arc is created.
AddAttribute	AcadAttribute	Creates an AcadAttribute object as specified by the parameters. Parameters: Height As Double, Mode As AcadAttributeMode, Prompt As String, InsertionPoint As Variant (an array of three Doubles representing 3-D WCS coordinates), Tag As String, and Value As String. For the values of the Mode parameter, see Appendix B.
AddBox	Acad3DSolid	Creates an Acad3DSolid object representing a solid box with center at the given origin and with edges parallel to the axes of the WCS. Parameters: Origin As Variant (an array of three Doubles as 3-D WCS coordinates), Length As Double, Width As Double, and Height As Double.

Continued

Name	Returns	Description
AddCircle	AcadCircle	Creates an AcadCircle object as specified by the parameters. Parameters: Center As Variant (an array of three Doubles representing 3-D WCS coordinates) and Radius As Double.
AddCone	Acad3DSolid	Creates an Acad3DSolid object representing a solid cone with a base on the XY plane of the WCS. Parameters: Center As Variant (an array of three Doubles representing 3-D WCS coordinates), BaseRadius As Double, and Height As Double.
AddCustomObject	Object	Creates a custom object. Parameter: ClassName As String. ClassName must be an ARX file prior to the call.
AddCylinder	Acad3DSolid	Creates an Acad3DSolid object representing a cylinder whose base is on the XY plane of the WCS. Parameters: Center As Variant (an array of three Doubles representing 3-D WCS coordinates), Radius As Double, and Height As Double.
AddDim3PointAngular	AcadDim3PointAngular	Creates an AcadDim3PointAngular object as specified by the parameters. Parameters: AngleVertex As Variant, FirstEndPoint As Variant, SecondEndPoint As Variant, and TextPoint As Variant. All four parameters are arrays of three Doubles representing 3-D WCS coordinates.
AddDimAligned	AcadDimAligned	Creates an AcadDimAligned object as specified by the parameters. Parameters: ExtLine1Point As Variant, ExtLine2Point As Variant, and TextPosition As Variant. All three parameters are arrays of three Doubles representing 3-D WCS coordinates.
AddDimAngular	AcadDimAngular	Creates an AcadDimAngular object representing the angular dimension for an arc, two lines, or a circle. Parameters: AngleVertex As Variant, FirstEndPoint As Variant, SecondEndPoint As Variant, and TextPoint As Variant. All four parameters are arrays of three Doubles representing 3-D WCS coordinates.
AddDimDiametric	AcadDimDiametric	Creates a diametric dimension for a circle or arc given the two points on the diameter and the length of the leader line. Parameters: ChordPoint As Variant, FarChordPoint As Variant, and LeaderLength As Double. The first two parameters are arrays of three Doubles representing 3-D WCS coordinates.
AddDimOrdinate	AcadDimOrdinate	Creates an ordinate dimension given the definition point and leader end point. Parameters: DefinitionPoint As Variant, LeaderEndPoint As Variant, and UseXAxis As Integer. The first two parameters are arrays of three Doubles representing 3-D WCS coordinates.
AddDimRadial	AcadDimRadial	Creates a radial dimension for the selected object at the given location. Parameters: Center As Variant, ChordPoint As Variant, and LeaderLength As Double. The first two parameters are arrays of three Doubles representing 3-D WCS coordinates.

Name	Returns	Description
AddDimRotated	AcadDimRotated	Creates an AcadDimRotated object as specified by the parameters. Parameters: ExtLine1Point As Variant, ExtLine2Point As Variant, DimLineLocation As Variant, and RotationAngle As Double. The first three parameters are arrays of three Doubles representing 3-D WCS coordinates.
AddEllipse	AcadEllipse	Creates an AcadEllipse object representing an ellipse in the XY plane of the WCS as specified by the parameters. Parameters: Center As Variant, MajorAxis As Variant, and RadiusRatio As Double. The first two parameters are arrays of three Doubles representing 3-D WCS coordinates.
AddEllipticalCone	Acad3DSolid	Creates an Acad3DSolid object representing an elliptical cone in the XY plane of the WCS given the center, major radius, minor radius, and height. Parameters: Center As Variant (an array of three Doubles representing 3-D WCS coordinates), MajorRadius As Double, MinorRadius As Double, and Height As Double.
AddEllipticalCylinder	Acad3DSolid	Creates an Acad3DSolid object representing an elliptical cylinder as specified by the parameters with its base in the XY plane. Parameters: Center As Variant (an array of three Doubles representing 3-D WCS coordinates), MajorRadius As Double, MinorRadius As Double, and Height As Double.
AddExtrudedSolid	Acad3DSolid	Creates an Acad3DSolid object representing an extruded solid as specified by the parameters. Parameters: Profile As AcadRegion, Height As Double, and TaperAngle As Double. The Height parameter specifies the height of the extrusion along the Z-axis of the object's coordinate system. The TaperAngle parameter must be provided in radians. The range of the taper angle is from -90 to +90 degrees. Positive angles taper in from the base; negative angles taper out. The default angle, 0, extrudes a 2-D object perpendicular to its plane. Only 2-D planar regions can be extruded.
AddExtrudedSolid AlongPath	Acad3DSolid	Creates an Acad3DSolid object representing an extruded solid, given the profile and an extrusion path. Only 2-D planar regions can be extruded, and the path shouldn't lie on the same plane as the profile, and it shouldn't have areas of high curvature. Parameters: Profile As AcadRegion and Path As Object.
AddHatch	AcadHatch	Creates an AcadHatch object as specified by the parameters. Parameters: PatternType As AcPatternType, PatternName As String, and Associativity As Boolean. For a list of possible values for the AcPatternType enumerated type, see Appendix B. The Associativity parameter is True if the hatch is to be associative and False otherwise. Don't forget to create the outer boundary for the hatch and to append it using Obj.AppendOuterLoop(outerLoop), in which Obj is the newly created hatch and outerLoop is an array of drawing objects forming the hatch boundary. (See the section on the AcadHatch object for details.)

Continued

Name	Returns	Description
AddLeader	AcadLeader	Creates an AcadLeader object as specified by the parameters. Parameters: PointsArray As Variant, Annotation As AcadEntity, and Type As AcLeaderType. The PointsArray parameter must be an array of 3-D WCS coordinates (Doubles) and has a minimum of two points (six elements). The Annotation parameter must be an AcadTolerance, AcadMText, or AcadBlockReference object or NULL. See Appendix B for the values of the AcLeaderType enumeration.
AddLightweightPolyline	AcadLWPolyline	Creates an AcadLWPolyline object as specified by the parameter. Parameter: VerticesList As Variant. VerticesList is an array of 2-D OCS coordinates specifying the vertices of the polyline. A minimum of two vertices or four elements is required. You can convert between OCS and WCS using the TranslateCoordinates method (see the section on the AcadUtility object for details). Arcs are added to vertices using the SetBulge method (see the section on the AcadLWPolyline object for details).
AddLine	AcadLine	Creates an AcadLine object representing a line passing through the given points. Parameters: StartPoint As Variant and EndPoint As Variant. Both parameters are arrays of three Doubles representing 3-D WCS coordinates.
AddInsertBlock	AcadMInsertBlock	Inserts an array of AcadBlockReferences from an original block definition of a given Name, as specified by the parameters. Parameters: InsertionPoint As Variant, Name As String, Xscale As Double, Yscale As Double, Zscale As Double, Rotation As Double, NumRows As Integer, NumColumns As Integer, RowSpacing As Integer, and ColumnSpacing As Integer. The InsertionPoint parameter is a set of 3-D WCS coordinates. The rotation angle is in radians.
AddMLine	AcadMLine	Creates an AcadMLine representing a set of lines passing through the points specified in the parameter. Parameter: VertexList As Variant. VertexList is an array of the 3-D WCS coordinates specifying the vertices for the multiline.
AddMText	AcadMText	Creates an AcadMText object as specified by the parameters. Parameters: InsertionPoint As Variant, Width As Double, and Text As String. The InsertionPoint parameter is a set of 3-D WCS coordinates.
AddPoint	AcadPoint	Creates an AcadPoint object as specified by the 3-D WCS coordinates specified. Parameter: Point As Variant.
AddPolyfaceMesh	AcadPolyfaceMesh	Creates an AcadPolyfaceMesh object from a list of vertices. Parameters: VertexList As Variant and Facelist As Variant. VertexList is an array of 3-D WCS coordinates used to create the polyface mesh vertices. At least four points (12 elements) are required for constructing a polyface mesh object. Facelist is an array of integers representing the vertex numbers for each face. Faces are defined in groups of four vertex index values.

Name	Returns	Description
AddPolyline	AcadPolyline	Creates an AcadPolyline object from a list of vertices. The polyline vertices are an array of OCS coordinates. At least two points (six values) are required to form a polyline. Parameter: VerticesList As Variant. You can convert between OCS and WCS using the TranslateCoordinates method (see the section on the AcadUtility object for details). Arcs are added to vertices using the SetBulge method (see the section on the AcadPolyline object for details).
AddRaster	AcadRasterImage	Creates an AcadRasterImage object based on the given image file. Parameters: ImageFileName As String, InsertionPoint As Variant, ScaleFactor As Double, and RotationAngle As Double. The InsertionPoint parameter is a set of 3-D WCS coordinates. The default value for the ScaleFactor parameter is 1. The RotationAngle is in radians.
AddRay	AcadRay	Creates an AcadRay object representing a ray passing through two points. Parameters: Point1 As Variant and Point2 As Variant. Both parameters are arrays of three Doubles representing 3-D WCS coordinates.
AddRegion	Variant	Creates a region from an array of AcadEntity objects. The array must consist of AcadLine, AcadArc, AcadCircle, AcadEllipse, AcadLWPolyline, and AcadSpline objects, which must form a closed coplanar region. Parameter: ObjectList As Variant.
AddRevolvedSolid	Acad3DSolid	Creates an Acad3DSolid object representing a revolved solid as specified by the parameters. Parameters: Profile As AcadRegion, AxisPoint As Variant, AxisDir As Variant, and Angle As Double. The AxisPoint parameter specifies the start point of the axis of revolution and is defined with an array of 3-D WCS coordinates. The AxisDir parameter specifies the direction of the axis of revolution. The Angle parameter is the angle of revolution in radians. Enter 6.28 for a full circle revolution.
AddShape	AcadShape	Creates an AcadShape object based on a template identified by name, at the given insertion point, scale factor, and rotation. Parameters: Name As String, InsertionPoint As Variant, ScaleFactor As Double, and RotationAngle As Double. The InsertionPoint parameter is a set of 3-D WCS coordinates.
AddSolid	AcadSolid	Creates an AcadSolid object representing a 2-D solid polygon. The first two points define one edge of the polygon. For a four-sided solid, the third point is defined to be diagonally opposite from the second point. If the third and fourth points are equal, then a triangle is created. The AcadSolid is filled when the FILLMODE system variable is set to 1. Parameters: Point1 As Variant, Point2 As Variant, Point3 As Variant, and Point4 As Variant.

Continued

Name	Returns	Description
AddSphere	Acad3DSolid	Creates an Acad3DSolid object representing a sphere as specified by the parameters. Parameters: Center As Variant (an array of three Doubles representing 3-D WCS coordinates) and Radius As Double.
AddSpline	AcadSpline	Creates an AcadSpline object as specified by the parameters, the first one being an array of 3-D WCS coordinates defining the spline curve. The StartTangent and EndTangent parameters specify the tangents (a 3-D vector) at the two ends of the spline. At least two points (six elements) are required for constructing an AcadSpline object. Parameters: PointsArray As Variant, StartTangent As Variant, and EndTangent As Variant.
AddText	AcadText	Creates an AcadText object representing a single line of text. Parameters: TextString As String, InsertionPoint As Variant, and Height As Double. The InsertionPoint parameter is a set of 3-D WCS coordinates.
AddTolerance	AcadTolerance	Creates an AcadTolerance object from the supplied parameters, the second one being a set of 3-D WCS coordinates and the last a 3-D vector. Parameters: Text As String, InsertionPoint As Variant, and Direction As Variant.
AddTorus	Acad3DSolid	Creates an Acad3DSolid object representing a torus from the parameters. Parameters: Center As Variant, TorusRadius As Double, and TubeRadius As Double. The first parameter is a set of 3-D WCS coordinates, and the values for both radii must be positive.
AddTrace	AcadTrace	Creates an AcadTrace object from an array of points. Parameter: PointsArray As Variant (an array of three Doubles representing 3-D WCS coordinates).
AddWedge	Acad3DSolid	Creates an Acad3DSolid object representing a wedge with edges parallel to the axes given the length, width, and height. Parameters: Center As Variant (an array of three Doubles representing 3-D WCS coordinates), Length As Double, Width As Double, and Height As Double.
AddXline	AcadXline	Creates an AcadXline object representing an infinite line passing through the given points. Parameters: Point1 As Variant and Point2 As Variant. Both parameters are arrays of three Doubles representing 3-D WCS coordinates. The width and fill mode are set using the TRACEWID and FILLMODE system variables.

Name	Returns	Description
AttachExternalReference	AcadExternalReference	Attaches an AcadExternalReference (Xref) object to the AcadBlock, given the full path name of the external drawing, the name of the Xref, and other parameters. Note that attached AcadExternalReference objects can be nested. Parameters: PathName As String, Name As String, InsertionPoint As Variant, Xscale As Double, Yscale As Double, Zscale As Double, Rotation As Double, and Overlay As Boolean. The InsertionPoint parameter is a set of 3-D WCS coordinates. The Rotation parameter specifies the angle of rotation of the Xref, in radians. The Overlay parameter specifies whether the Xref is to be an overlay (True) or an attachment (False). An overlaid Xref will be ignored when the container drawing is referenced at its turn.
Bind		Binds an AcadExternalReference (Xref) object as a nested block, complete with all its named dependent symbols (dimension styles, linetypes, etc.). This operation makes the Xref a permanent part of the drawing and not an externally referenced file. Named objects from the Xref can be used in the current drawing. Parameter: PrefixName As Boolean. If this parameter is True, then the symbols named in the Xref are prefixed with <blockname>\$x\$, where x is an integer that is automatically incremented to avoid overriding existing block functions. If this parameter is False, no prefix is used. If there is a clash of names, AutoCAD uses the symbols already defined in the drawing.
Detach		Detaches an AcadExternalReference (Xref) object from the block, including all copies of the Xref and the definition itself.
InsertBlock	AcadBlockReference	Inserts an AcadBlockReference object representing a file or a block that has been defined in the current drawing. Parameters: InsertionPoint As Variant, Name As String, Xscale As Double, Yscale As Double, Zscale As Double, and Rotation As Double. The InsertionPoint parameter is a set of 3-D WCS coordinates. If the Name is a file name, include any path information necessary for AutoCAD to find the file and the .dwg extension.
Item	AcadEntity	Gets the member object at a given index in the block. Parameter: Index As Variant (String or Integer).
Reload		Reloads the most recently saved version of the external reference (Xref) or an unloaded Xref.
Unload		Unloads an external reference (Xref). This means that the drawing opens faster and uses less memory. The Xref and its table of symbols don't appear in the drawing.

AcadBlock Object Properties

In addition to the properties inherited from the AcadEntity object and the common Application property, the AcadBlock object supports the following properties.

Name	Returns	Description
Count	Integer	Gets the number of items in the block. This property is read-only.
IsLayout	Boolean	Determines whether the block is a layout block. This property's value is read-only. If the values for IsLayout and IsXref are both False, then the block is a simple block.
IsXref	Boolean	Determines whether the block is an Xref block. This property's value is read-only.
Layout	AcadLayout	Gets the plot settings (AcadLayout object) for the block. This property's value is read-only.
Name	String	Gets the name of the block.
Origin	Variant	Gets or sets the origin of the block in WCS coordinates.
XrefDatabase	AcadDatabase	Gets the AcadDatabase object that defines the contents of the block. Available only if the IsXref property for the block is True. This property's value is read-only.

Note Because this object inherits from AcadObject, it supports the Modified event.

AcadBlockReference Object

The AcadBlockReference object represents an instance of a block definition inserted into a drawing. The creation of a new block definition is done automatically when an external block file is inserted with the InsertBlock method. To add or delete items from the block reference, you may first use the Explode method to break it into its component objects (subentities). However, the block definition still remains in the drawing's block symbol table with all its constituents.

AcadBlockReference Object Methods

The AcadBlockReference object inherits all the methods of the AcadEntity and AcadObject objects. It also supports the following methods.

Name	Returns	Description
Explode	Variant	Explodes the block and returns the subentities as an array of Objects
GetAttributes	Variant	Returns an array of editable attribute references attached to the block reference, only if the HasAttributes property is set to True
GetConstantAttributes	Variant	Returns an array of constant attributes for the block

AcadBlockReference Object Properties

The AcadBlockReference object inherits all the properties of the AcadEntity and AcadObject objects, as well as the common Application property. It also supports the following properties.

Name	Returns	Description
HasAttributes	Boolean	Specifies if the block has any attributes in it. This property's value is read-only.
InsertionPoint	Variant	Gets or sets an insertion point in the block as a set of 3-D WCS coordinates. It returns a three-element array of Doubles.
Name	String	Gets or sets the name of the block.
Normal	Variant	Gets or sets the 3-D (Z-axis) normal unit vector for the block.
Rotation	Double	Gets or sets the rotation angle (in radians) for the block, relative to the X-axis with positive values going counterclockwise when viewed along the Z-axis toward the origin.
XScaleFactor	Double	Gets or sets the X scale factor for the block.
YScaleFactor	Double	Gets or sets the Y scale factor for the block.
ZScaleFactor	Double	Gets or sets the Z scale factor for the block.

Note Because this object inherits from AcadObject, it supports the Modified event.

AcadBlocks Collection

The AcadBlocks collection contains all the blocks in the drawing. The collection contains two special collections: the AcadModelSpace collection for model space entities, and the AcadPaperSpace collection for paper space entities. Although the AcadBlocks collection inherits a Delete method, you can't actually delete the collection. There's no limit to the number of blocks you can create in your drawing. However, there can be only one instance of the AcadBlocks collection predefined for each drawing. You can make multiple references to it by using the Blocks property of the AcadDocument object.

AcadBlocks Collection Methods

In addition to the methods inherited from the AcadObject object, the AcadBlocks collection supports the following methods.

Name	Returns	Description
Add	AcadBlock	Creates an AcadBlock object and adds it to the collection. Parameters: InsertionPoint As Variant and Name As String. The InsertionPoint parameter is a set of 3-D WCS coordinates that specify where the AcadBlock object will be added. Once the AcadBlock has been created, you can add subentities to it.
Item	AcadBlock	Returns the object at the given index in the collection. Parameter: Index As Variant (Integer or String). If the Index value is a String, it must match an existing block definition.

AcadBlocks Collection Properties

The AcadBlocks collection supports the Count property, the common Application property, and those properties inherited from the AcadObject object. It supports no other properties.

Note Because this collection inherits from AcadObject, it supports the Modified event.

AcadCircle Object

The AcadCircle object represents a full circle. It is created using the AddCircle method of the AcadBlock, AcadModelSpace, or AcadPaperSpace object.

AcadCircle Object Methods

The AcadCircle object inherits all the methods of the AcadEntity and AcadObject objects. It also supports the following method.

Name	Returns	Description
Offset	Variant	Creates a new circle by offsetting the current circle by a specified distance, which must be nonzero. If the offset is positive, a larger circle is created with the same origin. If the offset is negative, a smaller circle is created. Parameter: Distance As Double.

AcadCircle Object Properties

The AcadCircle object inherits all the properties of the AcadEntity and AcadObject objects, as well as the common Application property. It also supports the following properties.

Name	Returns	Description
Area	Double	Gets the area of the circle in square drawing units. This property's value is read-only.
Center	Variant	Gets or sets the center of the circle as a set of 3-D coordinates.
Circumference	Double	Gets or sets the circumference of the circle.
Diameter	Double	Gets or sets the diameter of the circle.
Normal	Variant	Gets or sets the 3-D (Z-axis) normal unit vector for the object.
Radius	Double	Gets or sets the radius of the circle.
Thickness	Double	Gets or sets the distance the AcadCircle object is extruded above or below its elevation, in a Z-axis direction. The default is 0.0.

Note Because this object inherits from AcadObject, it supports the Modified event.

AcadDatabase Object

The AcadDatabase object represents the contents of an external reference (Xref) block. This object provides access to the contents of an external reference block. It's available only on blocks whose IsXref property is equal to True.

AcadDatabase Object Methods

The AcadDatabase object supports the following methods. These are identical to three of the methods supported by the AcadDocument object, except they act upon the contents of an Xref block, not the active document.

Name	Returns	Description
CopyObjects	Variant	Duplicates an object along with other objects owned and referenced by it. This is called <i>deep cloning</i> . Parameters: Objects As Variant, [Owner As Variant], and [IDPairs As Variant]. The Objects parameter is array of primary objects to be copied. They must all have the same owner. The Owner parameter is the new owner for the copied objects. If unspecified, the original owner is assumed. The IDPairs parameter gives information on what happened during the copy and translation process. Input is an empty Variant. Output is an array of AcadIDPair objects. You can't execute this method while simultaneously iterating through a collection.
HandleToObject	Object	Returns the object that corresponds to the given handle. The object must be in the current Xref drawing. Parameter: Handle As String. A handle is persistent. It can be stored for reuse with this method for the lifetime of the object.
ObjectIDToObject	Object	Returns the object that corresponds to the given object ID. The object must be in the current Xref drawing. Parameter: ObjectID As Integer. An object ID isn't persistent. Its lifetime is limited to a drawing session.

AcadDatabase Object Properties

As well as the common Application property, the AcadDatabase object supports the following properties. These are identical to the corresponding properties of the AcadDocument object, except they refer to the contents of an Xref block, not the active document.

Name	Returns	Description
Blocks	AcadBlocks	Gets the AcadBlocks collection for the Xref drawing. This property's value is read-only.
Dictionaries	AcadDictionaries	Gets the AcadDictionaries collection for the Xref drawing. This property's value is read-only.
DimStyles	AcadDimStyles	Gets the AcadDimStyles collection for the Xref drawing. This property's value is read-only.
ElevationModelSpace	Double	Gets or sets the elevation setting in the model space. The current elevation is the Z value that is used whenever a 3-D point is expected but only the X and Y values are supplied.

Continued

Name	Returns	Description
ElevationPaperSpace	Double	Gets or sets the elevation setting in the paper space. The current elevation is the Z value that is used whenever a 3-D point is expected but only the X and Y values are supplied.
Groups	AcadGroups	Gets the AcadGroups collection for the Xref drawing. This property's value is read-only.
Layers	AcadLayers	Gets the AcadLayers collection for the Xref drawing. This property's value is read-only.
Layouts	AcadLayouts	Gets the AcadLayouts collection for the Xref drawing. This property's value is read-only.
Limits	Variant	Gets or sets the Xref drawing limits in the XY plane. Two sets of WCS coordinates specify the lower-left (LIMMIN) and upper-right corners (LIMMAX) of the plane.
Linetypes	AcadLineTypes	Gets the AcadLinetypes collection for the Xref drawing. This property's value is read-only.
ModelSpace	AcadModelSpace	Gets the AcadModelSpace collection for the Xref drawing. This property's value is read-only.
PaperSpace	AcadPaperSpace	Gets the AcadPaperSpace collection for the Xref drawing. This property's value is read-only.
PlotConfigurations	AcadPlotConfigurations	Gets the AcadPlotConfigurations collection for the Xref drawing. This property's value is read-only.
Preferences	AcadDatabasePreferences	Gets the AcadPreferences object for the Xref drawing. The AcadPreferences object holds the options from the Options dialog box that resides in the Registry. Options that reside in the drawing can be found through the AcadDatabasePreferences object. This property's value is read-only.
RegisteredApplications	AcadRegisteredApplications	Gets the AcadRegisteredApplications collection for the Xref drawing. This property's value is read-only.
TextStyles	AcadTextStyles	Gets the AcadTextStyles collection for the Xref drawing. This property's value is read-only.
UserCoordinateSystems	AcadUCSs	Gets the AcadUCSs collection for the Xref drawing. This property's value is read-only.
Viewports	AcadViewports	Gets the AcadViewports collection for the Xref drawing. This property's value is read-only.
Views	AcadViews	Gets the AcadViews collection for the Xref drawing. This property's value is read-only.

AcadDatabasePreferences Object

The AcadDatabasePreferences object specifies the current AutoCAD drawing-specific settings, that is, all the options from the Options dialog box that reside in a drawing. (Options that reside in the Registry can be found through the main AcadPreferences object.) This object has no methods.

AcadDatabasePreferences Object Properties

As well as the common Application property, the AcadDatabasePreferences object supports the following properties.

Name	Returns	Description
AllowLongSymbolNames	Boolean	Determines whether symbol names are extended (True) or restricted (False). ("Symbol names" refers to the names of non-graphical objects such as linetypes and layers.) Extended symbol names can be up to 255 characters and can use any symbol not reserved by Microsoft or AutoCAD, in addition to the letters, numerals, and the space. This is the default. Restricted symbol names can only be up to 31 letters long and are restricted to the letters and numerals, the underscore (_), dollar (\$), and hyphen (-). The value for this property is stored in the EXTNames system variable.
ContourLinesPerSurface	Integer	Gets or sets the number of contour lines per surface on an object. The valid range is 0 to 2047. The number is stored in the ISOLINES system variable.
DisplaySilhouette	Boolean	Determines whether silhouette curves of solid objects are displayed in wireframe mode: True for silhouette curves and False for isolines (default). The value is stored in the DISPSILH system variable.
Lineweight	AcLineweight	Gets or sets the default lineweight for the drawing. For a list of possible values for the AcLineweight enumerated type, see Appendix B.
LineweightDisplay	Boolean	Determines whether lineweights are displayed in the model space. True is the default. This property is read/write.
MaxActiveViewports	Integer	Gets or sets the maximum number of active viewports. The range is 2 to 48. The value of is stored in the MAXACTVP system variable.
ObjectSortByPlotting	Boolean	Specifies whether objects are sorted by their plotting order. True is the default. The values of this property and the five that follow are stored in the SORTENTS system variable. Note that setting these additional ObjectSortBy... properties to True slows your application.
ObjectSortByPSOutput	Boolean	Specifies whether objects are sorted by their PostScript output order. True is the default.
ObjectSortByRedraws	Boolean	Specifies whether objects are sorted by their redraw order. False is the default.
ObjectSortByRegens	Boolean	Specifies whether objects are sorted by their regeneration order. False is the default.
ObjectSortBySelection	Boolean	Specifies whether drawing objects are sorted by object selection. False is the default.
ObjectSortBySnap	Boolean	Gets or sets whether drawing objects are sorted by object snap. False is the default.
OLELaunch	Boolean	Specifies whether the parent application is launched when OLE objects are plotted. The value of this property is stored in the OLESTARTUP system variable. The initial value of this property is False. This gives a high-quality plot but at the cost of speed.

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Name	Returns	Description
RenderSmoothness	Double	Gets or sets the smoothness of shaded, rendered, and hidden line-removed objects. The value of this property is stored in the FACETRES system variable. The initial value for this property is 0.5. Values less than or equal to 1 increase performance.
SegmentPerPolyline	Integer	Gets or sets the number of line segments for each polyline curve. The value of this property is stored in the SPLINESEGS system variable. The initial value is 8.
SolidFill	Boolean	Specifies whether multilines, traces, solids, hatches, and wide polylines are filled in. The value of this property is stored in the FILLMODE system variable. The default value is True.
TextFrameDisplay	Boolean	Specifies whether the frames for text objects are displayed rather than the text itself. The value of this property is stored in the QTEXTMODE system variable. The default is False.
XrefEdit	Boolean	Determines whether the current drawing can be edited in place while being referenced by another user. The value of this property is stored in the XEDIT system variable. The default is True.
XrefLayerVisibility	Boolean	Specifies whether Xref-dependent layers are visible and whether nested Xref path changes are saved. If True, Xref-dependent layer changes made in the current drawing take precedence (default). If False, the layer table as stored in the reference drawing (Xref) takes precedence. The value of this property is stored in the VISRETAIN system variable.

AcadDictionaries Collection

The AcadDictionaries collection contains all the dictionaries in the drawing. Although this collection inherits a Delete method, you can't actually delete it. If you need to delete a specific dictionary, use the Delete method found in the AcadDictionary object. There's no limit to the number of dictionaries you can create in your drawing. However, there can be only one instance of the AcadDictionaries collection, which is predefined for each drawing. You can make multiple references to it by using the Dictionaries property.

AcadDictionaries Collection Methods

In addition to the methods inherited from the AcadObject object, the AcadDictionaries collection supports the following methods.

Name	Returns	Description
Add	AcadDictionary	Creates a member object and adds it to the collection. Parameter: Name As String.
Item	AcadObject	Gets the member object at a given index in a collection. Parameter: Index As Variant (an Integer or a String). If the value for Index is a String, it must match an existing object name in the collection.

AcadDictionaries Collection Properties

The AcadDictionaries collection supports the Count property, the common Application property, and the properties inherited from the AcadObject object. It supports no other properties.

Note Because this collection inherits from `AcadObject`, it supports the `Modified` event.

AcadDictionary Object

The `AcadDictionary` object represents a container object for storing and retrieving objects, with associated string keywords by which they are referenced. A dictionary can contain any type of object, including other dictionaries. You can create new dictionaries, add entries to an existing dictionary, and get the keyword for a given object or the object for a given keyword. You can change the object associated with a given keyword or rename the keyword for a given object. However, it doesn't perform type checking of entries.

Note that this object doesn't represent the spell-checking dictionary, which is specified in the `AcadPreferences` object.

AcadDictionary Object Methods

In addition to the methods inherited from the `AcadObject` object, the `AcadDictionary` object supports the following properties.

Name	Returns	Description
<code>AddObject</code>	<code>AcadObject</code>	Adds an object to a named dictionary. Note that an entry replaces one already existing with the same name. Parameters: <code>Keyword As String</code> and <code>ObjectName As String</code> . The first parameter is the keyword to be listed in the dictionary for this object. The second parameter is the name of the object to be stored. The ARX application defining the object must first be loaded with the <code>LoadArx</code> method.
<code>AddXRecord</code>	<code>AcadXRecord</code>	Creates an <code>XRecord</code> object in any dictionary. Parameter: <code>Keyword As String</code> .
<code>GetName</code>	<code>String</code>	Gets the keyword of an object in a dictionary. The parameter is the object's name. Parameter: <code>Object As AcadObject</code> .
<code>GetObject</code>	<code>AcadObject</code>	Gets the object in a dictionary, given the keyword for the object. Parameter: <code>Name As String</code> .
<code>Item</code>	<code>AcadObject</code>	Gets the member object at a given index in the dictionary. Parameter: <code>Index As Variant (Integer or String)</code> . If the value for <code>Index</code> is a <code>String</code> , it must match an existing object name in the collection.
<code>Remove</code>	<code>AcadObject</code>	Removes a named object from the dictionary, given its keyword. Parameter: <code>Name As String</code> .
<code>Rename</code>		Renames the keyword of an item in the dictionary. Parameters: <code>OldName As String</code> and <code>NewName As String</code> .
<code>Replace</code>		Replaces an item in the dictionary by a given item. This changes the object but retains the keyword. Parameters: <code>OldName As String</code> and <code>Obj As AcadObject</code> .

AcadDictionary Object Properties

In addition to the properties inherited from the `AcadObject` object and the common `Application` property, the `AcadDictionary` object supports the following properties.

Name	Returns	Description
Count	Integer	Gets the number of items in the dictionary. This property's value is read-only.
Name	String	Gets or sets the name of the dictionary.

Note Because this object inherits from `AcadObject`, it supports the `Modified` event.

AcadDim3PointAngular Object

The `AcadDim3PointAngular` object represents a dimension of the angular distance defined by three points. When creating a three-point angular dimension, AutoCAD draws the dimension line arc between the extension lines, which are drawn from the angle end points to the intersection of the dimension line arc. AutoCAD uses the location of the dimension line arc to choose between the minor and major angles specified by the angle vertex and extension lines. To create a three-point angular dimension, use the `AddDim3PointAngular` method of the `AcadBlock`, `AcadModelSpace`, or `AcadPaperSpace` object.

AcadDim3PointAngular Object Methods

The `AcadDim3PointAngular` object inherits all the methods of the `AcadEntity` and `AcadObject` objects. It supports no other methods.

AcadDim3PointAngular Object Properties

The `AcadDim3PointAngular` object inherits all the properties of the `AcadEntity` object, the `AcadObject` object, and the `AcadDimension` object, as well as the common `Application` property. It also supports the following properties.

Name	Returns	Description
AngleFormat	AcAngleUnits	Gets or sets the unit format for angles. The initial value is <code>acDegrees</code> . For the values of the <code>AcAngleUnit</code> enumerated type, see Appendix B.
AngleVertex	Variant	Gets or sets a 3-D set of coordinates representing the vertex.
Arrowhead1Block Arrowhead2Block	String	Gets or sets the block used as the custom arrowhead at the first/second end of the dimension line. It overrides the value of the <code>DIMBLK1</code> or <code>DIMBLK2</code> system variable.
Arrowhead1Type Arrowhead2Type		Gets or sets the type of arrowhead at the first/second end of the dimension line. For the values of the <code>AcDimArrowheadType</code> enumerated type, see Appendix B.

Name	Returns	Description
ArrowheadSize	Double	Gets or sets the size of the dimension line arrowheads, leader line arrowheads, and hook lines. The initial value for this property is 0.1800.
DimensionLineColor	AcColor	Gets or sets the color of the dimension lines. Use a color index number from 0 to 256 or one of the constants listed here: <code>acByBlock</code> (where AutoCAD draws objects in the default color) or <code>acByLayer</code> (where AutoCAD draws objects in the color specified for the layer). For a list of possible values for the <code>AcColor</code> enumerated type, see Appendix B.
DimensionLineWeight	AcLineWeight	Gets or sets the lineweight of the dimension lines. It overrides the value of the <code>DIMLWD</code> system variable. For a list of possible values for the <code>AcLineWeight</code> enumerated type, see Appendix B.
DimLine1Suppress DimLine2Suppress	Boolean	Gets or sets whether the first/second dimension line is suppressed. It overrides the value of the <code>DIMSD1</code> or <code>DIMSD2</code> system variable. The initial value for this property is <code>False</code> . When this property is set to <code>True</code> , the display of the dimension line and arrowhead between the first extension line and the text is suppressed.
DimLineInside	Boolean	Gets or sets whether the dimension lines are only displayed inside the extension lines. It overrides the value of the <code>DIMSOXD</code> system variable. The initial value for this property is <code>False</code> .
ExtensionLineColor	AcColor	Gets or sets the color of the extension lines. Use a color index number from 0 to 256 or one of the constants listed here: <code>acByBlock</code> (where AutoCAD draws objects in the default color), <code>acByLayer</code> (where AutoCAD draws objects in the color specified for the layer), or the <code>AcColor</code> enumerated type (see Appendix B). It overrides the value of the <code>DIMCLRE</code> system variable.
ExtensionLineExtend	Double	Gets or sets the distance that the extension line extends beyond the dimension line. It overrides the value of the <code>DIMEXE</code> system variable. The initial value for this property is 0.1800.
ExtensionLineOffset	Double	Gets or sets the distance that the extension lines are offset from the origin points. It overrides the value of the <code>DIMEXO</code> system variable. The initial value for this property is 0.0625.
ExtensionLineWeight	AcLineWeight	Gets or sets the lineweight of the extension lines. It overrides the value of the <code>DIMLWE</code> system variable. For a list of possible values for the <code>AcLineWeight</code> enumerated type, see Appendix B.
ExtLine1Point ExtLine2Point	Variant	Gets or sets the end point of the first/second extension line.
ExtLine1Suppress ExtLine2Suppress	Boolean	Gets or sets whether the first/second extension line is suppressed. It overrides the value of the <code>DIMSE1</code> or <code>DIMSE2</code> system variable. The initial value for this property is <code>False</code> .

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Name	Returns	Description
Fit	AcDimFit	Gets or sets whether text and arrowheads are placed inside or outside extension lines given the availability of space between the extension lines. This property sets priorities for moving text and arrowheads when space isn't available to fit both within the extension lines. It overrides the value of the DIMAFIT system variable. The initial value for this property is acBestFit. For a list of the possible values, see Appendix B.
ForcelineInside	Boolean	Gets or sets whether a dimension line is drawn between the extension lines when the text is outside the extension lines. It overrides the value of the DIMTOFL system variable. The initial value for this property is False.
FractionFormat		Gets or sets the horizontal position of the text. For a list of possible values for the AcDimHorizontalJustification enumerated type, see Appendix B.
Measurement	Double	Gets the measurement for the dimension. For angular dimensions, this property overrides the value for the DIMAUNIT system variable. This property's value is read-only.
TextInside	Boolean	Gets or sets whether the text appears inside the extension lines. It overrides the value of DIMTIX system variable. The initial value for this property is True. If False, AutoCAD places it inside if there is room.
TextInsideAlign	Boolean	Gets or sets the position of any text inside/outside the extension lines. The values for these properties override the values of the DIMTIH or DIMTOH system variable, respectively. The initial value for this property is True. The TextInsideAlign property is available only when the TextInside property is set to True.
TextPrecision	AcDimPrecision	Gets or sets the precision of the text. For the possible values for the AcDimPrecision enumerated type, see Appendix B.

Note Because this object inherits from AcadObject, it supports the Modified event.

AcadDimAligned Object

The AcadDimAligned object represents a linear dimension, measuring the distance between two points, that's displayed parallel to the points being measured. In aligned dimensions, the dimension line is parallel to the extension line origins. The extension line origins are specified using the ExtLine1Point and ExtLine2Point properties. The AcadDimAligned object is created using the AddDimAligned method of the AcadBlock, AcadModelSpace, or AcadPaperSpace object.

AcadDimAligned Object Methods

The AcadDimAligned object inherits all the methods of the AcadEntity and AcadObject objects. It supports no other methods.

AcadDimAligned Object Properties

The AcadDimAligned object inherits all the properties of the AcadEntity object, the AcadObject object, and the AcadDimension object, as well as the common Application property. It also supports the following properties.

Name	Returns	Description
AltRoundDistance	Double	Gets or sets the rounding of alternative units, if the AltUnits property is turned on. This property, and the two that follow, all override the value of the DIMALTZ system variable.
AltSuppressLeadingZeros	Boolean	Gets or sets whether leading zeros in alternative dimension values are suppressed. The initial value for this property is False.
AltSuppressTrailingZeros	Boolean	Gets or sets whether trailing zeros in alternative dimension values are suppressed. The initial value for this property is False.
AltSuppressZeroFeet AltSuppressZeroInches	Boolean	Gets or sets whether a zero foot/inches measurement is suppressed in alternative dimension values. Both properties change the value of the DIMALTZ system variable. The initial value for this property is False.
AltTextPrefix AltTextSuffix	String	Gets or sets a prefix/suffix for the alternative dimension measurement. Both properties override the value of the DIMAPOST system variable. The initial value for this property is a NULL string. To turn off an established prefix, set this property equal to a single period (.).
AltTolerancePrecision	AcDimPrecision	Gets or sets the precision of tolerance values in alternative dimensions. The initial value for this property is acDimPrecisionTwo. This property is available only when the ToleranceDisplay property is set to any value other than acTolNone, and it overrides the value of the DIMTDEC system variable. For a list of the values for the AcDimPrecision enumerated type, see Appendix B.
AltToleranceSuppressLeadingZeros	Boolean	Gets or sets whether leading zeros are suppressed in alternative dimension values. This property, and the three that follow, all override the DIMALTZ system variable. The initial value for this property is False.
AltToleranceSuppressTrailingZeros	Boolean	Gets or sets whether trailing zeros are suppressed in alternative dimension values. The initial value for this property is False.

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Name	Returns	Description
AltToleranceSuppressLeadingZeros	Boolean	Gets or sets whether a zero foot measurement is suppressed in alternative tolerance values. The initial value for this property is False.
AltSuppressZeroInches	Boolean	Gets or sets whether a zero inch measurement is suppressed in alternative tolerance values. The initial value for this property is False.
AltUnits	Boolean	Gets or sets whether alternative units dimensioning is enabled. It overrides the value of the DIMALT system variable.
AltUnitsFormat	AcDimUnits	Gets or sets the units format for alternative units. The initial value for this property is <code>acDimDecimal</code> . It overrides the value of the DIMALTU system variable. For a list of possible values for the <code>AcDimUnits</code> enumerated type, see Appendix B.
AltUnitsPrecision	AcDimPrecision	Gets or sets the number of decimal places in alternative units. It overrides the value of the DIMALTD system variable. For a list of possible values for the <code>AcDimPrecision</code> enumerated type, see Appendix B.
AltUnitsScale	Double	Gets or sets the scale factor for alternative units. If the <code>AltUnits</code> property is turned on, this property multiplies linear dimensions by a factor to produce a value in an alternate system of measurement. The initial value represents the number of millimeters in an inch. It overrides the value of the DIMALTF system variable.
Arrowhead1Block Arrowhead2Block	String	Gets or sets the block used as the custom arrowhead at the first/second end of the dimension line. It overrides the value of the DIMBLK1 or DIMBLK2 system variable.
Arrowhead1Type Arrowhead2Type	AcDimArrowheadType	Gets or sets the type of arrowhead at the first/second end of the dimension line. For a list of possible values for the <code>AcDimArrowheadType</code> enumerated type, see Appendix B.
ArrowheadSize	Double	Gets or sets the size of the dimension line arrowheads, leader line arrowheads, and hook lines. It overrides the value of the DIMASZ system variable. The initial value for this property is 0.1800.
DimensionLineColor	AcColor	Gets or sets the color of the dimension lines. Use a color index number from 0 to 256 or one of the constants listed here: <code>acByBlock</code> (where AutoCAD draws objects in the default color) or <code>acByLayer</code> (where AutoCAD draws objects in the color specified for the layer). For a list of possible values for the <code>AcColor</code> enumerated type, see Appendix B.

Name	Returns	Description
DimensionLineExtend	Double	Gets or sets the distance the dimension line extends beyond the extension line when oblique strokes rather than arrowheads are used. It overrides the value of the DIMDLE system variable. The initial value for this property is 0.0000.
DimensionLineWeight	AcLineWeight	Gets or sets the lineweight of the dimension lines. It overrides the value of the DIMLWD system variable. For a list of possible values for the AcLineWeight enumerated type, see Appendix B.
DimLine1Suppress DimLine2Suppress	Boolean	Gets or sets whether the first/second dimension line is suppressed. It overrides the value of the DIMSD1 or DIMSD2 system variable. The initial value for this property is False. When this property is set to True, the display of the dimension line and arrowhead between the first extension line and the text is suppressed.
DimLineInside	Boolean	Gets or sets whether the dimension lines are displayed only inside the extension lines. It overrides the value of the DIMSOXD system variable. The initial value for this property is False.
ExtensionLineColor	AcColor	Gets or sets the color of the extension lines. Use a color index number from 0 to 256 or one of the constants listed here: acByBlock (where AutoCAD draws objects in the default color), acByLayer (where AutoCAD draws objects in the color specified for the layer), or the AcColor enumerated type (see Appendix B). It overrides the value of the DIMCLRE system variable.
ExtensionLineExtend	Double	Gets or sets the distance that the extension line extends beyond the dimension line. It overrides the value of the DIMEXE system variable. The initial value for this property is 0.1800.
ExtensionLineOffset	Double	Gets or sets the distance that the extension lines are offset from the origin points. It overrides the value of the DIMEXO system variable. The initial value for this property is 0.0625.
ExtensionLineWeight	AcLineWeight	Gets or sets the lineweight of the extension lines. It overrides the value of the DIMLWE system variable. For a list of possible values for the AcLineWeight enumerated type, see Appendix B.
ExtLine1Point ExtLine2Point	Variant	Gets or sets the end point of the first/second extension line.
ExtLine1Suppress ExtLine2Suppress	Boolean	Gets or sets whether the first/second extension line is suppressed. It overrides the value of the DIMSE1 or DIMSE2 system variable. The initial value for this property is False.
Fit	AcDimFit	Gets or sets whether text and arrowheads are placed inside or outside extension lines given the availability of space between the extension lines. This property sets priorities for moving text and arrowheads when space isn't available to fit both within the extension lines. It overrides the value of the DIMAFIT system variable. The initial value for this property is acBestFit. For a list of the possible values, see Appendix B.
ForcelineInside	Boolean	Gets or sets whether a dimension line is drawn between the extension lines when the text is outside the extension lines. It overrides the value of the DIMTOFL system variable. The initial value for this property is False.

Continued

Name	Returns	Description
FractionFormat	AcDimFractionType	Gets or sets the formats of fractional values. It overrides the value of the DIMFRAC system variable. This property is used when the UnitsFormat property is set to acDimLArchitectural or acDimLFractional. Possible values: acHorizontal, acDiagonal, and acNotStacked.
HorizontalTextPosition	AcDimHorizontalJustification	Gets or sets the horizontal position of the text. It overrides the value of the DIMJUST system variable. For a list of possible values for the AcDimHorizontalJustification enumerated type, see Appendix B.
LinearScaleFactor	Double	Gets or sets the global scale factor for linear dimensioning measurements. It overrides the value of the DIMLFAC system variable. The initial value for this property is 1.0000. When LinearScaleFactor is assigned a negative value, the factor is applied only in paper space.
Measurement	Double	Gets the measurement for the dimension. This property overrides the value for the DIMLUNIT system variable. This property's value is read-only.
PrimaryUnitsPrecision	AcDimPrecision	Gets or sets the number of decimal places displayed for the primary units. For a list of possible values for the AcDimPrecision enumerated type, see Appendix B.
RoundDistance	Double	Gets or sets the rounding of units. It overrides the value of the DIMRND system variable.
SuppressZeroFeet	Boolean	Gets or sets whether a zero foot measurement is suppressed. It overrides the value of the DIMZIN system variable. The initial value for this property is False.
SuppressZeroInches	Boolean	Gets or sets whether a zero inch measurement is suppressed. It overrides the value of the DIMZIN system variable. The initial value for this property is False.
TextInside	Boolean	Gets or sets whether the text appears inside the extension lines. It overrides the value of the DIMITX system variable. The initial value for this property is True. If False, AutoCAD places it inside if there is room.

Name	Returns	Description
TextInsideAlign	Boolean	Gets or sets the position of any text inside/outside the extension lines. The values for these properties override the values of the DIMTIH or DIMTOH system variable, respectively. The initial value for this property is True. The TextInsideAlign property is available only when the TextInside property is set to True.
ToleranceSuppressZeroFeet	Boolean	Gets or sets whether a zero foot measurement is suppressed in tolerance values. It overrides the value of the DIMZIN system variable. The initial value for this property is False.
ToleranceSuppressZeroInches	Boolean	Gets or sets whether a zero inch measurement is suppressed in tolerance values. It overrides the value of the DIMZIN system variable. The initial value for this property is False.
UnitsFormat	AcDimLUnits	Gets or sets the unit format. The initial value for this property is acDimLDecimal, in which the format specified by the DecimalSeparator and PrimaryUnitsPrecision properties will be used to format the decimal value. For a list of possible values for the AcDimLUnits enumerated type, see Appendix B.

Note Because this object inherits from `AcadObject`, it supports the `Modified` event.

AcadDimAngular Object

The `AcadDimAngular` object represents a dimension of the angular distance between two lines or the angle of a circular arc. If you need extension lines, they'll be added automatically. The end points provided will be used as origin points for the extension lines. The `AcadDimAngular` object is created using the `AddDimAngular` method of the `AcadBlock`, `AcadModelSpace`, or `AcadPaperSpace` object.

AcadDimAngular Object Methods

The `AcadDimAngular` object inherits all the methods of the `AcadEntity` and `AcadObject` objects. It supports no other methods.

AcadDimAngular Object Properties

The `AcadDimAngular` object inherits all the properties of the `AcadEntity` object, the `AcadObject` object, and the `AcadDimension` object, as well as the common `Application` property. It also supports all the other properties supported by the `AcadDim3PointAngular` object except for the `AngleVertex` property. This object, however, has the following two extra properties.

Name	Returns	Description
ExtLine1StartPoint ExtLine2StartPoint	Variant	Gets or sets the start point of the first/second extension line.

Note Because this object inherits from `AcadObject`, it supports the `Modified` event.

AcadDimArcLength Object

This is a circular-style dimension measuring the length of an arc.

AcadDimArcLength Object Methods

The `AcadDimArcLength` object inherits all the methods of the `AcadEntity` and `AcadObject` objects. It supports no other methods.

AcadDimArcLength Object Properties

The `AcadDimArcLength` object inherits all the properties of the `AcadEntity` object, the `AcadObject` object, and the `AcadDimension` object, as well as the common `Application` property. It also supports the following properties.

Name	Returns	Description
AltRoundDistance	Double	Gets or sets the rounding of alternative units, if the <code>AltUnits</code> property is turned on. This property, and the two that follow, all override the value of the <code>DIMALTZ</code> system variable.
AltSuppressLeadingZeros	Boolean	Gets or sets whether leading zeros in alternative dimension values are suppressed. The initial value for this property is <code>False</code> .
AltSuppressTrailingZeros	Boolean	Gets or sets whether trailing zeros in alternative dimension values are suppressed. The initial value for this property is <code>False</code> .
AltSuppressZeroFeet AltSuppressZeroInches	Boolean	Gets or sets whether a zero foot/inches measurement is suppressed in alternative dimension values. Both properties change the value of the <code>DIMALTZ</code> system variable. The initial value for this property is <code>False</code> .
AltTextPrefix AltTextSuffix	String	Gets or sets a prefix/suffix for the alternative dimension measurement. Both properties override the value of the <code>DIMAPOST</code> system variable. The initial value for this property is a <code>NULL</code> string. To turn off an established prefix, set this property equal to a single period (<code>.</code>).

Name	Returns	Description
AltTolerancePrecision	AcDimPrecision	Gets or sets the precision of tolerance values in alternative dimensions. The initial value for this property is <code>acDimPrecisionTwo</code> . This property is available only when the <code>ToleranceDisplay</code> property is set to any value other than <code>acTolNone</code> , and it overrides the value of the <code>DIMTDEC</code> system variable. For a list of the values for the <code>AcDimPrecision</code> enumerated type, see Appendix B.
AltToleranceSuppressLeadingZeros	Boolean	Gets or sets whether leading zeros are suppressed in alternative dimension values. This property, and the three that follow, all override the <code>DIMALTTZ</code> system variable. The initial value for this property is <code>False</code> .
AltToleranceSuppressTrailingZeros	Boolean	Gets or sets whether trailing zeros are suppressed in alternative dimension values. The initial value for this property is <code>False</code> .
AltToleranceSuppressZeroFeet	Boolean	Gets or sets whether a zero foot measurement is suppressed in alternative tolerance values. The initial value for this property is <code>False</code> .
AltToleranceSuppressZeroInches	Boolean	Gets or sets whether a zero inch measurement is suppressed in alternative tolerance values. The initial value for this property is <code>False</code> .
AltUnits	Boolean	Gets or sets whether alternative units dimensioning is enabled. This overrides the value of the <code>DIMALT</code> system variable.
AltUnitsFormat	AcDimUnits	Gets or sets the units format for alternative units. The initial value for this property is <code>acDimDecimal</code> . This overrides the value of the <code>DIMALTU</code> system variable. For a list of possible values for the <code>AcDimUnits</code> enumerated type, see Appendix B.
AltUnitsPrecision	AcDimPrecision	Gets or sets the number of decimal places in alternative units. This overrides the value of the <code>DIMALTD</code> system variable. For a list of possible values for the <code>AcDimPrecision</code> enumerated type, see Appendix B.
AltUnitsScale	Double	Gets or sets the scale factor for alternative units. If the <code>AltUnits</code> property is turned on, this property multiplies linear dimensions by a factor to produce a value in an alternate system of measurement. The initial value represents the number of millimeters in an inch. This overrides the value of the <code>DIMALTF</code> system variable.
ArcEndParam	Double	Gets or sets the end parameter of the arc as a 3-D coordinate.
ArcPoint	Variant	Gets or sets a point on the arc as a 3-D coordinate.
ArcStartParam	Double	Gets or sets the start parameter of the arc as a 3-D coordinate.

Continued

Name	Returns	Description
Arrowhead1Block Arrowhead2Block	String	Gets or sets the block used as the custom arrowhead at the first/second end of the dimension line. This overrides the value of the DIMBLK1 or DIMBLK2 system variable.
Arrowhead1Type Arrowhead2Type	AcDimArrowheadType	Gets or sets the type of arrowhead at the first/second end of the dimension line. For a list of possible values for the AcDimArrowheadType enumerated type, see Appendix B.
ArrowheadSize	Double	Gets or sets the size of the dimension line arrowheads, leader line arrowheads and hook lines. This overrides the value of the DIMASZ system variable. The initial value for this property is 0.1800.
CenterPoint	Variant	Gets or sets the center as a set of 3-D coordinates.
DimensionLineColor	AcColor	Gets or sets the color of the dimension lines. Use a color index number from 0 to 256 or one of the constants listed here: acByBlock (where AutoCAD draws objects in default color) or acByLayer (where AutoCAD draws objects in the color specified for the layer). For a list of possible values for the AcColor enumerated type, see Appendix B.
DimensionLineExtend	Double	Gets or sets the distance the dimension line extends beyond the extension line when oblique strokes rather than arrowheads are used. This overrides the value of the DIMDLE system variable. The initial value for this property is 0.0000.
DimensionLineWeight	AcLineWeight	Gets or sets the line weight of the dimension lines. This overrides the value of the DIMLWD system variable. For a list of possible values for the AcLineWeight enumerated type, see Appendix B.
Dimline1Suppress Dimline2Suppress	Boolean	Gets or sets whether the first/second dimension line is suppressed. This overrides the value of the DIMSD1 or DIMSD2 system variable. The initial value for this property is False. When this property is set to True, the display of the dimension line and arrowhead between the first extension line and the text is suppressed.
DimLineInside	Boolean	Gets or sets whether the dimension lines are displayed only inside the extension lines. This overrides the value of the DIMSOXD system variable. The initial value for this property is False.
ExtensionLineColor	AcCmColor	Gets or sets the color of the extension lines. Use a color index number from 0 to 256 or one of the constants listed here: acByBlock (where AutoCAD draws objects in default color), acByLayer (where AutoCAD draws objects in the color specified for the layer), and the AcColor enumerated type (see Appendix B). This overrides the value of the DIMCLRE system variable.
ExtensionLineExtend	Double	Gets or sets the distance that the extension line extends beyond the dimension line. This overrides the value of the DIMEXE system variable. The initial value for this property is 0.1800.

Name	Returns	Description
ExtensionLineOffset	Double	Gets or sets the distance that the extension lines are offset from the origin points. This overrides the value of the DIMEXO system variable. The initial value for this property is 0.0625.
ExtensionLineWeight	AcLineWeight	Gets or sets the line weight of the extension lines. This overrides the value of the DIMLWE system variable. For a list of possible values for the AcLineWeight enumerated type, see Appendix B.
ExtLine1Point ExtLine2Point	Variant	Gets or sets the endpoint of the first/second extension line.
ExtLine1Suppress ExtLine2Suppress	Boolean	Gets or sets whether the first/second extension line is suppressed. This overrides the value of the DIMSE1 or DIMSE2 system variable. The initial value for this property is False.
Fit	AcDimFit	Gets or sets whether text and arrowheads are placed inside or outside extension lines given the availability of space between the extension lines. This property sets priorities for moving text and arrowheads when space is not available to fit both within the extension lines. This overrides the value of the DIMAFIT system variable. The initial value for this property is acBestFit. For a list of the possible values, see Appendix B.
ForceLineInside	Boolean	Gets or sets whether a dimension line is drawn between the extension lines when the text is outside the extension lines. This overrides the value of the DIMTOFL system variable. The initial value for this property is False.
FractionFormat	AcDimFractionType	Gets or sets the formats of fractional values. This overrides the value of the DIMFRAC system variable. This property is used when the UnitsFormat property is set to acDimLArchitectural or acDimLFractional. Possible values: acHorizontal, acDiagonal and acNotStacked.
HasLeader	Boolean	Specifies whether the dimension has a leader. The initial value for this property is False.
HorizontalTextPosition	AcDimHorizontalJustification	Gets or sets the horizontal position of the text. This overrides the value of the DIMJUST system variable. For a list of possible values for the AcDimHorizontalJustification enumerated type, see Appendix B.
IsPartial	Boolean	Gets or sets whether the dimension is for a partial arc. The initial value for this property is False.
Leader1Point	Variant	Gets or sets the origin for Leader 1 as a set of 3-D coordinates.

Continued

Name	Returns	Description
Leader2Point	Variant	Gets or sets the origin for Leader 2 as a set of 3-D coordinates.
LinearScaleFactor	Double	Gets or sets the global scale factor for linear dimensioning measurements. This overrides the value of the DIMLFAC system variable. The initial value for this property is 1.0000. When LinearScaleFactor is assigned a negative value, the factor is applied only in paper space.
Measurement	Double	Gets the measurement for the dimension. This property overrides the value for the DIMLUNIT system variable. This is read-only.
PrimaryUnitsPrecision	AcDimPrecision	Gets or sets the number of decimal places displayed for the primary units. For a list of possible values for the AcDimPrecision enumerated type, see Appendix B.
RoundDistance	Double	Gets or sets the rounding of units. This overrides the value of the DIMRND system variable.
SuppressZeroFeet	Boolean	Gets or sets whether a zero foot measurement is suppressed. This overrides the value of the DIMZIN system variable. The initial value for this property is False.
SuppressZeroInches	Boolean	Gets or sets whether a zero inch measurement is suppressed. This overrides the value of the DIMZIN system variable. The initial value for this property is False.
SymbolPosition	AcDimArcLengthSymbol	Gets and sets placement of the arc length dimension symbol. For a list of possible values for the AcDimArcLengthSymbol enumerated type, see Appendix B.
TextInside	Boolean	Gets or sets whether the text appears inside the extension lines. This overrides the value of DIMTIX system variable. The initial value for this property is True. If False, AutoCAD places it inside if there is room.
TextInsideAlign	Boolean	Gets or sets the position of any text inside/outside the extension lines. The values for these properties override the values of the DIMTIH or DIMTOH system variable, respectively. The initial value for this property is True. The TextInsideAlign property is available only when the TextInside property is set to True.
TextOutsideAlign	Boolean	Gets or sets the position of any text inside/outside the extension lines. The values for these properties override the values of the DIMTIH or DIMTOH system variables, respectively. The initial value for this property is True. The TextInsideAlign property is available only when the TextInside property is set to True.
ToleranceSuppressZeroFeet	Boolean	Gets or sets whether a zero foot measurement is suppressed in tolerance values. This overrides the value of the DIMZIN system variable. The initial value for this property is False.

Name	Returns	Description
ToleranceSuppressZeroInches	Boolean	Gets or sets whether a zero inch measurement is suppressed in tolerance values. This overrides the value of the DIMZIN system variable. The initial value for this property is False.
TrueColor	AcCmColor	Gets or sets an object's color. Colors are identified by an AcCmColor object. This object can hold an RGB value, an ACI number (an integer from 1 to 255), or a named color. Using an RGB value, you can choose from millions of colors.
UnitsFormat	AcDimLUnits	Gets or sets the unit format. The initial value for this property is acDimLDecimal, where the format specified by the DecimalSeparator and PrimaryUnitsPrecision properties will be used to format the decimal value. For a list of possible values for the AcDimLUnits enumerated type, see Appendix B.

Note Because this object inherits from AcadObject, it supports the Modified event.

AcadDimDiametric Object

The AcadDimDiametric object represents a dimension of the diameter of a circle or arc. The position of the text, set in the TextPosition property, determines the location of the dimension line. The AcadDimDiametric object is created using the AddDimDiametric method of the AcadBlock, AcadModelSpace, or AcadPaperSpace object.

AcadDimDiametric Object Methods

The AcadDimDiametric object inherits all the properties of the AcadEntity and AcadObject objects. It supports no other methods.

AcadDimDiametric Object Properties

The AcadDimDiametric object supports all the properties inherited from the AcadEntity object, the AcadObject object, and the AcadDimension object, as well as the common Application property. It also supports the following properties.

Name	Returns	Description
AltRoundDistance	Double	Gets or sets the rounding of alternative units if the AltUnits property is turned on. This property and the two that follow override the value of the DIMALTZ system variable.
AltSuppressLeadingZeros	Boolean	Gets or sets whether leading zeros in alternative dimension values are suppressed. The initial value for this property is False.
AltSuppressTrailingZeros	Boolean	Gets or sets whether trailing zeros in alternative dimension values are suppressed. The initial value for this property is False.

Continued

Name	Returns	Description
AltSuppressZeroFeet AltSuppressZeroInches	Boolean	Gets or sets whether a zero foot/inches measurement is suppressed in alternative dimension values. Both properties change the value of the DIMALTZ system variable. The initial value for this property is False.
AltTextPrefix AltTextSuffix	String	Gets or sets a prefix/suffix for the alternative dimension measurement. Both properties override the value of the DIMAPOST system variable. The initial value for this property is a NULL string. To turn off an established prefix, set this property equal to a single period (.).
AltTolerancePrecision	AcDimPrecision	Gets or sets the precision of tolerance values in alternative dimensions. The initial value for this property is acDimPrecisionTwo. This property is available only when the ToleranceDisplay property is set to any value other than acTolNone, and it overrides the value of the DIMTDEC system variable. For a list of the values for the AcDimPrecision enumerated type, see Appendix B.
AltToleranceSuppressLeadingZeros	Boolean	Gets or sets whether leading zeros are suppressed in alternative dimension values. This property and the three that follow override the DIMALTTZ system variable. The initial value for this property is False.
AltToleranceSuppressTrailingZeros	Boolean	Gets or sets whether trailing zeros are suppressed in alternative dimension values. The initial value for this property is False.
AltToleranceSuppressZeroFeet	Boolean	Gets or sets whether a zero foot measurement is suppressed in alternative tolerance values. The initial value for this property is False.
AltToleranceSuppressZeroInches	Boolean	Gets or sets whether a zero inch measurement is suppressed in alternative tolerance values. The initial value for this property is False.
AltUnits	Boolean	Gets or sets whether alternative units dimensioning is enabled. It overrides the value of the DIMALT system variable.
AltUnitsFormat	AcDimUnits	Gets or sets the units format for alternative units. The initial value for this property is acDimDecimal. It overrides the value of the DIMALTU system variable. For a list of possible values for the AcDimUnits enumerated type, see Appendix B.
AltUnitsPrecision	AcDimPrecision	Gets or sets the number of decimal places in alternative units. It overrides the value of the DIMALTD system variable. For a list of possible values for the AcDimPrecision enumerated type, see Appendix B.

Name	Returns	Description
AltUnitsScale	Double	Gets or sets the scale factor for alternative units. If the AltUnits property is turned on, this property multiplies linear dimensions by a factor to produce a value in an alternate system of measurement. The initial value represents the number of millimeters in an inch. It overrides the value of the DIMALT system variable.
Arrowhead1Block/Arrowhead2Block	String	Gets or sets the block used as the custom arrowhead at the first/second end of the dimension line. It overrides the value of the DIMBLK1 or DIMBLK2 system variable.
AcDimArrowheadType		Gets or sets the type of arrowhead at the first/second end of the dimension line. For a list of possible values for the AcDimArrowheadType enumerated type, see Appendix B.
ArrowheadSize	Double	Gets or sets the size of the dimension line arrowheads, leader line arrowheads, and hook lines. It overrides the value of the DIMASZ system variable. The initial value for this property is 0.1800.
CenterMarkSize	Double	Gets or sets the size of the center mark. It overrides the value of the DIMCEN system variable. The initial value for this property is 0.0900. This property isn't available if the CenterType property is set to acCenterNone.
CenterType	AcDimCenterType	Gets or sets the type of the center mark. The center mark is visible only if you place the dimension line outside the circle or arc. It overrides the value of the DIMCEN system variable. For a list of possible values for the AcDimCenterType enumerated type, see Appendix B.
DimensionLineColor	AcColor	Gets or sets the color of the dimension lines. Use a color index number from 0 to 256 or one of the constants listed here: acByBlock (where AutoCAD draws objects in the default color) or acByLayer (where AutoCAD draws objects in the color specified for the layer). For a list of possible values for the AcColor enumerated type, see Appendix B.
DimensionLineWeight	AcLineWeight	Gets or sets the lineweight of the dimension lines. It overrides the value of the DIMLWD system variable. For a list of possible values for the AcLineWeight enumerated type, see Appendix B.
DimLine1Suppress DimLine2Suppress	Boolean	Gets or sets whether the first/second dimension line is suppressed. It overrides the value of the DIMSD1 or DIMSD2 system variable. The initial value for this property is False. When this property is set to True, the display of the dimension line and arrowhead between the first extension line and the text is suppressed.

Continued

Name	Returns	Description
Fit	AcDimFit	Gets or sets whether text and arrowheads are placed inside or outside extension lines given the availability of space between the extension lines. This property sets priorities for moving text and arrowheads when space isn't available to fit both within the extension lines. It overrides the value of the DIMAFIT system variable. The initial value for this property is acBestFit. For a list of the possible values, see Appendix B.
ForceLineInside	Boolean	Gets or sets whether a dimension line is drawn between the extension lines when the text is outside the extension lines. It overrides the value of the DIMTOFL system variable. The initial value for this property is False.
FractionFormat	AcDimFractionType	Gets or sets the formats of fractional values. It overrides the value of the DIMFRAC system variable. This property is used when the UnitsFormat property is set to acDimLArchitectural or acDimLFractional. Possible values: acHorizontal, acDiagonal, and acNotStacked.
HorizontalTextPosition	AcDimHorizontalJustification	Gets or sets the horizontal position of the text. For a list of possible values for the AcDimHorizontalJustification enumerated type, see Appendix B.
LeaderLength	Double	Sets the length of the leader. This property is write-only; it's used when the object is created and isn't stored.
LinearScaleFactor	Double	Gets or sets the global scale factor for linear dimensioning measurements. It overrides the value of the DIMLFAC system variable. The initial value for this property is 1.0000. When LinearScaleFactor is assigned a negative value, the factor is applied only in paper space.
Measurement	Double	Gets the measurement for the dimension. This property overrides the value for the DIMLUNIT system variable. This property's value is read-only.
PrimaryUnitsPrecision	AcDimPrecision	Gets or sets the number of decimal places displayed for the primary units. For a list of possible values for the AcDimPrecision enumerated type, see Appendix B.
RoundDistance	Double	Gets or sets the rounding of units. It overrides the value of the DIMRND system variable.
SuppressZeroFeet	Boolean	Gets or sets whether a zero foot measurement is suppressed. It overrides the value of the DIMZIN system variable. The initial value for this property is False.

Name	Returns	Description
SuppressZeroInches	Boolean	Gets or sets whether a zero inch measurement is suppressed. It overrides the value of the DIMZIN system variable. The initial value for this property is False.
TextInside	Boolean	Gets or sets whether the text appears inside the extension lines. It overrides the value of DIMTIX system variable. The initial value for this property is True. If False, AutoCAD places it inside if there's room.
TextInsideAlign	Boolean	Gets or sets the position of any text inside/outside the extension lines. The values for these properties override the values of the DIMTIH and DIMTOH system variables, respectively. The initial value for this property is True. The TextInsideAlign property is available only when the TextInside property is set to True.
ToleranceSuppressZeroFeet	Boolean	Gets or sets whether a zero foot measurement is suppressed in tolerance values. It overrides the value of the DIMZIN system variable. The initial value for this property is False.
ToleranceSuppressZeroInches	Boolean	Gets or sets whether a zero inch measurement is suppressed in tolerance values. It overrides the value of the DIMZIN system variable. The initial value for this property is False.
UnitsFormat	AcDimLUnits	Gets or sets the unit format. The initial value for this property is acDimLDecimal, where the format specified by the DecimalSeparator and PrimaryUnitsPrecision properties will be used to format the decimal value. For a list of possible values for the AcDimLUnits enumerated type, see Appendix B.

Note Because this object inherits from AcadObject, it supports the Modified event.

AcadDimOrdinate Object

The AcadDimOrdinate object represents a dimension of the absolute X or Y position of a point from the origin. Ordinate dimensions display the X or Y UCS coordinate of an object along with a simple leader line. The absolute value of the coordinate is used according to the prevailing standards for ordinate dimensions. To change the extension lines of an ordinate dimension, change the Rotation property and/or the TextPosition property. The extension lines will be recalculated to fit the new requirements. The AcadDimOrdinate object is created using the AddDimOrdinate method of the AcadBlock, AcadModelSpace, or AcadPaperSpace object.

AcadDimOrdinate Object Methods

The AcadDimOrdinate object inherits all the methods of the AcadEntity and AcadObject objects. It supports no other methods.

AcadDimOrdinate Object Properties

The AcadDimOrdinate object inherits all the properties of the AcadEntity object, the AcadObject object, and the AcadDimension object, as well as the common Application property. It also supports the following properties.

Name	Returns	Description
AltRoundDistance	Double	Gets or sets the rounding of alternative units if the AltUnits property is turned on. This property and the two that follow override the value of the DIMALTZ system variable.
AltSuppressLeadingZeros	Boolean	Gets or sets whether leading zeros in alternative dimension values are suppressed. The initial value for this property is False.
AltSuppressTrailingZeros	Boolean	Gets or sets whether trailing zeros in alternative dimension values are suppressed. The initial value for this property is False.
AltSuppressZeroFeet AltSuppressZeroInches	Boolean	Gets or sets whether a zero foot/inches measurement is suppressed in alternative dimension values. Both properties change the value of the DIMALTZ system variable. The initial value for this property is False.
AltTextPrefix AltTextSuffix	String	Gets or sets a prefix/suffix for the alternative dimension measurement. Both properties override the value of the DIMAPOST system variable. The initial value for this property is a NULL string. To turn off an established prefix, set this property equal to a single period (.).
AltTolerancePrecision	AcDimPrecision	Gets or sets the precision of tolerance values in alternative dimensions. The initial value for this property is acDimPrecisionTwo. This property is available only when the ToleranceDisplay property is set to any value other than acTolNone, and it overrides the value of the DIMTDEC system variable. For a list of the values for the AcDimPrecision enumerated type, see Appendix B.
AltToleranceSuppressLeadingZeros	Boolean	Gets or sets whether leading zeros are suppressed in alternative dimension values. This property and the three that follow override the DIMALTZ system variable. The initial value for this property is False.
AltToleranceSuppressTrailingZeros	Boolean	Gets or sets whether trailing zeros are suppressed in alternative dimension values. The initial value for this property is False.
AltToleranceSuppressZeroFeet	Boolean	Gets or sets whether a zero foot measurement is suppressed in alternative tolerance values. The initial value for this property is False.
AltToleranceSuppressZeroInches	Boolean	Gets or sets whether a zero inch measurement is suppressed in alternative tolerance values. The initial value for this property is False.

Name	Returns	Description
AltUnits	Boolean	Gets or sets whether alternative units dimensioning is enabled. It overrides the value of the DIMALT system variable.
AltUnitsFormat	AcDimUnits	Gets or sets the units format for alternative units. The initial value for this property is <code>acDimDecimal</code> . It overrides the value of the DIMALTU system variable. For a list of possible values for the <code>AcDimUnits</code> enumerated type, see Appendix B.
AltUnitsPrecision	AcDimPrecision	Gets or sets the number of decimal places in alternative units. It overrides the value of the DIMALTD system variable. For a list of possible values for the <code>AcDimPrecision</code> enumerated type, see Appendix B.
AltUnitsScale	Double	Gets or sets the scale factor for alternative units. If the <code>AltUnits</code> property is turned on, this property multiplies linear dimensions by a factor to produce a value in an alternate system of measurement. The initial value represents the number of millimeters in an inch. It overrides the value of the DIMALTF system variable.
ArrowheadSize	Double	Gets or sets the size of the dimension line arrowheads, leader line arrowheads, and hook lines. It overrides the value of the DIMASZ system variable. The initial value for this property is 0.1800.
ExtensionLineColor	AcColor	Gets or sets the color of the extension lines. Use a color index number from 0 to 256 or one of the constants listed here: <code>acByBlock</code> (where AutoCAD draws objects in the default color), <code>acByLayer</code> (where AutoCAD draws objects in the color specified for the layer), or the <code>AcColor</code> enumerated type (see the <code>AcadEntity</code> object in Appendix B). It overrides the value of the DIMCLRE system variable.
ExtensionLineOffset	Double	Gets or sets the distance that the extension lines are offset from the origin points. It overrides the value of the DIMEXO system variable. The initial value for this property is 0.0625.
ExtensionLineWeight	AcLineWeight	Gets or sets the lineweight of the extension lines. It overrides the value of the DIMLWE system variable. For a list of possible values for the <code>AcLineWeight</code> enumerated type, see Appendix B.
FractionFormat	AcDimFractionType	Gets or sets the formats of fractional values. It overrides the value of the DIMFRAC system variable. This property is used when the <code>UnitsFormat</code> property is set to <code>acDimLArchitectural</code> or <code>acDimLFractional</code> . Possible values: <code>acHorizontal</code> , <code>acDiagonal</code> , and <code>acNotStacked</code> .
LinearScaleFactor	Double	Gets or sets the global scale factor for linear dimensioning measurements. It overrides the value of the DIMLFAC system variable. The initial value for this property is 1.0000. When <code>LinearScaleFactor</code> is assigned a negative value, the factor is applied only in paper space.
Measurement	Double	Gets the measurement for the dimension. This property overrides the value for the DIMLUNIT system variable. This property's value is read-only.
PrimaryUnitsPrecision	AcDimPrecision	Gets or sets the number of decimal places displayed for the primary units. For a list of possible values for the <code>AcDimPrecision</code> enumerated type, see Appendix B.

Continued

Name	Returns	Description
RoundDistance	Double	Gets or sets the rounding of units. It overrides the value of the DIMRND system variable.
SuppressZeroFeet	Boolean	Gets or sets whether a zero foot measurement is suppressed. It overrides the value of the DIMZIN system variable. The initial value for this property is False.
SuppressZeroInches	Boolean	Gets or sets whether a zero inch measurement is suppressed. It overrides the value of the DIMZIN system variable. The initial value for this property is False.
TextInside	Boolean	Gets or sets whether the text appears inside the extension lines. It overrides the value of the DIMTIX system variable. The initial value for this property is True. If False, AutoCAD places it inside if there's room.
TextInsideAlign	Boolean	Gets or sets the position of any text inside/ outside the extension lines. The values for these properties override the values of the DIMTIH and DIMTOH system variables, respectively. The initial value for this property is True. The TextInsideAlign property is available only when the TextInside property is set to True.
ToleranceSuppressZeroFeet	Boolean	Gets or sets whether a zero foot measurement is suppressed in tolerance values. It overrides the value of the DIMZIN system variable. The initial value for this property is False.
ToleranceSuppressZeroInches	Boolean	Gets or sets whether a zero inch measurement is suppressed in tolerance values. It overrides the value of the DIMZIN system variable. The initial value for this property is False.
UnitsFormat	AcDimLUnits	Gets or sets the unit format. The initial value for this property is acDimLDecimal, where the format specified by the DecimalSeparator and PrimaryUnitsPrecision properties will be used to format the decimal value. For a list of possible values for the AcDimLUnits enumerated type, see Appendix B.

Note Because this object inherits from AcadObject, it supports the Modified event.

AcadDimRadial Object

The AcadDimRadial object represents a dimension of the radius of a circle or arc. The position of the text, set in the TextPosition property, determines the location of the dimension line. The AcadDimRadial object is created using the AddDimRadial method of the AcadBlock, AcadModelSpace, or AcadPaperSpace object.

AcadDimRadial Object Methods

The AcadDimRadial object inherits all the methods of the AcadEntity and AcadObject objects. It supports no other methods.

AcadDimRadial Object Properties

The AcadDimRadial object inherits all the properties of the AcadEntity object, the AcadObject object, and the AcadDimension object, as well as the common Application property. It also supports the following properties.

Name	Returns	Description
AltRoundDistance	Double	Gets or sets the rounding of alternative units if the AltUnits property is turned on. This property and the two that follow override the value of the DIMALTZ system variable.
AltSuppressLeadingZeros	Boolean	Gets or sets whether leading zeros in alternative dimension values are suppressed. The initial value for this property is False.
AltSuppressTrailingZeros	Boolean	Gets or sets whether trailing zeros in alternative dimension values are suppressed. The initial value for this property is False.
AltSuppressZeroFeet AltSuppressZeroInches	Boolean	Gets or sets whether a zero foot/inches measurement is suppressed in alternative dimension values. Both properties change the value of the DIMALTZ system variable. The initial value for this property is False.
AltTextPrefix AltTextSuffix	String	Gets or sets a prefix/suffix for the alternative dimension measurement. Both properties override the value of the DIMAPOST system variable. The initial value for this property is a NULL string. To turn off an established prefix, set this property equal to a single period (.).
AltTolerancePrecision	AcDimPrecision	Gets or sets the precision of tolerance values in alternative dimensions. The initial value for this property is acDimPrecisionTwo. This property is available only when the ToleranceDisplay property is set to any value other than acTolNone, and it overrides the value of the DIMTDEC system variable. For a list of the values for the AcDimPrecision enumerated type, see Appendix B.
AltToleranceSuppressLeadingZeros	Boolean	Gets or sets whether leading zeros are suppressed in alternative dimension values. This property and the three that follow override the DIMALTTZ system variable. The initial value for this property is False.
AltToleranceSuppressTrailingZeros	Boolean	Gets or sets whether trailing zeros are suppressed in alternative dimension values. The initial value for this property is False.
AltToleranceSuppressZeroFeet	Boolean	Gets or sets whether a zero foot measurement is suppressed in alternative tolerance values. The initial value for this property is False.
AltToleranceSuppressZeroInches	Boolean	Gets or sets whether a zero inch measurement is suppressed in alternative tolerance values. The initial value for this property is False.

Continued

Name	Returns	Description
AltUnits	Boolean	Gets or sets whether alternative units dimensioning is enabled. It overrides the value of the DIMALT system variable.
AltUnitsFormat	AcDimUnits	Gets or sets the units format for alternative units. The initial value for this property is <code>acDimDecimal</code> . It overrides the value of the DIMALTU system variable. For a list of possible values for the <code>AcDimUnits</code> enumerated type, see Appendix B.
AltUnitsPrecision	AcDimPrecision	Gets or sets the number of decimal places in alternative units. It overrides the value of the DIMALTD system variable. For a list of possible values for the <code>AcDimPrecision</code> enumerated type, see Appendix B.
AltUnitsScale	Double	Gets or sets the scale factor for alternative units. If the <code>AltUnits</code> property is turned on, this property multiplies linear dimensions by a factor to produce a value in an alternate system of measurement. The initial value represents the number of millimeters in an inch. It overrides the value of the DIMALTF system variable.
ArrowheadBlock	String	Gets or sets the block used as the custom arrowhead.
ArrowheadSize	Double	Gets or sets the size of the dimension line arrowheads, leader line arrowheads, and hook lines. It overrides the value of the DIMASZ system variable. The initial value for this property is 0.1800.
ArrowheadType	AcDimArrowheadType	Gets or sets the type of the arrowhead. It overrides the value of the DIMLDRBLK system variable. For a list of possible values for the <code>AcDimArrowheadType</code> enumerated type, see Appendix B.
CenterMarkSize	Double	Gets or sets the size of the center mark. The initial value for this property is 0.0900. This property isn't available if the <code>CenterType</code> property is set to <code>acCenterNone</code> .
CenterType	AcDimCenterType	Gets or sets the type of the center mark. The center mark is visible only if you place the dimension line outside the circle or arc. It overrides the value of the DIMCEN system variable. For a list of possible values for the <code>AcDimCenterType</code> enumerated type, see Appendix B.
DimensionLineColor	AcColor	Gets or sets the color of the dimension lines. Use a color index number from 0 to 256 or one of the constants listed here: <code>acByBlock</code> (where AutoCAD draws objects in the default color) or <code>acByLayer</code> (where AutoCAD draws objects in the color specified for the layer). For a list of possible values for the <code>AcColor</code> enumerated type, see Appendix B.
DimensionLineWeight	AcLineWeight	Gets or sets the lineweight of the dimension lines. It overrides the value of the DIMLWD system variable. For a list of possible values for the <code>AcLineWeight</code> enumerated type, see Appendix B.
DimLineSuppress	Boolean	Specifies whether the dimension line is suppressed. It overrides the value of the DIMSD2 system variable. The initial value for this property is <code>False</code> . When this property is set to <code>True</code> , the display of the dimension line and arrowhead between the extension line and the text is suppressed.

Name	Returns	Description
Fit	AcDimFit	Gets or sets whether text and arrowheads are placed inside or outside extension lines given the availability of space between the extension lines. This property sets priorities for moving text and arrowheads when space isn't available to fit both within the extension lines. It overrides the value of the DIMAFIT system variable. The initial value for this property is acBestFit. For a list of the possible values, see Appendix B.
ForcelineInside	Boolean	Gets or sets whether a dimension line is drawn between the extension lines when the text is outside the extension lines. It overrides the value of the DIMTOFL system variable. The initial value for this property is False.
FractionFormat	AcDimFractionType	Gets or sets the formats of fractional values. It overrides the value of the DIMFRAC system variable. This property is used when the UnitsFormat property is set to acDimLArchitectural or acDimLFractional. Possible values: acHorizontal, acDiagonal, and acNotStacked.
HorizontalTextPosition	AcDimHorizontalJustification	Gets or sets the horizontal position of the text. For a list of possible values for the AcDimHorizontalJustification enumerated type, see Appendix B.
LeaderLength	Double	Sets the length of the leader. This property is write-only; it's used when the object is created and isn't stored.
LinearScaleFactor	Double	Gets or sets the global scale factor for linear dimensioning measurements. It overrides the value of the DIMLFAC system variable. The initial value for this property is 1.0000. When LinearScaleFactor is assigned a negative value, the factor is applied only in paper space.
Measurement	Double	Gets the measurement for the dimension. This property overrides the value for the DIMLUNIT system variable. This property's value is read-only.
PrimaryUnitsPrecision	AcDimPrecision	Gets or sets the number of decimal places displayed for the primary units. For a list of possible values for the AcDimPrecision enumerated type, see Appendix B.
RoundDistance	Double	Gets or sets the rounding of units. It overrides the value of the DIMRND system variable.
SuppressZeroFeet	Boolean	Gets or sets whether a zero foot measurement is suppressed. It overrides the value of the DIMZIN system variable. The initial value for this property is False.

Continued

Name	Returns	Description
SuppressZeroInches	Boolean	Gets or sets whether a zero inch measurement is suppressed. It overrides the value of the DIMZIN system variable. The initial value for this property is False.
TextInside	Boolean	Gets or sets whether the text appears inside the extension lines. It overrides the value of the DIMTIX system variable. The initial value for this property is True. If False, AutoCAD places it inside if there's room.
TextInsideAlign	Boolean	Gets or sets the position of any text inside/outside the extension lines. The values for these properties override the values of the DIMTIH and DIMTOH system variables, respectively. The initial value for this property is True. The TextInsideAlign property is available only when the TextInside property is set to True.
ToleranceSuppressZeroFeet	Boolean	Gets or sets whether a zero foot measurement is suppressed in tolerance values. It overrides the value of the DIMZIN system variable. The initial value for this property is False.
ToleranceSuppressZeroInches	Boolean	Gets or sets whether a zero inch measurement is suppressed in tolerance values. It overrides the value of the DIMZIN system variable. The initial value for this property is False.
UnitsFormat	AcDimUnits	Gets or sets the unit format. The initial value for this property is acDimLDecimal, where the format specified by the DecimalSeparator and PrimaryUnitsPrecision properties will be used to format the decimal value. For a list of possible values for the AcDimUnits enumerated type, see Appendix B.

Note Because this object inherits from AcadObject, it supports the Modified event.

AcadDimRadialLarge Object

This is a dimension measuring a large radius, also known as a *jagged radius dimension*.

AcadDimRadialLarge Object Methods

The AcadDimRadialLarge object inherits all the methods of the AcadEntity and AcadObject objects. It doesn't support any other methods.

AcadDimRadialLarge Object Properties

The AcadDimRadialLarge object inherits all the properties of the AcadEntity object, the AcadObject object, and the AcadDimension object, as well as the common Application property. It also supports the following properties.

Name	Returns	Description
AltRoundDistance	Double	Gets or sets the rounding of alternative units, if the AltUnits property is turned on. This property, and the two that follow, all override the value of the DIMALTZ system variable.
AltSuppressLeadingZeros	Boolean	Gets or sets whether leading zeros in alternative dimension values are suppressed. The initial value for this property is False.
AltSuppressTrailingZeros	Boolean	Gets or sets whether trailing zeros in alternative dimension values are suppressed. The initial value for this property is False.
AltSuppressZeroFeet AltSuppressZeroInches	Boolean	Gets or sets whether a zero foot/inches measurement is suppressed in alternative dimension values. Both properties change the value of the DIMALTZ system variable. The initial value for this property is False.
AltTextPrefix AltTextSuffix	String	Gets or sets a prefix/suffix for the alternative dimension measurement. Both properties override the value of the DIMAPOST system variable. The initial value for this property is a NULL string. To turn off an established prefix, set this property equal to a single period (.).
AltTolerancePrecision	AcDimPrecision	Gets or sets the precision of tolerance values in alternative dimensions. The initial value for this property is acDimPrecisionTwo. This property is available only when the ToleranceDisplay property is set to any value other than acTolNone, and it overrides the value of the DIMTDEC system variable. For a list of the values for the AcDimPrecision enumerated type, see Appendix B.
AltToleranceSuppressLeadingZeros	Boolean	Gets or sets whether leading zeros are suppressed in alternative dimension values. This property, and the three that follow, all override the DIMALTTZ system variable. The initial value for this property is False.
AltToleranceSuppressTrailingZeros	Boolean	Gets or sets whether trailing zeros are suppressed in alternative dimension values. The initial value for this property is False.
AltToleranceSuppressZeroFeet	Boolean	Gets or sets whether a zero foot measurement is suppressed in alternative tolerance values. The initial value for this property is False.
AltToleranceSuppressZeroInches	Boolean	Gets or sets whether a zero inch measurement is suppressed in alternative tolerance values. The initial value for this property is False.
AltUnits	Boolean	Gets or sets whether alternative units dimensioning is enabled. This overrides the value of the DIMALT system variable.
AltUnitsFormat	AcDimUnits	Gets or sets the units format for alternative units. The initial value for this property is acDimDecimal. This overrides the value of the DIMALTU system variable. For a list of possible values for the AcDimUnits enumerated type, see Appendix B.

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Name	Returns	Description
AltUnitsPrecision	AcDimPrecision	Gets or sets the number of decimal places in alternative units. This overrides the value of the DIMALTD system variable. For a list of possible values for the AcDimPrecision enumerated type, see Appendix B.
AltUnitsScale	Double	Gets or sets the scale factor for alternative units. If the AltUnits property is turned on, this property multiplies linear dimensions by a factor to produce a value in an alternate system of measurement. The initial value represents the number of millimeters in an inch. This overrides the value of the DIMALTF system variable.
ArrowheadBlock	String	Gets or sets the block used as the custom arrowhead at the end of the dimension line. This overrides the value of the DIMBLK1 or DIMBLK2 system variable.
ArrowheadSize	Double	Gets or sets the size of the dimension line arrowheads, leader line arrowheads, and hook lines. This overrides the value of the DIMASZ system variable. The initial value for this property is 0.1800.
ArrowheadType	AcDimArrowheadType	Gets or sets the type of arrowhead at the end of the dimension line. For a list of possible values for the AcDimArrowheadType enumerated type, see Appendix B.
Center	Variant	Gets or sets the center of the circle as a set of 3-D coordinates.
CenterMarkSize	Double	Gets or sets the size of the center mark. This overrides the value of the DIMCEN system variable. The initial value for this property is 0.0900. This property is not available if the CenterType property is set to acCenterNone.
CenterType	AcDimCenterType	Gets or sets the type of the center mark. The center mark is visible only if you place the dimension line outside the circle or arc. This overrides the value of the DIMCEN system variable. For a list of possible values for the AcDimCenterType enumeration, see Appendix B.
ChordPoint	Variant	Gets or sets the chord point for the arc as a set of 3-D coordinates.
DimensionLineColor	AcColor	Gets or sets the color of the dimension lines. Use a color index number from 0 to 256 or one of the constants listed here: acByBlock (where AutoCAD draws objects in default color) or acByLayer (where AutoCAD draws objects in the color specified for the layer). For a list of possible values for the AcColor enumerated type, see Appendix B.
DimensionLineWeight	AcLineWeight	Gets or sets the line weight of the dimension lines. This overrides the value of the DIMLWD system variable. For a list of possible values for the AcLineWeight enumerated type, see Appendix B.
DimLineSuppress	Boolean	Gets or sets whether the dimension line is suppressed. This overrides the value of the DIMSD1 or DIMSD2 system variable. The initial value for this property is False. When this property is set to True, the display of the dimension line and arrowhead between the first extension line and the text is suppressed.

Name	Returns	Description
Fit	AcDimFit	Gets or sets whether text and arrowheads are placed inside or outside extension lines given the availability of space between the extension lines. This property sets priorities for moving text and arrowheads when space is not available to fit both within the extension lines. This overrides the value of the DIMAFIT system variable. The initial value for this property is acBestFit. For a list of the possible values, see Appendix B.
ForcelineInside	Boolean	Gets or sets whether a dimension line is drawn between the extension lines when the text is outside the extension lines. This overrides the value of the DIMTOFL system variable. The initial value for this property is False.
FractionFormat	AcDimFractionType	Gets or sets the formats of fractional values. This overrides the value of the DIMFRAC system variable. This property is used when the UnitsFormat property is set to acDimLArchitectural or acDimLFractional. Possible values: acHorizontal, acDiagonal and acNotStacked.
HorizontalTextPosition	AcDimHorizontalJustification	Gets or sets the horizontal position of the text. This overrides the value of the DIMJUST system variable. For a list of possible values for the AcDimHorizontalJustification enumeration, see Appendix B.
JogAngle	ACAD_ANGLE	Gets or sets the jog angle. For a list of possible values for the ACAD_ANGLE enumeration, see Appendix B.
JogLocation	Variant	Gets or sets the jog location or user pick point as a set of 3-D coordinates.
LinearScaleFactor	Double	Gets or sets the global scale factor for linear dimensioning measurements. This overrides the value of the DIMLFAC system variable. The initial value for this property is 1.0000. When LinearScaleFactor is assigned a negative value, the factor is applied only in paper space.
Measurement	Double	Gets the measurement for the dimension. This property overrides the value for the DIMLUNIT system variable. It is read-only.
OverrideCenter	Variant	Gets or sets the override center or pick point as a set of 3-D coordinates.
PrimaryUnitsPrecision	AcDimPrecision	Gets or sets the number of decimal places displayed for the primary units. For a list of possible values for the AcDimPrecision enumerated type, see Appendix B.

Continued

Name	Returns	Description
RoundDistance	Double	Gets or sets the rounding of units. This overrides the value of the DIMRND system variable.
SuppressZeroFeet	Boolean	Gets or sets whether a zero foot measurement is suppressed. This overrides the value of the DIMZIN system variable. The initial value for this property is False.
SuppressZeroInches	Boolean	Gets or sets whether a zero inch measurement is suppressed. This overrides the value of the DIMZIN system variable. The initial value for this property is False.
TextInside	Boolean	Gets or sets whether the text appears inside the extension lines. This overrides the value of DIMTIX system variable. The initial value for this property is True. If False, AutoCAD places it inside if there is room.
TextInsideAlign	Boolean	Gets or sets the position of any text inside/outside the extension lines. The values for these properties override the values of the DIMTIH or DIMTOH system variable, respectively. The initial value for this property is True. The TextInsideAlign property is available only when the TextInside property is set to True.
TextOutsideAlign	Boolean	Gets or sets the position of any text inside/outside the extension lines. The values for these properties override the values of the DIMTIH or DIMTOH system variable, respectively. The initial value for this property is True. The TextInsideAlign property is available only when the TextInside property is set to True.
ToleranceSuppressZeroFeet	Boolean	Gets or sets whether a zero foot measurement is suppressed in tolerance values. This overrides the value of the DIMZIN system variable. The initial value for this property is False.
ToleranceSuppressZeroInches	Boolean	Gets or sets whether a zero inch measurement is suppressed in tolerance values. This overrides the value of the DIMZIN system variable. The initial value for this property is False.
TrueColor	AcCmColor	Gets or sets an object's color. Colors are identified by an AcCmColor object. This object can hold an RGB value, an ACI number (an integer from 1 to 255), or a named color. Using an RGB value, you can choose from millions of colors.
UnitsFormat	AcDimLUnits	Gets or sets the unit format. The initial value for this property is acDimLDecimal, where the format specified by the DecimalSeparator and PrimaryUnitsPrecision properties will be used to format the decimal value. For a list of possible values for the AcDimLUnits enumeration, see Appendix B.

Note Because this object inherits from AcadObject, it supports the Modified event.

AcadDimRotated Object

The AcadDimRotated object represents a dimension that measures the distance between two points and is displayed at a given rotation. To change the extension lines of a rotated dimension, change the Rotation property and/or the TextPosition property. The extension lines will be recalculated to fit the new requirements. The AcadDimRotated object is created using the AddCircle method of the AcadBlock, AcadModelSpace, or AcadPaperSpace object.

AcadDimRotated Object Methods

The AcadDimRotated object inherits all the methods of the AcadEntity and AcadObject objects. It supports no other methods.

AcadDimRotated Object Properties

The AcadDimRadial object inherits all the properties of the AcadEntity object, the AcadObject object, and the AcadDimension object, as well as the common Application property. It also supports the following properties.

Name	Returns	Description
AltRoundDistance	Double	Gets or sets the rounding of alternative units if the AltUnits property is turned on. This property and the two that follow override the value of the DIMALTZ system variable.
AltSuppressLeadingZeros	Boolean	Gets or sets whether leading zeros in alternative dimension values are suppressed. The initial value for this property is False.
AltSuppressTrailingZeros	Boolean	Gets or sets whether trailing zeros in alternative dimension values are suppressed. The initial value for this property is False.
AltSuppressZeroFeet AltSuppressZeroInches	Boolean	Gets or sets whether a zero foot/inches measurement is suppressed in alternative dimension values. Both properties change the value of the DIMALTZ system variable. The initial value for this property is False.
AltTextPrefix AltTextSuffix	String	Gets or sets a prefix/suffix for the alternative dimension measurement. Both properties override the value of the DIMAPOST system variable. The initial value for this property is a NULL string. To turn off an established prefix, set this property equal to a single period (.).
AltTolerancePrecision	AcDimPrecision	Gets or sets the precision of tolerance values in alternative dimensions. The initial value for this property is acDimPrecisionTwo. This property is available only when the ToleranceDisplay property is set to any value other than acToINone, and it overrides the value of the DIMTDEC system variable. For a list of the values for the AcDimPrecision enumerated type, see Appendix B.

Continued

Name	Returns	Description
AltToleranceSuppressLeadingZeros	Boolean	Gets or sets whether leading zeros are suppressed in alternative dimension values. This property and the three that follow override the DIMALTTZ system variable. The initial value for this property is False.
AltToleranceSuppressTrailingZeros	Boolean	Gets or sets whether trailing zeros are suppressed in alternative dimension values. The initial value for this property is False.
AltToleranceSuppressZeroFeet	Boolean	Gets or sets whether a zero foot measurement is suppressed in alternative tolerance values. The initial value for this property is False.
AltToleranceSuppressZeroInches	Boolean	Gets or sets whether a zero inch measurement is suppressed in alternative tolerance values. The initial value for this property is False.
AltUnits	Boolean	Gets or sets whether alternative units dimensioning is enabled. It overrides the value of the DIMALT system variable.
AltUnitsFormat	AcDimUnits	Gets or sets the units format for alternative units. The initial value for this property is acDimDecimal. It overrides the value of the DIMALTU system variable. For a list of possible values for the AcDimUnits enumerated type, see Appendix B.
AltUnitsPrecision	AcDimPrecision	Gets or sets the number of decimal places in alternative units. It overrides the value of the DIMALTD system variable. For a list of possible values for the AcDimPrecision enumerated type, see Appendix B.
AltUnitsScale	Double	Gets or sets the scale factor for alternative units. If the AltUnits property is turned on, this property multiplies linear dimensions by a factor to produce a value in an alternate system of measurement. The initial value represents the number of millimeters in an inch. It overrides the value of the DIMALTF system variable.
ArrowheadBlock	String	Gets or sets the block used as the custom arrowhead at the first/second end of the dimension line. It overrides the value of the DIMBLK1 or DIMBLK2 system variable.
ArrowheadSize	Double	Gets or sets the size of the dimension line arrowheads, leader line arrowheads, and hook lines. It overrides the value of the DIMASZ system variable. The initial value for this property is 0.1800.
ArrowheadType	AcDimArrowheadType	Gets or sets the type of arrowhead at the first/second end of the dimension line. For a list of possible values for the AcDimArrowheadType enumerated type, see Appendix B.

Name	Returns	Description
DimensionLineColor	AcColor	Gets or sets the color of the dimension lines. Use a color index number from 0 to 256 or one of the constants listed here: <code>acByBlock</code> (where AutoCAD draws objects in the default color) or <code>acByLayer</code> (where AutoCAD draws objects in the color specified for the layer). For a list of possible values for the <code>AcColor</code> enumerated type, see Appendix B.
DimensionLineExtend	Double	Gets or sets the distance the dimension line extends beyond the extension line when oblique strokes rather than arrowheads are used. It overrides the value of the <code>DIMDLE</code> system variable. The initial value for this property is 0.0000.
DimLineInside	Boolean	Gets or sets whether the dimension lines are displayed only inside the extension lines. It overrides the value of the <code>DIMSOXD</code> system variable. The initial value for this property is <code>False</code> .
DimensionLineWeight	AcLineWeight	Gets or sets the lineweight of the dimension lines. It overrides the value of the <code>DIMLWD</code> system variable. For a list of possible values for the <code>AcLineWeight</code> enumerated type, see Appendix B.
DimLineSuppress	Boolean	Gets or sets whether the first/second dimension line is suppressed. It overrides the value of the <code>DIMSD1</code> or <code>DIMSD2</code> system variable. The initial value for this property is <code>False</code> . When this property is set to <code>True</code> , the display of the dimension line and arrowhead between the first extension line and the text is suppressed.
ExtensionLineColor	AcColor	Gets or sets the color of the extension lines. Use a color index number from 0 to 256 or one of the constants listed here: <code>acByBlock</code> (where AutoCAD draws objects in the default color), <code>acByLayer</code> (where AutoCAD draws objects in the color specified for the layer), or the <code>AcColor</code> enumerated type (see Appendix B). It overrides the value of the <code>DIMCLRE</code> system variable.
ExtensionLineExtend	Double	Gets or sets the distance that the extension line extends beyond the dimension line. It overrides the value of the <code>DIMEXE</code> system variable. The initial value for this property is 0.1800.
ExtensionLineOffset	Double	Gets or sets the distance that the extension lines are offset from the origin points. It overrides the value of the <code>DIMEXO</code> system variable. The initial value for this property is 0.0625.
ExtensionLineWeight	AcLineWeight	Gets or sets the lineweight of the extension lines. It overrides the value of the <code>DIMLWE</code> system variable. For a list of possible values for the <code>AcLineWeight</code> enumerated type, see Appendix B.
ExtLine1Suppress ExtLine2Suppress	Boolean	Gets or sets whether the first/second extension line is suppressed. It overrides the value of the <code>DIMSE1</code> or <code>DIMSE2</code> system variable. The initial value for this property is <code>False</code> .
Fit	AcDimFit	Gets or sets whether text and arrowheads are placed inside or outside extension lines given the availability of space between the extension lines. This property sets priorities for moving text and arrowheads when space isn't available to fit both within the extension lines. It overrides the value of the <code>DIMAFIT</code> system variable. The initial value for this property is <code>acBestFit</code> . For a list of the possible values, see Appendix B.
ForceLineInside	Boolean	Gets or sets whether a dimension line is drawn between the extension lines when the text is outside the extension lines. It overrides the value of the <code>DIMTOFL</code> system variable. The initial value for this property is <code>False</code> .

Continued

Name	Returns	Description
FractionFormat	AcDimFractionType	Gets or sets the formats of fractional values. It overrides the value of the DIMFRAC system variable. This property is used when the UnitsFormat property is set to acDimLArchitectural or acDimLFractional. Possible values: acHorizontal, acDiagonal, and acNotStacked.
HorizontalTextPosition	AcDimHorizontalJustification	Gets or sets the horizontal position of the text. It overrides the value of the DIMJUST system variable. For a list of possible values for the AcDimHorizontalJustification enumerated type, see Appendix B.
LinearScaleFactor	Double	Gets or sets the global scale factor for linear dimensioning measurements. It overrides the value of the DIMLFAC system variable. The initial value for this property is 1.0000. When LinearScaleFactor is assigned a negative value, the factor is applied only in paper space.
Measurement	Double	Gets the measurement for the dimension. This property overrides the value for the DIMLUNIT system variable. This property's value is read-only.
PrimaryUnitsPrecision	AcDimPrecision	Gets or sets the number of decimal places displayed for the primary units. For a list of possible values for the AcDimPrecision enumerated type, see Appendix B.
RoundDistance	Double	Gets or sets the rounding of units. It overrides the value of the DIMRND system variable.
SuppressZeroFeet	Boolean	Gets or sets whether a zero foot measurement is suppressed. It overrides the value of the DIMZIN system variable. The initial value for this property is False.
SuppressZeroInches	Boolean	Gets or sets whether a zero inch measurement is suppressed. It overrides the value of the DIMZIN system variable. The initial value for this property is False.
TextInside	Boolean	Gets or sets whether the text appears inside the extension lines. It overrides the value of DIMTIX system variable. The initial value for this property is True. If False, AutoCAD places it inside if there's room.
TextInsideAlign	Boolean	Gets or sets the position of any text inside/outside the extension lines. The values for these properties override the values of the DIMTIH and DIMTOH system variables, respectively. The initial value for this property is True. The TextInsideAlign property is available only when the TextInside property is set to True.

Name	Returns	Description
ToleranceSuppressZeroFeet	Boolean	Gets or sets whether a zero foot measurement is suppressed in tolerance values. It overrides the value of the DIMZIN system variable. The initial value for this property is False.
ToleranceSuppressZeroInches	Boolean	Gets or sets whether a zero inch measurement is suppressed in tolerance values. It overrides the value of the DIMZIN system variable. The initial value for this property is False.
UnitsFormat	AcDimLUnits	Gets or sets the unit format. The initial value for this property is acDimLDecimal, where the format specified by the DecimalSeparator and PrimaryUnitsPrecision properties will be used to format the decimal value. For a list of possible values for the AcDimLUnits enumerated type, see Appendix B.

Note Because this object inherits from `AcadObject`, it supports the `Modified` event.

AcadDimStyle Object

The `AcadDimStyle` object represents a group of dimension settings that determine the appearance of new dimensions created in the drawing. It is created by the `Add` method of the `AcadDimStyles` collection and is activated through the `ActiveDimStyle` property of the `Document` object. To change the style of a dimension, use the `StyleName` property found on the dimension. To control the settings of the current document overrides, use the dimensioning system variables.

When you change a dimensioning system variable, you aren't changing the active dimension style itself. Rather, you're setting a document override for the dimension style, which means that all newly created dimensions will still be created with the active dimension style. They won't reflect the overrides from the system variables until the active dimension style is updated. To change the settings of any dimension style, use the `CopyFrom` method. Dimensions created via `ActiveX` use the active dimension style only. To apply the system overrides for dimensions created by `ActiveX`, use the `CopyFrom` method to copy the dimension style from the document to the active dimension style.

AcadDimStyle Object Methods

In addition to the methods inherited from the `AcadObject` object, the `AcadDimStyle` object supports the following method.

Name	Description
<code>CopyFrom</code>	Copies the dimension style data from a source object, which can be a document, dimension, or another dimension style. Parameter: <code>StyleSource As Object</code> . If <code>StyleSource</code> is an <code>AcadDocument</code> object, this method copies the active dimension style settings for the drawing plus any drawing overrides. If it's a dimension, an <code>AcadTolerance</code> object, or an <code>AcadLeader</code> object, this method copies the style for that object plus any object overrides. If <code>StyleSource</code> is an <code>AcadDimStyle</code> object, then the style data from that dimension style is copied.

AcadDimStyle Object Properties

In addition to the properties inherited from the AcadObject object and the common Application property, the AcadDimStyle object supports the following property.

Name	Returns	Description
Name	String	Gets or sets the object's name.

Note Because this object inherits from AcadObject, it supports the Modified event.

AcadDimStyles Collection

The AcadDimStyles collection contains all the dimension styles in the drawing. Although this collection inherits a Delete method, you can't actually delete it. If you need to delete a specific dimension style, use the Delete method found in the AcadDimStyle object. There's no limit to the number of dimension styles you can create in your drawing. However, there can be only one instance of the AcadDimStyles collection, which is predefined for each drawing. You can make multiple references to it by using the DimStyles property.

AcadDimStyles Collection Methods

In addition to the methods inherited from the AcadObject object, the AcadDimStyles collection supports the following methods.

Name	Returns	Description
Add	AcadDimStyle	Creates an AcadDimStyle object and adds it to the collection. Parameter: Name As String.
Item	AcadDimStyle	Gets the member object at a given index in the collection. Parameter: Index As Variant (Integer or String). If the Index value is a String, it must match an existing style definition.

AcadDimStyles Collection Properties

The AcadDimStyles collection supports the Count property, the common Application property, and the properties inherited from the AcadObject object. It supports no other properties.

Note Because this collection inherits from AcadObject, it supports the Modified event.

AcadDocument Object

The `AcadDocument` object represents an AutoCAD drawing. The active document can be accessed using `ThisDrawing` in VBA and the `ActiveDocument` property of the `AcadApplication` object in Visual Basic.

Certain objects in the drawing establish the format, location, or style that new objects will adopt. These include the `AcadDimStyle`, `AcadTextStyle`, `AcadLinetype`, `AcadViewport`, and `AcadLayer` objects. A drawing may contain many objects of these types, but only one can be active at a time. The `AcadDocument` object also contains a `link` property to all the collections, all of which are created when the document is created. You can thus iterate through every object in the drawing.

You control the model space and paper space settings from the `AcadDocument` object by using its `ActiveSpace` property, its `MSpace` property, and the `Display` method of the `AcadPViewport` object. By default, a drawing is opened in the model space with the tiled viewport setting.

Changes to most active objects will appear immediately. However, several objects must be reset for changes to appear: the active text style, the active UCS, and the active viewport. These are reset using the `ActiveTextStyle`, `ActiveUCS`, and `ActiveViewport` properties, respectively. An example syntax for resetting the active viewport is as follows:

```
ThisDrawing.ActiveViewport = modifiedViewportObj
```

AcadDocument Object Methods

The `AcadDocument` object supports the following methods.

Name	Returns	Description
Activate		Makes the specified drawing active.
AuditInfo		Evaluates the drawing's integrity. For every error detected, AutoCAD provides a description of the error and recommends corrective action. Parameter: <code>FixError</code> As Boolean. If the parameter is set to True, AutoCAD should attempt to fix any errors.
Close		Closes the specified drawing. If no drawing is specified, the drawing is closed only if it already has a valid file name or wasn't modified. Parameters: <code>SaveChanges</code> As Boolean and <code>FileName</code> As String. If the first parameter is set to True, changes are saved. The second parameter specifies the name to assign to the drawing. The document will be closed and no error will occur if the drawing hasn't been modified or if the first parameter is set to False. The document will be saved and closed if the first parameter is set to True (the default value) and the drawing was previously saved or was opened from an existing drawing file. In any other case, a valid name has to be specified.
CopyObjects	Variant	Copies objects, their child objects, and so on recursively (deep cloning). Parameters: <code>Objects</code> As Variant, <code>[Owner]</code> As Variant, and <code>[IDPairs]</code> As Variant. For more information on this method, see the <code>AcadDatabaseObject</code> section.
EndUndoMark		Marks the end of a block of operations. See also the <code>StartUndoMark</code> method.

Continued

Name	Returns	Description
Export		Exports the drawing to a .wmf, .sat, .eps, .dxf, or .bmp format. Parameters: FileName As String, Extension As String, and SelectionSet As AcadSelectionSet. When exporting to wmf, .sat, or .bmp format, a nonempty selection set must be provided. For the .sat format, only AcadRegion and Acad3DSolid objects can be exported. When exporting to .eps and .dxf formats, the selection set is ignored and the entire drawing is exported.
GetVariable	Variant	Returns the current setting of an AutoCAD system variable. For a list of all the AutoCAD system variables and their types, please refer to Appendix C. Parameter: Name As String.
HandleToObject	Object	Returns the object that corresponds to a handle. This method can return only objects in the current document. Parameter: Handle As String. A handle is persistent. It can be stored for reuse with this method for the lifetime of the object.
Import	Object	Imports a drawing in .wmf, .sat, .eps, or .dxf format. In the case of importing a .wmf file, an AcadBlockReference object is returned. In all other cases, the return value is NULL. Parameters: FileName As String, InsertionPoint As Variant, and ScaleFactor As Double.
LoadShapeFile		Loads an .shx (shape) file. This method makes all the shapes in the shape file available to the current drawing. Parameter: FullName As String.
New	AcadDocument	Creates a new document in SDI mode. It's good practice to immediately save the new document with a valid file name. This will simplify later using the save and close methods. Parameter: TemplateFileName As String.
ObjectIDToObject	Object	Returns the object that corresponds to an object ID. This method can return objects only in the current document. Parameter: ID As Long. An object ID isn't persistent. Its lifetime is limited to a drawing session.
Open		Opens an existing .dwg (drawing) file when working in SDI mode. The file becomes the active document. Parameters: Name As String and [ReadOnly As Boolean]. The Name parameter is an existing file name, including the path or the valid URL address of the drawing to open. The second parameter is True if the drawing is to be read-only and False otherwise (the default).
PurgeAll		Removes all unused named references from the document. Deleted objects remain in the document until they're purged using this method.
Regen		Regenerates the entire drawing and updates the screen coordinates and view resolution for all objects. Parameter: WhichViewports As AcRegenType (acActiveViewport or acAllViewports).
Save		Saves the document. When the user is saving to a secure URL, a dialog box will prompt the user for the necessary password information. Message boxes may also appear if the user hasn't suppressed this activity in the browser.

Name	Returns	Description
SaveAs		Saves the document as specified by the parameters. Parameters: FileName As String and FileType As AcMenuFileType. For a list of possible values for the AcMenuFileType enumeration, see Appendix B.
SendCommand		Sends a command string from a VB or VBA application to the document to be processed. If the drawing specified isn't active, it will be made active. This method processes any AutoCAD command-line function, including LISP expressions. This method is generally synchronous. Only when responding to user interaction or when called from an event handler will commands be processed asynchronously. You should never use this method to issue a command for which there is an ActiveX method available. Instead, use the LoadDVB method. Parameter: Command As String.
SetVariable		Sets the value of an AutoCAD system variable. For a list of all AutoCAD system variables and their types, please refer to Appendix C. Parameters: Name As String and Value As Variant.
StartUndoMark		Marks the beginning of a block of operations that will be treated as a unique operation in an Undo command. Unfortunately, the Undo command hasn't been wrapped for VBA but is available synchronously using the SendCommand method.
WBlock		Creates a new drawing file from the selection set. Parameter: FileName As String and SelectionSet As AcadSelectionSet.

Note The insertion point isn't specified. If it isn't already at the origin, each object within the selection set will have to be copied and moved before a call to WBlock.

AcadDocument Object Properties

As well as the common Application property, the AcadDocument object supports the following properties.

Name	Returns	Description
Active	Boolean	Specifies whether the document is active. This property's value is read-only.
ActiveDimStyle	AcadDimStyle	Gets or sets the active dimension style. This style will be applied to all newly created dimensions.
ActiveLayer	AcadLayer	Gets or sets the active layer. New objects are placed on the active layer as they're created.
ActiveLayout	AcadLayout	Gets or sets the active layout.
ActiveLinetype	AcadLinetype	Gets or sets the active linetype. The specified linetype must already exist in the drawing.
ActivePViewport	AcadPViewport	Gets or sets the active paper space viewport. A viewport must be set active using this property before you can see any changes to it.

Continued

Name	Returns	Description
ActiveSelectionSet	AcadSelectionSet	Gets or sets the active selection set.
ActiveSpace	AcActiveSpace	Gets or sets whether the paper space or the model space is active. The value is stored in the TILEMODE system variable. For model space and floating viewports, this value is set to acPaperSpace. Even though you have the ability to edit the model, when the MSpace property is set to True, you're still technically in paper space. Values for AcActiveSpace: acModelSpace or acPaperSpace.
ActiveTextStyle	AcadTextStyle	Gets or sets the active text style. New text added to the drawing will adopt this text style. Existing text that has no distinct text style specified will adopt the new style. Note that a call to the Regen method is necessary to see any changes.
ActiveUCS	AcadUCS	Gets or sets the active UCS. Changes made to the current active UCS will become visible only after that UCS is reset as the active UCS.
ActiveViewport	AcadViewport	Gets or sets the active viewport. Changes made to the current active viewport will become visible only after you reset the viewport as the active viewport.
Blocks	AcadBlocks	Gets the drawing's blocks collection. This property's value is read-only.
Database	AcadDatabase	Gets the database to which the document belongs. This property's value is read-only.
Dictionaries	AcadDictionaries	Gets the document's dictionaries collection. This property's value is read-only.
DimStyles	AcadDimStyles	Gets the document's dimension styles collection. This property's value is read-only.
ElevationModelSpace	Double	Gets or sets the elevation setting for the model space. The current elevation is the Z value that is used whenever a 3-D point is expected but only the X and Y values are supplied.
ElevationPaperSpace	Double	Gets or sets the elevation setting for the paper space. The current elevation is the Z value that is used whenever a 3-D point is expected but only the X and Y values are supplied.
FullName	String	Gets the document's name, including the path. It returns a null string for a new drawing. This property's value is read-only.
Groups	AcadGroups	Gets the document's groups collection. This property's value is read-only.
HWND	Long	Gets the window handle of the document's window frame. This property's value is read-only.
Layers	AcadLayers	Gets the document's layers collection.
Layouts	AcadLayouts	Gets the document's layouts collection.
Limits	Variant	Gets or sets the document's drawing limits as an array of four Double numbers: the X and Y coordinates of the lower-left limit, and the second pair of values define the X and Y coordinates of the upper-right limit. You can't specify limits for the Z-axis.
Linetypes	AcadLinetypes	Gets the document's linetypes collection. This property's value is read-only.

Name	Returns	Description
ModelSpace	AcadModelSpace	Gets the document's model space object. This property's value is read-only.
MSpace	Boolean	Specifies whether it is possible to edit the model from the floating paper space viewports.
Name	String	Gets the document's name but not the path. It returns a null string for a new drawing. This property's value is read-only.
ObjectSnapMode	Boolean	Specifies whether the document's snap mode is on or off. True indicates that it is on.
PaperSpace	AcadPaperSpace	Gets the document's paper space object. This property's value is read-only.
Path	String	Gets the document's path but not the file name. This property's value is read-only.
PickfirstSelectionSet	AcadSelectionSet	Gets the PickFirst selection set. This property's value is read-only.
Plot	AcadPlot	Gets the document's AcadPlot object. This property's value is read-only.
PlotConfigurations	AcPlotConfigurations	Gets the document's plot configurations collection. This property's value is read-only.
Preferences	AcadPreferences	Gets the document's AcadPreferences object. This property's value is read-only.
ReadOnly	Boolean	Specifies whether the document is read-only. This property's value is read-only.
RegisteredApplications	AcadRegisterApplications	Gets the document's registered applications collection. This property's value is read-only.
Saved	Boolean	Specifies whether the document has any unsaved changes. It's True if there are no unsaved changes and False if there are. This property's value is read-only.
SelectionSets	AcadSelectionSets	Gets the document's selection set collection. This property's value is read-only.
TextStyles	AcadTextStyles	Gets the document's text styles collection. This property's value is read-only.
UserCoordinateSystems	AcadUCSs	Gets the document's UCSs collection. This property's value is read-only.
Utility	AcadUtility	Gets the document's AcadUtility object. This property's value is read-only.
Viewports	AcadViewports	Gets the document's viewports collection. This property's value is read-only.
Views	AcadViews	Gets the document's views collection. This property's value is read-only.
WindowState	AcWindowState	Gets or sets the state of the document window. Values: acMin, acMax, and acNorm.
WindowTitle	String	Gets the title of the document window, which will be the same as for the Name property for a saved document. This property's value is read-only.

AcadDocument Object Events

The following events are triggered during various stages in the lifetime of the AcadDocument object. Note that no events will be fired while a modal dialog box is displayed.

Event	Description
Activate	Occurs when the document window is activated.
BeginClose	Occurs after a request to close a drawing is received.
BeginCommand	Occurs after a command is issued but before it's initiated. Parameter: Command As String.
BeginDoubleClick	Occurs after the user double-clicks anywhere in the drawing. Parameter: PickPoint As Variant (array of WCS coordinates).
BeginLISP	Occurs after a request to evaluate a LISP expression is received. Parameter: FirstLine As String. FirstLine won't have any case conversion of the alpha characters.
BeginPlot	Occurs after a request to print a drawing is received. Parameter: DrawingName As String.
BeginRightClick	Occurs after the user right-clicks anywhere in the drawing window. Parameter: PickPoint As Variant.
BeginSave	Occurs after a request to save a drawing is received. Parameter: FileName As String.
BeginShortcutMenuCommand	Occurs after the user right-clicks in the drawing window but before the menu is in command mode. Parameters: ShortcutMenu As AcadPopupMenu and Command As String.
BeginShortcutMenuDefault	Occurs after the user right-clicks in the drawing window but before the menu is in default mode. Parameter: ShortcutMenu As AcadPopupMenu.
BeginShortcutMenuEdit	Occurs after the user right-clicks in the drawing window but before the menu is in edit mode. Parameters: ShortcutMenu As AcadPopupMenu and SelectionSet As AcadSelectionSet.
BeginShortcutMenuGrip	Occurs after the user right-clicks in the drawing window but before the menu is in grip mode. Parameter: ShortcutMenu As AcadPopupMenu.
BeginShortcutMenuOSnap	Occurs after the user right-clicks in the drawing window but before the menu is in osnap mode. Parameter: ShortcutMenu As AcadPopupMenu.
Deactivate	Occurs when the drawing window is deactivated.
EndCommand	Occurs after a command is executed. Parameter: CommandName As String.
EndLISP	Occurs after a LISP expression is evaluated.
EndPlot	Occurs after a document is sent to the printer. Parameter: DrawingName As String.
EndSave	Occurs after a drawing has been saved. Parameter: FileName As String.
EndShortcutMenu	Occurs after a shortcut menu appears. Use this event to perform any cleanup work on the shortcut menu. Parameter: ShortcutMenu As AcadPopupMenu.
LayoutSwitched	Occurs after the user changes the layout. Parameter: LayoutName As String.
LISPCancelled	Occurs when the evaluation of a LISP expression is canceled.

Event	Description
ObjectAdded	Occurs after an object is added to the drawing. Parameter: Entity As AcadEntity.
ObjectErased	Occurs after an object is erased from the drawing. Parameter: ObjectID As Long.
ObjectModified	Occurs after an object in the drawing is modified. Parameter: Entity As AcadEntity.
SelectionChanged	Occurs when the current PickFirst selection set changes.
WindowChanged	Occurs when the drawing or document window is changed. Parameter: WindowState As acMin or acMax or acNorm. This event (and the one that follows) is helpful when implementing floating palettes, toolbars, or modeless dialog boxes that track with the application or document window. The VB or ObjectARX application can use the HwndFrame parameter to get the coordinates of the window, convert those coordinates to either screen or parent coordinates, and use this information to position other windows.
WindowMovedOrResized	Occurs after the drawing or document window is moved or resized. Parameters: HwndFrame As Long and Moved As Boolean. If Moved is set to True, the window was moved; otherwise, it was resized.

AcadDocuments Collection

The AcadDocuments collection contains all AutoCAD drawings open in the current session. To add a new member to the collection, use the Add or Open method. To select a specific document, use the Item method. To close all the documents, use the Close method similar to the one found in the AcadDocument object. To close all documents, use the Close method found in this collection. There's no limit to the number of documents you can create in your drawing. However, there can be only one instance of the AcadDocuments collection, which is predefined for each application. You can make multiple references to it by using the Documents property or the AcadDocuments object.

AcadDocuments Collection Methods

The AcadDocuments collection supports these four methods.

Name	Returns	Description
Add	AcadDocument	Creates an AcadDocument object and adds it to the collection. Optional parameter: Name As String. This method should not be called when AutoCAD runs in SDI mode.
Close		Closes all open drawings in MDI mode. Note that you can't call this method while in SDI mode. Use the Close method of the AcadDocument object.
Item	AcadDocument	Returns the object at the given index in the collection. Parameter: Index As Variant (Integer or String). If the Index value is a String, it must match an existing document in the collection.
Open		Opens an existing .dwg (drawing) file in MDI mode. The document then becomes the active document. When the user is accessing a secure URL, a dialog box will be posted that prompts the user for the necessary password information. Message boxes may also appear if the user hasn't suppressed this activity in the browser. Parameters: Name As String and [ReadOnly As Boolean]. The second parameter specifies whether the drawing is read-only.

AcadDocuments Collection Properties

The AcadDocuments collection supports the common Application property and the Count property but no other properties.

AcadDynamicBlockReferenceProperty Object

This is an object containing the properties of a dynamic block. Dynamic blocks cannot be created through VBA; however, properties of dynamic blocks may be queried.

AcadDynamicBlockReferenceProperty Methods

The AcadDynamicBlockReferenceProperty object does not support any methods.

AcadDynamicBlockReferenceProperty Properties

The AcadDynamicBlockReferenceProperty object inherits no properties. However, it does support the following properties.

Name	Returns	Description
AllowedValuesVariant		Gets the allowed values for the property. Not all dynamic block properties have a restricted set of property values. Some properties are unrestricted, some specify a minimum or maximum value (or both), and some have a list of allowed values. This property supports only allowed property values that are defined by a list; minimum, maximum, and unrestricted property types are not supported. Read-only.
Description	String	Gets the description of an object. The description for a dynamic block reference property is set by the block author and cannot be edited.
PropertyName	String	Gets the name for the property. Property names are guaranteed to be unique among all properties on a given block reference. Read-only.
ReadOnly	Boolean	Gets if the document or property is read-only or read-write. The property is read-only when the ReadOnly property is set to True.
Show	Boolean	Gets whether the property is shown in the user interface. If this property is True, the dynamic block reference property is displayed in the Properties palette when the block reference containing it is selected. It is read-only.
UnitsType	AcDynamicBlockReferencePropertyUnitsType	Gets and sets the current display units type for the property. For a list of possible values for the AcDynamicBlockReferencePropertyUnitsType enumeration, see Appendix B.
Value	Variant	Gets and sets the current value for the property.

AcadEllipse Object

The AcadEllipse object represents an elliptical arc or full ellipse, drawing a true ellipse, not a polyline approximation. It is created using the AddEllipse method of the AcadBlock, AcadModelSpace, or AcadPaperSpace object.

AcadEllipse Object Methods

The AcadEllipse object inherits all the methods of the AcadEntity and AcadObject objects. It also supports the following method.

Name	Returns	Description
Offset	Variant	Creates a new ellipse by offsetting the current ellipse by a specified distance, which must be nonzero. If the offset is positive, a larger ellipse is created with the same origin. If the offset is negative, a smaller ellipse is created. Parameter: Distance As Double.

AcadEllipse Object Properties

The AcadEllipse object inherits all the properties of the AcadEntity and AcadObject objects, as well as the common Application property. It also supports the following properties.

Name	Returns	Description
Area	Double	Gets the area of the ellipse. This property's value is read-only.
Center	Variant	Gets or sets the center of the ellipse as an array of the Doubles.
EndAngle	Double	Gets or sets the end angle of the ellipse in radians. Use 6.28 radians to specify a closed ellipse.
EndParameter	Double	Gets or sets the end parameter for an ellipse. The valid range is 0 to 2*PI.
EndPoint	Variant	Gets the 3-D coordinates of the end point of the ellipse. This property's value is read-only.
MajorAxis	Variant	Gets or sets the direction of the major (i.e., longer) axis of the ellipse as a 3-D vector. The vector originates at the ellipse center.
MajorRadius	Double	Gets or sets the length of the major (i.e., longer) axis of the ellipse.
MinorAxis	Variant	Gets the direction of the minor (i.e., shorter) axis of the ellipse as a 3-D vector. This property's value is read-only.
MinorRadius	Double	Gets or sets the length of the minor (i.e., shorter) axis of the ellipse.
Normal	Variant	Gets or sets the 3-D (Z-axis) normal unit vector for the ellipse.
RadiusRatio	Double	Gets or sets the major to minor axis ratio of an ellipse. A value of 1.0 denotes a circle.
StartAngle	Double	Gets or sets the start angle of the ellipse in radians. Use 6.28 radians to specify a closed ellipse.
StartParameter	Double	Gets or sets the start parameter for an ellipse. The valid range is 0 to 2*PI.
StartPoint	Variant	Gets the X, Y, Z coordinate of the start point of the ellipse. This property's value is read-only.

Note Because this object inherits from `AcadObject`, it supports the `Modified` event.

AcadExternalReference Object

The `AcadExternalReference` object represents an instance of an external reference (Xref) inserted into a drawing (i.e., where a second drawing is linked into the current drawing). It differs from the `AcadBlockReference` object in the following respect: if a block is inserted into a drawing, its definition and all associated geometry are stored in the current drawing database and aren't updated if the original drawing changes. However, when you insert a drawing as an Xref, changes are updated in the externally referenced file. Like a block reference, an Xref is displayed in the current drawing as a single object. However, it doesn't significantly increase the file size of the current drawing and can't be exploded. The `AcadExternalReference` object is created using the `AttachExternalReference` method of the `AcadBlock`, `AcadModelSpace`, or `AcadPaperSpace` object. Note that many methods and properties of the `AcadBlock` and `AcadDatabase` objects are specific to Xrefs.

AcadExternalReference Object Methods

The `AcadExternalReference` object inherits all the methods of the `AcadEntity` and `AcadObject` objects. It also supports the following method.

Name	Returns	Description
<code>GetConstantAttributes</code>	Variant	Returns an array of constant attributes for the external reference.

AcadExternalReference Object Properties

The `AcadExternalReference` object inherits all the properties of the `AcadEntity` and `AcadObject` objects, as well as the common `Application` property. It also supports the following properties.

Name	Returns	Description
<code>HasAttributes</code>	Boolean	Specifies whether the external reference has any attributes. This property's value is read-only.
<code>InsertionPoint</code>	Variant	Gets or sets the insertion point for the external reference as a set of 3-D WCS coordinates. It returns a three-element array of Doubles.
<code>Name</code>	String	Gets or sets the name of the external reference.
<code>Normal</code>	Variant	Gets or sets the 3-D (Z-axis) normal unit vector for the external reference.
<code>Path</code>	String	Gets or sets the path of the external reference. This doesn't include the file name.
<code>Rotation</code>	Double	Specifies the rotation angle (in radians) for the block, relative to the X-axis with positive values going counterclockwise when viewed along the Z-axis toward the origin.
<code>XScaleFactor</code>	Double	Gets or sets the X scale factor for the block.
<code>YScaleFactor</code>	Double	Gets or sets the Y scale factor for the block.
<code>ZScaleFactor</code>	Double	Gets or sets the Z scale factor for the block.

Note Because this object inherits from `AcadObject`, it supports the `Modified` event.

AcadFileDependency Object

The `AcadFileDependency` object describes an entry in the File Dependency List. The files on which a drawing file depends are in its File Dependency List, which is the `AcadFileDependencies` collection object. This object has no methods, but it supports the following properties.

Name	Returns	Description
<code>AffectsGraphics</code>	Boolean	If True, affects; if False, doesn't affect.
<code>Feature</code>	String	Feature name (e.g., "Acad:Xref").
<code>FileName</code>	String	File name of current file.
<code>FileSize</code>	Long	File size in bytes. This property's value is read-only.
<code>FingerprintGUID</code>	String	The GUID fingerprint of the file.
<code>FoundPath</code>	String	Alternate path found in (if any).
<code>FullFileName</code>	String	Saved full file name.
<code>IsModified</code>	Boolean	If True, modified; if False, not modified.
<code>ReferenceCount</code>	Long	Current item count of reference.
<code>TimeStamp</code>	Long	Current time/date in seconds since 1/1/1980.
<code>VersionGUID</code>	String	GUID version stamp.

AcadFileDependencies Object

The `AcadFileDependencies` object creates, updates, retrieves, and removes items in the File Dependency List. The File Dependency List is a list of the files on which the drawing file depends. You can perform operations on the File Dependency List using the `FileDependencies` collection.

AcadFileDependencies Object Methods

In addition to the methods inherited from the `AcadDocument` object, the following methods are available.

Name	Returns	Description
<code>CreateEntry</code>	Long	An index into the File Dependency List
<code>IndexOf</code>	Long	The index of an entry in the File Dependency List
<code>RemoveEntry</code>		Removes an entry from the File Dependency List
<code>UpdateEntry</code>		Updates an entry in the File Dependency List

AcadGroup Object

The `AcadGroup` object represents a named `AcadSelectionSet` object. It is created using the `Add` method of the `AcadGroups` collection and can be accessed by the `Item` method of the `AcadGroups` collection. See the `AcadSelectionSet` object for more information on selection sets.

AcadGroup Object Methods

In addition to the methods inherited from the `AcadObject` object, the `AcadGroup` object supports the following methods.

Name	Returns	Description
<code>AppendItems</code>		Appends one or more entities to the specified group. Parameter: <code>Objects As Variant</code> .
<code>Highlight</code>		Determines whether the objects in a given group are to be highlighted. Parameter: <code>HighlightFlag As Boolean</code> .
<code>Item</code>	<code>AcadEntity</code>	Gets the member object at a given index in a group. Parameter: <code>Index As Variant (Integer or String)</code> . If the <code>Index</code> value is a <code>String</code> , it must match an existing object name in the collection.
<code>RemoveItems</code>		Removes specified items from the group. Note that items removed from a group remain in the drawing; they're no longer associated with the group. Parameter: <code>Objects As Variant</code> .
<code>Update</code>		Updates the object to the drawing screen.

AcadGroup Object Properties

In addition to the properties inherited from the `AcadObject` object, the `AcadGroup` object supports the following properties.

Name	Returns	Description
<code>Color</code>	<code>AcColor</code>	Gets or sets the color of the group object. Use a color index number from 0 to 256 or one of the constants listed here: <code>acByBlock</code> (where AutoCAD draws objects in the default color) or <code>acByLayer</code> (where AutoCAD draws objects in the color specified for the layer). For a list of possible values for the <code>AcColor</code> enumerated type, see Appendix B.
<code>Count</code>	<code>Integer</code>	Gets the number of items in the group. This property's value is read-only.
<code>Layer</code>	<code>String</code>	Gets or sets the layer for the group.
<code>Linetype</code>	<code>String</code>	Gets or sets the linetype of the group.
<code>LinetypeScale</code>	<code>Double</code>	Gets or sets the linetype scale of the group. The default value is 1.0.
<code>Lineweight</code>	<code>AcLineweight</code>	Gets or sets the lineweight of the group. For a list of possible values for the <code>AcLineweight</code> enumerated type, see Appendix B.
<code>Name</code>	<code>String</code>	Gets or sets the name of the group.
<code>PlotStyleName</code>	<code>String</code>	Gets or sets the group's plot style name.
<code>Visible</code>	<code>Boolean</code>	Specifies whether the group is visible.

Note Because this object inherits from `AcadObject`, it supports the `Modified` event.

AcadGroups Collection

The `AcadGroups` collection contains all the groups in the drawing. Although this collection inherits a `Delete` method, you can't actually delete it. If you need to delete a specific group, use the `Delete` method found in the `AcadGroup` object. There is no limit to the number of groups you can create in your drawing. However, there can be only one instance of the `AcadGroups` collection, which is predefined for each drawing. You can make multiple references to it by using the `Groups` property.

AcadGroups Collection Methods

In addition to the methods inherited from the `AcadObject` object, the `AcadGroups` collection supports the following methods.

Name	Returns	Description
<code>Add</code>	<code>AcadGroup</code>	Creates a member object and adds it to the collection. Parameter: <code>Name As String</code> .
<code>Item</code>	<code>AcadGroup</code>	Gets the member object at a given index in a group. Parameter: <code>Index As Variant</code> (an <code>Integer</code> or a <code>String</code>). If the <code>Index</code> value is a <code>String</code> , it must match an existing group in the collection.

AcadGroups Collection Properties

The `AcadGroups` collection supports the `Count` property, the common `Application` property, and the properties inherited from the `AcadObject` object. It supports no other properties.

Note Because this collection inherits from `AcadObject`, it supports the `Modified` event.

AcadHatch Object

The `AcadHatch` object represents an area fill consisting of a pattern of lines. It is created using the `AddHatch` method of the `AcadBlock`, `AcadModelSpace`, or `AcadPaperSpace` object. After creating the hatch, what you must do next is add the outer loop using the `AppendOuterLoop` method. You have to do this for the hatch to be a valid AutoCAD object; otherwise, AutoCAD will enter an unpredictable state. Once you've successfully created the outer loop, you can add any inner loops using the `AppendInnerLoop` method or any other operation carried out on the hatch.

AcadHatch Object Methods

In addition to the methods inherited from the `AcadObject` object, the `AcadHatch` object supports the following methods.

Name	Description
AppendInnerLoop	Appends an inner loop to the hatch. You must add the outer loop before you can add any inner loop. Parameter: <code>ObjectArray As Variant</code> . The parameter is an array of objects forming a closed boundary, which can consist of one or more objects. If more than one object is used, their end points must coincide for the loop to be created properly. The loop may contain the following types of objects: <code>AcadLine</code> , <code>AcadPolyline</code> , <code>AcadCircle</code> , <code>AcadEllipse</code> , <code>AcadSpline</code> , and <code>AcadRegion</code> . It's important that the object array used in the call to <code>AppendInnerLoop</code> forms a closed boundary.
AppendOuterLoop	Appends an outer loop to the hatch. The outer loop must be closed and must be created before any inner loops can be added. Parameter: <code>ObjectArray As Variant</code> . For a description of the parameter, see the <code>AppendInnerLoop</code> method. It's important that the object array used in the call to <code>AppendOuterLoop</code> forms a closed boundary.
Evaluate	Evaluates the hatch lines or solid fill using the specified hatch pattern. For regular hatch patterns, this method performs intersection calculations between pattern definition lines and hatch boundary curves to form hatch lines. For solid fill hatch patterns, this method performs triangulation of the hatch area and fills in the triangular meshes with the given color. AutoCAD may not succeed in evaluating an <code>AcadHatch</code> object when either the inner or outer loop is too small for the hatch patterns and will return an ambiguous output error.
GetLoopAt	Gets the objects used to define a loop at the given index of the hatch. Parameters: <code>Index As Integer</code> and <code>ObjectArray As Variant</code> . The <code>Index</code> parameter can take any value from 0 (the index for the outer loop) to the index of the last loop created with <code>AppendInnerLoop</code> . The second parameter is an object or an array of objects that makes up the loop.
InsertLoopAt	Inserts a loop at a given index of a hatch. Parameters: <code>Index As Integer</code> , <code>LoopType As AcLoopType</code> , and <code>ObjectArray As Variant</code> . Values for the <code>AcLoopType</code> enumeration are given in Appendix B. For a description of the third parameter, see the <code>AppendInnerLoop</code> method.
SetPattern	Sets the pattern type and name of the hatch. Note that the integer value for the pattern type is supplied. Parameters: <code>PatternType As AcPatternType</code> and <code>PatternName As String</code> . See the <code>AcPatternType</code> enumeration table in Appendix B.

AcadHatch Object Properties

In addition to the properties inherited from the `AcadEntity` object, the `AcadObject` object, and the common `Application` property, the `AcadHatch` object supports the following properties.

Name	Returns	Description
AssociativeHatch	Boolean	Specifies whether the hatch is associative. An associative hatch is updated when its boundaries are modified.
Elevation	Double	Gets or sets the current elevation of the hatch object.
Gradient1	<code>AcCmColor</code>	Gets or sets the start color of the gradient. It must be one of the constants <code>acColorMethodByACI</code> or <code>acColorMethodByBlock</code> .
Gradient2	<code>AcCmColor</code>	Gets or sets the end color of the gradient. It must be one of the constants <code>acColorMethodByACI</code> or <code>acColorMethodByBlock</code> .

Name	Returns	Description
GradientCentered	Boolean	Gets or sets whether the gradient is centered.
GradientName	String	Gets or sets the pattern name. The pattern name for the gradient may be LINEAR, CYLINDER, INVCYLINDER, SPHERICAL, HEMISPHERICAL, CURVED, INVSPHERICAL, INVHEMISPHERICAL, or INVCURVED.
HatchObjectType	AcHatchObject	Gets or sets the hatch object type to be hatch, classic, or gradient. The default value of this property is 0, acHatchObject, which creates a classic hatch. If the value of this property is 1, acGradientObject, a gradient is created by the AddHatch method for hatch creation. If a gradient is created, then PatternType should be AcGradientPatternType and PatternName should contain a gradient pattern name of LINEAR, CYLINDER, INVCYLINDER, SPHERICAL, HEMISPHERICAL, CURVED, INVSPHERICAL, INVHEMISPHERICAL, or INVCURVED.
HatchStyle	AcHatchStyle	Gets or sets the hatch style. For a list of possible values for the AcHatchStyle enumerated type, see Appendix B.
ISOPenWidth	AcISOPenWidth	Gets or sets the ISO pen width of an ISO hatch pattern. For a list of possible values for the AcISOPenWidth enumerated type, see Appendix B. When you query a hatch and receive a value of acPenWidthUnk, use the PatternScale property to obtain the nonstandard value. Setting the pen width of any hatch to acPenWidthUnk has no effect.
Normal	Variant	Gets or sets the 3-D (Z-axis) normal unit vector for the hatch object.
NumberOfLoops	Integer	Gets the number of loops in the hatch boundary. This property's value is read-only.
PatternAngle	Double	Gets or sets the angle of the hatch pattern in radians. The valid range is 0 to 6.28. The value is stored in the HPANG system variable.
PatternDouble	Boolean	Specifies whether the user-defined hatch is double-hatched. The value of this property is stored in the HPDOUBLE system variable. If the PatternType property is set to acHatchPatternTypePreDefined or acHatchPatternTypeCustomDefined, this property isn't used.
PatternName	String	Gets or sets the pattern name of the hatch, which can be up to 34 characters long but can't have spaces (by default). To remove the default setting, enter a period (.), and a NULL string is returned. If the PatternType property is set to acHatchPatternTypePreDefined, then this property refers to the name of a hatch pattern in acad.pat. If the PatternType property is set to acHatchPatternTypeCustomDefined, then this property refers to the name of a hatch pattern in a custom .pat file. If the PatternType property is set to acHatchPatternTypeUserDefined, then this property isn't used. The value of this property is stored in the HPNAME system variable. This property's value is read-only.

Continued

Name	Returns	Description
PatternScale	Double	Gets or sets the hatch pattern scale. The value of this property must not be zero, and it is stored in the HPSCALE system variable. If the PatternType property is set to acHatchPatternTypeUserDefined, then this property isn't used.
PatternSpace	Double	Gets or sets the user-defined hatch pattern spacing. This value is also controlled by the HPSPACE system variable. If the PatternType property is set to acHatchPatternTypePreDefined or acHatchPatternTypeCustomDefined, then this property isn't used.
PatternType	AcPatternType	Gets the pattern type of the hatch. For a list of possible values for the AcPatternType enumerated type, see Appendix B. This property's value is read-only.

Note Because this object inherits from AcadObject, it supports the Modified event.

AcadHyperlink Object

The AcadHyperlink object represents a URL and URL description, which is stored within the XData of the object's corresponding entity and not in the drawing itself. This means a hyperlink object is created every time a URL is requested, and the name and description is read from the XData. In this manner, it's similar to the AcadUtility object. You should take care to not create multiple hyperlink objects referencing the same URL. If one such object is updated, the others won't be.

AcadHyperlink Object Method

The AcadHyperlink object doesn't inherit from AcadEntity or AcadObject. It supports only a single method.

Name	Description
Delete	Deletes a specified hyperlink object

AcadHyperlink Object Properties

In addition to the common Application property, the AcadHyperlink object supports the following properties.

Name	Returns	Description
URL	String	Gets or sets the URL for the hyperlink object. The file pointed to by the URL can be stored locally on a network drive or via an Internet connection.
URLDescription	String	Gets or sets the URL description for the hyperlink object. This is useful if the URL name itself isn't very helpful in identifying the contents of the file.
URLNamedLocation	String	Gets or sets the named location for the hyperlink object. This can be another AutoCAD drawing or a completely different application, such as a word processing program. If you specify a named view to jump to in an AutoCAD drawing, AutoCAD restores that view when the hyperlink is opened.

AcadHyperlinks Collection

The AcadHyperlinks collection contains all the hyperlinks for a given entity. To add a new member to the collection, use the Add method. To select a specific hyperlink, use the Item method. It is accessed via the Hyperlinks property on all AcadEntity objects.

AcadHyperlinks Collection Methods

The AcadHyperlinks collection doesn't inherit from AcadObject. It supports the following methods.

Name	Returns	Description
Add	AcadHyperlink	Creates a named AcadHyperlink object and adds it to the collection. Parameters: Name As String, [Description As Variant], and [NamedLocation As Variant].
Item	AcadHyperlink	Gets the member object at a given index in a collection, group, or selection set. Parameter: Index As Integer (An Integer of a String). If the value for Index is a String, it must match an existing object name in the collection.

AcadHyperlinks Collection Properties

The AcadHyperlinks collection supports the common Application and Count properties. It supports no other properties.

AcadIDPair Object

The AcadIDPair object is a transient object used as an optional parameter in the CopyObjects method of the AcadDatabase and AcadDocument objects. It contains the object IDs of both the source and destination objects. See the AcadDatabase object for more details. The CopyObjects method populates the AcadIDPair object with the object IDs of the source objects and the newly copied objects, thus creating a map between all the objects copied and created. This object also contains information about the copy and translation process.

Note that this object is used for advanced cloning functions. General users should avoid using this optional functionality.

The AcadIDPair object has no methods.

AcadIDPair Object Properties

The AcadIDPair object supports the common Application property and the following properties.

Name	Returns	Description
IsCloned	Boolean	Determines whether the source object in a CopyObjects operation has been cloned. This property's value is read-only.
IsOwnerXlated	Boolean	Determines whether the owning object in a CopyObjects operation has been translated. This property's value is read-only.
IsPrimary	Boolean	Determines whether the source object in a CopyObjects operation was part of the primary set of objects being copied or if it was simply owned by a member in the primary set. This property's value is read-only.
Key	Integer	The object ID of the source object in the CopyObjects operation. This property's value is read-only.
Value	Integer	The object ID of the newly created cloned object in the CopyObjects operation. This property's value is read-only.

AcadLayer Object

The AcadLayer object represents a logical grouping of data, similar to transparent acetate overlays on a drawing. It is created using the Add method of the AcadLayers collection and can be accessed by the Item method of the AcadLayers collection. All new objects are added to the active layer. To make a layer active, use the ActiveLayer property of the Document object.

AcadLayer Object Methods

The AcadLayer object supports the methods inherited from the AcadObject object and no other methods. Note that layer 0 can't be deleted and layer DEFPOINTS shouldn't. This restriction applies also to layers referenced in a block definition.

AcadLayer Object Properties

In addition to the properties inherited from the AcadObject object, and the common Application property, the AcadLayer object supports the following properties.

Name	Returns	Description
Color	AcColor	Gets or sets the color of the layer object, which will determine the acByLayer color of the entities on the specific layer. Use a color index number from 0 to 256 or the AcColor enumeration (see Appendix B).
Freeze	Boolean	Specifies the freeze status of a layer: True if frozen, False if thawed. Freezing layers makes them invisible and excludes them from regeneration and plotting. Thawing a layer enables these capabilities. You can't freeze the active layer or make a frozen layer active.
LayerOn	Boolean	Specifies whether a layer is on or off. Layers that are "off" aren't displayed or plotted, though they're regenerated when the drawing is regenerated.
Linetype	String	Gets or sets the linetype for the layer. The default linetype for any new entity will become the linetype of the layer (ByLayer).

Name	Returns	Description
Lineweight	AcLineWeight	Gets or sets the lineweight for the layer. For a list of possible values for the AcLineWeight enumerated type, see Appendix B.
Lock	Boolean	Specifies whether a layer is locked. You can't edit objects on a locked layer, though they're still visible if the layer is on and thawed. You can activate locked layers and add objects to them, apply snap modes, freeze and turn them off, and change their associated colors.
Name	String	Gets or sets the name of the layer. Layer 0 can't be renamed. Layer DEFPOINTS is used for dimensioning shouldn't be renamed either.
PlotStyleName	String	Gets or sets the layer's plot style name.
Plottable	Boolean	Specifies whether the layer is plottable.
ViewportDefault	Boolean	Specifies whether the layer is to be frozen in new viewports.

Note Because this object inherits from AcadObject, it supports the Modified event.

AcadLayers Collection

The AcadLayers collection contains all the layers in the drawing. Although this collection inherits a Delete method, you can't actually delete this collection. If you need to delete a specific layer, use the Delete method found in the AcadLayer object. There is no limit to the number of layers you can create in your drawing. However, there can be only one instance of the AcadLayers collection, which is predefined for each drawing. You can make multiple references to it by using the Layers property.

AcadLayers Collection Methods

In addition to the methods inherited from the AcadObject object, the AcadLayers collection supports the following methods.

Name	Returns	Description
Add	AcadLayer	Creates a member object and adds it to the collection. Parameter: Name As String.
Item	AcadLayer	Gets the member object at a given index in a collection, group, or selection set. Parameter: Index As Variant (an Integer or a String). If the Index value is a String, it must match an existing object name in the collection.

AcadLayers Collection Properties

The AcadLayers collection supports the Count property, the common Application property, and the properties inherited from the AcadObject object. It supports no other properties.

Note Because this object inherits from AcadObject, it supports the Modified event.

AcadLayout and AcadPlotConfiguration Objects

The AcadLayout object contains the plot settings and visual properties of a model space or paper space block, as displayed in the AutoCAD user interface. Each AcadLayout object is associated with a single AcadBlock object, which can be accessed through the Block property. Alternatively, the AcadLayout object can be accessed through the Item property of the AcadLayouts collection, the Layout property of the AcadBlock (when IsLayout is set to True), the AcadModelSpace and AcadPaperSpace objects, or the ActiveLayout property of the Document object.

The AcadPlotConfiguration object is similar to the AcadLayout object in that it contains the same plot settings and visual properties. The difference between the two is as follows: whereas a layout is defined only for a particular block, a plot configuration isn't specific to any one block; it's a named collection of plot settings that can be applied to any geometry.

As the two objects are all but identical, I will present their methods and properties together.

AcadLayout and AcadPlotConfiguration Objects: Methods

The AcadLayout and AcadPlotConfiguration objects support all the methods inherited from the AcadObject object. They also support the following methods.

Name	Returns	Description
CopyFrom		Copies the settings from the given plot configuration. Parameter: PlotConfig As AcadPlotConfiguration. This method is specific to the AcadPlotConfiguration object.
GetCanonicalMediaNames	Variant	Returns all available canonical media names for the specified plot device. You should call RefreshPlotDeviceInfo before calling this method. This method returns the names as an array of strings.
GetCustomScale		Gets the custom scale for a layout or plot configuration, in millimeters or inches, depending on the value of the PaperUnits property. Parameters: Numerator As Double and Denominator As Double.
GetLocaleMediaName	String	Given a canonical media name, this method returns the localized version of the canonical media name. Parameter: Name As String.
GetPaperMargins		Gets the margins offsets for the layout or plot configuration, the units of which depend on the values of the PaperUnits property. The print origin is offset from the edge of the paper by the values returned by this method. Parameters: LowerLeft As Variant and UpperRight As Variant. (Both parameters are two-element arrays of Doubles.)
GetPaperSize		Gets the width and height of the configured paper using the values specified by the PaperUnits property. Parameters: Width As Double and Height As Double.
GetPlotDeviceNames	Variant	Returns all available plot device names. You should call RefreshPlotDeviceInfo before calling this method. This method returns the names as an array of strings.
GetPlotStyleTableNames	Variant	Returns all available plot style table names. You should call RefreshPlotDeviceInfo before calling this method. This method returns the names as an array of strings.

Name	Returns	Description
GetWindowToPlot		Gets the coordinates that define the portion of the layout to plot, the units of which depend on the values of the PaperUnits property. The coordinates of the window are taken from the origin. For these coordinates to be used for the plot, the PlotType property must be set to acWindows. Parameters: LowerLeft As Variant and UpperRight As Variant. (Both parameters are two-element arrays of Doubles.)
RefreshPlotDeviceInfo		Updates the plot, canonical media, and plot-style table information to reflect the current system state. You should call this method once before calling the GetCanonicalMediaNames, GetPlotDeviceNames, or GetPlotStyleTableNames method.
SetCustomScale		Sets the custom scale for a layout or plot configuration, in millimeters or inches depending on the value of the PaperUnits property. Parameters: Numerator As Double and Denominator As Double. Note that the parameters must be greater than zero.
SetWindowToPlot		Sets the coordinates that define the portion of the layout to plot, the units of which depend on the values of the PaperUnits property. The coordinates of the window are taken from the origin. For these coordinates that can be used for the plot, the PlotType property must be set to acWindows. Parameters: LowerLeft As Variant and UpperRight As Variant.

AcadLayout and AcadPlotConfiguration Objects: Properties

The AcadLayout and AcadPlotConfiguration objects inherit all the properties inherited from the AcadObject object, as well as the common Application property. They also support the following properties.

Name	Returns	Description
Block	AcadBlock	This property is specific to the AcadLayout object. It gets the block associated with the layout. This property's value is read-only.
CanonicalMediaName	String	Gets or sets the name of the media that specifies the paper size. Note that changes to this property won't be visible until the drawing is regenerated.
CenterPlot	Boolean	Specifies whether there is centering of the plot on the media. Note that changes to this property won't be visible until the drawing is regenerated. Note also that this property can't be set to True if the PlotType property is set to acLayout.
ConfigName	String	Gets or sets the name of the PC3 file or print device to be used by the layout or plot configuration. Note that this property takes only the file name for the configuration file. Use the PrinterConfigPath property to specify the path for printer configuration files.

Continued

Name	Returns	Description
ModelType	Boolean	Specifies whether the plot configuration applies only to model space or to all layouts. If <code>True</code> , it applies only to the model space layout; if <code>False</code> , it applies to all layouts. This property's value is read-only.
Name	String	Gets or sets the name of the object.
PaperUnits	AcPlotPaperUnits	Gets or sets the units for the display of the layout or plot configuration in the user interface. Note that changes to this property won't be visible until the drawing is regenerated. Possible values for the <code>AcPlotPaperUnits</code> enumeration are <code>acInches</code> , <code>acMillimeters</code> , and <code>acPixels</code> .
PlotHidden	Boolean	Specifies whether objects in the paper space are to be hidden during a plot. This is <code>True</code> if objects are to be hidden and <code>False</code> otherwise. Note that this property does <i>not</i> affect objects inside floating model space viewports.
PlotOrigin	Variant	Gets or sets the origin of the UCS, block, layout, or raster image in WCS coordinates. It returns a two-element array of <code>Doubles</code> representing the XY coordinates of the origin. The origin is offset from the media edge by the paper margin and is given in millimeters. Note that changes to this property won't be visible until the drawing is regenerated.
PlotRotation	AcPlotRotation	Gets or sets the rotation angle for the layout or plot configuration relative to the X-axis, with positive values going counterclockwise when viewed along the Z-axis toward the origin. Note that changes to this property won't be visible until the drawing is regenerated. Possible values for the <code>AcPlotRotation</code> enumeration are <code>ac0degrees</code> , <code>ac90degrees</code> , <code>ac180degrees</code> , and <code>ac270degrees</code> .
PlotType	AcPlotType	Gets or sets the type of layout or plot configuration. Note that changes to this property won't be visible until the drawing is regenerated. Note also that the <code>ViewToPlot</code> property or <code>SetWindowToPlot</code> method needs to be called before you can set the <code>PlotType</code> to <code>acView</code> or <code>acWindow</code> , respectively. For a list of possible values for the <code>AcPlotType</code> enumerated type, see Appendix B.
PlotViewportBorders	Boolean	Specifies whether the viewport borders are to be plotted.
PlotViewportsFirst	Boolean	Specifies whether all geometry in paper space viewports is plotted first rather than the geometry in the paper space. This is <code>True</code> if geometry in the paper space viewports is plotted first and <code>False</code> otherwise.
PlotWithLineweights	Boolean	Specifies whether objects are plotted with the line-weights assigned from a plot style or from a drawing file. This is <code>True</code> if the plot uses the lineweights assigned from a plot style and <code>False</code> otherwise.
PlotWithPlotStyles	Boolean	Specifies whether objects are plotted with the plot configuration assigned from a plot style or from a drawing file. This is <code>True</code> if the plot uses the plot configuration assigned from a plot style and <code>False</code> otherwise.

Name	Returns	Description
ScaleLineweights	Boolean	Specifies whether the lineweight is scaled with the rest of the geometry when a layout is printed. The value of this property is stored in the LWSCALE system variable. Note that this property is disabled for the model space layout.
ShowPlotStyles	Boolean	Specifies whether plot styles are to be used in the plot (True) or use the plot styles assigned to objects in the drawing (False).
StandardScale	AcPlotScale	Gets or sets the standard scale for the layout, viewport, or plot configuration. For a list of possible values for the AcPlotScale enumerated type, see Appendix B.
StyleSheet	String	Gets or sets the name of the style sheet for the layout or plot configuration.
TabOrder	Integer	Gets or sets the tab order of a layout. Controls the order in which the layouts are displayed in the tab control. The tab order should be unique and sequential among all layouts in the database. The model space tab must have a tab order of 0. Paper space tabs must have a tab order of 1 or greater. This applies to the AcadLayout object only.
UseStandardScale	Boolean	Specifies whether the plot is to use a standard or custom scale. This is True if the standard scale is to be used and False otherwise. See the SetCustomScale method.
ViewToPlot	String	Gets or sets the name of the view to be plotted. Note that the PlotType property for the layout or plot configuration has to be set to acView.

Note Because these objects inherit from AcadObject, they both support the Modified event.

AcadLayouts and AcadPlotConfigurations Collections

The AcadLayouts collection contains all the AcadLayout objects in the drawing. It is accessed by the Layouts property of the Document object. It is created when the document is created and can't be deleted (even though the collection inherits a Delete method from AcadObject). To remove a layout definition, use the Delete method of the AcadLayout object.

The AcadPlotConfigurations collection is similar to the AcadLayouts collection, except it contains all the AcadPlotConfiguration objects for a project. Whereas a layout is specific to a particular block, a plot configuration contains a named collection of plot settings that can be applied to any geometry. Once again, this collection inherits a Delete method that you can't use. To remove a plot configuration, use the Delete method of the AcadPlotConfiguration object.

As the two collections are all but identical, I will present their methods and properties together.

AcadLayouts and AcadPlotConfigurations Collections: Methods

The AcadLayouts and AcadPlotConfigurations collections support all the methods inherited from the AcadObject object. They also support the following methods.

Name	Returns	Description
Add		Creates a named member object and adds it to the appropriate collection. Parameter: Name As String. Additional optional parameter (for the AcadPlotConfigurations collection only): ModelType As Variant (this is True if the plot configuration applies to a model space tab and False otherwise).
Item		Gets the member object at a given index in a collection. Parameter: Index As Variant (an Integer or a String). If the Index value is a String, it must match an existing object name in the collection.

AcadLayouts and AcadPlotConfigurations Collections: Properties

The AcadLayouts and AcadPlotConfigurations collections support all the properties inherited from the AcadObject object, the Count property, and the common Application property. They support no other properties.

Note Because these collections inherit from AcadObject, they both support the Modified event.

AcadLeader Object

A leader line is an object that can be composed of an arrowhead attached to either splines or straight-line segments. In addition, a short horizontal line (hook line) connects descriptive text and/or feature control frames to the leader line. The AcadLeader object specifies such a complex leader line, and the Leader property of the AcadDimDiametric and AcadDimRadial objects specifies a simple leader line.

The AcadLeader object is created using the AddLeader method of the AcadBlock, AcadModelSpace, or AcadPaperSpace object. Any annotation attached to the leader line is defined in the Annotation property of the AcadLeader object.

AcadLeader Object Methods

The AcadLeader object inherits all the methods of the AcadEntity and AcadObject objects. It also supports the following method.

Name	Description
Evaluate	Evaluates the relative position of the leader line to its associated annotation and updates the leader line's geometry if necessary.

AcadLeader Object Properties

The AcadLeader object inherits all the properties of the AcadEntity and AcadObject objects, as well as the common Application property. It also supports the following properties.

Name	Returns	Description
Annotation	AcadEntity	Gets or sets the annotation object for a leader, which can be a tolerance object, a multiline text object, or a block reference. Note that setting the Annotation property to a new annotation object modifies the drawing but returns an invalid input error.
ArrowheadBlock	String	Gets or sets the name of the block to use as the custom arrowhead for a leader line.
ArrowheadSize	Double	Gets or sets the size of the leader line arrowheads and hook lines. The initial value for this property is 0.1800.
ArrowheadType	AcDimArrowheadType	Gets or sets the type of arrowhead used for the leader line. The initial value for this property is acArrowDefault. Note that if you use the ArrowheadBlock property to specify a block to use as a custom arrowhead, this property will be set to acArrowUserDefined. For a list of possible values for the AcDimArrowheadType enumerated type, see Appendix B.
Coordinate	Variant	Gets or sets the coordinate of a single vertex in the object. This will replace the existing vertex for the specified object. Use standard array-handling techniques to process the values contained in this property. It returns a three-element array of Doubles containing 3-D coordinates in WCS. Note that the Z coordinate will default to 0 on the active UCS. Parameter: Index As Integer (the index in the zero-based array for the vertex you want to set or query).
Coordinates	Variant	Gets or sets the coordinates for each vertex in the object. This will replace any existing coordinates for the specified object. Use standard array-handling techniques to process the coordinates contained in this property. Note that you can't change the number of coordinates in the object by using this property; you can change only the location of existing coordinates. It returns a three-element array of Doubles containing 3-D coordinates in WCS. Note also that the Z coordinate will always default to 0 on the active UCS.
DimensionLineColor	AcColor	Gets or sets the color of the leader object. Use a color index number from 0 to 256 or one of the constants listed here: acByBlock (where AutoCAD draws objects in the default color), acByLayer (where AutoCAD draws objects in the color specified for the layer), or the AcColor enumeration (see Appendix B). This property overrides the value of the DIMCLRD system variable for the given leader object.
DimensionLineWeight	AcLineWeight	Gets or sets the lineweight for the leader lines. See Appendix B for a list of the values of the AcLineWeight enumeration.
Normal	Variant	Gets or sets the 3-D (Z-axis) normal unit vector for the leader line.

Continued

Name	Returns	Description
ScaleFactor	Double	Gets or sets the scale factor for the object. It has to be a value greater than 0.0. A scale factor greater than 1 enlarges the hook line. A scale factor from 0 to 1 shrinks the hook line. The initial value for this property is 1.0000. This property overrides the DIMSCALE system variable.
StyleName	String	Gets or sets the name of the style used with the object, the default being the current style. Use the TextStyle object to change the attributes of a given text style. Note that the name given must already be defined in the drawing.
TextGap	Double	Gets or sets the gap between the annotation and a hook line on a leader object.
Type	AcLeaderType	Gets or sets type of a leader line. For a list of possible values for the AcLeaderType enumerated type, see Appendix B.
VerticalTextPosition	AcDimVerticalJustification	Gets or sets the vertical position of text in relation to the leader line. This property overrides the value of the DIMTAD system variable for the given dimension, leader, or tolerance object. The initial value for this property is acVertCentered. (For a list of the values of the AcDimVerticalJustification enumerated type, see Appendix B.)

Note Because this object inherits from AcadObject, it supports the Modified event.

AcadLine Object

The AcadLine object represents a single line segment. In AutoCAD, lines can be one segment or a series of connected segments, but each is a separate AcadLine object. You use this object if you want to edit individual segments. If, however, you need to draw a series of line segments as a single object, use the AcadLWPolyline object. The AcadLine object is created using the AddLine method of the AcadBlock, AcadModelSpace, or AcadPaperSpace object.

AcadLine Object Methods

The AcadLine object inherits all the methods of the AcadEntity and AcadObject objects. It also supports the following method.

Name	Returns	Description
Offset	Variant	Creates a new line by offsetting the current line by a specified distance, which can be positive or negative but can't be zero. If the offset is negative, this means that the line is drawn closer to the WCS origin. Parameter: Distance As Double.

AcadLine Object Properties

The AcadLine object inherits all the properties of the AcadEntity and AcadObject objects, as well as the common Application property. It also supports the following properties.

Name	Returns	Description
Angle	Double	Gets the angle of the line in degrees, starting at the X-axis and proceeding counterclockwise. This property's value is read-only.
Delta	Variant	Gets the delta of the line. It returns a three-element array of Doubles representing the deltaX, deltaY, and deltaZ values for the line. This property's value is read-only.
EndPoint	Variant	Gets or sets a set of 3-D coordinates representing the end point of the line.
Length	Double	Gets the length of the line. This property's value is read-only.
Normal	Variant	Gets or sets the 3-D (Z-axis) normal unit vector for the line object.
StartPoint	Variant	Gets or sets the set of 3-D WCS coordinates representing the start point for the line.
Thickness	Double	Gets or sets the distance the AcadLine object is extruded above or below its elevation, in a Z-axis direction. The default is 0.0.

Note Because this object inherits from AcadObject, it supports the Modified event.

AcadLinetype Object

The AcadLinetype object contains line definitions or components consisting of combinations of dashes, dots, spaces, text, and shapes. You can create default linetypes programmatically only using the Add method of the AcadLinetypes collection. However, you can load an existing linetype definition to your drawing by using the Load method of the AcadLinetypes collection. Although there is no limit to the number of linetypes defined in any one drawing, only one can be active at a time. New objects are created using the linetype specified for the active layer. If no such linetype exists for the active layer, new objects are created with the active linetype. Linetypes are activated using the ActiveLinetype property.

AcadLinetype Object Methods

The AcadLinetype object inherits all the methods of the AcadObject object. It supports no other methods.

AcadLinetype Object Properties

The AcadLinetype object inherits all the properties of the AcadObject object, as well as the common Application property. It also supports the following properties.

Name	Returns	Description
Description	String	Gets or sets a text description of the linetype object. This can be up to 47 characters long and can contain text or a series of characters to represent the linetype pattern itself (using underscores, dots, dashes, and spaces).
Name	String	Gets or sets the name of the linetype definition.

Note Because this object inherits from `AcadObject`, it supports the `Modified` event.

AcadLineTypes Collection

The `AcadLineTypes` collection contains all the linetypes defined for a drawing. You use the `Add` method to add a new linetype definition to the collection, and you can select a particular linetype using the `Item` method. You can load an existing linetype definition using the `Load` method. The collection also inherits a `Delete` method from `AcadObject`, but you can't use it, as the collection can't be deleted. To remove a linetype, use the `Delete` method of the `AcadLinetype` object. There is no limit to the number of linetypes you can create in your drawing. However, there can be only one instance of the `AcadLineTypes` collection, which is predefined for each drawing. You can make multiple references to it by using the `Linetypes` property.

AcadLineTypes Collection Methods

The `AcadLineTypes` collection inherits all the methods of the `AcadObject` object. It also supports the following methods.

Name	Returns	Description
Add	AcadLinetype	Creates a named <code>AcadLinetype</code> object and adds it to the collection. Parameter: Name As String.
Item	AcadLinetype	Gets the <code>AcadLinetype</code> object at a given index in the collection. Parameter: Index As Variant (an Integer or a String). If the Index value is a String, it must match an existing linetype definition.
Load		Loads a definition of a named linetype from a library (.lin) file. Parameters: Name As String and FileName As String.

AcadLineTypes Collection Properties

The `AcadLineTypes` collection inherits all the properties of the `AcadObject` object and supports the `Count` property and the common `Application` property. It supports no other properties.

Note Because this collection inherits from `AcadObject`, it supports the `Modified` event.

AcadLWPolyline Object

The AcadLWPolyline object represents a 2-D line of adjustable width that's composed of line and arc segments. It's created using the AddLWPolyline method of the AcadBlock, AcadModelSpace, or AcadPaperSpace object.

AcadLWPolyline Object Methods

The AcadLWPolyline object inherits all the methods of the AcadEntity and AcadObject objects. It also supports the following methods.

Name	Returns	Description
AddVertex		Adds a vertex to the lightweight polyline, a vertex being the end point for a new line segment. See the SetBulge method for creating an arc. Each vertex is added to a three-element array of Doubles. Parameters: Index As Integer and Vertex As Variant. (Index refers to the position in the array where the new vertex is added. Vertex refers to the 3-D OCS coordinates where the vertex is to be created.)
Explode	Variant	Explodes the lightweight polyline and returns the subentities as an array of Objects (AcadLine and AcadArc).
GetBulge	Double	Returns the bulge value at a given index of the lightweight polyline. A bulge is the tangent of 1/4 of the included angle for the arc between the selected vertex and the next vertex in the polyline's vertex list. A negative bulge value indicates that the arc goes clockwise from the selected vertex to the next vertex. A bulge of 0 indicates a straight segment, and a bulge of 1 is a semicircle. Parameter: Index As Integer.
GetWidth		Returns segment width at a given index of the lightweight polyline. Parameters: Index As Integer, StartWidth As Double, and EndWidth As Double.
Offset	Variant	Creates a new lightweight polyline by offsetting the existing polyline by a given distance, which must be nonzero. If the offset is negative, this means that the line is drawn closer to the WCS origin. Parameter: Distance As Double.
SetBulge		Sets the vertex bulge of the lightweight polyline at a given index. See GetBulge for the definition of a bulge. Parameters: Index As Integer and bulge As Double.
SetWidth		Sets the segment width of the lightweight polyline for a given index. Parameters: Index As Integer, StartWidth As Double, and EndWidth As Double.

AcadLWPolyline Object Properties

The AcadLWPolyline object inherits all the properties of the AcadEntity and AcadObject objects, as well as the common Application property. It also supports the following properties.

Name	Returns	Description
Area	Double	Gets the area of a lightweight polyline in square drawing units. The area is computed as though a straight line connects the start and end points. This property's value is read-only.
Closed	Boolean	Specifies whether the lightweight polyline is closed (True) or open (False). Open is the default.
ConstantWidth	Double	Gets or sets a global width for all segments in a lightweight polyline.
Coordinate	Variant	Gets or sets the coordinate of a single vertex in the object. This will replace any existing vertices for the specified object. Use standard array-handling techniques to process the values contained in this property. This returns a two-element array of Doubles containing 2-D coordinates in OCS. These coordinates can be converted to and from other coordinate systems using the TranslateCoordinates method. Parameter: Index As Integer (the index in the zero-based array of vertices for the vertex you want to set or query).
Coordinates	Variant	Gets or sets the coordinates for each vertex in the object. This will replace any existing coordinates for the specified object. Use standard array-handling techniques to process the coordinates contained in this property. Note that you can't change the number of coordinates in the object by using this property; you can change only the location of existing coordinates. This returns the array point as a two-element array of Doubles containing 2-D points in OCS. These coordinates can be converted to and from other coordinate systems using the TranslateCoordinates method.
Elevation	Double	Gets or sets the current elevation of the lightweight polyline.
LinetypeGeneration	Boolean	Specifies the linetype generation of a lightweight polyline. If this is set to True, the linetype is generated in a continuous pattern through the polyline vertices; if it's set to False, the linetype is generated starting and ending with a dash at each polyline vertex.
Normal	Variant	Gets or sets the 3-D (Z-axis) normal unit vector for the lightweight polyline.
Thickness	Double	Gets or sets the distance the AcadLWPolyline object is extruded above or below its elevation, in a Z-axis direction. The default is 0.0.

Note Because this object inherits from AcadObject, it supports the Modified event.

AcadMenuBar Collection

The AcadMenuBar collection contains all the AcadPopupMenu objects displayed on the current AutoCAD menu bar. Any menu from a currently loaded menu group can be displayed on the menu bar. To display a menu on the menu bar, use the InsertInMenuBar method found on

the `AcadPopupMenu` object. To remove a menu from the current menu bar, use the `RemoveFromMenuBar` method also found on the `AcadPopupMenu` object.

AcadMenuBar Collection Methods

Just one method is supported.

Name	Returns	Description
Item	Object	Returns the object at the specified index in the collection. Parameter: Index As Variant (Integer or String). If the Index value is a String, it must match an existing object name in the collection.

AcadMenuBar Collection Properties

As well as the common `Application` and `Count` properties, the `AcadMenuBar` collection supports the following property.

Name	Returns	Description
Parent	AcadApplication	Returns the parent of the <code>AcadMenuBar</code> collection. This property's value is read-only.

AcadMenuGroup Object

The `AcadMenuGroup` object represents an AutoCAD menu group, which contains menus and toolbars, some or all of which may be currently displayed in AutoCAD. Each menu group contains an `AcadPopupMenu` collection and an `AcadToolbars` collection. The former collection contains all the menus within the menu group and can be accessed through the `Menus` property. The latter contains all the toolbars within the menu group and can be accessed through the `Toolbars` property. Menu groups are loaded into the AutoCAD session using the `Load` method, which adds the menu group to the `AcadMenuGroups` collection.

You can't create new menu groups. However, you can load an existing menu file containing a menu group and save it again with a new menu group name and to new menu file. The menus and toolbars thus become available for editing.

AcadMenuGroup Object Methods

Three methods are supported by the `AcadMenuGroup` object.

Name	Description
Save	Saves the menu group. Note that certain changes to the appearance of the AutoCAD user interface, such as changing system colors, will cause the menus and toolbars to be reloaded from the menu file. Be certain to save changes to your menu group often, using the <code>Save</code> method, to avoid losing your customizations. Parameter: <code>MenuFileType</code> As <code>AcMenuFileType</code> . For a list of possible values for the <code>AcMenuFileType</code> enumerated type, see Appendix B.
SaveAs	Saves the menu group and gives it the specified name using the full path and file name, or valid URL address, for the file. The active menu group takes on the new name. Parameters: <code>FileName</code> As String and <code>MenuFileType</code> As <code>AcMenuFileType</code> .
Unload	Unloads the menu group. When a menu group is unloaded from the drawing, any references to the menus and toolbars within that group become invalid. Always delete or set to NULL any references to toolbars and menus before unloading.

AcadMenuGroup Object Properties

As well as the common Application property, the AcadMenuGroup object supports the following properties.

Name	Returns	Description
MenuFileName	String	Gets the name of the file in which the menu group is located. This property's value is read-only.
Menus	AcadPopupMenu	Gets the AcadPopupMenu collection for the menu group.
Name	String	Gets the name of the menu group. The name is limited to 32 characters and can't contain spaces or punctuation marks. This property's value is read-only.
Parent	AcadMenuGroups	Gets the parent of the AcadMenuGroup object. This property's value is read-only.
Toolbars	AcadToolbars	Gets the AcadToolbars collection for the menu group.
Type	AcMenuGroupType	Gets the type of the menu group. This property's value is read-only. The possible values for the AcMenuGroupType enumeration are acBaseMenuGroup and acPartialMenuGroup.

AcadMenuGroups Collection

The AcadMenuGroups collection contains all the AcadMenuGroup objects, from which you can get a reference to all pop-up menus and toolbars loaded in the current AutoCAD session. To load a new menu group from a file into this collection, use the Load method. Note that you can't create new menu groups. However, you can load an existing menu group and save it out again with a new name and to new menu file. The menus and toolbars thus become available for editing. To unload a menu group from this collection, use the Unload method found on the AcadMenuGroup object.

AcadMenuGroups Collection Methods

Three methods are supported by the AcadMenuGroups object.

Name	Returns	Description
Item	AcadMenuGroup	Returns the object at the given index in the collection. Parameter: Index As Variant (Integer or String). If the Index value is a String, it must match an existing object name in the collection.
Load	AcadMenuGroup	Loads a menu group from a menu file. Parameters: MenuFileName As String and [BaseMenu As Variant]. The MenuFileName should have an .mnc, .mns, or .mnu extension. If the BaseMenu parameter is set to True, then the menu group will be loaded as a base menu, similar to the AutoCAD MENU command. If this parameter is omitted, the menu group will be loaded as a partial menu, similar to the AutoCAD MENULOAD command.

AcadMenuGroups Collection Properties

As well as the common Application property and the Count property, the AcadMenuGroups collection supports the following property.

Name	Returns	Description
Parent	AcadApplication	Gets the parent of the collection. This property's value is read-only.

AcadMInsertBlock Object

The AcadMInsertBlock object represents a rectangular array of block references. It is created using the AddMInsertBlock method of the AcadBlock, AcadModelSpace, or AcadPaperSpace object.

AcadMInsertBlock Object Methods

The AcadMInsertBlock object inherits all the methods of the AcadEntity and AcadObject objects. It also supports the following methods.

Name	Returns	Description
Explode	Variant	Explodes the block and returns the subentities as an array of AcadBlockReference objects.
GetAttributes	Variant	Returns an array of editable attribute references attached to the multiple block reference. After you edit one AcadAttribute object and update the AcadMInsertBlock object, changes will appear in each nested block because there's only one set of attribute references for the multiple inserted blocks.
GetConstantAttributes	Variant	Returns an array of constant attributes for the multiple blocks.

AcadMInsertBlock Object Properties

The AcadMInsertBlock object inherits all the properties of the AcadEntity and AcadObject objects, as well as the common Application property. It also supports the following properties.

Name	Returns	Description
Columns	Integer	Gets or sets the number of columns in a block array.
ColumnSpacing	Double	Gets or sets the spacing of the columns in a block array.
HasAttributes	Boolean	Indicates whether the block has any attributes in it. This property's value is read-only.
InsertionPoint	Variant	Represents an insertion point in the block as a set of 3-D WCS coordinates. It returns a three-element array of Doubles.
Name	String	Gets or sets the name of the multiple block, which is identical to the name of the block used in the AddMInsertBlock method.
Normal	Variant	Gets or sets the 3-D (Z-axis) normal unit vector for the block.

Continued

Name	Returns	Description
Rotation	Double	Gets or sets the rotation angle (in radians) for the block relative to the X-axis, with positive values going counterclockwise when viewed along the Z-axis toward the origin.
Rows	Integer	Gets or sets the number of rows in a block array.
RowSpacing	Double	Gets or sets the spacing of the rows in a block array.
XScaleFactor	Double	Gets or sets the X scale factor for the block.
YScaleFactor	Double	Gets or sets the Y scale factor for the block.
ZScaleFactor	Double	Gets or sets the Z scale factor for the block.

Note Because this object inherits from `AcadObject`, it supports the `Modified` event.

AcadMLine Object

The `AcadMLine` object represents multiple parallel lines. One `AcadMLine` object can comprise up to 16 lines. The object is created using the `AddMLine` method of the `AcadBlock`, `AcadModelSpace`, or `AcadPaperSpace` object.

AcadMLine Object Methods

The `AcadMLine` object inherits all the methods of the `AcadEntity` and `AcadObject` objects. It supports no other methods.

AcadMLine Object Properties

The `AcadMLine` object inherits all the properties of the `AcadEntity` and `AcadObject` objects, as well as the common `Application` property. It also supports the following properties.

Name	Returns	Description
Coordinates	Variant	Gets or sets the coordinates for each vertex in the object. This will replace any existing coordinates for the specified object. Use standard array-handling techniques to process the coordinates contained in this property. Note that you can't change the number of coordinates in the object by using this property; you can change only the location of existing coordinates. Returns a three-element array of Doubles containing 3-D coordinates in WCS. Note also that the Z coordinate will always default to 0 on the active UCS.
StyleName	String	Gets the MLine style name. This property's value is read-only.

Note Because this object inherits from `AcadObject`, it supports the `Modified` event.

AcadModelSpace Collection

The AcadModelSpace collection contains all the objects in model space. It is similar to the AcadBlock object, and it supports the same methods and properties. Moreover, objects contained in this collection can also be found in the blocks collection in a special block called *MODEL_SPACE, which is to say that both the block named *MODEL_SPACE and the AcadModelSpace collection actually point to the same data. The AcadModelSpace collection was created to provide a faster and easier means of accessing the data in model space. There is a single AcadModelSpace collection provided in the drawing. The collection can't be created or deleted. To add objects to the collection, you use the same Add* methods presented for the AcadBlock object.

Use model space for drafting and design work and to create 2-D drawings or 3-D models.

Methods and Properties

The AcadModelSpace collection supports all the methods and properties supported by the AcadBlock object, as well as the common Application property. See the AcadBlock object for more details.

Note Because this collection inherits from AcadObject, it supports the Modified event.

AcadMText Object

The AcadMText object represents a paragraph of alphanumeric characters that fits within a nonprinting text boundary. Each AcadMText object is a single object regardless of the number of lines it contains, and AutoCAD uses word wrap to break long lines into paragraphs. Text can be imported into an AcadMText object, but all embedded formatting will be lost. The text boundary definition specifies the width and alignment of the paragraph, and although it isn't plotted or printed, it remains part of the object's framework. AcadMText objects are created using the AddMText method of the AcadBlock, AcadModelSpace, or AcadPaperSpace object.

AcadMText Object Methods

The AcadMText object inherits all the methods of the AcadEntity and AcadObject objects. It supports no other methods.

AcadMText Object Properties

The AcadMText object inherits all the properties of the AcadEntity and AcadObject objects, as well as the common Application property. It also supports the following properties.

Name	Returns	Description
AttachmentPoint	AcAttachmentPoint	Gets or sets the attachment point for an AcadMText object. The option selected determines both the text justification and text spill in relation to the text boundary. When the property is changed, the position of the existing bounding box doesn't change. The text is simply rejustified within the bounding box. For a list of possible values for the AcAttachmentPoint enumerated type, see Appendix B.
DrawingDirection	AcDrawingDirection	Gets or sets the direction in which the MText paragraph is to be read. Possible values for the AcDrawingDirection enumeration are acLeftToRight and acTopToBottom. (Three other values are reserved for future use: acRightToLeft, acBottomToTop, and acByStyle.)
Height	Double	Sets the height of uppercase text, measured in the current units. This property is used as a scale factor for both the height and width of the text. \H also indicates a height change.
InsertionPoint	Variant	Sets the insertion point for the text and specifies the location for a corner of the text boundary. The AttachmentPoint property specifies which corner of the text boundary is to be positioned at this insertion point.
LineSpacingFactor	Double	Gets or sets the relative line spacing factor for the AcadMText object, defined as the vertical distance between the baseline of one text line and the baseline of the next text line. It's set as a multiple of single line spacing and should have a value between 2.5 and 4.0.
LineSpacingStyle	AcLineSpacingStyle	Gets or sets the line spacing style for the AcadMText object. For a list of possible values for the AcLineSpacingStyle enumerated type, see Appendix B.
Normal	Variant	Gets or sets the 3-D (Z-axis) normal unit vector for the AcadMText object.
Rotation	Double	Gets or sets the rotation angle (in radians) for MText relative to the X-axis, with positive values going counterclockwise when viewed along the Z-axis toward the origin.
StyleName	String	Gets or sets the name of the style used with the object, the default being the current style. Use the TextStyle object to change the attribute of a given text style. Note that the name given must already be defined in the drawing.
TextString	String	Gets or sets the text string for the entity. The text string may contain format codes; for example, you can underline text, add a line over text, and create stacked text.
Width	Double	Gets or sets the width of the text boundary in the current units. AutoCAD wraps the text within the text boundary; therefore, the width must be a positive number large enough to accommodate the text. If the width isn't large enough, the text may be difficult to read or may not be visible at all. \W also indicates a width change.

Note Because this object inherits from AcadObject, it supports the Modified event.

AcadPaperSpace Collection

The AcadPaperSpace collection contains all the objects in paper space layout, and if a new paper space is made active, the data in this collection will be changed to reflect the data in the new active layout. It is similar to the AcadBlock object, and it supports the same methods and properties, with just one additional method. Moreover, objects contained in this collection can also be found in the blocks collection in a special block called *PAPER_SPACE, which is to say that both the block named *PAPER_SPACE and the AcadPaperSpace collection actually point to the same data. The AcadPaperSpace collection was created to provide a faster and easier means of accessing the data in the active paper space layout.

Paper space layouts are used for creating a finished layout for printing or plotting the drawing and not doing drafting or design work, which is carried out in the model space. Although both 2-D and 3-D objects can exist in paper space, commands that render a 3-D viewpoint are disabled. The AcadPaperSpace collection can't be created or deleted.

Use paper space to create a finished layout of a drawing for printing or plotting.

AcadPaperSpace Collection Methods

The AcadPaperSpace collection inherits all the methods of the AcadBlock and AcadModelSpace collections. It also supports the following method.

Name	Returns	Description
AddPViewport	AcadPViewport	Adds a paper space viewport, given the coordinates for the center of the viewport, its height, and its width. Parameters: Center As Variant, Width As Double, and Height As Double.

AcadPaperSpace Collection Properties

The AcadPaperSpace collection inherits all the properties of the AcadBlock and AcadModelSpace collections, as well as the common Application property. It supports no other properties.

Note Because this collection inherits from AcadObject, it supports the Modified event.

AcadPlot Object

The AcadPlot object represents the set of methods and properties used for plotting specified layouts. You can display a plot preview, plot to a file or plot to a device, or batch-plot several plot commands. You can set the number of copies of the drawing to be plotted and specify quiet error mode so that the plot session will be uninterrupted.

However, it should be noted that you can't create a configured plotter (PC3) file using ActiveX. To create a PC3 file, use the Add-a-Plotter wizard in AutoCAD. For more information on PC3 files, see "Plotter-Specific Configuration Information" in the AutoCAD documentation.

AcadPlot Object Methods

The AcadPlot object supports the following methods.

Name	Returns	Description
DisplayPlotPreview		Displays the plot preview dialog box for the active layout. Parameter: Preview As AcPreviewMode. For a description of possible values for the AcPreviewMode enumerated type, see Appendix B. Note that there's no hourglass indicating the regeneration process and that the user needs to dismiss the preview for this method to return. Any VBA UserForm should be hidden during a plot preview.
PlotToDevice	Boolean	Plots a layout to a device and returns whether the plot was successfully sent. Parameter: plotConfig As String. Specifies the full path and file name of the PC3 file to use instead of the current configuration. If this parameter isn't provided, the current configuration will be used. A dialog window will indicate the printing progress. Any VBA UserForm should be hidden during a PlotToDevice operation.
PlotToFile	Boolean	Plots a layout to a file and returns whether the plot was successfully sent. Parameters: PlotFile As String and [PlotConfig As String]. PlotFile specifies the name of the file to which to plot the active layout. When you plot multiple layouts, the file name for each plot will be generated from the drawing and layout names. A dialog window will indicate the printing progress. Any VBA UserForm should be hidden during a PlotToFile operation. PlotConfig specifies the full path and file name of the PC3 file to use instead of the current configuration. If this parameter isn't provided, the current configuration will be used.
SetLayoutsToPlot		Sets the layout or layouts to be plotted. If you want a layout other than the active layout to be plotted, you must call this method. Parameter: LayoutList As Variant (an array of layout objects). If the LayoutList parameter is NULL or this method isn't called at all, the active layout is sent to the plot. Note that a call to this method has to precede each layout plotting in an iteration. Before you enter a batch operation, you have to specify the number of layouts to plot using the StartBatchMode method.
StartBatchMode		Invokes batch mode printing. This method allows the plot progress dialog box to show the progress of batch plots. This remains active until the batch is completed. Parameter: EntryCount As Long. The EntryCount parameter specifies the number of calls to either the PlotToFile method or the PlotToDevice method that must follow the StartBatchMode method.

AcadPlot Object Properties

As well as the common Application property, the AcadPlot object supports the following properties.

Name	Returns	Description
QuietErrorMode	Boolean	Specifies whether quiet error mode is enabled. Quiet error mode logs all plot-related error messages into a log file. This is essential for certain plotting applications, such as batch plotting, that require uninterrupted application execution. When the quiet error mode is disabled, errors during printing are displayed in alert boxes.
NumberOfCopies	Long	Gets or sets the number of copies to be plotted.
BatchPlotProgress	Boolean	Specifies whether the batch plot is in progress. Setting this property to False terminates the batch plot. You can achieve this programmatically or by clicking the Cancel button in the progress dialog box.

AcadPlotConfiguration Object

See the AcadLayout object.

AcadPlotConfigurations Collection

See the AcadLayouts collection.

AcadPoint Object

The AcadPoint object represents a point marker on a drawing appearing as a dot, square, circle, X, tick, plus sign, or as any combination of these. You can specify a full 3-D location for a point. However, the current elevation is assumed if you supply two coordinates. Points can also act as nodes onto which you can snap objects. The Coordinates property provides an array that can be used in any method requiring a point as an input parameter. AcadPoint objects are created using the AddPoint method of the AcadBlock, AcadModelSpace, or AcadPaperSpace object.

The PDMODE and PDSIZE system variables control the appearance of point objects. The values for PDMODE are as follows:

0	Draws point as a dot (.)
1	No symbol drawn
2	Draws point as a plus (+)
3	Draws point as a cross (x)
4	Draws point as a vertical dash (l)

You can add 32, 64, or 96 to any of the preceding values to specify a shape to draw around the point in addition to the point symbol itself. The values represent a circle, a square, and a circle within a square, respectively.

The PDSIZE variable controls the size of the point symbols for PDMODE values 2 and 4. A setting of 0 generates the point at 5 percent of the graphics area height. A positive value specifies an absolute size for the point figures; a negative value is interpreted as a percentage of the viewport size.

Any changes to the points (PDMODE or PDSIZE) are displayed when the drawing is regenerated.

AcadPoint Object Methods

The AcadPoint object inherits all the methods of the AcadEntity and AcadObject objects. It supports no other methods.

AcadPoint Object Properties

The AcadPoint object inherits all the properties of the AcadEntity and AcadObject objects, as well as the common Application property. It also supports the following properties.

Name	Returns	Description
Coordinates	Variant	Gets or sets the coordinates. It returns a three-element array of Doubles containing 3-D coordinates in WCS.
Normal	Variant	Gets or sets the 3-D (Z-axis) normal unit vector for the mesh.
Thickness	Double	Gets or sets the distance an AcadPoint object is extruded above or below its elevation, in a Z-axis direction. The default is 0.0.

Note Because this object inherits from AcadObject, it supports the Modified event.

AcadPolyfaceMesh Object

The AcadPolyfaceMesh object represents a 3-D polyface mesh. It is created using the AddPolyfaceMesh method of the AcadBlock, AcadModelSpace, or AcadPaperSpace object.

AcadPolyfaceMesh Object Methods

The AcadPolyfaceMesh object inherits all the methods of the AcadEntity and AcadObject objects. It supports no other methods.

AcadPolyfaceMesh Object Properties

The AcadPolyfaceMesh object inherits all the properties of the AcadEntity and AcadObject objects, as well as the common Application property. It also supports the following properties.

Name	Returns	Description
Coordinate	Variant	Gets or sets the coordinate of a single vertex in the object. This will replace an existing vertex for the specified object. Use standard array-handling techniques to process the values contained in this property. It returns a three-element array of Doubles containing 3-D coordinates in WCS. Note that the Z coordinate will default to 0 on the active UCS. Parameter: Index As Integer (the index in the zero-based array of vertices for the vertex you want to set or query).

Name	Returns	Description
Coordinates	Variant	Gets or sets the coordinates for each vertex in the object. This will replace any existing coordinates for the specified object. Use standard array-handling techniques to process the coordinates contained in this property. Note that you can't change the number of coordinates in the object by using this property; you can change only the location of existing coordinates. It returns a three-element array of Doubles containing 3-D coordinates in WCS. Note that the Z coordinate will always default to 0 on the active UCS.
NumberOfFaces	Integer	Gets the number of faces in the mesh. This property's value is read-only.
NumberOfVertices	Integer	Gets the number of vertices in the mesh. This property's value is read-only.

Note Because this object inherits from `AcadObject`, it supports the `Modified` event.

AcadPolygonMesh Object

The `AcadPolygonMesh` object simulates the geometric representation of a mesh laid out on a 3-D surface. It is specified as an $M \times N$ mesh, where M represents the number of vertices in a row of the mesh and N represents the number of vertices in a column. A mesh can be open or closed in either or both the M and N directions. A mesh that is closed in a particular direction is considered to be continuous from the last row or column on to the first row or column. If it is open, it is treated as discontinuous between the last row and the first row. The `AcadPolygonMesh` object is created using the `AddPolygonMesh` method of the `AcadBlock`, `AcadModelSpace`, or `AcadPaperSpace` object.

AcadPolygonMesh Object Methods

The `AcadPolygonMesh` object inherits all the methods of the `AcadEntity` and `AcadObject` objects. It also supports the following methods.

Name	Returns	Description
AppendVertex		Appends a vertex to the polygon mesh, which is an array of 3-D WCS coordinates specifying the appended row of vertices. Note that appending a vertex is the equivalent of adding an extra row to the polygon mesh matrix. Parameter: <code>Vertex As Variant</code> (a $3 \times M$ array of Doubles). Note that an array of the wrong size will result in an error, because a polygon mesh is a regular $M \times N$ matrix.
Explode	Variant	Explodes the polygon mesh and returns the subentities as an array of Objects (<code>Acad3DFace</code> objects).

AcadPolygonMesh Object Properties

The AcadPolygonMesh object inherits all the properties of the AcadEntity and AcadObject objects, as well as the common Application property. It also supports the following properties.

Name	Returns	Description
Coordinate	Variant	Gets or sets the coordinate of a single vertex in the object. This will replace an existing vertex for the specified object. Use standard array-handling techniques to process the values contained in this property. For simple polylines, this property specifies simple vertices. For splines or curve-fit polylines, this property specifies control point vertices. It returns a three-element array of Doubles containing 3-D coordinates in WCS. Note that the Z coordinate will default to 0 on the active UCS. Parameter: Index As Integer (the index in the zero-based array of vertices for the vertex you want to set or query).
Coordinates	Variant	Gets or sets the coordinates for each vertex in the object. This will replace any existing coordinates for the specified object. Use standard array-handling techniques to process the coordinates contained in this property. For simple polylines, this property specifies simple vertices. For splines or curve-fit polylines, this property specifies control point vertices. Note that you can't change the number of coordinates in the object by using this property; you can change only the location of existing coordinates. It returns a 3×M×N-element array of Doubles containing 3-D coordinates in WCS.
MClose	Boolean	Specifies whether the polygon mesh is closed in the M direction. The document has to be regenerated for the changes to appear.
MDensity	Integer	Gets or sets the smooth surface density of a polygon mesh in the M direction, the value of which is an integer between 2 and 255. The initial value for this property is derived from the value in the SURFU system variable plus 1. The M surface density is the number of vertices in the M direction for AcadPolygonMesh objects of the following types: acQuadSurfaceMesh, acCubicSurfaceMesh, and acBezierSurfaceMesh.
MVertexCount	Integer	Gets the vertex count in the M direction of the polygon mesh. This property's value is read-only. This is the number of vertices that will be used to make up an M row in the polygon mesh if the Type property is acSimpleMesh. For any other polygon mesh type, the MDensity value will be used as the row size. Note that you can't change the number of vertices for a simple mesh. This property is necessary when appending new vertices.
NClose	Boolean	Specifies if the polygon mesh is closed in the N direction. The document has to be regenerated for the changes to appear.

Name	Returns	Description
NDensity	Integer	Gets or sets the smooth surface density of a polygon mesh in the N direction, the value of which is an integer between 2 and 255. The initial value for this property is derived from the value in the SURFV system variable plus 1. The N surface density is the number of vertices in the N direction for AcadPolygonMesh objects of the following types: acQuadSurfaceMesh, acCubicSurfaceMesh, and acBezierSurfaceMesh.
NVertexCount	Integer	Gets the vertex count in the N direction of the polygon mesh. This property's value is read-only. This is the number of vertices that will be used to make up an N row in the polygon mesh if the Type property is acSimpleMesh. For any other polygon mesh type, the MDensity value will be used as the row size. Note that you can't change the number of vertices for a simple mesh.
Type	AcPolymeshType	Gets or sets type of a polygon mesh object. For a list of possible values for the AcPolymeshType enumerated type, see Appendix B.

Note Because this object inherits from AcadObject, it supports the Modified event.

AcadPolyline Object

The AcadPolyline object represents a 2-D line of adjustable width or a 3-D line or mesh of non-adjustable width composed of line and arc segments. Use the AcadLWPolyline object to create polylines with an optimized format that saves memory and disk space. The AcadPolyline object is created using the AddPolyline method of the AcadBlock, AcadModelSpace, or AcadPaperSpace object.

AcadPolyline Object Methods

The AcadPolyline object inherits all the methods of the AcadEntity and AcadObject objects. It also supports the following methods.

Name	Returns	Description
AppendVertex		Appends a vertex to the polyline, which is an array of 3-D coordinates. The X and Y coordinates are given in OCS; the Z coordinate is ignored. Parameter: vertex As Variant (a three-element array of doubles).
Explode	Variant	Explodes the polyline and returns the subentities as an array of Objects (AcadLine and AcadArc).

Continued

Name	Returns	Description
GetBulge	Double	Returns the vertex bulge value at a given index of the polyline. A bulge is the tangent of $\frac{1}{4}$ of the included angle for the arc between the selected vertex and the next vertex in the polyline's vertex list. A negative bulge value indicates that the arc goes clockwise from the selected vertex to the next vertex. A bulge of 0 indicates a straight segment, and a bulge of 1 is a semicircle. Note that this method will fail if the polyline Type property isn't <code>acSimplePoly</code> . Parameter: Index As Integer.
GetWidth		Returns segment width at a given index of the polyline. Parameters: Index As Integer, StartWidth As Double, and EndWidth As Double.
Offset	Variant	Creates a new polyline by offsetting the polyline by a given distance, which must be nonzero. If the offset is negative, this means that the line is drawn closer to the WCS origin. Parameter: Distance As Double.
SetBulge		Sets the vertex bulge of the polyline at a given index. See GetBulge for the definition of a bulge. Note that this method will fail if the polyline Type property isn't <code>acSimplePoly</code> . Parameters: Index As Integer and bulge As Double.
SetWidth		Sets the segment width of the polyline for a given index. Parameters: Index As Integer, StartWidth As Double, and EndWidth As Double.

AcadPolyline Object Properties

The `AcadPolyline` object inherits all the properties of the `AcadEntity` and `AcadObject` objects, as well as the common `Application` property. It also supports the following properties.

Name	Returns	Description
Area	Double	Gets the area of the polyline in square drawing units. The area is computed as though a straight line connects the start and end points. This property's value is read-only.
Closed	Boolean	Specifies whether the polyline is closed (True) or open (False). Open is the default.
ConstantWidth	Double	Gets or sets a global width for all segments in a polyline.
Coordinate	Variant	Gets or sets the coordinate of a single vertex in the object. This will replace an existing vertex for the specified object. Use standard array-handling techniques to process the values contained in this property. For simple polylines, this property specifies simple vertices. For splines or curve-fit polylines, this property specifies control point vertices. It returns a three-element array of Doubles containing 3-D coordinates in OCS (the Z coordinate is ignored). These coordinates can be converted to and from other coordinate systems using the <code>TranslateCoordinates</code> method. Parameter: Index As Integer (the index in the zero-based array of vertices for the vertex you want to set or query).

Name	Returns	Description
Coordinates	Variant	Gets or sets the coordinates for each vertex in the object. This will replace any existing coordinates for the specified object. Use standard array-handling techniques to process the coordinates contained in this property. For simple polylines, this property specifies simple vertices. For splines or curve-fit polylines, this property specifies control point vertices. Note that you can't change the number of coordinates in the object by using this property; you can change only the location of existing coordinates. It returns the array point as a three-element array of Doubles containing 3-D points in OCS (the Z coordinate is ignored). These coordinates can be converted to and from other coordinate systems using the TranslateCoordinates method.
Elevation	Double	Gets or sets the current elevation of the polyline.
LinetypeGeneration	Boolean	Specifies the linetype generation of a polyline. If it's set to True, the linetype is generated in a continuous pattern through the polyline vertices; if it's set to False, the linetype is generated starting and ending with a dash at each polyline vertex.
Normal	Variant	Gets or sets the 3-D (Z-axis) normal unit vector for the polyline.
Thickness	Double	Gets or sets the distance an AcadPolyline object is extruded above or below its elevation, in a Z-axis direction. The default is 0.0.
Type	AcPolylineType	Gets or sets type of a polyline object. For a list of possible values for the AcPolylineType enumerated type, see Appendix B.

Note Because this object inherits from AcadObject, it supports the Modified event.

AcadPopupMenu Object

The AcadPopupMenu object represents an AutoCAD cascading menu of which there are two types: pull-down menus, which are accessed from the AutoCAD menu bar, and shortcut menus, which are accessed by pressing Shift and right-clicking. A pop-up menu is a collection of menu items that includes AcadPopupMenuItems, separators, and submenus, which can be added to the collection using the AddMenuItem, AddSeparator, and AddSubmenu methods.

To create a new pull-down menu, use the Add method to add a new AcadPopupMenu object to the AcadPopupMenu collection, and then add the menu to the menu bar using the InsertInMenuBar method. To create a new shortcut menu, you must first delete any existing shortcut menu, as there can be only one shortcut menu per menu group. If there is no other shortcut menu in a menu group, you can then add a menu with the label POP0, which will be recognized as the new shortcut menu. This label will tell AutoCAD that you want to create a cursor menu. Consult the AcadPreferenceUser object for the list of properties to customize the shortcut menu.

AcadPopupMenu Object Methods

The AcadPopupMenu object supports the following methods.

Name	Returns	Description
AddMenuItem	AcadPopupMenuItem	Adds an item to the pop-up menu. Parameters: Index As Variant (Integer or String), Label As String, and Macro As String. The Index parameter for this property and the four that follow must be between 0 and N minus 1 (if it's an Integer), where N is the number of objects in the pop-up menu. The new item will be added immediately before the specified index location. To add the new menu item to the end of a menu, set the index to be greater than N. If a String is specified and the indexed item doesn't exist, then the new menu item is added at the end of the menu. The Label parameter may contain DIESEL string expressions. Labels also identify the accelerator keys (keyboard key sequences) that correspond to the menu item by placing an ampersand (&) in front of the accelerator character.
AddSeparator	AcadPopupMenuItem	Adds a separator to the pop-up menu. The first item in a menu can't be a separator. You can't add a separator immediately next to another separator. Parameter: Index As Variant (Integer or String).
AddSubmenu	AcadPopupMenu	Adds a submenu to the pop-up menu. Parameters: Index As Variant (Integer or String) and Label As String. The Label parameter may contain DIESEL string expressions. Labels also identify the accelerator keys (keyboard key sequences) that correspond to the menu item by placing an ampersand (&) in front of the accelerator character.
InsertInMenuBar		Inserts the pop-up menu into the menu bar at the location specified by the index. Parameter: Index As Variant.
Item	AcadPopupMenuItem	Returns the member of the collection specified by the index. Parameter: Index As Variant (Integer or String). If the Index value is a String, it must match an existing object name in the collection. To determine if the returned item is a menu item, a separator, or a submenu, use the Type property of the returned AcadPopupMenuItem object.
RemoveFromMenuBar		Removes the pop-up menu from the menu bar.

AcadPopupMenu Object Properties

As well as the common Application property and the Count property, the AcadPopupMenu object supports the following properties.

Name	Returns	Description
Name	String	Gets or sets the name of the AcadPopupMenu object.
NameNoMnemonic	String	Gets the name of the menu without the underscore mnemonic.
OnMenuBar	Boolean	Specifies whether the menu is on the menu bar. This property's value is read-only.
Parent	Object	Gets the parent of the AcadPopupMenu object. This property's value is read-only. The valid object types are AcadMenuBar and AcadPopupMenu.
ShortcutMenu	Boolean	Specifies whether the menu is a shortcut menu.
TagString	String	Gets the menu's tag string, which can consist of alphanumeric and underscore (_) characters. This string uniquely identifies the item within a given menu file. This string is automatically assigned when the object is created and is used internally by AutoCAD for toolbar and menu identification. This property's value is read-only.

AcadPopupMenuItem Object

The AcadPopupMenuItem object represents a single menu item on an AutoCAD pull-down menu. There are three types: a simple menu item, a separator, and a submenu item. They are created using the AddMenuItem, AddSeparator, or AddSubmenu method of the AcadPopupMenu object.

Simple menu items have an index, a label, a tag, and a macro. The index is the position on the menu where the individual menu item is located. The label is a string that defines the content and formatting of the menu item as it appears to the user. The tag is a string consisting of alphanumeric and underscore (_) characters, which uniquely identifies the menu item within a given menu. The macro is a series of commands that execute specific actions when a menu item is selected.

Separators have only an index, which specifies where on the menu the separator is to appear. Submenu items have an index, a label, and a tag, but no macro. Instead, they have an entire AcadPopupMenu object associated with them, which appears when the user selects the submenu item. To find an AcadPopupMenu object associated with a submenu item, use the SubMenu property.

AcadPopupMenuItem Object Methods

The AcadPopupMenuItem object supports a single method.

Name	Description
Delete	Deletes the item from the menu

AcadPopupMenuItem Object Properties

As well as the common Application property, the AcadPopupMenuItem object supports the following properties.

Name	Returns	Description
Caption	String	Gets the text seen by the user for the menu item. This property is derived from the Label property by removing any DIESEL string expressions. This property's value is read-only.
Check	Boolean	Specifies the check status for the menu item (the tick to the left of the menu item caption).
Enable	Boolean	Specifies whether the item is enabled (grayed out).
EndSubMenuLevel	Integer	Gets or sets the last item in a submenu. The default value for this property is 0, specifying that this menu item isn't the last item in the submenu. A value of 1 indicates that this item is the last item in a submenu. A value of 2 indicates that this item is the last item of a submenu and also of its parent menu. This value should be incremented for each parent menu for which this item terminates.
HelpString	String	Gets or sets the help string for the menu item, which appears in the AutoCAD status line when a user highlights a menu item.
Index	Integer	Gets the index for the menu item. The first position in the index is 0. This property's value is read-only.
Label	String	Gets or sets the content and formatting of the menu item as they appear to the user.
Macro	String	Gets or sets the name of the macro associated with the menu item.
Parent	AcadPopupMenu	Gets the parent of the object. This property's value is read-only.
SubMenu	AcadPopupMenu	Gets the submenu, if the item is associated with one. This information can be obtained from the object Type property. This property's value is read-only.
TagString	String	Gets or sets the item's tag string, which can consist of alphanumeric and underscore (_) characters. This string uniquely identifies the item within a given menu file. This string is automatically assigned when the object is created and is used internally by AutoCAD for toolbar and menu identification.
Type	AcMenuItemType	Gets the item's type. This property's value is read-only. Possible values for the AcMenuItemType enumeration are acMenuItem, acMenuSeparator, and acMenuSubMenu.

AcadPopupMenu Collection

The AcadPopupMenu collection contains all the AcadPopupMenu objects loaded in the menu group, some or all of which may be currently displayed in AutoCAD. To load an existing menu into the current session, use the Load method of the AcadMenuGroups collection. To access this collection, use the Menus property on the specific AcadMenuGroup. To create a new menu, use the Add method. To display a menu on the AutoCAD menu bar, use the InsertMenuInMenuBar method. To remove a menu from the menu bar, use the RemoveMenuFromMenuBar method.

AcadPopupMenu Collection Methods

The AcadPopupMenu collection supports four methods.

Name	Returns	Description
Add	AcadPopupMenu	Creates an AcadPopupMenu object and adds it to the collection. Parameter: Name As String.
InsertMenuInMenuBar		Inserts a menu into the menu bar immediately before the position specified by the index. Parameters: MenuName As String and Index As Variant (Integer or String). If the Index is an Integer, it must be between 0 and N minus 1, where N is the number of pop-up menus in the menu bar. If the Index is a String, the name of the pop-up menu must contain any underscore (_) that appears in the menu. This method is equivalent to the InsertInMenuBar method of the AcadPopupMenu object.
Item	AcadPopupMenu	Returns the item in the collection specified by the index. Parameter: Index As Variant (Integer or String).
RemoveMenuFromMenuBar		Removes the menu specified by the index from the menu bar. Parameter: Index As Variant (Integer or String). If the Index is an Integer, it must be between 0 and N minus 1, where N is the number of pop-up menus in the menu bar. If it is a String, the name of the pop-up menu must contain any underscore (_) that appears in the menu, which corresponds to the TagString property of the AcadPopupMenu object.

AcadPopupMenu Collection Properties

As well as the common Application property, the AcadPopupMenu collection supports the following property.

Name	Description
ParentAcadMenuGroup	Gets the collection's parent object. This property's value is read-only.

AcadPreferences Object

The AcadPreferences object specifies the current AutoCAD setting, holding all the options from the Options dialog box that reside in the Registry. Consult this dialog box from the Tools menu for a list of all the current settings, most of them being accessed using the AcadPreferences object. Options that reside in the drawing can be found through the AcadDatabasePreferences object. The AcadPreferences object can be referenced from the Preferences property on the AcadApplication object. The AcadPreferences object has no methods.

AcadPreferences Object Properties

In addition to the common Application property, the AcadPreferences object supports the following properties.

Name	Returns	Description
Drafting		Gets the AcadPreferencesDrafting object for the application. This property's value is read-only.
Display		Gets the AcadPreferencesDisplay object for the application. This property's value is read-only.
Files		Gets the AcadPreferencesFiles object for the application. This property's value is read-only.
OpenSave		Gets the AcadPreferencesOpenSave object for the application. This property's value is read-only.
Output		Gets the AcadPreferencesOutput object for the application. This property's value is read-only.
Profiles		Gets the AcadPreferencesProfiles object for the application. This property's value is read-only.
Selection		Gets the AcadPreferencesSelection object for the application. This property's value is read-only.
System		Gets the AcadPreferencesSystem object for the application. This property's value is read-only.
User		Gets the AcadPreferencesUser object for the application. This property's value is read-only.

AcadPreferencesDisplay Object

The AcadPreferencesDisplay object contains the options from the Display tab on the Options dialog box. The object has no methods.

AcadPreferencesDisplay Object Properties

In addition to the common Application property, the AcadPreferencesDisplay object supports the following properties.

Name	Returns	Description
AutoTrackingVecColor	VBA.ColorConstants	Gets or sets the color of the autotracking vector. The default color for this property is vbWhite. Possible values: any member of VBA.ColorConstants.
CursorSize	Integer	Gets or sets the size of the crosshairs as a percentage of the size of the screen. The default value for this property is 5 percent. The allowable range is 1 to 100 percent of the total screen. At 100 percent, the ends of the crosshairs aren't visible. The value for this property is stored in the CURSOR_SIZE system variable.
DisplayLayoutTabs	Boolean	Specifies whether the Model and Layout tabs appear in the drawing editor. True is the initial value.

Name	Returns	Description
DisplayScreenMenu	Boolean	Specifies whether the screen menu appears on the right side of the drawing window. False is the initial value.
DisplayScrollBars	Boolean	Specifies whether scrollbars appear at the bottom and right sides of the drawing window. False is the initial value.
DockedVisibleLines	Integer	Gets or sets the number of lines of text visible in the command window. The initial value for this property is 3. Note that while it's still available, this property has no effect in AutoCAD 2004 products.
GraphicsWinLayout-BackgrndColor	VBA.ColorConstants	Gets or sets the background color of the paper space layouts. The initial value for this property is a color halfway between the Windows standard background color and the Windows standard active button color. Possible values: any member of VBA.ColorConstants.
GraphicsWinModel-BackgrndColor	VBA.ColorConstants	Gets or sets the background color of the model space window. The default color for this property is vbBlack. Possible values: any member of VBA.ColorConstants.
HistoryLines	Integer	Gets or sets the number of lines of text from the text window kept in the memory. The initial value is 400.
ImageFrameHighlight	Boolean	Specifies how the raster image is displayed when the image is selected. If True, only the raster image is displayed. If False, the raster image content is displayed. The value for this property is stored in the IMAGEHLT system variable.
LayoutCreateViewport	Boolean	Specifies whether a viewport is automatically created for each new layout. True is the initial value.
LayoutCrosshairColor	VBA.ColorConstants	Gets or sets the color of the crosshairs and text in paper space layouts. The initial value for this property is the Windows standard text color. Possible values: any member of VBA.ColorConstants.
LayoutDisplayMargins	Boolean	Specifies whether margins are displayed in layouts. True is the initial value.
LayoutDisplayPaper	Boolean	Specifies whether the paper background is displayed in layouts. True is the initial value.
LayoutDisplayPaperShadow	Boolean	Specifies whether the paper background shadow is displayed in layouts. True is the initial value.
LayoutShowPlotSetup	Boolean	Specifies whether the plot setup dialog box is displayed when a new layout is created. True is the initial value.

Continued

Name	Returns	Description
MaxAutoCADWindow	Boolean	Specifies whether AutoCAD occupies the entire screen at the start of the session.
ModelCrosshairColor	VBA.ColorConstants	Gets or sets the color of the crosshairs and of the text for the model space. The initial value of this property is vbWhite. Possible values: any member of VBA.ColorConstants.
ShowRasterImage	Boolean	Specifies how raster images are displayed during real-time pan and zooms. If True, the image content is displayed. If False, only the image outline is displayed (the default). If the dragging display is turned on and you enable ShowRasterImage, a copy of the image moves with the cursor as you reposition the original image. The value for this property is stored in the RTDISPLAY system variable.
TextFontSize	Integer	Gets or sets the font size for new text. The initial value for this property is 10.
TextFont	String	Gets or sets the font for new text. The initial value for this property is Courier.
TextFontStyle	AcTextFontStyle	Gets or sets the font style for new text. For a list of possible values for the AcTextFontStyle enumerated type, see Appendix B.
TextWinBackgrndColor	VBA.ColorConstants	Gets or sets the text window's background color. The default color for this property is the Windows standard window color. Possible values: any member of VBA.ColorConstants.
TextWinTextColor	VBA.ColorConstants	Gets or sets the text window's text color. The default color for this property is the Windows standard window text color. Possible values: any member of VBA.ColorConstants.
TrueColorImages	Boolean	Gets or sets how the colors in raster images and render images are displayed. If True, the highest possible color resolution is used. If False, palette colors are used (the default).
XrefFadeIntensity	Long	Gets or sets the percentage of fade for external references. The value must be between 1 and 90. The default value is 50.

AcadPreferencesDrafting Object

The AcadPreferencesDrafting object contains the options from the Drafting tab on the Options dialog box. The object has no methods.

AcadPreferencesDrafting Object Properties

In addition to the common Application property, the AcadPreferencesDrafting object supports the following properties.

Name	Returns	Description
AlignmentPointAcquisition	AcAlignmentPointAcquisition	Gets or sets how autoalignment points are acquired. For a list of possible values for the <code>AcAlignmentPointAcquisition</code> enumerated type, see Appendix B.
AutoSnapAperture	Boolean	Specifies whether the autosnap aperture is displayed in the center of the cross-hairs when you snap to an object. This option is available only when the autosnap marker, magnet, or tooltip is enabled. The value for this property is stored in the <code>APBOX</code> system variable. The initial value for this property is <code>True</code> .
AutoSnapApertureSize	Long	Gets or sets whether the size of the autosnap aperture in pixels. This option is available only when the autosnap marker, magnet, or tooltip is enabled. The value for this property is stored in the <code>APERTURE</code> system variable. The initial value is 10.
AutoSnapMagnet	Boolean	Specifies whether the autosnap magnet is enabled. The value for this property is stored in the <code>AUTOSNAP</code> system variable. <code>True</code> is the initial value.
AutoSnapMarker	Boolean	Specifies whether the autosnap marker is enabled. The value for this property is stored in the <code>AUTOSNAP</code> system variable. <code>True</code> is the initial value.
AutoSnapMarkerColor	AcColor	Gets or sets the autosnap marker's color. For a list of possible values for the <code>AcColor</code> enumerated type, see Appendix B.
AutoSnapMarkerSize	Long	Gets or sets the autosnap marker's size in pixels. The value must be between 1 and 20.
AutoSnapToolTip	Boolean	Specifies whether the autosnap tooltips are enabled. The value for this property is stored in the <code>AUTOSNAP</code> system variable. <code>True</code> is the initial value.
AutoTrackToolTip	Boolean	Specifies whether the autotrack tooltips are displayed. When it's enabled, the user sees a tooltip whenever the cursor is over an alignment path. When it's disabled, the tooltip doesn't appear. The value for this property is stored in the <code>AUTOSNAP</code> system variable. <code>True</code> is the initial value.
FullScreenTrackingVector	Boolean	Specifies whether full-screen tracking vectors are displayed. The value for this property is stored in the <code>TRACKPATH</code> system variable. <code>True</code> is the initial value.
PolarTrackingVector	Boolean	Specifies whether polar tracking vectors are displayed. The value for this property is stored in the <code>TRACKPATH</code> system variable. <code>True</code> is the initial value.

AcadPreferencesFiles Object

The AcadPreferencesFiles object contains the options from the Files tab on the Options dialog box.

AcadPreferencesFiles Object Methods

The AcadPreferencesFiles object supports just two methods.

Name	Returns	Description
GetProjectFilePath	String	Returns the directory where external reference files are sought. This name is also controlled by the PROJECTNAME system variable. Parameter: <code>ProjectName As String</code> .
SetProjectFilePath		Sets the directory where external reference files are sought. This name is also controlled by the PROJECTNAME system variable. Parameters: <code>ProjectName As String</code> and <code>ProjectFilePath As String</code> .

AcadPreferencesFiles Object Properties

In addition to the common Application property, the AcadPreferencesFiles object supports the following properties.

Name	Returns	Description
AltFontFile	String	Gets or sets the location of the font file used if the original font can't be located and an alternative font isn't specified in the font-mapping file.
AltTabletMenuFile	String	Gets or sets the path for an alternative menu to swap with the AutoCAD standard tablet menu. This property is applicable only if you use a digitizer.
AutoSavePath	String	Gets or sets the path for the file that is created if automatic saving is enabled with the AutoSaveInterval property of the AcadPreferencesOpenSave object. The value for this property is stored in the SAVEFILEPATH system variable.
ConfigFile	String	Gets the location of the configuration file where information about the hardware device driver is stored. This value is read-only and can be changed only using the /c command-line switch, which is added to the acad.exe command line at startup.
CustomDictionary	String	Gets or sets the custom dictionary for the AutoCAD session. This property contains its drive, path, and file name.
DefaultInternetURL	String	Gets or sets the default Internet address. The value for this property is stored in the INETLOCATION system variable.
DriversPath	String	Gets or sets the directory where to look for ADI device drivers for the video display, pointing devices, printers, and plotters. Specify a drive letter and path for each path. Separate multiple directory listing with a semicolon (;).
FontFileMap	String	Gets or sets the location of the file that defines how fonts are converted if they can't be located.
HelpFilePath	String	Gets or sets the location of the AutoCAD help file. This property specifies its drive, path, and file name.

Name	Returns	Description
LicenseServer	String	Gets a list of client license servers available to the network license manager program. The value for this property is stored in the ACADSERVER environment variable. This property's value is read-only.
LogFilePath	String	Gets or sets the location of the log file. The value for this property is stored in the LOGFILEPATH system variable. Use the LogFileOn property of the AcadPreferencesOpenSave object to enable or disable log file capabilities.
MainDictionary	String	Gets or sets the dictionary used for spell checking.
MenuFile	String	Gets or sets the location of the menu file for the AutoCAD session. This property contains the drive, path, and file name for an .mnu, .mns, or .mnc file.
ObjectARXPath	String	Gets or sets the location for ObjectARX applications.
PostScriptPrologFile	String	Gets or sets a name for a customized prolog section in the acad.psf file. The value for this property is stored in the PSPROLOG system variable.
PrinterConfigPath	String	Gets or sets the location of the printer configuration files. A semicolon is used to separate multiple directories but shouldn't be used at the end of the string.
PrinterDescPath	String	Gets or sets the location of the printer description files. A semicolon is used to separate multiple directories but shouldn't be used at the end of the string.
PrinterStyleSheetPath	String	Gets or sets the location of the printer style sheet files. A semicolon is used to separate multiple directories but shouldn't be used at the end of the string.
PrintFile	String	Gets or sets an alternative name for the temporary plot file. The default is the name of the drawing followed by the extension .plt.
PrintSpoolerPath	String	Gets or sets the directory of the print spool files. The plot is written to this location.
PrintSpoolExecutable	String	Gets or sets the application used for print spooling including the drive, the path, and the application name of the print spooling executable used with autospool.
SupportPath	String	Gets or sets the directories to look for support files (support, fonts, help, and bonus files). A semicolon is used to separate multiple directories but shouldn't be used at the end of the string.
TempFilePath	String	Gets or sets the directory used to store temporary files.
TemplateDWGPath	String	Gets or sets the path for the template files used by the startup wizards.
TempXrefPath	String	Gets or sets the location of the external reference files. This location is used for the copy of the external reference if you choose acEnableWithCopy demand loading on the XrefDemandLoad property of the AcadPreferencesOpenSave object.
TextEditor	String	Gets or sets the name of the text editor for the MTEXT command.
TextureMapPath	String	Gets or sets the directory where to look for rendering texture maps. Multiple paths can be entered by using a semicolon (;) to paths.
WorkspacePath	String	Gets or sets the path for the database workspace file.

AcadPreferencesOpenSave Object

The AcadPreferencesOpenSave object contains the options from the Open and Save tab on the Options dialog box. The object has no methods.

AcadPreferencesOpenSave Object Properties

In addition to the common Application property, the AcadPreferencesOpenSave object supports the following properties.

Name	Returns	Description
AutoEdit	Boolean	Specifies whether an audit should be performed after a DXFIN or DXBIN interchange command is rendered.
AutoSaveInterval	Integer	Gets or sets an interval for automatic saving. The value must be between 0 and 600. A value of 0 indicates that no automatic saving is required. The timer starts as soon as you make a change to a drawing. It is reset and restarted whenever the drawing is saved. The value for this property is stored in the SAVETIME system variable. The initial setting for this property is 120.
CreateBackup	Boolean	Specifies whether a backup file is used. The value of this property is stored in the ISAVEBAK system variable. True is the initial value.
DemandLoadARXApp	AcARXDemandLoad	Gets and sets when a third-party application is loaded if a drawing contains custom objects created in that application. The value for this property is stored in the DEMANDLOAD system variable. For a list of possible values for the AcARXDemandLoad enumerated type, see Appendix B.
FullCRCValidation	Boolean	Specifies whether a cyclic redundancy check (CRC) should be performed whenever an object is read into the drawing. False is the initial value. If your drawings are being corrupted and you suspect a hardware problem or an AutoCAD error, set this property to True.
IncrementalSavePercent	Integer	Gets or sets the wasted space allowed in a file as a percentage. When the specified percentage is reached, a full save occurs instead of an incremental save. Full saves eliminate wasted space. If this property is 0, every save is a full save. Incremental saves increase the size of the drawing, but full saves take longer. The value of this property is stored in the ISAVEPERCENT system variable. The initial value of this property is 50. Autodesk recommends a setting of zero (0) when problems occur during file saves in network environments.
LogFileOn	Boolean	Specifies whether the contents of the text window are written to a log file. The value for this property is stored in the LOGFILEMODE system variable. True is the initial value.
MRUNumber	Long	Gets the number of most recently used files appearing in the File menu. The default value for this property is 4. This property's value is read-only.
ProxyImage	AcProxyImage	Gets or sets how objects are displayed in a drawing that has been created in a third-party application. Possible values for the AcProxyImage enumeration are acProxyNotShow, acProxyShow (default), and acProxyBoundingBox.

Name	Returns	Description
SaveAsType	AcSaveAsType	Gets or sets the drawing type the drawing is saved as. The initial value for this property is <code>ac2004_dwg</code> . For a list of possible values for the <code>AcSaveAsType</code> enumerated type, see Appendix B.
SavePreviewThumbnail	Boolean	Specifies whether .bmp preview images are saved with the drawing. The value for this property is saved in the <code>RASTERPREVIEW</code> system variable. True is the initial value.
ShowProxyDialogBox	Boolean	Specifies whether a warning message is displayed when a drawing that contains custom objects is opened. The value of this property is stored in the <code>PROXYNOTICE</code> system variable. True is the initial value.
TempFileExtension	String	Gets or sets the extension for temporary files. The string can contain up to three characters. The default value is <code>"ac\$"</code> .
XrefDemandLoad	AcXrefDemandLoad	Gets or sets demand loading of external references. The value for this property is stored in the <code>XLOADCTL</code> system variable. Demand loading improves performance by loading only the parts of the referenced drawing needed to regenerate the current drawing. The initial value for this property is <code>acDemandLoadEnabled</code> . For a list of possible values for the <code>AcXrefDemandLoad</code> enumerated type, see Appendix B.

AcadPreferencesOutput Object

The `AcadPreferencesOutput` object contains the options from the Output tab on the Options dialog box. The object has no methods.

AcadPreferencesOutput Object Properties

As well as the common `Application` property, the `AcadPreferencesOutput` object supports the following properties.

Name	Returns	Description
DefaultOutputDevice	String	Gets or sets the default output device for new layouts and model space. If there are no devices on the system, <code>None</code> will be returned.
DefaultPlotStyleForLayer	String	Gets or sets the default plot style for new drawings or for drawings created with earlier releases that haven't been saved in AutoCAD 2000 format. This property is available only when <code>PlotPolicy</code> is set to <code>acPolicyNamed</code> . The initial value of this property is <code>Normal</code> . The value for this property is stored in the <code>DEFPLSTYLE</code> system variable.
DefaultPlotStyleForObjects	String	Gets or sets the default plot style name for newly created objects. This property is available only if <code>PlotPolicy</code> is set to <code>acPolicyNamed</code> . The initial value is <code>ByLayer</code> . The value for this property is stored in the <code>DEFPLSTYLE</code> system variable.

Continued

Name	Returns	Description
OLEQuality	AcOleQuality	Gets or sets the plot quality of OLE objects. The initial value is acOQText. The value for this property is stored in the OLEQUALITY system variable. Possible values for the AcOleQuality enumeration are acOQLineArt, acOQText (default), acOQGraphics, acOQPhoto, and acOQHighPhoto.
PlotLegacy	Boolean	Gets or sets whether legacy plot scripts can run. The value for this property is stored in the PLOTLEGACY system variable. The initial value is False.
PlotPolicy	AcPlotPolicy	Gets or sets whether an object's color property is associated with its plot style name when a new drawing is created. The value for this property is stored in the PSTYLEPOLICY system variable. The initial value for this property is acPolicyNamed. For a list of possible values for the AcPlotPolicy enumerated type, see Appendix B.
PrinterPaperSizeAlert	Boolean	Specifies whether the user is alerted when a layout is configured with a paper size that is different from the default setting for the PC3 file. The value for this property is stored in the PAPER ALERT system variable. The initial value is True.
PrinterSpoolAlert	AcPrinterSpoolAlert	Gets or sets whether the user is alerted when output needs to be spooled through a system printer because of a conflict with an I/O port. The value for this property is stored in the PSPPOOLALERT system variable. The initial value is acPrinterAlwaysAlert. For a list of possible values for the AcPrinterSpoolAlert enumerated type, see Appendix B.
UseLastPlotSettings	Boolean	Specifies whether the settings of the last successful plot are applied.

AcadPreferencesProfiles Object

The AcadPreferencesProfiles object contains the options from the Profiles tab on the Options dialog box.

AcadPreferencesProfiles Object Methods

This object supports the following methods.

Name	Description
CopyProfile	Copies a profile in the Registry. Parameters: oldProfileName As String and newProfileName As String.
DeleteProfile	Deletes a profile from the Registry. Parameter: ProfileName As String.
ExportProfile	Exports a profile to a file so that other users can use it. Parameters: Profile As String and RegFile As String. The extension of the RegFile should be .arg.
GetAllProfileNames	Gets all profiles available to the system. Parameter: PNames As Variant.

Name	Description
ImportProfile	Imports a profile from a file created by another user. Parameters: Profile As String, RegFile As String, and IncludePathInfo As Boolean. The imported Registry file specified by the RegFile parameter should have an .arg extension. The IncludePathInfo parameter is True if the path information in the Registry file will be preserved and False otherwise.
RenameProfile	Renames a profile. Parameters: OrigProfileName As String and NewProfileName As String.
ResetProfile	Resets the value in a profile to its default values. The specified profile must be the currently active profile. Parameter: Profile As String.

AcadPreferencesProfiles Object Properties

As well as the common Application property, the AcadPreferencesProfiles object supports the following property.

Name	Returns	Description
ActiveProfile	String	Gets or sets the active profile for the session. The available profile names are returned by the GetAllProfileNames method.

AcadPreferencesSelection Object

The AcadPreferencesSelection object contains the options from the Selection tab on the Options dialog box. The object has no methods.

AcadPreferencesSelection Object Properties

As well as the common Application property, the AcadPreferencesSelection object supports the following properties.

Name	Returns	Description
DisplayGrips	Boolean	Specifies whether selection set grips are displayed for the stretch, move, rotate, scale, and mirror grip modes. The value for this property is stored in the GRIPS system variable. The initial value is True.
DisplayGripsWithinBlocks	Boolean	Specifies whether grips are assigned within blocks. The value for this property is stored in the GRIPBLOCK system variable. The initial value is False.
GripColorSelected	AcColor	Gets or sets the color of selected grips. The value for this property is stored in the GRIPHOT system variable. The initial value is acRed. For a list of possible values for the AcColor enumerated type, see Appendix B.
GripColorUnselected	AcColor	Gets or sets the color of grips that aren't selected. The value for this property is stored in the GRIPCOLOR system variable. The initial value is acBlue. For a list of possible values for the AcColor enumerated type, see Appendix B.

Continued

Name	Returns	Description
GripSize	Long	Gets or sets the size of grips in pixels. The value for this property is stored in the GRIPSIZE system variable. The value may be between 1 and 255, but a size greater than 20 isn't recommended. The initial value is 3.
PickAdd	Boolean	Specifies whether objects are added to the selection set by using the Shift key. The value for this property is stored in the PICKADD system variable. The initial value is False.
PickAuto	Boolean	Specifies whether windowing is automatic at the select objects prompt. The value for this property is stored in the PICKAUTO system variable. The initial value is True.
PickBoxSize	Long	Gets or sets the size of the object selection target. The value for this property is stored in the PICKBOX system variable. The initial value is 3.
PickDrag	Boolean	Specifies how a selection window is drawn. The value for this property is stored in the PICKDRAG system variable. If True, the selection window is drawn by dragging. If False, the selection window is drawn by clicking the corners. The initial value is False.
PickFirst	Boolean	Specifies whether objects are selected before a command is issued. The value for this property is stored in the PICKFIRST system variable. The initial value is True.
PickGroup	Boolean	Specifies whether picking an object in a group selects the entire group. The initial value is False.

Note The GRIPHOVER system variable added in AutoCAD 2004 isn't exposed through ActiveX in any API.

AcadPreferencesSystem Object

The AcadPreferencesSystem object contains the options from the System tab on the Options dialog box. The object has no methods.

AcadPreferencesSystem Object Properties

As well as the common Application property, the AcadPreferencesSystem object supports the following properties.

Name	Returns	Description
BeepOnError	Boolean	Specifies whether an invalid entry causes an alarm to beep. The initial value is False.
DisplayOLEScale	Boolean	Specifies whether the OLE scaling dialog box is displayed when an OLE object is inserted into a drawing. The initial value is True.
EnableStartupDialog	Boolean	Specifies whether the startup dialog box is displayed at the beginning of the session. The initial value is True.

Name	Returns	Description
LoadAcadLspInAllDocuments	Boolean	Specifies when acad.lsp is loaded. The value for this property is stored in the ACADLSPASDOC system variable. If True, it's loaded with each drawing. If False, it's loaded at startup.
ShowWarningMessages	Boolean	Specifies whether all dialog boxes with a Don't Display This Warning Again check box are displayed again. The initial value is False.
SingleDocumentMode	Boolean	Specifies whether AutoCAD runs in single-document mode rather than in multiple-document mode. The value for this property is stored in the SDI system variable. The initial is False.
StoreSQLIndex	Boolean	Specifies whether the SQL index is stored in the drawing. The initial value is True.
TablesReadOnly	Boolean	Specifies whether database tables are opened in read-only mode rather than read-write mode. The initial value is False.

AcadPreferencesUser Object

The AcadPreferencesUser object contains the options from the User tab on the Options dialog box. The object has no methods.

AcadPreferencesUser Object Properties

As well as the common Application property, the AcadPreferencesUser object supports the following properties.

Name	Returns	Description
ADCInsertUnits-DefaultSource	AcInsertUnits	Gets or sets the units that are automatically used for objects in the AutoCAD Design Center for a source drawing without assigned insert units. The value of this property is stored in the INSUNITSDEFSOURCE system variable. The initial value is acInsertUnitsUnitless. For a list of possible values for the AcInsertUnits enumerated type, see Appendix B.
ADCInsertUnits-DefaultTarget	AcInsertUnits	Gets or sets the units that are automatically used for objects in the AutoCAD Design Center for a target drawing without assigned insert units. The value of this property is stored in the INSUNITSDEFTARGET system variable. The initial value is acInsertUnitsUnitless. For a list of possible values for the AcInsertUnits enumerated type, see Appendix B.
HyperlinkDisplayCursor	Boolean	Specifies whether the hyperlink cursor and shortcut menu are displayed. When it's enabled, the user sees a small bitmap (just below and to the right of the crosshairs) whenever the cursor is over an object that contains a hyperlink. The initial value is True.

Continued

Name	Returns	Description
HyperlinkDisplayTooltip	Boolean	Specifies whether the hyperlink tooltips are displayed. When it's enabled, the user sees a tooltip whenever the cursor is over an object that contains a hyperlink. This property is automatically disabled when the HyperlinkDisplayCursor property is disabled. The initial value is True.
KeyboardAccelerator	AcKeyboardAccelerator	Gets or sets the keyboard used. The initial value is acPreferenceCustom. For a list of possible values for the AcKeyboardAccelerator enumerated type, see Appendix B.
KeyboardPriority	AcKeyboardPriority	Gets or sets the response to the input of coordinate data. The value of this property is stored in the OSNAPCOORD system variable. The initial value is acKeyboardEntryExceptScripts. For a list of possible values for the AcKeyboardPriority enumerated type, see Appendix B.
SCMCommandMode	AcDrawingAreaSCMCommand	Gets or sets the right-click functionality in the drawing area while in command mode (that is, while a command is currently in progress). The value of this property (and the three that follow) is stored in the SHORTCUTMENU system variable. This property can be set only when the ShortcutMenuDisplay property is set to True. The initial value is acEnableSCMOptions. For a list of possible values for the AcDrawingAreaSCMCommand enumerated type, see Appendix B.
SCMDefaultMode	AcDrawingAreaSCMDefault	Gets or sets the right-click functionality in the drawing area while in default mode (that is, while no objects are selected and no commands are in progress). This property can be set only when the ShortcutMenuDisplay property is set to True. The initial value is acSCM. The initial value is acEnableSCMOptions. For a list of possible values for the AcDrawingAreaSCMDefault enumerated type, see Appendix B.
SCMEditMode	AcDrawingAreaSCMEdit	Gets or sets the right-click functionality in the drawing area while in edit mode (that is, while one or more objects are selected and no commands are in progress). This property can be set only when the ShortcutMenuDisplay property is set to True. The initial value is acEdSCM. For a list of possible values for the AcDrawingAreaSCMEdit enumerated type, see Appendix B.
ShortCutMenuDisplay	Boolean	Specifies whether right-clicking in the drawing area displays a shortcut. If False, a right-click will be interpreted as an Enter. The initial value is True.

AcadPViewport Object

The AcadPViewport object represents rectangular objects created in paper space that display views. The functionality of this object is slightly different from what is used in the AutoCAD user interface. In

ActiveX Automation, the `ActiveSpace` property (of the Document object) is used to control the `TILEMODE` system variable. Also, the `MSpace` property (also of the Document object) is the equivalent of both AutoCAD's `MSPACE` and `PSPACE` commands. In addition, the ActiveX Automation user is required to use the `Display` method before setting the `MSpace` property to `acOn`. It's good practice to activate the display using the `Display` method for at least one `AcadPViewport` object before setting the `MSpace` property, or else the latter will throw an exception.

The `AcadPViewport` object is created using the `AddPViewport` method of the `AcadPaperSpace` object.

AcadPViewport Object Methods

The `AcadPViewport` object inherits all the methods of the `AcadEntity` and `AcadObject` objects. It also supports the following methods.

Name	Description
<code>Display</code>	Determines whether viewport is on or off. The display control must be on before the <code>MSpace</code> property can be used to activate the model space editing capabilities. Use the <code>ViewportOn</code> property to determine whether a paper space viewport display has already been turned on with this method. Parameter: <code>Status As Boolean</code> .
<code>GetGridSpacing</code>	Gets the grid spacing for the viewport. Parameters: <code>XSpacing As Double</code> and <code>YSpacing As Double</code> .
<code>GetSnapSpacing</code>	Gets the snap spacing for the viewport. Parameters: <code>XSpacing As Double</code> and <code>YSpacing As Double</code> .
<code>SetGridSpacing</code>	Sets the grid spacing for the viewport. Parameters: <code>XSpacing As Double</code> and <code>YSpacing As Double</code> .
<code>SetSnapSpacing</code>	Sets the snap spacing for the viewport. Parameters: <code>XSpacing As Double</code> and <code>YSpacing As Double</code> .

AcadPViewport Object Properties

The `AcadPViewport` object inherits all the properties of the `AcadEntity` and `AcadObject` objects, as well as the common `Application` property. It also supports the following properties.

Name	Returns	Description
<code>ArcSmoothness</code>	Integer	Gets or sets the smoothness of circles, arcs, and ellipses. The initial value for this property is 100. The higher the number, the smoother the object, but AutoCAD needs longer to regenerate it. Performance can be improved by setting this property to a low value for drawing and increasing the value for rendering. The valid range is 1 to 20,000.
<code>Center</code>	Variant	Gets or sets the center of the viewport as a set of 2-D coordinates (as Doubles). The default center is (0,0).
<code>Clipped</code>	Boolean	Determines whether the viewport has been clipped. This property's value is read-only.
<code>CustomScale</code>	Double	Gets or sets the custom scale factor for the viewport. To set the viewport to a custom scale, first set the <code>StandardScale</code> property to <code>acVpCustomScale</code> , and then use this property to define the custom scale value.

Continued

Name	Returns	Description
Direction	Variant	Gets or sets the viewing direction for a 3-D visualization of the drawing. This property puts the viewer in a position to look at the drawing as if looking back at the origin (0, 0, 0) from a specified point in space. It's similar to AutoCAD's VPOINT command. It returns a three-element array of Doubles.
DisplayLocked	Boolean	Determines whether the viewport is locked, that is, whether it can or can't be modified (scaled, rotated, or moved). It's True if it is locked and False otherwise.
GridOn	Boolean	Specifies the status of the viewport grid. It's True if it's on and False otherwise. The value of this property is stored in the GRIDMODE system variable. Note that the grid is used only for visual reference. It isn't plotted and it isn't considered part of the drawing. The grid origin is set by the SnapBasePoint method. When you turn the grid on or off on a viewport, the visibility of the grid won't change until the viewport is made active or is reset (using the ActiveViewport property).
Height	Double	Use to set the height of the viewport, which is the Y-axis measurement of the viewport frame. The height also determines the magnification parameter in a ZoomCenter.
LensLength	Double	Gets or sets the lens length (in millimeters) used in perspective viewing, the value of which is stored in the LENSLENGTH system variable.
RemoveHiddenLines	Boolean	Specifies whether hidden lines are to be plotted on a paper space viewport.
SnapBasePoint	Variant	Gets or sets the snap base point for the viewport, the value of which is stored in the SNAPBASE system variable. The snap base point can't be changed for the active paper space viewport. Any changes to this property aren't reflected in the display until the drawing is regenerated.
SnapOn	Boolean	Specifies the status of snap mode. The value of this property is stored in the SNAPMODE system variable, and its style is changed using the SNAPSTYLE system variable. This property activates snap mode using the current snap grid resolution, rotation, and style. Note that the snap grid is invisible. Use the GridOn property to activate a separate visible grid.
SnapRotationAngle	Double	Gets or sets the snap rotation angle (in radians) of the viewport relative to the current UCS, the value of which is stored in the SNAPANG system variable. Changes to this property aren't reflected in the drawing until it's updated. The valid range is 0 to 6.28.
StandardScale	AcViewportScale	Gets or sets the standard scale for the viewport. Changes to this property won't be visible until the drawing is regenerated. For a list of possible values for the AcViewportScale enumerated type, see Appendix B.
StyleSheet	String	Gets or sets the named style sheet for the viewport.

Name	Returns	Description
Target	Variant	Gets or sets the target point for the viewport as a set of 3-D WCS coordinates representing the target point. The line of sight is drawn from the center to the target point. It returns a three-element array of Doubles.
TwistAngle	Double	Gets or sets the twist angle (in radians) for the viewport. This method twists or tilts the view around the line of sight, measured counterclockwise.
UCSIconAtOrigin	Boolean	Specifies whether the UCS icon is displayed at the origin.
UCSIconOn	Boolean	Specifies whether the UCS icon is on.
UCSPerViewport	Boolean	Specifies whether the UCS is saved with the viewport.
ViewportOn	Boolean	Specifies the display status of the viewport. This property is set by the Display method.
Width	Double	Sets the width of the viewport, which is the X-axis measurement of the viewport frame.

Note Because this object inherits from `AcadObject`, it supports the Modified event. There is no direct programmatic means to create or manipulate polygonal `PViewport` objects.

AcadRasterImage Object

The `AcadRasterImage` object represents an image consisting of a rectangular grid, or *raster*, of small squares or dots known as pixels. It is created using the `AddRaster` method of the `AcadBlock`, `AcadModelSpace`, or `AcadPaperSpace` object.

AcadRasterImage Constant Methods

The `AcadRasterImage` object inherits all the methods of the `AcadEntity` and `AcadObject` objects. It also supports the following method.

Name	Description
ClipBoundary	Sets the clipping boundary for a raster image as an array of 2-D WCS coordinates. This must be a closed and visible polygon or rectangle completely contained within the boundaries of the image. You can use this method to define a region of an image for display or plotting. Multiple instances of the same image can have different boundaries. Parameter: <code>Boundary As Variant</code> (an array of two-element array of Doubles).

AcadRasterImage Constant Properties

The `AcadRasterImage` object inherits all the properties of the `AcadEntity` and `AcadObject` objects, as well as the common `Application` property. It also supports the following properties.

Name	Returns	Description
Brightness	Integer	Gets or sets the current brightness value of an image, with values between 0 and 100. The default is 50. You adjust this value to darken or lighten an image. Note that you can't use this property with bitonal images.
ClippingEnabled	Boolean	Enables or disables the clipping boundary, defined with the <code>ClipBoundary</code> method.
Contrast	Integer	Gets or sets the current contrast value of an image, with values between 0 and 100. The default is 50. You adjust this value to make a dull image clearer. Note that you can't use this property with bitonal images.
Fade	Integer	Gets or sets the current fade value of an image, with values of 0 to 100. You adjust this value to make vectors easier to see over images and to create a watermark effect in your plotted output. Note that you can't use this property with bitonal images.
Height	Double	Gets the height of the raster image in pixels. This property's value is read-only.
ImageFile	String	Gets or sets the full path and file name of the raster image file. This property can be used to load a new raster image into an existing raster entity.
ImageHeight	Double	Gets or sets the height of the raster image in current units. For uniform stretching/shrinking of the image, use the <code>ScaleFactor</code> property.
ImageVisibility	Boolean	Specifies whether image visibility is on or off. Image redrawing speed is increased by turning off image visibility. Hidden images aren't displayed or plotted only the image boundary. Images can be hidden regardless of the current viewport coordinate system or when it isn't aligned with the current viewport coordinate system.
ImageWidth	Double	Gets or sets the width of the raster image in current units. For uniform stretching/shrinking of the image, use the <code>ScaleFactor</code> property.
Name	String	Gets or sets the name of the raster object, but not its path (see the <code>ImageFile</code> property).
Origin	Variant	Gets or sets the origin of the raster image in WCS coordinates, relative to the lower-left corner. Returned an array of Doubles.
Rotation	Double	Gets or sets the rotation angle (in radians) for the object, relative to the X-axis of the object's WCS with positive angles going counterclockwise when viewed down from the Z-axis toward the origin.
ScaleFactor	Double	Gets or sets the scale factor for the object. It has to be a value greater than 0. A scale factor greater than 1 enlarges the object. A scale factor between 0 and 1 shrinks the object. The initial value for this property is 1.0000.
ShowRotation	Boolean	Determines whether a raster image is displayed at the value of its <code>Rotation</code> property.

Name	Returns	Description
Transparency	Boolean	Specifies whether the transparency for a particular bitonal image is on or off. Attached bitonal images inherit current layer settings for background and foreground color. In addition to the modifications you can make to any attached image, you can modify bitonal images by turning the transparency of the background on and off. Note that bitonal images and bitonal image boundaries are always the same color.
Width	Double	Gets the width of the raster image in pixels. This property's value is read-only.

Note Because this object inherits from `AcadObject`, it supports the `Modified` event.

AcadRay Object

The `AcadRay` object represents a semi-infinite line and is commonly used as a construction line. A ray has a finite starting point and extends to infinity. The `AcadRay` object is created using the `AddRay` method of the `AcadBlock`, `AcadModelSpace`, or `AcadPaperSpace` object.

AcadRay Object Methods

The `AcadRay` object inherits all the methods of the `AcadEntity` and `AcadObject` objects. It supports no other methods.

AcadRay Object Properties

The `AcadRay` object inherits all the properties of the `AcadEntity` and `AcadObject` objects, as well as the common `Application` property. It also supports the following properties.

Name	Returns	Description
BasePoint	Variant	Gets or sets the point through which the ray passes. It returns a set of 3-D coordinates as a three-element array of Doubles.
DirectionVector	Variant	Gets or sets the direction for the ray through a vector. It returns a three-element array of Doubles.
SecondPoint	Variant	Gets or sets the second point of the ray. It returns a set of 3-D WCS coordinates as a three-element array of Doubles. Note that you can't set the second point of a ray equal to its base point.

Note Because this object inherits from `AcadObject`, it supports the `Modified` event.

AcadRegion Object

The AcadRegion object represents a bounded planar face consisting of lines, circles, arcs, elliptical arcs, and spline curves. All objects in the region retain their layer, linetype, and color. AutoCAD deletes the original objects after converting them to regions and doesn't hatch the regions by default. The AcadRegion object is created using the AddRegion method of the AcadBlock, AcadModelSpace, or AcadPaperSpace object. Regions are mostly used as loops for AcadHatch objects to calculate the area of a composite figure or to find its Centroid and other physical characteristics.

AcadRegion Object Methods

The AcadRegion object inherits all the methods of the AcadEntity and AcadObject objects. It also supports the following methods.

Name	Returns	Description
Boolean		Performs a Boolean operation against another region, allowing you to create composite regions from the intersection, union, or subtraction of one region from another. Parameters: Operation As AcBooleanType and Object As AcadRegion. For a list of AcBooleanType constants, see Appendix B.
Explode	Variant	Explodes the region and returns the subentities as an array of Objects. The individual loops of the regions are converted to lines, circular or elliptical arcs, or spline curves.

AcadRegion Object Properties

The AcadRegion object inherits all the properties of the AcadEntity and AcadObject objects, as well as the common Application property. It also supports the following properties.

Name	Returns	Description
Area	Double	Gets the enclosed area of a region, which equals the combined area for all the objects in the region. The area of the object is specified in square drawing units. This property's value is read-only.
Centroid	Variant	Gets the center of area or mass for a region or solid as a 2-D coordinate. It returns as a two-element array of Doubles. This property's value is read-only.
MomentOfInertia	Variant	Gets the moment of inertia for the region as a 3-D coordinate. It returns as a three-element array of Doubles. This property's value is read-only.
Normal	Variant	Gets the 3-D (Z-axis) normal unit vector for the region. This property's value is read-only.
Perimeter	Double	Gets the total length of the inner and outer region loops in drawing units. This property's value is read-only.
PrincipalDirections	Variant	Gets the principal directions of the region as X, Y, and Z coordinates calculated on the current coordinate system. This property's value is read-only.
PrincipalMoments	Variant	Gets the principal moments property of the region as X, Y, and Z coordinates calculated on the current coordinate system. This property's value is read-only.

Name	Returns	Description
ProductOfInertia	Double	Gets the product of inertia of the region, as X, Y, and Z coordinates calculated on the current coordinate system. This property's value is read-only.
RadiiOfGyration	Variant	Gets the radius of gyration of the region, as X, Y, and Z coordinates calculated on the current coordinate system. This property's value is read-only.

Note Because this object inherits from `AcadObject`, it supports the `Modified` event.

AcadRegisteredApplication Object

The `AcadRegisteredApplication` object represents an external application that has been added to the drawing. To be recognized by AutoCAD, an application must register the name or names that it uses. This happens only once per drawing. These application names are then saved along with the extended data (XData) of each entity that uses them. Every instance of extended data referenced in a drawing must have its application registered in the drawing. Each `AcadRegisteredApplication` object is created using the `Add` method of the `AcadRegisteredApplications` collection and can be accessed using the `Item` method of the `AcadRegisteredApplications` collection.

AcadRegisteredApplication Object Methods

The `AcadRegisteredApplication` object inherits all the methods of the `AcadObject` object. It supports no other methods.

AcadRegisteredApplication Object Properties

The `AcadRegisteredApplication` object inherits all the properties of the `AcadObject` object, as well as the common `Application` property. It also supports the following property.

Name	Returns	Description
Name	String	Gets or sets the name of the registered application

Note Because this object inherits from `AcadObject`, it supports the `Modified` event.

AcadRegisteredApplications Collection

The `AcadRegisteredApplications` collection contains a collection of all registered applications in the drawing. To add a new member to the collection, use the `Add` method. To select a specific registered application, use the `Item` method. Although this collection inherits a `Delete`

method, you can't actually delete this collection. If you need to delete a specific registered application, use the Delete method found in the AcadRegisteredApplication object. There is no limit to the number of registered applications you can create in your drawing. However, there can be only one instance of the AcadRegisteredApplications collection, which is predefined for each drawing. You can make multiple references to it by using the RegisteredApplications property.

AcadRegisteredApplications Collection Methods

The AcadRegisteredApplications collection inherits all the methods of the AcadObject object. It also supports the following methods.

Name	Returns	Description
Add		Creates a member object and adds it to the collection. Parameter: Name As String.
Item		Gets the member object at a given index in a collection. Parameter: Index As Variant (an Integer or a String). If the Index value is a String, it must match an existing object name in the collection.

AcadRegisteredApplications Collection Properties

The AcadRegisteredApplications collection inherits all the properties of the AcadObject object, the Count property, and the common Application property. It supports no other properties.

Note Because this collection inherits from AcadObject, it supports the Modified event.

AcadSelectionSet Object

The AcadSelectionSet object represents a group of one or more AutoCAD objects specified for processing as a single unit. It is created by the Add method of the AcadSelectionSets collections and can be accessed by the Item method of the AcadSelectionSets collection. Four methods can be used to remove selection sets or their contents: Clear, RemoveItems, Erase, and Delete.

AcadSelectionSet Object Methods

The AcadSelectionSet object supports the following methods. Note that it does not inherit from AcadObject.

Name	Returns	Description
AddItems		Adds an object or objects to the selection set. Parameter: Items As Variant. The parameter must be an array containing a unique or multiple objects.
Clear		The Clear method removes all the items from the selection set. The objects still exist, but they no longer reside in the selection set.

Name	Returns	Description
Delete		The Delete method deletes a selection set object, but not the objects that were contained within it. The key (Name) is removed from the collection and is available for further use.
Erase		The Erase method deletes all items in a selection set. The selection set still exists but the objects removed do not.
Highlight		Sets the highlight status for all objects in the selection set. Once the highlight flag for an object has been set, a call to the Update or Regen method is required to view the change. Parameter: HighlightFlag As Boolean. The parameter is True if the object is highlighted and False otherwise.
Item	Object	Gets the member of the selection set specified by the index. Parameter: Index As Variant (Integer or String). If the Index value is a String, it must match an existing object name in the collection.
RemoveItems		The RemoveItems method removes one or more items from a selection set. These objects still exist but they no longer reside in the selection set. Parameter: Objects As Variant. The parameter must be an array containing a unique or multiple objects.
Select		Adds objects to the selection set. Parameters: Mode As AcSelect, [Point1 As Variant, Point2 As Variant], and [FilterType As Integer, FilterData As Variant]. For a list of possible values for the AcSelect enumeration (the Mode parameter), see Appendix B. The second and third parameters are three-element arrays of Doubles representing 3-D WCS coordinates. For this method and the three that follow, FilterType is a DXF group code specifying the type of filter to use. FilterData is the value to filter on. All four Select methods support the filtering mechanism.
SelectAtPoint		Adds an object containing a point to the selection set. Parameters: Point As Variant and [FilterType As Integer, FilterData As Variant]. Point is a three-element array of Doubles representing 3-D WCS coordinates.
SelectByPolygon		Adds any objects inside a polygon to the selection set. Parameters: Mode As AcSelect, PointsList As Variant, and [FilterType As Integer, FilterData As Variant]. For a list of possible values for the AcSelect enumeration (the Mode parameter), see Appendix B. PointsList is a three-element array of Doubles representing 3-D WCS coordinates.
SelectOnScreen		Prompts the user to select an object from the screen. Parameters: [FilterType As Integer, FilterData As Variant].
Update		Updates the selection set.

AcadSelectionSet Object Properties

As well as the common Application property, the AcadSelectionSet object supports the following properties.

Name	Returns	Description
Count	Integer	Gets the number of items in the selection set.
Name	String	Gets name of the selection set. This property's value is read-only.

AcadSelectionSets Collection

The AcadSelectionSets collection contains all the selection sets in the drawing. To add a new member to the collection, use the Add method. To select a specific selection set, use the Item method. To delete a specific selection set, use the Delete method of the AcadSelectionSet object. AutoCAD allows you up to 128 selection sets objects. However, there can be only one instance of the AcadSelectionSets collection, which is predefined for each drawing. You can make multiple references to it by using the SelectionSets property of the AcadDocument object.

AcadSelectionSets Collection Methods

The AcadSelectionSets collection supports just two methods. Note that it does not inherit from AcadObject.

Name	Returns	Description
Add	AcadSelectionSet	Creates an AcadSelectionSet object and adds it to the collection. Parameter: Name As String.
Item	AcadSelectionSet	Returns the member of the collection specified by the index. Parameter: Index As Variant (Integer or String). If the Index value is a String, it must match an existing object name in the collection.

AcadSelectionSets Collection Properties

The AcadSelectionSets collection supports the Application property and the Count property but no others.

AcadShape Object

The AcadShape object represents an object comprising lines, arcs, and circles defined in an .shx file. Note that before inserting a shape, you must load the file containing the desired shape file by using the LoadShapeFile method. The AcadShape object is created using the AddShape method of the AcadBlock, AcadModelSpace, or AcadPaperSpace object.

AcadShape Object Methods

The AcadShape object inherits all the methods of the AcadEntity and AcadObject objects. It supports no other methods.

AcadShape Object Properties

The AcadShape object inherits all the properties of the AcadEntity and AcadObject objects, as well as the common Application property. It also supports the following properties.

Name	Returns	Description
Height	Double	Sets the height of the shape object in drawing units.
InsertionPoint	Variant	Gets or sets the insertion point for a shape as a set of 3-D WCS coordinates. It returns a three-element array of Doubles.
Name	String	Gets or sets the name of the object.

Name	Returns	Description
Normal	Variant	Gets or sets the 3-D (Z-axis) normal unit vector for the shape object.
ObliqueAngle	Double	Gets or sets the oblique angle of the attribute and is measured from the vertical axis. The units are radians within the range of -85 to +85 degrees. A positive value denotes a lean toward the right.
Rotation	Double	Gets or sets the rotation angle (in radians) for the shape, relative to the X-axis with positive values going counterclockwise when viewed along the Z-axis toward the origin.
ScaleFactor	Double	Gets or sets the scale factor for the object. It has to be a value greater than 0.0. A scale factor greater than 1 enlarges the object. A scale factor between 0 and 1 shrinks the object. The initial value for this property is 1.0000.
Thickness	Double	Gets or sets the distance the AcadShape object is extruded above or below its elevation, in a Z-axis direction. The default is 0.0.

Note Because this object inherits from AcadObject, it supports the Modified event.

AcadSolid Object

The AcadSolid object represents a solid-filled polygon. Solid objects are filled only when the FILLMODE system variable is on, which can be set or queried by the SetVariable and GetVariable methods. The AcadSolid object is created using the AddSolid method of the AcadBlock, AcadModelSpace, or AcadPaperSpace object.

AcadSolid Object Methods

The AcadSolid object inherits all the methods of the AcadEntity and AcadObject objects. It supports no other methods.

AcadSolid Object Properties

The AcadSolid object inherits all the properties of the AcadEntity and AcadObject objects, as well as the common Application property. It also supports the following properties.

Name	Returns	Description
Coordinate	Variant	Gets or sets the coordinate of a single vertex in the object. This will replace an existing vertex for the specified object. Use standard array-handling techniques to process the values contained in this property. It returns a three-element array of Doubles containing 3-D coordinates in WCS. Note that the Z coordinate will default to 0 on the active UCS. Parameter: Index As Integer (the index in the zero-based array of vertices for the vertex you want to set or query).

Continued

Name	Returns	Description
Coordinates	Variant	Gets or sets the coordinates for each vertex in the object. This will replace any existing coordinates for the specified object. Use standard array-handling techniques to process the coordinates contained in this property. Note that you can't change the number of coordinates in the object by using this property, you can only change the location of existing coordinates. It returns a three-element array of Doubles containing 3-D coordinates in WCS.
Normal	Variant	Gets or sets the 3-D (Z-axis) normal unit vector for the solid object.
Thickness	Double	Gets or sets the distance the AcadSolidobject is extruded above or below its elevation, in a Z-axis direction. The default is 0.0.

Note Because this object inherits from AcadObject, it supports the Modified event.

AcadSortEntsTable Object

This object contains and manipulates draw order information.

AcadSortEntsTable Methods

The AcadSortEntsTable object does not support any methods.

AcadSortEntsTable Properties

The AcadSortEntsTable object inherits no properties. However, it does support the following properties.

Name	Returns	Description
Block	AcadBlock	Returns the block of a SortentsTable object.
GetFullDrawOrder		Returns all of the objects in a block, sorted by draw order with the bottom object first. Parameters: Objects As Variant, HonorSortentsSysVar As Boolean. If True, uses the setting in the DRAWORDERCTL system variable.
GetRelativeDrawOrder		Returns the objects specified, sorted by the draw order, with the bottom object first. Parameters: Objects As Variant, HonorSortentsSysVar As Boolean. If True, uses the setting in the DRAWORDERCTL system variable.
MoveAbove		Moves objects above the target in the draw order. Parameters: Objects As Variant, Target As AcadEntity.
MoveBelow		Moves objects below the target in the draw order. Parameters: Objects As Variant, Target As AcadEntity.
MoveToBottom		Moves objects to the bottom of the draw order. Parameters: Objects As Variant.
MoveToTop		Moves objects to the top of the draw order. Parameters: Objects As Variant.

Name	Returns	Description
SetRelativeDrawOrder		Sets the relative draw order specified, with the bottom objects first. Parameters: Objects As Variant.
SwapOrder		Swaps the draw order positions of two objects. Parameters: Object1 As AcadEntity, Object2 As AcadEntity.

AcadSpline Object

A spline is a smooth curve passing through a given set of points. AutoCAD uses a particular type of spline known as a nonuniform rational B-spline (NURBS) curve, which produces a smooth curve between control points. Splines are useful for creating irregularly shaped curves, for example, drawing contour lines for geographic information system applications or automobile design. Splines are created by specifying coordinate points and can form a closed loop if the start and end points and tangents are coincident. You can also alter the spline-fitting tolerance. Fit tolerance refers to how closely the spline fits the set of given coordinates. The lower the tolerance, the more closely the spline fits the points. Zero tolerance means that the spline passes through all the points.

The AcadSpline object represents a NURBS curve and is created using the AddSpline method of the AcadBlock, AcadModelSpace, or AcadPaperSpace object.

AcadSpline Object Methods

The AcadSpline object inherits all the methods of the AcadEntity and AcadObject objects. It also supports the following methods.

Name	Returns	Description
AddFitPoint		Adds the fit point (a set of 3-D WCS coordinates) to the spline at a given index. If the index is negative, then the point is added to the beginning of the spline. If the index exceeds the maximum value of the index, then the point is added to the end of the spline. The fit point is added to the spline and is refitted through the new set of points. The changes can be viewed only if the drawing is regenerated. Parameters: Index As Integer and FitPoint As Variant.
DeleteFitPoint		Deletes the fit point of the spline at a given index, which must be a positive integer between 0 and N minus 1, where N is the total number of fit points for the spline. The fit point is removed, and the spline refitted through the remaining points. There must be at least three fit points in the spline for this method to succeed. The changes can be viewed only if the drawing is regenerated. Parameter: Index As Integer.
ElevateOrder		Elevates the order of the spline. The range of values is one above the current order (which is the value of the Degree property plus 1) to a maximum of 26. When a spline is elevated, it is converted from an interpolated (fit) spline to a control point spline. Note that, once elevated, the spline order can't be reduced. The StartTangent and EndTangent properties are no longer used or accessible, the spline being editable by the mean of its control points. Parameter: Order As Integer.

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Name	Returns	Description
GetControlPoint	Variant	Returns the 3-D WCS coordinates of the control point of the spline at a given index as an array of Doubles. Control points fine-tune a spline definition by adding weight to a portion of the spline curve. Parameter: Index As Integer.
GetFitPoint	Variant	Returns the 3-D WCS coordinates of the fit point of the spline at a given index as an array of Doubles. Parameter: Index As Integer.
GetWeight	Double	Returns the weight of the spline at a given control point index. Parameter: Index As Integer.
Offset	Variant	Creates a new spline by offsetting the current spline by a specified distance, which must be nonzero. If the offset is negative, this means that the spline is drawn closer to the WCS origin. Parameter: Distance As Double.
PurgeFitData		Purges the fit data of the spline.
Reverse		Reverses the direction of the spline.
SetControlPoint		Sets the indexed control point of the spline at a specified point. Parameters: Index As Integer and ControlPoint As Variant.
SetFitPoint		Sets the indexed fit point of the spline at a specified point. Parameters: Index As Integer and FitPoint As Variant.
SetWeight		Sets the weight of the spline for a given control point index. Parameters: Index As Integer and Weight As Double.

AcadSpline Object Properties

The AcadSpline object inherits all the properties of the AcadEntity and AcadObject objects, as well as the common Application property. It also supports the following properties.

Name	Returns	Description
Area	Double	Gets the enclosed area of the spline in square drawing units. The area is computed as though a straight line connects the start and end points. This property's value is read-only.
Closed	Boolean	Determines whether the spline is closed. When you close a spline, you make it tangent-continuous at both ends. If the spline's start and end points are already the same, closing the spline makes it tangent-continuous at both points. When you open a closed spline, if the start and end points were the same before the close, this property returns the spline to its original state. The start and end points remain the same but lose their tangent continuity. If the spline's start and end points were not the same before the close, opening the spline returns it to its original state and removes tangent continuity. This property's value is read-only.
ControlPoints	Variant	Gets or sets the control points of a spline storing the 3-D WCS coordinates as an array of Doubles. Control points fine-tune a spline definition by adding weight or density to a portion of the spline curve.

Name	Returns	Description
Degree	Integer	Gets the degree of the spline's polynomial representation. The return value must be in the range 1 to 25. This property's value is read-only. To set a higher value for the degree, use the <code>ElevateOrder</code> method. The higher the degree is, the higher the control points are.
EndTangent	Variant	Gets or sets the end tangent of the spline as a directional vector. This property is inaccessible to control points.
FitPoints	Variant	Gets or sets the fit points of a spline, which define the path of the spline.
FitTolerance	Double	Refits the spline to the existing points with new tolerance values. If this value is 0, the spline passes through all fit points. A value greater than zero allows the curve to pass through the fit points within the specified tolerance.
IsPeriodic	Boolean	Determines whether the given spline is periodic. This property's value is read-only.
IsPlanar	Boolean	Determines whether the spline is planar. This property's value is read-only.
IsRational	Boolean	Determines whether the given spline is rational. This property's value is read-only.
Knots	Variant	Gets the knot vector for a spline.
NumberOfControlPoints	Integer	Gets the number of control points of the spline. This property's value is read-only.
NumberOfFitPoints	Integer	Gets the number of fit points of the spline. This property's value is read-only.
StartTangent	Variant	Gets or sets the start tangent for the spline as a directional vector. This property is inaccessible to control points for an elevated spline.
Weights	Variant	Gets the weight vector for spline.

Note Because this object inherits from `AcadObject`, it supports the `Modified` event.

AcadState Object

The `AcadState` object is a special object for use in monitoring the state of AutoCAD from out-of-process applications. It is a transient object that is returned from the `GetAcadState` method of the `AcadApplication` object and is used to check for AutoCAD quiescence from out-of-process applications. This object has no methods.

AcadState Object Properties

Other than the common `Application` property, the `AcadState` object supports just one property.

Name	Returns	Description
IsQuiescent	Boolean	Returns <code>True</code> if AutoCAD is idle and <code>False</code> if AutoCAD is busy. This property's value is read-only.

AcadSummaryInfo Object

This object accesses drawing properties such as the Title, Subject, Author, and Keywords properties.

AcadSummaryInfo Methods

The AcadSummaryInfo object does not inherit any methods. However, it does support the following methods.

Name	Returns	Description
AddCustomInfo		Adds a custom property field at the end of the existing list of fields. Parameters: Key As String, Value As String.
GetCustomByIndex	String	Returns the custom property name (key) and value that correspond to an index value. Parameters: Index As Long, Key As String, Value As String.
GetCustomByKey	String	Returns the custom property value that corresponds to a name (key) value. Parameters: Key As String, Value As String.
NumCustomInfo	Long	Returns the number of custom property fields that have been set.
RemoveCustomByIndex		Deletes the custom property name (key) and value that correspond to an index value. Parameter: Index As Long.
RemoveCustomByKey		Deletes the custom property value that corresponds to a name (key) value. Parameter: Key As String.
SetCustomByIndex		Sets the custom property name (key) and value that correspond to an index value. Parameters: Index As Long, Key As String, Value As String.
SetCustomByKey		Sets the custom property value that corresponds to a name (key) value. Parameters: Key As String, Value As String.

AcadSummaryInfo Properties

The AcadSummaryInfo object inherits no properties. However, it does support the following properties.

Name	Returns	Description
Author	String	Gets or sets the author of the drawing.
Comments	String	Gets or sets the comments you want to use to locate the drawing.
HyperlinkBase	String	Gets or sets the base address used for all relative links inserted within the drawing. An Internet address or path to a folder on a network drive may be specified as well.
Keywords	String	Gets or sets the keywords you want to use to locate the drawing.
LastSavedBy	String	Gets or sets the name of the last person who modified the file.
RevisionNumber	String	Gets or sets the drawing revision number.
Subject	String	Gets or sets the subject of the drawing.
Title	String	Gets or sets the title used to locate the drawing.

AcadTable Object

A table contains data in rows and columns. You can use the Table object to add and modify tables in a drawing.

To add a table to a drawing, use the `AddTable` method. After you add a table, you can modify the number of columns, the number of rows, and other settings.

All methods that change the Table object do the following:

1. Open the Table object in write mode.
2. Modify the Table object based on the input parameters.
3. Close the Table object, which recomputes the Table.

Recomputing large tables consumes a lot of time and memory because the Table object is reconstructed from scratch. If you are making multiple calls to update a table, you can improve performance by setting `RegenerateTableSuppressed` before and after making the modifications.

AcadTable Methods

The `AcadTable` object inherits all the methods of the `AcadEntity` and `AcadObject` objects. It also supports the following methods.

Name	Returns	Description
<code>ClearSubSelection</code>		Removes a subselection set of cells from a table.
<code>ClearTableStyleOverrides</code>		Clears table style overrides. Parameter: <code>Flag As Long</code> . 0 deletes all table and cell overrides. 1 deletes all table overrides but retains cell overrides. 2 deletes all cell overrides but retains table overrides.
<code>DeleteCellContent</code>		Deletes the cell content in the specified row and column. Parameters: <code>Row As Long</code> , <code>Column As Long</code> .
<code>DeleteColumns</code>		Deletes columns from a table. Parameters: <code>Row As Long</code> , <code>Column As Long</code> .
<code>DeleteRows</code>		Deletes rows from a table. Parameters: <code>Row As Long</code> , <code>Rows As Long</code> .
<code>GenerateLayout</code>		Generates the layout of a table. This method updates the Table object according to the current table style. This method computes the table geometry, including gridlines, text content, block content, and background fills.
<code>GetAlignment</code>	<code>AcRowType</code>	Returns the cell alignment for a row type. For a list of the values for the <code>AcRowType</code> enumeration, see Appendix B.
<code>GetAttachmentPoint</code>		Sets the attachment point for the specified row and column. Parameters: <code>Row As Long</code> , <code>Column As Long</code> .
<code>GetAutoScale</code>	<code>Boolean</code>	Returns the value specifying whether auto scale is used in the specified row and column. Parameters: <code>Row As Long</code> , <code>Column As Long</code> .
<code>GetBackgroundColor</code>	<code>AcCmColor</code>	Returns the background color value for the specified row type. Parameter: <code>RowType As AcRowType</code> . For a list of possible values for the <code>AcRowType</code> enumeration, see Appendix B.

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Name	Returns	Description
GetBackgroundColorNone	Boolean	Returns the value specifying that there is no background color for the specified row type. For a list of the values for the <code>AcRowType</code> enumeration, see Appendix B.
GetBlockAttributeValue	String	Returns the attribute value of the specified cell for the attribute definition object contained in the block. Parameters: Row As Long, Column As Long, AttDefId As Long.
GetBlockRotation	Double	Returns the block rotation angle for the specified row and column. Parameters: Row As Long, Column As Long.
GetBlockScale	Double	Returns the block scale value for the specified row and column. Parameters: Row As Long, Column As Long.
GetBlockTableRecordId	Long	Returns the block table record ID of the cell. Parameters: Row As Long, Column As Long.
GetCellAlignment	AcCellAlignment	Returns the alignment for the cell in the specified row and column. Parameters: Row As Long, Column As Long. For a list of possible values for the <code>AcCellAlignment</code> enumeration, see Appendix B.
GetCellBackgroundColor	AcCmColor	Returns the background true color value of a cell in the specified row and column. Parameters: Row As Long, Column As Long. For a list of possible values for the <code>AcCmColor</code> enumeration, see Appendix B.
GetCellBackgroundColorNone	Boolean	Returns the value specifying that there is no background color for the specified row and column. Parameters: Row As Long, Column As Long.
GetCellContentColor	AcCmColor	Returns the true color value for the content of the specified row and column. Parameters: Row As Long, Column As Long. For a list of possible values for the <code>AcCmColor</code> enumeration, see Appendix B.
GetCellExtents		Returns the cell extents for the specified row and column. Parameters: Row As Long, Column As Long, OuterCell As Boolean. The specified cell is an outer cell when the <code>OuterCell</code> parameter is True, an inner cell when False.
GetCellGridColor	AcCmColor	Returns the grid color value for an edge of the specified row and column. Parameters: Row As Long, Column As Long, Edge As <code>AcCellEdgeMask</code> . For a list of possible values for the <code>AcCmColor</code> and <code>AcCellEdgeMask</code> enumerations, see Appendix B.
GetCellGridLineWeight	Acad_LWeight	Returns the grid lineweight value for an edge of the specified row and column. Parameters: Row As Long, Column As Long, Edge As <code>AcCellEdgeMask</code> . For a list of possible values for the <code>Acad_LWeight</code> and <code>AcCellEdgeMask</code> enumerations, see Appendix B.
GetCellGridVisibility	Boolean	Returns the grid visibility value for an edge of the specified row and column. Parameters: Row As Long, Column As Long, Edge As <code>AcCellEdgeMask</code> . For a list of possible values for the <code>AcCellEdgeMask</code> enumeration, see Appendix B.
GetCellStyleOverrides		Returns the cell style overrides. Parameters: Row As Long, Column As Long.

Name	Returns	Description
GetCellTextHeight	Double	Returns the text height for the specified row and column. Parameters: Row As Long, Column As Long.
GetCellTextStyle	String	Returns the text style name for the specified row and column. Parameters: Row As Long, Column As Long.
GetCellType	AcCellType	Returns the cell type for the specified row and column. Parameters: Row As Long, Column As Long. For a list of possible values for the AcCellType enumeration, see Appendix B.
GetColumnWidth	Double	Returns the column width for the column at the specified column index in the table. Parameters: Column As Long.
GetContentColor	AcCmColor	Returns the true color value for the specified row type. Parameters: RowType As AcRowType. For a list of possible values for the AcCmColor and AcRowType enumerations, see Appendix B.
GetFieldId	Long	Returns the field object ID of the specified cell. Parameters: Row As Long, Column As Long.
GetGridColor	AcCmColor	Returns the grid color value for a grid linetype and row type. Parameters: GridLineType As AcGridLineType, RowType As AcRowType. For a list of possible values for the AcCmColor, AcGridLineType, and AcRowType enumerations, see Appendix B.
GetGridLineWeight	Acad_LWeight	Returns a grid lineweight value for a grid linetype and row type. Parameters: GridLineType As AcGridLineType, RowType As AcRowType. For a list of possible values for the Acad_LWeight, AcGridLineType, and AcRowType enumerations, see Appendix B.
GetGridVisibility	Boolean	Returns a grid visibility value for the specified grid linetype and row type. Parameters: GridLineType As AcGridLineType, RowType As AcRowType. For a list of possible values for the AcGridLineType, and AcRowType enumerations, see Appendix B.
GetMinimumColumnWidth	Double	Gets the minimum column width for the column at the specified column index in the table. Parameter: Column As Long.
GetMinimumRowHeight	Double	Returns the minimum row height for the specified row. Parameter: Row As Long.
GetRowHeight	Double	Returns the row height for the row at the specified row index in the table. Parameter: Row As Long.
GetRowType	AcRowType	Returns the row type for the specified row. Parameter: Row As Long. For a list of possible values for the AcRowType enumeration, see Appendix B.
GetSubSelection		Returns the row and column indexes of the cells in a subselection set. This method gets the row and column indices of the cells in the subselection set. The total number of selected cells is equal to (rowMax - rowMin + 1) * (colMax - colMin + 1). Parameters: RowMin As Long, RowMax As Long, ColumnMin As Long, ColumnMax As Long.
GetText	String	Returns the text value for the specified row and column. Parameters: Row As Long, Column As Long.
GetTextHeight	Double	Returns the text height for the specified row type. Parameter: RowType As AcRowType. For a list of possible values for the AcRowType enumeration, see Appendix B.

Continued

Name	Returns	Description
GetTextRotation	AcRotationAngle	Returns the text rotation for the specified row and column. Parameters: Row As Long, Column As Long. For a list of possible values for the AcRotationAngle enumeration, see Appendix B.
GetTextStyle	String	Returns the text style name for the specified row type. Parameter: RowType As AcRowType. For a list of possible values for the AcRowType enumeration, see Appendix B.
HitTest	Boolean	Returns the cell at the specified location. This function performs a hit test by specifying a point and viewing direction. The cell hit by the ray is returned. Parameters: InputPickPoint As Variant, ViewVector As Variant, ResultRowIndex As Long, ResultColumnIndex As Long.
InsertColumns		Inserts columns in a table. Parameters: StartColumn As Long, Width As Double, NumberOfColumns As Long.
InsertRows		Inserts rows in a table. Parameters: StartRow As Long, Height As Double, NumberOfRows As Long.
IsMergedCells	Boolean	Returns the merge status of a cell. Parameters: Row As Long, Column As Long, MinRow As Long, MaxRow As Long, MinColumn As Long, MaxColumn As Long.
MergedCells		Merges cells in a table. This method merges a rectangular region of cells. The total number of cells to be merged is equal to $(\text{maxRow} - \text{minRow} + 1) * (\text{maxCol} - \text{minCol} + 1)$. Parameters: MinRow As Long, MaxRow As Long, MinColumn As Long, MaxColumn As Long.
RecomputeTableBlock		Updates a table block. This function updates the table block record that the Table object references to match changes made to the Table object since the most recent update of the table block record. Parameter: ForceUpdate As Boolean. An update will occur when the parameter value is True.
ReselectSubRegion		Returns subentities of a selection set in a table.
Select		Selects a cell in a table. This function selects a cell in the table by specifying a point, viewing direction, and orientation. Parameters: InputPickPoint As Variant, ViewVector As Variant, ViewOrientationVector As Variant, ApertureBoxWidth As Double, ApertureBoxHeight As Double, AllowOutside As Boolean, ResultRowIndex As Long, ResultColumnIndex As Long.
SelectSubRegion		Selects a set of cells in a table. Parameters: WindowPoint1 As Variant, WindowPoint2 As Variant, ViewVector As Variant, ViewOrientationVector As Variant, Type As AcSelectType, IncludeCurrentSelection As Boolean, MinRow As Long, MaxRow As Long, MinColumn As Long, MaxColumn As Long.
SetAlignment		Sets a cell alignment for the specified row types. Parameters: RowTypes As Long, CellAlignment As AcCellAlignment. For a list of possible values for the AcCellAlignment enumeration, see Appendix B.
SetAutoScale		Sets the value specifying whether auto scale is used in the specified row and column. Parameters: Row As Long, Column As Long, AutoScale As Boolean. Autoscale is used when True.

Name	Returns	Description
SetBackgroundColor		Sets a background color value for the specified row types. Parameters: RowTypes As Long, Color As AcCmColor. For a list of possible values for the AcCmColor enumeration, see Appendix B.
SetBackgroundColorNone		Sets the value specifying whether there is no background color for the specified row types. Parameters: RowTypes As Long, Value As Boolean. No background color used when Value is True.
SetBlockAttributeValue		Sets the attribute value of the specified cell for the attribute definition object that is contained in the block. Parameters: Row As Long, Column As Long, AttDefId As Long, AttributeValue As String.
SetBlockRotation		Sets the block rotation angle for the specified row and column. Parameters: Row As Long, Column As Long, Rotation As Double.
SetBlockScale		Sets the block scale value for the specified row and column. Parameters: Parameters: Row As Long, Column As Long, Scale As Double.
SetBlockTableRecordId		Sets the block table record ID of the cell. Parameters: Row As Long, Column As Long, BlockId As Long, AutoFit As Boolean.
SetCellAlignment		Sets the cell alignment for the specified row and column. Parameters: Row As Long, Column As Long, CellAlignment As AcCellAlignment. For a list of possible values for the AcCellAlignment enumeration, see Appendix B.
SetCellBackgroundColor		Sets the background true color value for the specified row and column. Parameters: Row As Long, Column As Long, Color As AcCmColor. For a list of possible values for the AcCmColor enumeration, see Appendix B.
SetCellBackgroundColorNone		Sets the value specifying that there is no background color for the specified row and column. Parameters: Row As Long, Column As Long, Value As Boolean. No background color used when Value is True.
SetCellContentColor		Sets the true color value for the content of the specified row and column. Parameters: Row As Long, Column As Long, Color As AcCmColor.
SetCellGridColor		Sets the grid color value for an edge of the specified row and column. Parameters: Row As Long, Column As Long, Color As AcCmColor.
SetCellGridLineWeight		Sets the grid lineweight value for an edge of the specified row and column. Parameters: Row As Long, Column As Long, Edges As Long, Lineweight As AcadLWeight. For a list of possible values for the AcadLWeight enumeration, see Appendix B.
SetCellGridVisibility		Sets the grid visibility value for an edge of the specified row and column. Parameters: GridLineTypes As Long, RowTypes As Long, Value As Boolean. Grid cell will be visible when value is True.
SetCellTextHeight		Sets the text height for the specified row and column. Parameters: Row As Long, Column As Long, TextHeight As Double.

Continued

Name	Returns	Description
SetCellTextStyle		Sets the text style name for the specified row and column. Parameters: Row As Long, Column As Long, TextStyle As String.
SetCellType		Sets the cell type for the specified row and column. Parameters: Row As Long, Column As Long, CellType As AcCellType. For a list of possible values for the AcCellType enumeration, see Appendix B.
SetColumnWidth		Sets the column width for the column at the specified column index in the table. Parameters: Column As Long, Width As Double.
SetContentColor		Sets the true color value for the specified row types. Parameters: RowTypes As Long, Color As AcCmColor. For a list of possible values for the AcCmColor enumeration, see Appendix B.
SetFieldId		Sets the field object ID of the specified cell. Parameters: Row As Long, Column As Long, FieldId As Long.
SetGridColor		Sets the grid color value for the specified grid linetypes and row types. Parameters: GridLineTypes As Long, RowTypes As Long, Color As AcCmColor. For a list of possible values for the AcCmColor enumeration, see Appendix B.
SetGridLineWeight		Sets the grid lineweight value for the specified grid linetypes and row types. Parameters: GridLineTypes As Long, RowTypes As Long, Lineweight As AcadLWeight. For a list of possible values for the AcadLWeight enumeration, see Appendix B.
SetGridVisibility		Sets the grid visibility value for the specified grid linetype and row types. Parameters: GridLineTypes As Long, RowTypes As Long, Value As Boolean. Grid will be visible when value is True.
SetRowHeight		Sets the height for the row at the specified row index in the table. Parameters: Row As Long, Height As Double.
SetSubSelection		Sets the row and column indices of the cells in a subselection set. This method sets the row and column indices of the cells in the subselection set. The total number of selected cells is equal to (rowMax - rowMin + 1) * (colMax - colMin + 1). Parameters: RowMin As Long, RowMax As Long, ColumnMin As Long, ColumnMax As Long.
SetText		Sets the text value for the specified row and column. Row As Long, Column As Long, Text As String.
SetTextHeight		Sets the text height for the specified row types. Parameters: RowTypes As Long, Height As Double.
SetTextRotation		Sets the text rotation for the specified row and column. Parameters: Row As Long, Column As Long, Rotation As AcRotationAngle. For a list of possible values for the AcRotationAngle enumeration, see Appendix B.
SetTextStyle		Sets the text style name for the specified row types. Parameters: RowTypes As Long, TextStyle As String.
UnmergeCells		Splits merged cells in a table. Parameters: RowMin As Long, RowMax As Long, ColumnMin As Long, ColumnMax As Long.

AcadTable Properties

The AcadTable object inherits all the properties of the AcadEntity and AcadObject objects as well as the common Application property. It also supports the following properties.

Name	Returns	Description
Columns	Long	Gets or sets the number of columns in the table.
ColumnWidth	Double	Gets or sets the uniform width of all columns in the table.
Direction	Variant	Gets or sets the direction vector of the table. The vector (X-axis, in WCS coordinates) defines the horizontal plane containing the table.
FlowDirection	AcTableDirection	Gets or sets whether the title and header rows are at the bottom or the top of the table. For a list of possible values for the AcTableDirection enumeration, see Appendix B.
HasSubSelection	Boolean	Gets whether the table contains a subselection set. Read-only.
HeaderSuppressed	Boolean	Gets or sets whether the header of the table is suppressed.
Height	Double	Gets or sets the height of the table.
HorzCellMargin	Double	Gets or sets the value of the horizontal margin of cells.
InsertionPoint	Variant	Gets or sets insertion point for a table
MinimumTableHeight	Double	Gets the minimum height of the table. Read-only.
MinimumTableWidth	Double	Gets the minimum width of the table. Read-only.
RegenerateTableSuppressed	Boolean	Enables or disables regeneration of the table block. All methods that change the Table object do the following: open the Table object in write mode, modify the Table object based on the input parameters, close the Table object, which recomputes the Table. Recomputing large tables consumes a lot of time and memory because the Table object is reconstructed from scratch. To avoid performance problems, you should temporarily disable the regeneration of the Table object when modifying large tables through the VBA APIs and then apply the modifications and re-enable regeneration. To access this property, use the IAcadTable2 interface.
RowHeight	Double	Gets or sets a uniform height for all rows in the table.
Rows	Long	Gets or sets the number of rows in the table.
StyleName	String	Gets or sets the name of the table style.
TableStyleOverrides	Variant	Gets the table style overrides. Read-only.
TitleSuppressed	Boolean	Gets or sets whether the title of the table is suppressed.
VertCellMargin	Double	Gets or sets the value of the vertical margin of cells.
Width	Double	Gets or sets the width of the table.

Note Because this object inherits from AcadObject, it supports the Modified event.

AcadTableStyle Object

You can use the TableStyle object to store table formatting, such as grid visibility, lineweight, and color. The object controls the initial formatting of a newly created Table object.

AcadTableStyle Methods

The AcadTableStyle object inherits all the methods of the AcadObject object. It also supports the following methods.

Name	Returns	Description
GetAlignment	AcRowType	Returns the cell alignment for a row type. For a list of the values for the AcRowType enumeration, see Appendix B.
GetBackgroundColor	AcCmColor	Returns the background color value for the specified row type. Parameter: RowType As AcRowType. For a list of possible values for the AcRowType enumeration, see Appendix B.
GetBackgroundColorNone	Boolean	Returns the value specifying that there is no background color for the specified row type. For a list of the values for the AcRowType enumeration, see Appendix B.
GetColor	AcCmColor	Returns the true color value for the specified row type. Parameter: RowType As AcRowType. For a list of the values for the AcCmColor and AcRowType enumerations, see Appendix B.
GetGridColor	AcCmColor	Returns the grid color value for a grid linetype and row type. Parameters: GridLineType As AcGridLineType, RowType As AcRowType. For a list of possible values for the AcCmColor, AcGridLineType, and AcRowType enumerations, see Appendix B.
GetGridLineWeight	Acad_LWeight	Returns a grid lineweight value for a grid linetype and row type. Parameters: GridLineType As AcGridLineType, RowType As AcRowType. For a list of possible values for the Acad_LWeight, AcGridLineType, and AcRowType enumerations, see Appendix B.
GetGridVisibility	Boolean	Returns a grid visibility value for the specified grid linetype and row type. Parameters: GridLineType As AcGridLineType, RowType As AcRowType. For a list of possible values for the AcGridLineType, and AcRowType enumerations, see Appendix B.
GetTextHeight	Double	Returns the text height for the specified row type. Parameter: RowType As AcRowType. For a list of possible values for the AcRowType enumeration, see Appendix B.
GetTextStyle	String	Returns the text style name for the specified row type. Parameter: RowType As AcRowType. For a list of possible values for the AcRowType enumeration, see Appendix B.
SetAlignment		Sets a cell alignment for the specified row types. Parameters: RowTypes As Long, CellAlignment As AcCellAlignment. For a list of possible values for the AcCellAlignment enumeration, see Appendix B.
SetBackgroundColor		Sets a background color value for the specified row types. Parameters: RowTypes As Long, Color As AcCmColor. For a list of possible values for the AcCmColor enumeration, see Appendix B.

Name	Returns	Description
SetBackgroundColorNone		Sets the value specifying whether there is no background color for the specified row types. Parameters: RowTypes As Long, Value As Boolean. No background color used when Value is True.
SetColor		Sets the true color value for the specified row types. Parameters: RowTypes As Long, Color As AcCmColor. For a list of possible values for the AcCmColor enumeration, see Appendix B.
SetGridColor		Sets the grid color value for the specified grid linetypes and row types. Parameters: GridLineTypes As Long, RowTypes As Long, Color As AcCmColor. For a list of possible values for the AcCmColor enumeration, see Appendix B.
SetGridLineWeight		Sets the grid lineweight value for the specified grid linetypes and row types. Parameters: GridLineTypes As Long, RowTypes As Long, Lineweight As AcadLWeight. For a list of possible values for the AcadLWeight enumeration, see Appendix B.
SetGridVisibility		Sets the grid visibility value for the specified grid linetype and row types. Parameters: GridLineTypes As Long, RowTypes As Long, Value As Boolean. Grid will be visible when value is True.
SetTextHeight		Sets the text height for the specified row types. Parameters: RowTypes As Long, Height As Double.
SetTextStyle		Sets the text style name for the specified row types. Parameters: RowTypes As Long, TextStyle As String.

AcadTableStyle Properties

The AcadTableStyle object inherits all the properties of the AcadObject objects as well as the common Application property. It also supports the following properties.

Name	Returns	Description
BitFlags	Long	Gets and sets the bit flag values of a table style.
Description	String	Gets and sets the description of an object.
FlowDirection	AcTableDirection	Gets or sets whether the title and header rows are at the bottom or the top of the table. For a list of possible values for the AcTableDirection enumeration, see Appendix B.
HeaderSuppressed	Boolean	Gets or sets whether the header of the table is suppressed.
HorzCellMargin	Double	Gets or sets the value of the horizontal margin of cells.
Name	String	Gets or sets the name of the table style.
TitleSuppressed	Boolean	Gets or sets whether the title of the table is suppressed.
VertCellMargin	Double	Gets or sets the value of the vertical margin of cells.

Note Because this object inherits from AcadObject, it supports the Modified event.

AcadText Object

The AcadText object represents a single line of alphanumeric characters. This object differs from the AcadMText object in that it creates only a single line of text, whereas the AcadMText object creates a paragraph of text. The AcadText object is created using the AddText method of the AcadBlock, AcadModelSpace, or AcadPaperSpace object.

AcadText Object Methods

The AcadText object inherits all the methods of the AcadEntity and AcadObject objects. It supports no other methods.

AcadText Object Properties

The AcadText object inherits all the properties of the AcadEntity and AcadObject objects, as well as the common Application property. It also supports the following properties.

Name	Returns	Description
Alignment	AcAlignment	Gets or sets both the vertical and horizontal alignments for the text. For a full list of the values of the AcAlignment enumerated type, see Appendix B. See also the InsertPoint and TextAlignmentPoint properties.
Backward	Boolean	Specifies the direction of text. True is for backward; False is for forward.
Height	Double	Use to set the height of uppercase text, measured in the current units. This property is used as a scale factor for both the height and width of the text.
HorizontalAlignment	AcHorizontalAlignment	Gets or sets the horizontal alignment for the text. For a full list of the values of the AcHorizontalAlignment enumerated type, see Appendix B. See also the InsertPoint and TextAlignmentPoint properties.
InsertionPoint	Variant	Gets or sets the insertion point for text. This property's value is read-only except for text whose Alignment property is set to acAlignmentLeft, acAlignmentAligned, or acAlignmentFit. To position text whose justification is other than left, aligned, or fit, use the TextAlignmentPoint property. Also, text with the HorizontalAlignment property set to acHorizontalAlignmentLeft, acHorizontalAlignmentAligned, or acHorizontalAlignmentFit uses this property to position the text. All other settings for HorizontalAlignment use the TextAlignmentPoint property.
Normal	Variant	Gets or sets the 3-D (Z-axis) normal unit vector for the text object.
ObliqueAngle	Double	Gets or sets the oblique angle of the attribute, and is measured from the vertical axis. The units are radians within the range of -85 to +85 degrees. A positive value denotes a lean toward the right.

Name	Returns	Description
Rotation	Double	Gets or sets the rotation angle (in radians) for the text object, relative to the X-axis with positive values going counterclockwise when viewed along the Z-axis toward the origin. The rotation angle is read-only for text whose Alignment property is set to acAlignmentAligned or acAlignmentFit.
ScaleFactor	Double	Gets or sets the scale factor for the object. It has to be a value greater than 0.0. A scale factor greater than 1 enlarges the object. A scale factor between 0 and 1 shrinks the object. The initial value for this property is 1.0000.
StyleName	String	Gets or sets the name of the style used with the object, the default being the current style. Use the TextStyle object to change the attribute of a given text style. Note that the name given must already be defined in the drawing.
TextAlignmentPoint	Variant	Gets or sets the alignment point for text, returning a three-element array of Doubles. If the Alignment property is set to acAlignmentLeft, this property will be reset to (0,0,0) and will become read-only. Also, you can't use this property to position text with the HorizontalAlignment property set to acHorizontalAlignmentLeft. In both cases, use the InsertPoint property.
TextGenerationFlag		Gets or sets the attribute text generation flag. Possible values: acTextFlagBackward and acTextFlagUpsideDown.
TextString	String	Gets or sets the text string for the entity.
Thickness	Double	Gets or sets the distance the AcadText object is extruded above or below its elevation, in a Z-axis direction. The default is 0.0.
UpsideDown	Boolean	Specifies the direction of text. True is for upside down; False is for the right way up.
VerticalAlignment	AcVerticalAlignment	Gets or sets vertical alignment for the text. For a list of the values of the AcVerticalAlignment enumerated type, see Appendix B.

■ **Note** Because this object inherits from AcadObject, it supports the Modified event.

AcadTextStyle Object

The AcadTextStyle object represents a named, saved collection of settings that determine the appearance of text characters. The ActiveTextStyle property sets the active text style, which is used to format new text created in the drawing and also existing text that has no individual text style specified. If changes are made to the active text style, the new AcadTextStyle object must be reset as the active text style, and the drawing must be regenerated for the changes to

appear. To specify a distinct individual text style for an object so that it won't change along with the active text style, use the `StyleName` property for that particular object.

AcadTextStyle Object Methods

The `AcadTextStyle` object inherits all the methods of the `AcadObject` object. It also supports the following methods.

Name	Description
<code>GetFont</code>	Gets the definition data of the font for the text style. Parameters: <code>TypeFace As String</code> , <code>Bold As Boolean</code> , <code>Italic As Boolean</code> , <code>Charset As Integer</code> , and <code>PitchAndFamily As Integer</code> . The <code>TypeFace</code> parameter specifies the font name of the text style. The <code>Bold</code> parameter is set to <code>True</code> for text to be bold. The <code>Italic</code> parameter is set to <code>True</code> for text to be italic. The <code>Charset</code> parameter specifies the character set for the font. The default value is 1. The <code>PitchAndFamily</code> parameter combines several constants (by a Boolean OR operation) to give the pitch and family values for the font. The default value is 0. A full list of constants can be found in the <code>win32api.txt</code> file supplied with the Visual Basic IDE.
<code>SetFont</code>	Sets the definition data of the font for the text style. Parameters: <code>TypeFace As String</code> , <code>Bold As Boolean</code> , <code>Italic As Boolean</code> , <code>Charset As Integer</code> , and <code>PitchAndFamily As Integer</code> .

AcadTextStyle Object Properties

The `AcadTextStyle` object inherits all the properties of the `AcadObject` object, as well as the common `Application` property. It also supports the following properties.

Name	Returns	Description
<code>BigFontFile</code>	String	Gets or sets the name of the Asian language big font file associated with the text or attribute. The only valid file type is <code>.shx</code> , and the property can't be set to <code>NULL</code> or contain an empty string.
<code>FontFile</code>	String	Gets or sets the primary font file path and name. Fonts define the shapes of the text characters that make up each character set. Note that once this property has been set, you have to regenerate the drawing to see the changes to the text.
<code>Height</code>	Double	Sets the height of the text in current units.
<code>LastHeight</code>	Double	Gets or sets the last text height value used.
<code>Name</code>	String	Gets the name of the object. This property's value is read-only.
<code>ObliqueAngle</code>	Double	Gets or sets the oblique angle of the attribute and is measured from the vertical axis. The units are radians within the range of -85 to $+85$ degrees. A positive value denotes a lean toward the right.
<code>TextGenerationFlag</code>	Integer	Gets or sets the attribute text generation flag. Possible values are <code>acTextFlagBackward</code> and <code>acTextFlagUpsideDown</code> .
<code>Width</code>	Double	Gets or sets the width of the text boundary and sets the character spacing. Entering a value of less than 1.0 condenses the text. Entering a value of greater than 1.0 expands it. The maximum value is 100.

Note Because this object inherits from `AcadObject`, it supports the `Modified` event.

AcadTextStyles Collection

The `AcadTextStyles` collection contains all the text styles in the drawing. To add a new member to the collection, use the `Add` method. To select a specific text style, use the `Item` method. Although this collection inherits a `Delete` method, you can't actually delete this collection. If you need to delete a specific text style, use the `Delete` method found in the `AcadTextStyle` object. There is no limit to the number of text styles you can create in your drawing. However, there can be only one instance of the `AcadTextStyles` collection, which is predefined for each drawing. You can make multiple references to it by using the `TextStyles` property.

AcadTextStyles Collection Methods

The `AcadTextStyles` collection inherits all the methods of the `AcadObject` object. It also supports the following methods.

Name	Returns	Description
<code>Add</code>	<code>AcadTextStyle</code>	Creates a member object and adds it to the collection. Parameter: <code>Name</code> As String.
<code>Item</code>	<code>AcadTextStyle</code>	Gets the member object at a given index in a collection. Parameter: <code>Index</code> As Variant (an Integer or a String). If the <code>Index</code> value is a String, it must match an existing text style definition.

AcadTextStyles Collection Properties

The `AcadTextStyles` collection inherits all the properties of the `AcadObject` object, the `Count` property, and the common `Application` property. It supports no other properties.

Note Because this collection inherits from `AcadObject`, it supports the `Modified` event.

AcadTolerance Object

The `AcadTolerance` object represents a geometric tolerance contained in a feature control frame. Tolerances are influenced by several system variables, which can be set and queried using the `SetVariable` and `GetVariable` methods. The color of the feature control frame is controlled by `DIMCLRD`. The color of the tolerance text is controlled by `DIMCLRT`. The gap between the feature control frame and the text is controlled by `DIMGAP`. The size of the tolerance text is controlled by `DIMTXT`. Finally, the style of the tolerance text is controlled by `DIMTXTSTY`.

The `AcadTolerance` object is created using the `AddTolerance` method of the `AcadBlock`, `AcadModelSpace`, or `AcadPaperSpace` objects.

AcadTolerance Object Methods

The AcadTolerance object inherits all the methods of the AcadEntity and AcadObject objects. It supports no other methods.

AcadTolerance Object Properties

The AcadTolerance object inherits all the properties of the AcadEntity and AcadObject objects, as well as the common Application property. It also supports the following properties.

Name	Returns	Description
Arrowhead1Type	AcDimArrowheadType	Gets or sets the type of arrowhead for the tolerance object. It overrides the value of the DIMBLK1 system variable. For the values of the AcDimArrowheadType enumerated type, see Appendix B.
DecimalSeparator	String	Gets or sets the character used as the decimal separator in tolerance values. The initial value for this property is the period (.), though any character will be accepted as a valid value for this property. It overrides the value of the DIMSEP system variable.
DimensionLineColor	AcColor	Gets or sets the color of the tolerance object. Use a color index number from 0 to 256 or one of the constants listed here: acByBlock (where AutoCAD draws objects in the default color) or acByLayer (where AutoCAD draws objects in the color specified for the layer). For a list of possible values for the AcColor enumerated type, see Appendix B. This property overrides the value of the DIMCLRD system variable for the given object.
DirectionVector	Variant	Gets or sets the direction for the tolerance through a vector. It returns a three-element array of Doubles.
FractionFormat	AcDimFractionType	Gets or sets the formats of fractional values. It overrides the value of the DIMFRAC system variable. Possible values are acHorizontal, acDiagonal, and acNotStacked.
InsertionPoint	Variant	Represents an insertion point in the tolerance object as a set of 3-D WCS coordinates. It returns a three-element array of Doubles.
Normal	Variant	Gets or sets the 3-D (Z-axis) normal unit vector for the tolerance object.
PrimaryUnitsPrecision	AcDimPrecision	Gets or sets the number of decimal places displayed for the primary units. For a list of possible values for the AcDimPrecision enumerated type, see Appendix B.
ScaleFactor	Double	Gets or sets the scale factor for the object. It has to be a value greater than 0.0. A scale factor greater than 1 enlarges the object. A scale factor between 0 and 1 shrinks the object. The initial value for this property is 1.0000. This property overrides the DIMSCALE system variable.
StyleName	String	Gets or sets the name of the style used with the object, the default being the current style. Use the TextStyle object to change the attribute of a given text style. Note that the name given must already be defined in the drawing.

Name	Returns	Description
TextColor	AcColor	Gets or sets the color of the tolerance text. Use a color index number from 0 to 256 or one of the constants listed here: <code>acByBlock</code> (where AutoCAD draws objects in the default color), <code>acByLayer</code> (where AutoCAD draws objects in the color specified for the layer), and the <code>AcColor</code> enumerated type (see Appendix B). This property overrides the value of the <code>DIMCLRD</code> system variable for the given object.
TextHeight	Double	Gets or sets the height for the tolerance text, the value of which overrides the value of the <code>DIMTXT</code> system variable. This property is ignored if the current text style has a fixed text height. The initial value is 0.1800.
TextString	String	Gets or sets the text string for the entity.
TextStyle	String	Gets or sets the text style for the tolerance text, the value of which overrides the value of the <code>DIMTXSTY</code> system variable. The initial value is <code>STANDARD</code> .

Note Because this object inherits from `AcadObject`, it supports the `Modified` event.

AcadToolbar Object

The `AcadToolbar` object represents an AutoCAD toolbar, which provides access to frequently used or custom commands, settings, and macros. You can add tools to or remove tools from the AutoCAD default toolbars and reposition them. You can also create your own toolbars.

You create a new toolbar with the `Add` method. `AddToolbarButton` inserts a new button to the toolbar, and `AddSeparator` adds a separator. Use the `ToolbarItem` object to add functionality to the toolbar button or to create flyout toolbar buttons. You can change the position of toolbars by docking them along the top, bottom, or sides of the AutoCAD screen, or you can float toolbars anywhere on the screen.

AcadToolbar Object Methods

The `AcadToolbar` object supports the following methods.

Name	Returns	Description
AddToolBarButton	AcadToolBarItem	Adds an item to the toolbar immediately before the position specified by the index. To associate a toolbar to a flyout button, use the AttachToolBarToFlyout method on the returned AcadToolBarItem object. Note that to add a button to a toolbar, it has to be visible. Parameters: Index As Variant (Integer or String), Name As String, HelpString As String, Macro As String, and [FlyoutButton As Variant]. The Index parameter for this property, and the one that follows, must be between 0 and N minus 1 (if it is an Integer), where N is the number of objects in the pop-up menu. The new item will be added immediately before the specified index location. To add the new menu item to the end of a menu, set the index to be greater than N. If a String is specified and the indexed item doesn't exist, then the new menu item is added at the end of the menu. The Name must be composed of alphanumeric characters with no punctuation other than a dash (-) or an underscore (_). This will show up as the tooltip. The FlyoutButton parameter is a Boolean variable determining whether the new button is to be a flyout button (True) (False). False is the default value returned if this option is ignored.
AddSeparator	AcadToolBarItem	Adds a separator to the toolbar immediately before the position specified by the index. The first item in a menu can't be a separator. You can't add a separator immediately next to another separator. Parameter: Index As Variant (Integer or String).
Delete		Deletes an item from the toolbar. Note that the item must have no Index parameter.
Dock		Docks the toolbar to its owning frame window. Parameter: Side As AcToolBarDockStatus. For a list of possible values for the AcToolBarDockStatus enumeration, see Appendix B.
Float		Floats the toolbar. Parameters: Top As Integer, Left As Integer, and NumberFloatRows As Integer. The Top and Left parameters specify the pixel location for the top and left edges of the toolbar, respectively, and are screen coordinates. Get the position of the document window to float in the drawing limits. The NumberFloatRows parameter designates the number of rows to create in the horizontal toolbar. The buttons on the toolbar will be distributed automatically across the number of rows designated.
Item	AcadToolBarItem	Returns the item from the collection specified by the index. Parameter: Index As Variant (Integer or String). If the Index value is a String, it must match an existing object name in the collection.

AcadToolBar Object Properties

As well as the common Application property and the Count property, the AcadToolBar object supports the following properties.

Name	Returns	Description
DockStatus	AcToolbarDockStatus	Gets whether and where the toolbar is docked. This property's value is read-only. For a list of possible values for the AcToolbarDockStatus enumerated type, see Appendix B.
FloatingRows	Integer	Gets or sets the number of rows if the toolbar is floating.
Height	Integer	Gets the height of the toolbar. This property's value is read-only.
HelpString	String	Gets or sets the toolbar's help string.
LargeButtons	Boolean	Specifies whether the toolbar buttons are large. This property's value is read-only.
Left	Integer	Gets or sets the position in pixels of left edge of the toolbar from the left side of the screen.
Name	String	Gets or sets the toolbar's name.
Parent	AcadToolbars	Gets the object to which the toolbar object belongs. This property's value is read-only.
TagString	String	Gets the menu's tag string, which can consist of alphanumeric and underscore () characters. This string uniquely identifies the item within a given menu file. This string is automatically assigned when the object is created and is used internally by AutoCAD for toolbar and menu identification. This property's value is read-only.
Top	Integer	Gets or sets the position in pixels of the top edge of the toolbar from the top of the screen.
Visible	Boolean	Specifies whether the toolbar is visible.
Width	Double	Gets the width of the toolbar. This property's value is read-only.

AcadToolbarItem Object

The `AcadToolbarItem` object represents a single button or separator on an AutoCAD toolbar. A toolbar button can contain a macro to be executed when it is selected by the user, or it can contain a nested toolbar called a flyout. Separators can't contain macros or flyouts. The `Macro` property is used to add or change the button's associated macro.

AcadToolbarItem Object Methods

The `AcadToolbarItem` object supports the following methods.

Name	Description
AttachToolbarToFlyout	Attaches the toolbar to a toolbar button defined as a flyout. To create a new button as a flyout, you must first use the <code>AddToolbarButton</code> method of the <code>AcadToolbar</code> object with the <code>FlyoutButton</code> parameter set to <code>True</code> . Only then can you attach the toolbar to the flyout using this method. Parameters: <code>MenuGroupName</code> As String and <code>ToolbarName</code> As String.
Delete	Deletes a specified toolbar item. Note that you can only add or remove toolbar items when the toolbar is visible.

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Name	Description
GetBitmaps	Gets the large and small bitmaps used as icons for the toolbar item. Parameters: SmallIconName As String and LargeIconName As String. The parameters require both the path and file name of the bitmaps.
SetBitmaps	Sets the large (24×22 pixels) and small (15×16 pixels) bitmaps used as icons for the toolbar item. Parameters: SmallIconName As String and LargeIconName As String. The parameters require both the path and file name of the bitmaps.

AcadToolBarItem Object Properties

As well as the common Application property, the AcadToolBarItem object supports the following properties.

Name	Returns	Description
Flyout	AcadToolBar	Gets the toolbar associated with the item, if it's a flyout toolbar item. This property's value is read-only.
HelpString	String	Gets or sets the help string for the menu item, which appears in the AutoCAD status line when a user highlights a menu item.
Index	Integer	Gets the index for the menu item. The first position in the index is 0. This property's value is read-only.
Macro	String	Gets or sets the macro associated with the toolbar item.
Name	String	Gets or sets the name of the toolbar item.
Parent	AcadToolBar	Gets the parent object of the toolbar item. This property's value is read-only.
TagString	String	Gets the menu's tag string, which can consist of alphanumeric and underscore () characters. This string uniquely identifies the item within a given menu file. This string is automatically assigned when the object is created and is used internally by AutoCAD for toolbar and menu identification.
Type	AcToolBarItemType	Gets the toolbar item's type. This property's value is read-only. Possible values for the AcToolBarItemType enumeration are acToolBarButton, acToolBarFlyout, and acToolBarControl.

AcadToolbars Collection

The AcadToolbars collection contains all the toolbars loaded in the current AutoCAD session, some or all of which may be currently displayed in AutoCAD. To load an existing toolbar into the current session, load the menu group that contains the toolbar using the Load method of the AcadMenuGroups collection. To display large buttons for all toolbars, set the LargeButtons property to True.

AcadToolbars Collection Methods

The AcadToolbars collection supports just two methods.

Name	Returns	Description
Add	AcadToolbar	Creates an AcadToolbar object and adds it to the collection. Parameter: Name As String.
Item	AcadToolbar	Returns the member of the collection specified by the index. Parameter: Index As Variant.

AcadToolbars Collection Properties

As well as the common Application property and the Count property, the AcadToolbars collection supports the following properties.

Name	Returns	Description
LargeButtons	Boolean	Specifies whether the displayed toolbar buttons are large.
ParentAcadMenuGroup		Gets the collection's parent object. This property's value is read-only.

AcadTrace Object

The AcadTrace object represents a 2-D solid line of specified width. The end points of a trace are always positioned on the centerline and are cut square. AutoCAD automatically calculates the correct bevels when connecting adjacent trace segments. Traces are solid-filled when the fill mode is on. Otherwise, only the outline of the trace appears. The fill mode uses the system variable FILLMODE, which is set using the SetVariable method and queried using the GetVariable method. The system variable TRACEWID stores the current width used for an AcadTrace object.

The AcadTrace object is created using the AddTrace method of the AcadBlock, AcadModelSpace, or AcadPaperSpace object.

AcadTrace Object Methods

The AcadTrace object inherits all the methods of the AcadEntity and AcadObject objects. It supports no other methods.

AcadTrace Object Properties

The AcadTrace object inherits all the properties of the AcadEntity and AcadObject objects, as well as the common Application property. It also supports the following properties.

Name	Returns	Description
Coordinate	Variant	Gets or sets the coordinate of a single vertex in the object. This will replace an existing vertex for the specified object. Use standard array-handling techniques to process the values contained in this property. It returns a three-element array of Doubles containing 3-D coordinates in WCS. Note that the Z coordinate will default to 0 on the active UCS. Parameter: Index As Integer (the index in the zero-based array of vertices for the vertex you want to set or query).
Coordinates	Variant	Gets or sets the coordinates for each vertex in the object. This will replace any existing coordinates for the specified object. Use standard array-handling techniques to process the coordinates contained in this property. Note that you can't change the number of coordinates in the object by using this property, you can only change the location of existing coordinates. It returns a three-element array of Doubles containing 3-D coordinates in WCS. Note that the Z coordinate will always default to 0 on the active UCS.
Normal	Variant	Gets or sets the 3-D (Z-axis) normal unit vector for the trace object.
Thickness	Double	Gets or sets the distance the AcadTrace object is extruded above or below its elevation, in a Z-axis direction. The default is 0.0.

Note Because this object inherits from AcadObject, it supports the Modified event.

AcadUCS Object

The AcadUCS object represents a User-defined Coordinate System (UCS) that determines the orientation of the X-, Y-, and Z-axes in 3-D space. With this object you can change the location of the origin and the orientation of the XY plane and the Z-axis, and the resulting UCS definition can be orientated anywhere within the 3-D layout. Many UCSs can be defined as required for each drawing. The GetUCSMatrix method returns the transformation matrix for a given UCS. To turn on the UCS icon for a given viewport, use the UCSIconOn property of the AcadPViewport and AcadViewport objects. Note that all coordinates in ActiveX Automation are entered in the WCS.

The AcadUCS object is created by the Add method of the AcadUCSs collection and accessed by the Item method of the AcadUCSs collection. UCSs can also be activated using the ActiveUCS property of the Document object. If any changes are made to the active UCS, the new UCS object must be reset as the active UCS for the changes to appear.

AcadUCS Object Methods

The AcadUCS object inherits all the methods of the AcadObject object. It also supports the following method.

Name	Returns	Description
GetUCSMatrix	Variant	Gets the transformation matrix consisting of UCS coordinate system data. It returns a 4×4 array of Doubles. This matrix is the passed as a parameter to the TransformBy method of a drawing object to apply a given UCS to that object.

AcadUCS Object Properties

The AcadUCS object inherits all the properties of the AcadObject object and the common Application property. It also supports the following properties.

Name	Returns	Description
Name	String	Gets or sets the name of the UCS object.
Origin	Variant	Gets or sets the origin of the UCS object in WCS coordinates, relative to the lower-left corner. It returns an array of Doubles.
XVector	Variant	Gets or sets the X direction of the given UCS. It returns a three-element array of Doubles, which is stored in the UCSXDIR system variable. If the X vector value is changed on the active UCS, you must reset the active UCS to see the change.
YVector	Variant	Gets or sets the Y direction of the given UCS. It returns a three-element array of Doubles, which is stored in the UCSYDIR system variable. If the Y vector value is changed on the active UCS, you must reset the active UCS to see the change.

Note Because this object inherits from AcadObject, it supports the Modified event.

AcadUCSs Collection

The AcadUCSs collection contains all the UCS definitions in the drawing. To add a new member to the collection, use the Add method. To select a specific UCS definition, use the Item method. Although this collection inherits a Delete method, you can't actually delete this collection. If you need to delete a specific UCS definition, use the Delete method found in the AcadUCS object. There is no limit to the number of UCS definitions you can create in your drawing. However, there can be only one instance of the AcadUCSs collection, which is predefined for each drawing. You can make multiple references to it by using the UserCoordinateSystems property.

AcadUCSs Collection Methods

The AcadUCSs collection inherits all the methods of the AcadObject object. It also supports the following methods.

Name	Returns	Description
Add	AcadUCS	Creates a member object and adds it to the collection. Parameters: Origin As Variant, XAxisPoint As Variant, YAxisPoint As Variant, and Name As String. The Origin parameter specifies where the UCS is to be added. The XAxisPoint parameter specifies a point on the positive X-axis of the UCS. The YAxisPoint parameter specifies a point on the positive Y-axis of the UCS. (The Z-axis follows by applying the right-hand rule.) These coordinates are all 3-D in WCS. Note that XAxisPoint and YAxisPoint together can't specify the same location as the Origin. The Name parameter specifies the name of the UCS to add to the collection.
Item	AcadUCS	Gets the member object at a given index in a collection. Parameter: Index As Variant (an Integer or a String). If the Index value is a String, it must match an existing object name in the collection.

AcadUCSs Collection Properties

The AcadUCSs collection inherits all the properties of the AcadObject object, the Count property, and the common Application property. It supports no other properties.

Note Because this collection inherits from AcadObject, it supports the Modified event.

AcadUtility Object

The AcadUtility object represents a series of methods provided for utility purposes.

AcadUtility Object Methods

The following methods are supported.

Name	Returns	Description
AngleFromXAxis	Double	Returns the angle (in radians) of a line passing through two points from the X-axis. Parameters: Point1 As Variant and Point2 As Variant. The parameters are three-element arrays of Doubles.
AngleToReal	Double	Converts an angle as a string to a real number (Double). Parameters: Angle As String and Unit As AcAngleUnits. For the possible values of the AcAngleUnits enumerated type, see Appendix B.
AngleToString	String	Converts an angle from a real number (Double) to a string. Parameters: Angle As Double, Unit As AcAngleUnits, and Precision As Integer. For the possible values of the AcAngleUnits enumerated type, see Appendix B. The range of values for the third parameter is 0 to 8.
CreateTypedArray		Creates an array that contains the values specified by the parameters. The resulting array can be passed into any AutoCAD method or property that accepts an array of numbers. Parameters: VarArr As Variant, Type As (Visual Basic Constant), Value1 As Type, [Value2 As Type], ..., and [ValueN As Type]. The values for Type are vbBoolean, vbInteger, vbLong, vbSingle, and vbDouble.
DistanceToReal	Double	Converts a distance from a string to a real number (Double). Parameters: Distance As String and Unit As AcUnits. For the possible values of the AcUnits enumerated type, see Appendix B.
GetAngle	Double	Asks the user to input an angle and then returns its value, taking into account the value in the ANGBASE system variable. The value of the angle returned is expressed in radians. The direction of angular increase is always counterclockwise. Parameters: [Point As Variant] and [Prompt As String]. The Point parameter in this method and the two that follow specifies 3-D WCS coordinates. The Prompt parameter in this method and the methods that follow specifies text string used to prompt the user for input.

Name	Returns	Description
GetCorner	Variant	Asks the user to input the corner of a rectangle and returns the point as an array of three Doubles. Note that a call to GetCorner will frequently follow a call to GetPoint, the first point selected being used as the first parameter for the GetCorner method. Parameters: Point As Variant and [Prompt As String].
GetDistance	Variant	Asks the user to input a linear distance from the keyboard or by specifying two locations on the graphics screen, and returns it as a Double number or as an array of Double numbers. Parameters: [Point As Variant] and [Prompt As String].
GetEntity		Asks the user to select a drawing object by picking a point on the screen. The object is then placed in the first parameter. Parameters: Object As Object, PickedPoint As Variant, and [Prompt As String]. The PickedPoint parameter specifies 3-D WCS coordinates.
GetInput	String	Asks the user to input text and returns it. A call to this method must follow a call to a user-input function returning the “User input is a keyword” error description. The input must be no longer than 511 characters.
GetInteger	Integer	Asks the user to input an integer and returns it. The AutoCAD user can enter any short integer in the range of -32,768 to +32,767. Parameter: [Prompt As String].
GetKeyword	String	Asks the user to input a keyword and returns it. The keyword must be no longer than 511 characters. If the user doesn't enter anything but presses the <i>Enter</i> key, this method returns an empty string. This method can be used only if InitializeUserInput is called first. Parameter: [Prompt As String].
GetOrientation	Double	Asks the user to input an angle and then returns its value in radians without regard to the values stored in the ANGBASE system variable. The 0 angle employed by this method is always to the right (east). Parameters: [Point As Variant] and [Prompt As String]. The Point parameter specifies 3-D WCS coordinates.
GetPoint	Variant	Asks the user to choose a point and returns it. Parameters: [Point As Variant] and [Prompt As String]. The Point parameter and the return value specify 3-D WCS coordinates.
GetReal	Double	Asks the user to input a real (Double) number and returns it. Parameter: [Prompt As String].
GetRemoteFile		Downloads the specified file from a URL. Parameters: URL As String, LocalFile As String, and IgnoreCache As Boolean. The third parameter specifies whether to download the file, even if it has already been transferred earlier in this session.
GetString	String	Asks the user to input a string and returns it. Parameters: HasSpaces As Boolean and [Prompt As String]. HasSpaces determines whether the return string can contain spaces. If True, the string can contain spaces but terminated by a carriage return only. If False, the string can't contain spaces. It can be terminated by a carriage return or a space.

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Name	Returns	Description
GetSubEntity		Asks the user to select a drawing object by specifying the object and the selected in the first two parameters. Parameters: Object As Object, PickedPoint As Variant, TransMatrix As Variant, ContextData As Variant, and [Prompt As String]. The PickedPoint parameter specifies 3-D WCS coordinates. The TransMatrix parameter represents the translation matrix applied to this entity. The Variant parameter is a 4×4 array of Doubles. The ContextData parameter is an array of Longs, containing the object IDs for any nested objects in the selected object. More specifically, this argument returns an array holding the object ID(s) of the subentity's container block references. The first array element is the object ID of the block reference containing the picked entity. The last array element is the object ID of the block reference inserted directly in the drawing. The array of object IDs is therefore ordered from the most deeply nested block reference that contains the subentity to the object ID of the block reference that would have been picked with the GetEntity function. If the picked entity isn't a subentity (i.e., it isn't nested in a block reference), then the GetSubEntity function returns an empty array for the ContextData parameter.
InitializeUserInput		Initializes the GetKeyword method. Parameter: Bits As Integer and [Keyword As String]. For the values for the Bits parameter and their explanations, see the following table. The second parameter specifies the keywords, delimited by spaces, that the GetKeyword method will recognize. This method must be called before GetKeyword.
IsRemoteFile	Boolean	Returns whether the specified file was downloaded from a remote location. This method is the inverse of the IsURL method and provides a mapping from a local file to the corresponding URL that the file was downloaded from. Parameters: LocalFile As String and URL As String.
IsURL	Boolean	Returns whether a string is a valid URL for the application. The default functionality of this method for AutoCAD supports the FTP, HTTP, HTTPS, and FILE protocols. Parameter: URL As String.
LaunchBrowserDialog	Boolean	Runs the web browser dialog box that allows the user to navigate to any URL and select a URL. It returns whether the operation was successful. Parameters: SelectedURL As String, DialogTitle As String, OpenButtonCaption As String, StartPageURL As String, RegistryRootKey As String, and OpenButtonAlwaysEnabled As Boolean. The first four parameters specify the selected URL, the title to be displayed for the browser dialog box, the caption for the OK/Open button, and the URL that the web browser should use as its start page, respectively. The fifth parameter is used to specify where in the Registry persistent web browser dialog box information is stored. Input an empty string to disregard this functionality. The sixth parameter determines whether the user can select HTML links in addition to downloadable file. If True, the Open button is enabled, allowing a file or link to be selected. If False, the Open button is disabled and is only enabled when the user selects a file for download.

Name	Returns	Description
PolarPoint	Variant	Returns a point at the specified angle (in radians) and distance (in current units) from the specified point. Parameters: Point As Variant, Angle As Double, and Distance As Double. The value for Point and the return value are both 3-D WCS coordinates.
Prompt		Sends a message to the command line. Parameter: Message As String.
PutRemoteFile		Puts a local file at the location specified by a URL. This method is designed to complement GetRemoteFile method. When accessing a secure URL, a dialog box will be posted prompting the user for the necessary password information. Message boxes may also appear if the user hasn't suppressed this activity in the browser. Parameter: URL As String.
RealToString	String	Converts a real number into a properly formatted string. Parameters: Value As Double, Unit As AcUnits, and Precision As Integer. For a list of possible values for the AcUnits enumerated type, see Appendix B. The range of values for the third parameter is 0 to 8.
TranslateCoordinates	Variant	Changes a point from one coordinate system to another. You can't directly translate a set of coordinates from one OCS to another OCS. If you want to do this, you must first translate the OCS coordinates to WCS and then change these coordinates to the second OCS. Parameters: OriginalPoint As Variant, From As AcCoordinateSystem, To As AcCoordinateSystem, Disp As Boolean, and [OCSTNormal As Variant]. OriginalPoint specifies the original 3-D WCS coordinates to be translated. This parameter can be treated as a point or a displacement vector depending on the value of Disp. For a list of possible values for the AcCoordinateSystem enumeration (the From and To parameters), see Appendix B. Disp is a displacement vector flag. If True, OriginalPoint is treated as a displacement vector. Otherwise it is treated as a point. The optional last parameter represents the normal for the OCS.

AcadUtility Object Properties

The AcadUtility object inherits the common Application property. It supports no other properties.

AcadView Object

The AcadView object represents a graphical representation of a 2-D drawing or 3-D model from a specific location (viewpoint) in space. The line of sight is drawn from the viewpoint, represented by the Center property, to the target point. The Height and Width properties crop the view to fit into the viewport.

AcadView Object Methods

The AcadView object inherits all the methods of the AcadObject object. It supports no other methods.

AcadView Object Properties

The AcadView object inherits all the properties of the AcadObject object, as well as the common Application property. It also supports the following properties.

Name	Returns	Description
Center	Variant	Gets or sets the center of the view as a set of 2D coordinates (as Doubles). The default center is (0,0).
Direction	Variant	Gets or sets the viewing direction for a 3-D visualization of the drawing. This property puts the viewer in a position to look at the drawing as if looking back at the origin (0, 0, 0) from a specified point in space. It is similar to AutoCAD's VPOINT command. It returns a three-element array of Doubles.
Height	Double	Gets or sets the height of the view, which is the Y-axis measurement of the area within a viewport that is used to display the model.
Name	String	Gets or sets the name of the view.
Target	Variant	Gets or sets the target point for the view as a set of 3-D WCS coordinates representing the target point. The line of sight is drawn from the center to the target point. It returns a three-element array of Doubles.
Width	Double	Gets or sets the width of the view, which is the X-axis measurement of the area within a viewport that is used to display the model.

Note Because this object inherits from AcadObject, it supports the Modified event.

AcadViewport Object

The AcadViewport object represents a bounded area that displays some portion of a drawing's model space. You can activate it by using the ActiveViewport property of the Document object. The ActiveSpace property, which is equivalent to the TILEMODE system variable, determines the type of viewport used. You can make changes to the view only while the viewport is active, and you can see any changes only once the viewport has been reactivated.

AcadViewport Object Methods

The AcadViewport object inherits all the methods of the AcadObject object. It also supports the following methods.

Name	Description
GetGridSpacing	Gets the grid spacing for the viewport. Parameters: XSpacing As Double and YSpacing As Double.
GetSnapSpacing	Gets the snap spacing for the viewport. Parameters: XSpacing As Double and YSpacing As Double.
SetGridSpacing	Sets the grid spacing for the viewport. Parameters: XSpacing As Double and YSpacing As Double.
SetSnapSpacing	Sets the snap spacing for the viewport. Parameters: XSpacing As Double and YSpacing As Double.
SetView	Sets the view in a viewport to a saved view in the AcadViews collection. Parameter: View As AcadView.
Split	Splits a viewport into the given number of views. Parameter: NumWins As AcViewportSplitType. For a list of possible values for the AcViewportSplitType enumerated type, see Appendix B.

AcadViewport Object Properties

The AcadViewport object inherits all the properties of the AcadObject object, as well as the common Application property. It also supports the following properties. Note that many of these properties are identical to those of the AcadPViewport object. See the AcadPViewport object for more details for the ArcSmoothness, Center, Direction, GridOn, SnapBasePoint, SnapOn, and SnapRotationAngle properties.

Name	Returns	Description
ArcSmoothness	Integer	Gets or sets the smoothness of circles, arcs, and ellipses. The valid range of values is from 1 to 20000. The initial value for this property is 100.
Center	Variant	Gets or sets the center of the viewport as a set of 2-D coordinates (as Doubles). The default center is (0,0).
Direction	Variant	Gets or sets the viewing direction for a 3-D visualization of the drawing. This property puts the viewer in a position to look at the drawing as if looking back at the origin (0, 0, 0) from a specified point in space.
GridOn	Boolean	Specifies the status of the viewport grid. The value of this property is stored in the GRIDMODE system variable.
Height	Double	Gets or sets the height of the viewport, which is the Y-axis measurement of the area within a viewport that is used to display the model.
LowerLeftCorner	Variant	Gets the lower-left corner of the current active viewport, which with the UpperRightCorner property represents the graphic placement of the viewport on the display. It returns a set of 2-D coordinates as an array of Doubles. This property's value is read-only.
Name	String	Gets or sets the name of the viewport.
OrthoOn	Boolean	Specifies status of the Ortho mode for the viewport. You can use Ortho mode to restrict the cursor to the horizontal or vertical axis, which helps you draw parallel lines or move objects.

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Name	Returns	Description
SnapBasePoint	Variant	Gets or sets the snap base point for the viewport, the value of which is stored in the SNAPBASE system variable. The snap base point can't be changed for the active paper space viewport. Any changes to this property aren't reflected in the display until the drawing is regenerated.
SnapOn	Boolean	Specifies the status of snap mode. The value of this property is stored in the SNAPMODE system variable.
SnapRotationAngle	Double	Gets or sets the snap rotation angle (in radians) of the viewport relative to the current UCS. The value of this property is stored in the SNAPANG system variable. The valid range is 0 to 6.28.
Target	Variant	Gets or sets the target point for the viewport as a set of 3-D WCS coordinates representing the target point. The line of sight is drawn from the center to the target point. It returns a three-element array of Doubles.
UCSIconAtOrigin	Boolean	Specifies whether the UCS icon is displayed at the origin or at the WCS coordinate defined by the UCSORG system variable.
UCSIconOn	Boolean	Specifies whether the UCS icon is on.
UpperRightCorner	Variant	Gets the upper-right corner of the current active viewport, which with the LowerLeftCorner property represents the graphic placement of the viewport on the display. It returns a set of 2-D coordinates as an array of Doubles. This property's value is read-only.
Width	Double	Gets or sets the width of the viewport, which is the X-axis measurement of the area within a viewport that is used to display the model.

Note Because this object inherits from AcadObject, it supports the Modified event. There is no programmatic means to directly create or manipulate polygonal viewports.

AcadViewports Collection

The AcadViewports collection contains all the viewports in the drawing. Although this collection inherits a Delete method, you can't actually delete this collection. If you need to delete a specific viewport, use the DeleteConfiguration method or the Delete method of the AcadViewport object. The number of viewports you can have active at one time is controlled by the MAXACTVP system variable. However, there can be only one instance of the AcadViewports collection, which is predefined for each drawing. You can make multiple references to it by using the Viewports property of the Document object.

AcadViewports Collection Methods

The AcadViewports collection inherits all the methods of the AcadObject object. It also supports the following methods.

Name	Returns	Description
Add	AcadViewport	Creates a member object and adds it to the collection. Parameter: Name As String.
DeleteConfiguration		Deletes a viewport configuration, which consists of a single viewport that has been split using the Split method. Once a viewport has been split, the resulting viewports are considered a viewport configuration and have the same name as the original viewport before the split. Parameter: Name As String.
Item	AcadViewport	Gets the member object at a given index in a collection. Parameter: Index As Variant (an Integer or a String). If the Index value is a String, it must match an existing viewport in the collection.

AcadViewports Collection Properties

The AcadViewports collection inherits all the properties of the AcadObject object, the Count property, and the common Application property. It supports no other properties.

Note Because this collection inherits from AcadObject, it supports the Modified event.

AcadViews Collection

The AcadViews collection contains all the views in the drawing. Although this collection inherits a Delete method, you can't actually delete this collection. If you need to delete a specific view, use the Delete method of the AcadView object. There is no limit to the number of views you can create in your drawing. However, there can be only one instance of the AcadViews collection, which is predefined for each drawing. You can make multiple references to it by using the Views property of the Document object.

AcadViews Collection Methods

The AcadViews collection inherits all the methods of the AcadObject object. It supports no other methods.

AcadViews Collection Properties

The AcadViews collection inherits all the properties of the AcadObject object, the Count property, and the common Application property. It supports no other properties.

Note Because this collection inherits from AcadObject, it supports the Modified event.

AcadXline Object

The AcadXline object represents a construction line that is infinite in both directions, unlike the AcadRay object, which is infinite in only one direction. It is created using the AddXline method of the AcadBlock, AcadModelSpace, or AcadPaperSpace object.

AcadXline Object Methods

The AcadXline inherits all the methods of the AcadEntity and AcadObject objects. It also supports the following method.

Name	Returns	Description
Offset	Variant	Creates a new line by offsetting the current Xline by a specified distance, which can be positive or negative but can't be zero. If the offset is negative, this means that the line is drawn closer to the WCS origin. Parameter: Distance As Double.

AcadXline Object Properties

The AcadXline object inherits all the properties of the AcadEntity and AcadObject objects, as well as the common Application property. It also supports the following properties.

Name	Returns	Description
BasePoint	Variant	Gets or sets the point through which the Xline passes. It returns a set of 3-D coordinates as a three-element array of Doubles.
DirectionVector	Variant	Gets or sets the direction for the Xline through a vector. It returns a three-element array of Doubles.
SecondPoint	Variant	Gets or sets a second point on the Xline. It returns a set of 3-D WCS coordinates as a three-element array of Doubles.

Note Because this object inherits from AcadObject, it supports the Modified event.

AcadXRecord Object

The AcadXRecord object is used to store and manage arbitrary data. XRecords are similar in concept to XData but are not limited by size or order. They use standard AutoCAD group codes, the values of which are all below 1000, which means in addition to all the normally used data types, XRecords are capable of storing object IDs, allowing ownership of many other objects. For a list of supported group codes, see the AutoCAD documentation.

XRecord objects are saved with the drawing and can be directly accessed by other ObjectARX and LISP programs. For this reason, you should take care when storing security information in XRecords. XRecords are created using the AddXRecord method of the Dictionary object and accessed programmatically through the Item method of the Dictionary object.

The only basic operations that can be performed on XRecords are Add, Get, and Delete. There is no means to update (modify) an XRecord other than to delete and re-create it.

AcadXRecord Object Methods

The AcadXRecord object inherits all the methods of the AcadObject object. It also supports the following methods.

Name	Description
GetXRecordData	Gets the extended record data associated with a dictionary. Parameters: XRecordDataType As Variant and XRecordDataValue As Variant.
SetXRecordData	Sets the extended record data associated with a dictionary. Parameters: XRecordDataType As Variant and XRecordDataValue As Variant.

AcadXRecord Object Properties

The AcadXRecord object inherits all the properties of the AcadObject object, as well as the common Application property. It also supports the following properties.

Name	Returns	Description
Name	String	Gets or sets the name of the object within the dictionary. This name doesn't represent the class name of the object.
TranslateIDs	Boolean	Specifies whether translation of any contained object IDs occurs during deepClone or wblockClone operations.

Note Because this object inherits from AcadObject, it supports the Modified event.

SecurityParams Object

The SecurityParams object stores settings for drawing security. The SecurityParams object encrypts and digitally signs drawing files.

The object properties included provide information about cryptography providers and other settings for drawing security. Additional information about cryptography providers is available on MSDN (<http://msdn.microsoft.com>).

Name	Returns	Description
Action	Long	Gets or sets the desired security operation to be performed. Use one or more of the following constants for drawing encryption, drawing properties encryption, a digital signature, or a timestamp: ACADSECURITYPARAMS_ENCRYPT_DATA = 0x00000001 ACADSECURITYPARAMS_ENCRYPT_PROPS = 0x00000002 ACADSECURITYPARAMS_SIGN_DATA = 0x00000010 ACADSECURITYPARAMS_ADD_TIMESTAMP = 0x00000020

Continued

Name	Returns	Description
Algorithm	Long	Gets or sets the identifier of the encryption algorithm. This value can only be the following constant: ACADSECURITYPARAMS_ALGID_RC4 = 0x00006801
Comment	String	Gets or sets the comment to be included with a digital signature.
Issuer	String	Gets or sets the issuer of a digital certificate.
KeyLength	Long	Gets or sets the length of the encryption key, which is based upon the value in the ProviderName property.
Password	String	Gets or sets the encryption password.
ProviderName	String	Gets or sets the encryption provider name. More information about cryptography providers is available on MSDN.
ProviderType	Long	Gets or sets the encryption provider type. More information about cryptography providers is available on MSDN.
SerialNumber	String	Gets or sets the serial number of the digital certificate.
Subject	String	Gets or sets the subject name of the digital certificate.
TimeServer	String	Gets or sets the name of the time server to be used for a digital signature. If you don't set this property, the time from the local machine is used as the timestamp for the digital signature.

AcCmColor Object

The **AcCmColor** object represents colors within AutoCAD. You use the **AcCmColor** object to set colors and perform other color-related operations on objects.

AcCmColor Object Methods

Unlike the other AutoCAD objects, the **AcCmColor** object doesn't inherit methods from any other AutoCAD objects. It supports the following methods.

Name	Description
SetColorBookColor	Sets the color name from an existing color book. Parameters: ColorName As String and ColorBook As String.
SetNames	Sets the color name and book name of the color. Parameters: ColorName As String and BookName As String.
SetRGB	Sets the RGB values of the True Color. Parameters: Red As Long, Green As Long, and Blue As Long.

AcCmColor Object Properties

Unlike the other AutoCAD objects, the **AcCmColor** object doesn't inherit properties from any other AutoCAD objects. It supports the following properties.

Name	Returns	Description
Blue	Long	Gets the blue component, from 0 to 255, of <code>AcCmColor</code> . This property's value is read-only.
BookName	String	Gets the name of the color book that the color came from. This property's value is read-only.
ColorIndex	<code>acColor</code> enum	Gets or sets the default color designation in <code>acByLayer</code> . Use a color index number from 0 to 256 or one of the constants listed here: <code>acByBlock</code> <code>acByLayer</code> (not valid for the <code>Layer</code> object) <code>acRed</code> <code>acYellow</code> <code>acGreen</code> <code>acCyan</code> <code>acBlue</code> <code>acMagenta</code> <code>acWhite</code>
ColorMethod	<code>acColorMethod</code> enum	Gets or sets the default color method in <code>acColorMethodByLayer</code> . To directly determine how a color is set, use the constants listed here: <code>acColorMethodByACI</code> <code>acColorMethodByBlock</code> <code>acColorMethodByLayer</code> <code>acColorMethodByRGB</code> <code>acColorMethodForeground</code>
ColorName	String	Gets the name (if any) of the color. This property's value is read-only.
EntityColor	Long	Gets or sets the 32-bit <code>AcCmEntityColor</code> portion of the color.
Green	Long	Gets the green component, from 0 to 255, of <code>AcCmColor</code> . This property's value is read-only.
Red	Long	Gets the red component, from 0 to 255, of <code>AcCmColor</code> . This property's value is read-only.

LayerStateManager Object

The `LayerStateManager` object provides a set of functions for working with saved layer settings. The `LayerStateManager` object manipulates `XRecord` objects that define the properties of a layer. These `XRecords` are stored in the `ACAD_LAYERSTATE` dictionary, which is an extension dictionary in the drawing's `Layers` collection. (An extension dictionary is a mechanism for attaching data to objects. Every AutoCAD object can have an extension dictionary.)

After you retrieve the `LayerStateManager` object and associate a database with it, then you can access the object's methods. Use the `SetDatabase` method to associate a database with the `LayerStateManager` object.

LayerStateManager Object Methods

Unlike the other AutoCAD objects, the `LayerStateManager` object doesn't inherit methods from any other AutoCAD objects. It supports the following methods.

Name	Description
Delete	Name representing the layer state to be deleted. Parameter: Name As String.
Export	Exports an AutoCAD drawing or a group of saved layer settings to a file. Parameters: Filename As String, Extension As String, and SelectionSet As SelectionSet.
Import	Imports a drawing or a group of saved layer settings from a file. Parameters: Filename As String, InsertionPoint As Variant, and ScaleFactor As Double.
Rename	Renames a set of saved layer settings. Parameters: OldName As String and NewName As String.
Restore	Restores a group of layer property settings. Parameter: Name As String.
Save	Saves a group of layer property settings. Parameters: Name As String and Mask (set the Mask property as described in the next table).
SetDatabase	Gets AutoCAD database associated with the LayerStateManager object.

LayerStateManager Object Properties

Unlike the other AutoCAD objects, the LayerStateManager object doesn't inherit properties from any other AutoCAD objects. It supports the following properties.

Name	Returns	Description
Mask	Enum	Specifies the layer properties to be restored as a number representing the layer properties to be restored. The following table represents acceptable values for this property.

Constant	Layer Property	Numeric Value
AcLsAll	All layer properties	-65535
AcLsColor	Color	-32
AcLsFrozen	Frozen or thawed	-2
AcLsLineType	Linetype	-64
AcLsLineWeight	Lineweight	-128
AcLsLocked	Locked or unlocked	-4
AcLsNewViewport	New viewport layers frozen or thawed	-16
AcLsNone	None	-0
AcLsOn	On or off	-1
AcLsPlot	Plotting on or off	-8
AcLsPlotStyle	Plot style	-256



AutoCAD Constants Reference

This appendix lists all the enumerated types used in conjunction with the AutoCAD objects. The methods and properties (presented in Appendix A) that use them have been noted.

Ac3DPolylineType

The following constants can be values for the Type property of the Acad3DPolyline object.

Constant	Value	Description
acSimple3Dpoly	0	A simple polyline
acQuadSpline3Dpoly	1	A quadratic B-spline polyline
acCubicSpline3Dpoly	2	A cubic B-spline polyline

AcActiveSpace

The following constants can be values for the ActiveSpace property of the AcadDocument object.

Constant	Value	Description
acModelSpace	1	The active space is model space.
acPaperSpace	2	The active space is paper space.

ACAD_LWEIGHT

The following constants can be values for the GetGridLineWeight and SetGridLineWeight properties of an AcadTable and AcadTableStyle object. Additionally, the following constants can be values for the GetCell and SetCellGridLineWeight methods of an AcadTable object.

Constant	Value	Constant	Value
acLnWt000	0	acLnWt070	70
acLnWt005	5	acLnWt080	80
acLnWt009	9	acLnWt090	90
acLnWt013	13	acLnWt100	100
acLnWt015	15	acLnWt106	106
acLnWt018	18	acLnWt120	120
acLnWt020	20	acLnWt140	140
acLnWt025	25	acLnWt158	158
acLnWt030	30	acLnWt200	200
acLnWt035	35	acLnWt211	211
acLnWt040	40	acLnWtByLayer	-1
acLnWt050	50	acLnWtByBlock	-2
acLnWt053	53	acLnWtByLwDefault	-3
acLnWt060	60		

AcAlignment

The following constants can be values for the Alignment property of the AcadAttribute, AcadAttributeReference, and AcadText objects.

Constant	Value	Description
acAlignmentLeft	0	The attribute, attribute reference, or text is aligned with the insertion point.
acAlignmentCenter	1	The attribute, attribute reference, or text is aligned at the center.
AcAlignmentRight	2	The attribute, attribute reference, or text is aligned at the right.
acAlignmentAligned	3	The attribute, attribute reference, or text is aligned according to the points held in the InsertionPoint and TextAlignmentPoint properties.
acAlignmentMiddle	4	The attribute, attribute reference, or text is aligned at the middle, at a position slightly below middle center.
acAlignmentFit	5	The attribute, attribute reference, or text is aligned to fit to the points held in the InsertionPoint and TextAlignmentPoint properties.
acAlignmentTopLeft	6	The attribute, attribute reference, or text is aligned at the top left.
acAlignmentTopCenter	7	The attribute, attribute reference, or text is aligned at the top center.
acAlignmentTopRight	8	The attribute, attribute reference, or text is aligned at the top right.

Constant	Value	Description
acAlignmentMiddleLeft	9	The attribute, attribute reference, or text is aligned at the middle left.
acAlignmentMiddleCenter	10	The attribute, attribute reference, or text is aligned at the middle center.
acAlignmentMiddleRight	11	The attribute, attribute reference, or text is aligned at the middle right.
acAlignmentBottomLeft	12	The attribute, attribute reference, or text is aligned at the bottom left.
acAlignmentBottomCenter	13	The attribute, attribute reference, or text is aligned at the bottom center.
acAlignmentBottomRight	14	The attribute, attribute reference, or text is aligned at the bottom right.

AcAlignmentPointAcquisition

The following constants can be values for the `AlignmentPointAcquisition` property of the `AcadPreferencesDrafting` object.

Constant	Value	Description
acAlignPntAcquisitionAutomatic	0	Autoalignment points are acquired automatically.
acAlignPntAcquisitionShiftToAcquire	1	The user must use the Shift key to acquire autoalignment points.

AcAngleUnits

The following constants can be values for the `AngleFormat` property of the `AcadDim3PointAngular` object, and they can be supplied as the value for the second parameter of the `AngleToReal` and `AngleToString` methods of the `AcadUtility` object.

Constant	Value	Description
acDegrees	0	Angle units are degrees.
acDegreeMinuteSeconds	1	Angle units are degrees, minutes, and seconds.
acGrads	2	Angle units are grads.
acRadians	3	Angle units are radians.

AcARXDemandLoad

The following constants can be values for the DemandLoadARXApp property of the AcadPreferencesOpenSave object.

Constant	Value	Description
acDemandLoadDisable	0	Turns off demand loading.
acDemandLoadOnObjectDetect	1	Demand-loads the source application when you open a drawing that contains custom objects. This setting doesn't demand-load the application when you invoke one of the application's commands.
acDemandLoadCmdInvoke	2	Demand-loads the source application when you invoke one of the application's commands. This setting doesn't demand-load the application when you open a drawing that contains custom objects.

AcAttachmentPoint

The following constants can be values for the AttachmentPoint property of the AcadMText object.

Constant	Value	Description
acAttachmentPointTopLeft	1	Left-justified, spills down
acAttachmentPointTopCenter	2	Center-justified, spills down
acAttachmentPointTopRight	3	Right-justified, spills down
acAttachmentPointMiddleLeft	4	Left-justified, spills up and down
acAttachmentPointMiddleCenter	5	Center-justified, spills up and down
acAttachmentPointMiddleRight	6	Right-justified, spills up and down
acAttachmentPointBottomLeft	7	Left-justified, spills up
acAttachmentPointBottomCenter	8	Center-justified, spills up
acAttachmentPointBottomRight	9	Right-justified, spills up

AcAttributeMode

The following constants can be values for the Mode property of the AcadAttribute object, and they can also be supplied as the value for the second parameter of the AddAttribute method of the AcadBlock, AcadModelSpace, and AcadPaperSpace objects.

Constant	Value	Description
acAttributeModeNormal	0	The default mode. In other words, it lacks any of the other modes listed here.
acAttributeModeInvisible	1	When the block is inserted, the attribute's values won't be visible. The system variable ATTDISP overrides this mode setting.
acAttributeModeConstant	2	Attribute values will have a fixed value for each block inserted. When set, prompting for the attribute is disabled.
acAttributeModeVerify	4	When you insert the block, AutoCAD will ask you to verify that the attribute value is correct.
acAttributeModePreset	8	The block is inserted using its default attribute values. You can't edit these values.

AcBlockScaling

The following constants can be values for the BlockScaling property of an AcadBlock object. To access this property, use the IAcadBlock3 interface.

Constant	Value	Description
AcAny	0	The X, Y, Z scale values may be different.
acUniform	1	The X, Y, Z scale values are equal.

AcBooleanType

The following constants are values supplied as the value for the first parameter of the Boolean methods of the Acad3DSolid and AcadRegion objects.

Constant	Value	Description
acUnion	0	Performs a union operation
acIntersection	1	Performs an intersection operation
acSubtraction	2	Performs a subtraction operation

AcCellAlignment

The following constants are possible return values for the GetAlignment and GetCellAlignment methods. Also, they can be values for the CellAlignment parameter of the SetAlignment and SetCellAlignment methods.

Constant	Value	Description
acBottomCenter	8	Centers text at a point based on the bottom of the text
acBottomLeft	7	Left-justifies text at a point based on the bottom of the text
acBottomRight	9	Right-justifies text at a point based on the bottom of the text
acMiddleCenter	5	Centers text at a point based on the middle of the text
acMiddleLeft	4	Left-justifies text at a point based on the middle of the text
acMiddleRight	6	Right-justifies text at a point based on the middle of the text
acTopCenter	2	Centers text at a point based on the top of the text
acTopLeft	1	Left-justifies text at a point based on the top of the text
acTopRight	3	Right-justifies text at a point based on the top of the text

AcCellEdgeMask

The following constants can be values for the Edge parameter of the GetCellGridColor, GetCellGridLineWeight, and GetCellGridVisibility methods of an AcadTable object.

Constant	Value	Description
acBottomMask	4	Bottom-edge index of the cell
acLeftMask	8	Left-edge index of the cell
acRightMask	2	Right-edge index of the cell
acTopMask	1	Top-edge index of the cell

AcCellType

The following constants are possible return values for the GetCellType method of an AcadTable object. Also, they can be values for the CellType property or the SetCellType method of an AcadTable object.

Constant	Value
acBlockCell	2
acTextCell	1
acUnknownCell	0
AcColor (and the ColorIndex property of the AcCmColor object)	

The following constants can be values for these properties:

- The AutoSnapMarkerColor property of the AcadPreferencesDrafting object
- The Color property of the AcadEntity object (and all derived objects) and the AcadGroup and AcadLayer objects

- The **DimensionLineColor** property of the dimension objects that support it (all except **AcadDimOrdinate**) and the **AcadLeader** and **AcadTolerance** objects
- The **ExtensionLineColor** property of the dimension objects that support it (all except **AcadDimDiametric** and **AcadDimRadial**)
- The **GripColorSelected** and **GripColorUnselected** properties of the **AcadPreferencesSelection** object
- The **TextColor** property of the **AcadDimension** object (and all derived objects) and the **AcadTolerance** object

Constant	Value	Description
acByBlock	0	Objects are drawn in the default color until they're grouped into a block. When the block is inserted into a drawing, the objects inherit the value of the block's Color property.
acRed	1	Red
acYellow	2	Yellow
acGreen	3	Green
acCyan	4	Cyan
acBlue	5	Blue
acMagenta	6	Magenta
acWhite	7	White
acByLayer	256	Object assumes color of layer on which it is drawn.

AcColor

The following constants can be values for these properties:

- **AutoSnapMarkerColor** property of the **PreferencesDrafting** object
- The **Color** property of all **Drawing** Objects, **Group**, **Layer**, and **LabelTemplate** objects
- The **DimensionLineColor** property of the **Dim3PointAngular**, **DimAligned**, **DimAngular**, **DimArcLength**, **DimDiametric**, **DimRadial**, **DimRadialLarge**, **DimRotated**, **Leader**, and **Tolerance** objects
- The **ExtensionLineColor** property of the **Dim3PointAngular**, **DimAligned**, **DimAngular**, **DimArcLength**, **DimOrdinate**, and **DimRotated** objects
- The **GripColorSelected** and **GripColorUnselected** properties of the **PreferencesSelection** object
- The **TextColor** property of the **Dim3PointAngular**, **DimAligned**, **DimAngular**, **DimArcLength**, **DimDiametric**, **DimOrdinate**, **DimRadial**, **DimRadialLarge**, **DimRotated**, and **Tolerance** objects
- The **ColorIndex** property of the **AcCmColor** object

Constant	Value	Description
acByBlock	0	ByBlock
acRed	1	Red
acYellow	2	Yellow
acGreen	3	Green
acCyan	4	Cyan
acBlue	5	Blue
acMagenta	6	Magenta
acWhite	7	White
acByLayer	8	ByLayer

AcColorMethod

The following constants can be supplied as values for the `AcColorMethod` object as it relates to the `TrueColor` object. Only those constants marked with an asterisk (*), however, apply to the `GradientColor1` and `GradientColor2` properties for Hatch objects.

Constant	Value	Description
acColorMethodByACI*	195	Use traditional color index value.
acColorMethodByBlock*	193	
acColorMethodByLayer	192	
acColorMethodByRGB	194	Use RGB index values to define color.
acColorMethodForeground	197	

AcCoordinateSystem

You can supply the following constants as values for the second and third parameters of the `TranslateCoordinates` method of the `AcadUtility` object.

Constant	Value	Description
acWorld	0	World Coordinate System (WCS)
acUCS	1	User Coordinate System (UCS)
acDisplayDCS	2	Display Coordinate System (DCS)
acPaperSpaceDCS	3	Paper Space Display Coordinate System (PSDCS)
acOCS	4	Object Coordinate System (OCS)

AcDimArcLengthSymbol

The following constants can be values for the `SymbolPosition` property of an `AcadDimArcLength` object.

Constant	Value
<code>acSymAbove</code>	1
<code>acSymInFront</code>	0
<code>acSymNone</code>	2

AcDimArrowheadType

The following constants can be values for these properties:

- The `Arrowhead1Type` and `Arrowhead2Type` properties of the dimension objects that support them (all except `AcadDimOrdinate` and `AcadDimRadial`)
- The `Arrowhead1Type` property of the `AcadTolerance` object
- The `ArrowheadType` property of the `AcadDimRadial` and `AcadLeader` objects

Constant	Value	Description
<code>acArrowDefault</code>	0	The arrowhead is the default determined by the dimension style.
<code>acArrowClosedBlank</code>	1	The arrowhead is a closed empty triangle.
<code>acArrowClosed</code>	2	The arrowhead is a closed triangle with the dimension line inside.
<code>acArrowDot</code>	3	The arrowhead is a large, filled circle.
<code>acArrowArchTick</code>	4	The arrowhead is a bold, oblique line.
<code>acArrowOblique</code>	5	The arrowhead is an oblique line.
<code>acArrowOpen</code>	6	The arrowhead is a narrow, open arrow.
<code>acArrowOrigin</code>	7	The arrowhead is a circle with the dimension line inside.
<code>acArrowOrigin2</code>	8	The arrowhead comprises two circles with the dimension line inside.
<code>acArrowOpen90</code>	9	The arrowhead is an open arrow with an internal angle of 90 degrees.
<code>acArrowOpen30</code>	10	The arrowhead is an open arrow with an internal angle of 30 degrees.
<code>acArrowDotSmall</code>	11	The arrowhead is a small, filled circle.
<code>acArrowDotBlank</code>	12	The arrowhead is a large, empty circle.
<code>acArrowSmall</code>	13	The arrowhead is a small arrow.
<code>acArrowBoxBlank</code>	14	The arrowhead is an empty box.
<code>acArrowBoxFilled</code>	15	The arrowhead is a filled box.

Continued

Constant	Value	Description
acArrowDatumBlank	16	The arrowhead is a closed, empty, inward-pointing triangle.
acArrowDatumFilled	17	The arrowhead is a filled, inward-pointing triangle.
acArrowIntegral	18	The arrowhead is an integral sign.
acArrowNone	19	There is no arrowhead.
acArrowUserDefined	20	The arrowhead is defined by the user.

AcDimCenterType

The following constants can be values for the `CenterType` property of the `AcadDimDiametric` and `AcadDimRadial` objects.

Constant	Value	Description
acCenterMark	0	Marks the center with a small cross (+)
acCenterLine	1	Marks the center as the intersection of two perpendicular axes
acCenterNone	2	No center mark

AcDimFit

The following constants can be values for the `Fit` property of the dimension objects that support it (all except `AcadDimOrdinate`).

Constant	Value	Description
acTextAndArrows	0	Text and arrowheads are placed inside the extension lines.
acArrowsOnly	1	Arrowheads are placed inside the extension lines. The text is placed outside the arrowheads.
acTextOnly	2	Text is placed inside the extension lines. The arrowheads are placed outside the extension lines.
acBestFit	3	This option places the text and arrowheads in the best-fit location given the space available.

AcDimFractionType

The following constants can be values for the `FractionFormat` property of the `AcadDimension` object (and all derived objects) and the `AcadTolerance` object.

Constant	Value	Description
acHorizontal	0	Fraction displayed as $\frac{1}{2}$ (with a horizontal divider)
acDiagonal	1	Fraction displayed as $\frac{1}{2}$ (with a diagonal divider)
acNotStacked	2	Fraction displayed as "1/2"

AcDimHorizontalJustification

The following constants can be values for the `HorizontalTextPosition` property of the dimension objects that support it (all except `AcadDimOrdinate`).

Constant	Value	Description
<code>acHorzCentered</code>	0	The text is centered along the dimension line between the extension lines.
<code>acFirstExtensionLine</code>	1	The text is next to the first extension line.
<code>acSecondExtensionLine</code>	2	The text is next to the second extension line.
<code>acOverFirstExtension</code>	3	The text is above and aligned with the first extension line.
<code>acOverSecondExtension</code>	4	The text is above and aligned with the second extension line.

AcDimLUnits

The following constants can be values for the `UnitsFormat` property of the dimension objects that support it (all except `AcadDim3PointAngular` and `AcadDimAngular`).

Constant	Value	Description
<code>acDimLScientific</code>	1	Displays measurements in scientific notation
<code>acDimLDecimal</code>	2	Displays measurements in decimal notation
<code>acDimLEngineering</code>	3	Displays measurements in feet and decimal inches
<code>acDimLArchitectural</code>	4	Displays measurements in feet, inches, and fractional inches
<code>acDimLFractional</code>	5	Displays measurements in mixed-number (integer and fractional)
<code>acDimLWindowsDesktop</code>	6	Displays measurements in Windows Desktop format

AcDimPrecision

The following constants can be values for these properties:

- The `AltTolerancePrecision` and `AltUnitsPrecision` properties of the dimension objects that support them (all except `AcadDim3PointAngular` and `AcadDimAngular`)
- The `PrimaryUnitsPrecision` property of the dimension objects that support them (all except `AcadDim3PointAngular` and `AcadDimAngular`) and the `AcadTolerance` object
- The `TextPrecision` property of the `AcadDim3PointAngular` and `AcadDimAngular` objects
- The `TolerancePrecision` property of the `AcadDimension` object (and all derived objects)

Constant	Value	Description
acDimPrecisionZero	0	Accurate to no decimal places (0.)
acDimPrecisionOne	1	Accurate to one decimal place (0.0)
acDimPrecisionTwo	2	Accurate to two decimal places (0.00)
acDimPrecisionThree	3	Accurate to three decimal places (0.000)
acDimPrecisionFour	4	Accurate to four decimal places (0.0000)
acDimPrecisionFive	5	Accurate to five decimal places (0.00000)
acDimPrecisionSix	6	Accurate to six decimal places (0.000000)
acDimPrecisionSeven	7	Accurate to seven decimal places (0.0000000)
acDimPrecisionEight	8	Accurate to eight decimal places (0.00000000)

AcDimTextMovement

The following constants can be values for the `TextMovement` property of the `AcadDimension` object (and all derived objects).

Constant	Value	Description
acDimLineWithText	0	The dimension line follows the text.
acMoveTextAddLeader	1	The text moves independently of the dimension line, with a leader drawn from the text to the dimension line.
acMoveTextNoLeader	2	The text moves independently of the dimension line, with no leader drawn from the text to the dimension line.

AcDimToleranceJustify

The following constants can be values for the `ToleranceJustification` property of the `AcadDimension` object (and all derived objects).

Constant	Value	Description
acTolBottom	0	Displays the bottom tolerance value at the same level as the dimension text
acTolMiddle	1	Displays tolerance values half-size level with the dimension text
acTolTop	2	Displays the top tolerance value at the same level as the dimension text

AcDimToleranceMethod

The following constants can be values for the `ToleranceDisplay` property of the `AcadDimension` object (and all derived objects).

Constant	Value	Description
acTolNone	0	No tolerance values to be displayed with dimension text.
acTolSymmetrical	1	Use this constant to display tolerances when the positive and negative values are the same (e.g., 5.0 ± 0.2).
acTolDeviation	2	Use this constant to display tolerances when the positive and negative values are not the same. The positive value is displayed over the negative value.
acTolLimits	3	This constant incorporates tolerance values into the dimension value and displays the maximum dimension over the minimum.
acTolBasic	4	Basic draws a box around the dimension text, which is often used to indicate theoretically exact dimensions.

AcDimUnits

The following constants can be values for the `AltUnitsFormat` property of the dimension objects that support it (all except `AcadDim3PointAngular` and `AcadDimAngular`).

Constant	Value	Description
acDimScientific	1	Displays measurements in scientific notation
acDimDecimal	2	Displays measurements in decimal notation
acDimEngineering	3	Displays measurements in feet and decimal inches
acDimArchitecturalStacked	4	Displays measurements in feet, inches, and stacked (e.g., $\frac{1}{2}$) fractional inches
acDimFractionalStacked	5	Displays measurements in mixed-number notation, integers, and stacked (e.g., $\frac{1}{2}$) fractions
acDimArchitectural	6	Displays measurements in feet, inches, and fractional inches
acDimFractional	7	Displays measurements in mixed-number (integer and fractional) notation
acDimWindowsDesktop	8	Displays measurements in Windows Desktop format

AcDimVerticalJustification

The following constants can be values for the `VerticalTextPosition` property of the `AcadDimension` object (and all derived objects) and the `AcadLeader` object.

Constant	Value	Description
acVertCentered	0	Centers the dimension text between the extension lines
acAbove	1	Places the dimension text above the dimension line except when the dimension line isn't horizontal and text inside the extension lines is forced horizontal
acOutside	2	Places the dimension text on the side of the dimension line farthest from the defining points
acJIS	3	Places the dimension text to conform to Japanese Industrial Standards (JIS)

AcDrawingAreaSCMCommand

The following constants can be values for the `SCMCommandMode` property of the `AcadPreferencesUser` object.

Constant	Value	Description
<code>acEnter</code>	0	Disables the Command shortcut menu. As a result, right-clicking in the drawing area when a command is in progress issues Enter.
<code>acEnableSCMOptions</code>	1	Enables the Command shortcut menu only when options are currently available from the command-line prompt. In a command-line prompt, options are enclosed in square brackets. If no options are available, right-clicking issues Enter.
<code>acEnableSCM</code>	2	Enables the Command shortcut menu.

AcDrawingAreaSCMDefault

The following constants can be values for the `SCMDefaultMode` property of the `AcadPreferencesUser` object.

Constant	Value	Description
<code>acRepeatLastCommand</code>	0	Disables the Default shortcut menu. As a result, right-clicking in the drawing area when no objects are selected and no commands are in progress issues Enter, which repeats the last issued command.
<code>acSCM</code>	1	Enables the Default shortcut menu.

AcDrawingAreaSCMEdit

The following constants can be values for the `SCMEditMode` property of the `AcadPreferencesUser` object.

Constant	Value	Description
<code>acEdRepeatLastCommand</code>	0	Disables the Edit shortcut menu. As a result, right-clicking in the drawing area when one or more objects are selected and no commands are in progress issues Enter, which repeats the last issued command.
<code>acEdSCM</code>	1	Enables the Edit shortcut menu.

AcDrawingDirection

The following constants can be values for the DrawingDirection property of the AcadMText object.

Constant	Value	Description
acLeftToRight	1	Text written from left to right
acRightToLeft	2	Reserved for future use
acTopToBottom	3	Text written from top to bottom
acBottomToTop	4	Reserved for future use
acByStyle	5	Reserved for future use

AcDynamicBlockReferenceProperty

The following constants can be values for the UnitsType property of an AcadDynamicBlockReferenceProperty object.

Constant	Value
acAngular	1
acArea	3
acDistance	2
acNoUnits	0

AcEntityName

The following constants can be values for the EntityName property of the AcadEntity object.

Constant	Value	Description
ac3dFace	1	Acad3DFace object
ac3dPolyline	2	Acad3DPolyline object
ac3dSolid	3	Acad3DSolid object
acArc	4	AcadArc object
acAttribute	5	AcadAttribute object
acAttributeReference	6	AcadAttributeReference object
acBlockReference	7	AcadBlockReference object
acCircle	8	AcadCircle object
acDimAligned	9	AcadDimAligned object
acDimAngular	10	AcadDimAngular object
acDimDiametric	12	AcadDimDiametric object

Continued

Constant	Value	Description
acDimOrdinate	13	AcadDimOrdinate object
acDimRadial	14	AcadDimRadial object
acDimRotated	15	AcadDimRotated object
acEllipse	16	AcadEllipse object
acHatch	17	AcadHatch object
acLeader	18	AcadLeader object
acLine	19	AcadLine object
acMtext	21	AcadMText object
acPoint	22	AcadPoint object
acPolyline	23	AcadPolyline object
acPolylineLight	24	AcadLWPolyline object
acPolymesh	25	AcadPolygonMesh object
acRaster	26	AcadRasterImage object
acRay	27	AcadRay object
acRegion	28	AcadRegion object
acShape	29	AcadShape object
acSolid	30	AcadSolid object
acSpline	31	AcadSpline object
acText	32	AcadText object
acTolerance	33	AcadTolerance object
acTrace	34	AcadTrace object
acPViewport	35	AcadPViewport object
acXline	36	AcadXline object
acGroup	37	AcadGroup object
acMInsertBlock	38	AcadMInsertBlock object
acPolyfaceMesh	39	AcadPolyfaceMesh object
acMLine	40	AcadMLine object
acDim3PointAngular	41	AcadDim3PointAngular object
acExternalReference	42	AcadExternalReference object
acTable	43	AcadTable object
acDimArcLength	44	AcadDimArcLength object
acDimRadialLarge	45	AcadDimRadialLarge object

AcExtendOption

The following constants can be supplied as the value for the third parameter of the `IntersectWith` method of the `AcadEntity` object and those derived objects that support it (all except `AcadPolygonMesh` and `AcadPViewport`).

Constant	Value	Description
acExtendNone	0	Doesn't extend either object
acExtendThisEntity	1	Extends the base object
acExtendOtherEntity	2	Extends the object passed as an argument
acExtendBoth	3	Extends both objects

AcGradientPatternType

The following constants can be values for the HatchObjectType property of the Hatch object. They may also be used as the PatternType parameter value of an AddHatch method.

Constant	Value
acPredefinedGradient	0
acUserDefinedGradient	1

AcGridLineType

The following can be values for the GridLineType parameter of the GetGridColor, GetGridLineWeight, and GetGridVisibility methods of an AcadTable or AcadTableStyle object. Also, they can be values for the GridLineTypes parameter of the SetGridColor, SetGridLineWeight, and SetGridVisibility methods of an AcadTable or AcadTableStyle object.

Constant	Value	Description
acHorzBottom	4	Top or bottom horizontal grid line, based on the flow direction
acHorzInside	2	All horizontal grid lines, excluding the top and bottom lines
acHorzTop	1	Top or bottom horizontal grid line, based on the flow direction
acInvalidGridLine	0	An invalid grid line
acVertInside	16	All the vertical grid lines, excluding the farthest-left and farthest-right grid lines
acVertLeft	8	Farthest-left grid line
acVertRight	32	Farthest-right grid line

AcHatchObjectType

These constants specify the type of Hatch.

Constant	Value	Description
acHatchObject	0	Classic
acGradientObject	1	Gradient

AcHatchStyle

The following constants can be values for the HatchStyle property of the AcadHatch object.

Constant	Value	Description
acHatchStyleNormal	0	Specifies standard style or normal. This option hatches inward from the outermost area boundary. If AutoCAD encounters an internal boundary, it turns off hatching until it encounters another boundary.
acHatchStyleOuter	1	Fills the outermost areas only. This style also hatches inward from the area boundary, but it turns off hatching if it encounters an internal boundary and doesn't turn it back on again.
acHatchStyleIgnore	2	Ignores internal structure. This option hatches through all internal objects.

AcHorizontalAlignment

The following constants can be values for the HorizontalAlignment property of the AcadAttribute, AcadAttributeReference, and AcadText objects.

Constant	Value	Description
acHorizontalAlignmentLeft	0	Aligns the attribute, attribute reference, or text at the left, the bottom edge of the text being on the baseline.
acHorizontalAlignmentCenter	1	Aligns the attribute, attribute reference, or text at the center, the bottom edge of the text being on the baseline.
acHorizontalAlignmentRight	2	Aligns the attribute, attribute reference, or text at the right, the bottom edge of the text being on the baseline.
acHorizontalAlignmentAligned	3	The attribute, attribute reference, or text is aligned according to the points held in the InsertionPoint and TextAlignmentPoint properties.
acHorizontalAlignmentMiddle	4	Aligns text at the middle. This isn't the same as acHorizontalAlignmentCenter, because the middle is defined as the midpoint between the top and bottom limits, the bottom referring to a position below the baseline.
acHorizontalAlignmentFit	5	The attribute, attribute reference, or text is aligned to fit to the points held in the InsertionPoint and TextAlignmentPoint properties.

AcInsertUnits

The following constants can be values for the ADCInsertUnitsDefaultSource and ADCInsertUnitsDefaultTarget properties of the AcadPreferencesUser object.

Constant	Value	Description
acInsertUnitsUnitless	0	No units
acInsertUnitsInches	1	Inches
acInsertUnitsFeet	2	Feet
acInsertUnitsMiles	3	Miles
acInsertUnitsMillimeters	4	Millimeters
acInsertUnitsCentimeters	5	Centimeters
acInsertUnitsMeters	6	Meters
acInsertUnitsKilometers	7	Kilometers
acInsertUnitsMicroinches	8	Microinches
acInsertUnitsMils	9	Mils
acInsertUnitsYards	10	Yards
acInsertUnitsAngstroms	11	Angstrom units
acInsertUnitsNanometers	12	Nanometers
acInsertUnitsMicrons	13	Microns (micrometers)
acInsertUnitsDecimeters	14	Decimeters
acInsertUnitsDecameters	15	Decameters
acInsertUnitsHectometers	16	Hectometers
acInsertUnitsGigameters	17	Gigameters
acInsertUnitsAstronomicalUnits	18	Astronomical units
acInsertUnitsLightYears	19	Light years
acInsertUnitsParsecs	20	Parsecs

AcISOPenWidth

The following constants can be values for the ISOPenWidth property of the AcadHatch object.

Constant	Value	Description
acPenWidthUnk	-1	Unknown or nonstandard pen width
acPenWidth013	13	$\frac{13}{100}$ th of a millimeter
acPenWidth018	18	$\frac{18}{100}$ th of a millimeter
acPenWidth025	25	$\frac{25}{100}$ th of a millimeter
acPenWidth035	35	$\frac{35}{100}$ th of a millimeter
acPenWidth050	50	$\frac{50}{100}$ th of a millimeter
acPenWidth070	70	$\frac{70}{100}$ th of a millimeter
acPenWidth100	100	1 millimeter
acPenWidth140	140	1.4 millimeters
acPenWidth200	200	2 millimeters

AcKeyboardAccelerator

The following constants can be values for the KeyboardAccelerator property of the AcadPreferencesUser object.

Constant	Value	Description
acPreferenceClassic	0	Uses the AutoCAD classic keyboard
acPreferenceCustom	1	Uses the Windows standard keyboard

AcKeyboardPriority

The following constants can be values for the KeyboardPriority property of the AcadPreferencesUser object.

Constant	Value	Description
acKeyboardRunningObjSnap	0	Object snaps are strictly adhered to when typing in coordinates.
acKeyboardEntry	1	Keyboard entry values are strictly adhered to when typing in coordinates.
acKeyboardEntryExceptScripts	2	Keyboard entry values are strictly adhered to when typing in coordinates. However, when coordinates are entered through a script, running object snaps are adhered to.

AcLeaderType

The following constants can be values for the Type property of the AcadLeader object.

Constant	Value	Description
acLineNoArrow	0	A line with no arrow
acSplineNoArrow	1	A spline with no arrow
acLineWithArrow	2	A line with an arrow
acSplineWithArrow	3	A spline with an arrow

AcLineSpacingStyle

The following constants can be values for the LineSpacingStyle property of the AcadMText object.

Constant	Value	Description
acLineSpacingStyleAtLeast	1	Allows the spacing between different lines of text to adjust automatically, based on the height of the largest character in a line of text
acLineSpacingStyleExactly	2	Forces the line spacing to be the same size for all the lines in the text regardless of formatting overrides

AcLineWeight

The following constants can be values for these properties:

- The **Lineweight** property of the **AcadEntity** object (and all derived objects), the **Acad-DatabasePreferences**, **AcadGroup**, and **AcadLayer** objects
- The **DimensionLineweight** property of the dimension objects that support it (all except **AcadDimOrdinate**) and the **AcadLeader** object
- The **ExtensionLineweight** property of the dimension objects that support it (all except **AcadDimDiametric**, **AcadDimOrdinate**, and **AcadDimRadial**)

Constant	Value	Description
acLnWtByLwDefault	-3	Draws lines in the default lineweight.
acLnWtByBlock	-2	Draws lines in the lineweight specified for the block (initial value).
acLnWtByLayer	-1	Draws lines in the lineweight specified for the layer.
acLnWt000	0	Plots with the thinnest lineweight available for the specified plotting device. In model space, this is set with a width of 1 pixel.
acLnWt005	5	$\frac{5}{100}$ millimeter.
acLnWt009	9	$\frac{9}{100}$ millimeter or $\frac{1}{4}$ point.
acLnWt013	13	$\frac{13}{100}$ millimeter or $\frac{5}{1000}$ inch.
acLnWt015	15	$\frac{15}{100}$ millimeter.
acLnWt018	18	$\frac{18}{100}$ millimeter, $\frac{1}{2}$ point, or Pen Size 0000.
acLnWt020	20	$\frac{2}{10}$ millimeter or Pen Size 000.
acLnWt025	25	$\frac{1}{4}$ millimeter, $\frac{1}{100}$ inch, or $\frac{3}{4}$ point.
acLnWt030	30	$\frac{3}{10}$ millimeter or Pen Size 00.
acLnWt035	35	$\frac{35}{100}$ millimeter, 1 point, or Pen Size 0.
acLnWt040	40	$\frac{4}{10}$ millimeter.
acLnWt050	50	$\frac{1}{2}$ millimeter or Pen Size 1
acLnWt053	53	$\frac{53}{100}$ millimeter or 1 $\frac{1}{2}$ points.
acLnWt060	60	$\frac{6}{10}$ millimeter or Pen Size 2.
acLnWt070	70	$\frac{7}{10}$ millimeter, 2 $\frac{1}{4}$ point, or Pen Size 2 $\frac{1}{2}$.
acLnWt080	80	$\frac{8}{10}$ millimeter or Pen Size 3.
acLnWt090	90	$\frac{9}{10}$ millimeter.
acLnWt100	100	1 millimeter or Pen Size 3 $\frac{1}{2}$.
acLnWt106	106	3 point.
acLnWt120	120	1.2 millimeters or Pen Size 4.
acLnWt140	140	1.4 millimeters.
acLnWt158	158	4 $\frac{1}{4}$ point.
acLnWt200	200	2 millimeters.
acLnWt211	211	6 point.

AcLoopType

The following constants can be supplied as the value for the second parameter of the InsertLoopAt method of the AcadHatch object.

Constant	Value	Description
acHatchLoopTypeDefault	0	The loop type hasn't been specified.
acHatchLoopTypeExternal	1	The loop consists of external entities.
acHatchLoopTypePolyline	2	The hatch loop consists of a polyline.
acHatchLoopTypeDerived	4	The loop was derived by AutoCAD's boundary tracer from a picked point.
acHatchLoopTypeTextbox	8	The loop consists of a box around an existing text object.

AcMenuItemType

The following constants can be supplied as the value for the parameter of the Save method of the AcadMenuGroup object or the second parameter for the property of the SaveAs method of the AcadDocument and AcadMenuGroup objects.

Constant	Value	Description
acMenuItemCompiled	0	A compiled menu file (.mnc file type)
acMenuItemSource	1	A source menu file (.mns file type)

AcMenuGroupType

The following constants can be values for the Type property of the AcadMenuGroup object.

Constant	Value	Description
acBaseMenuGroup	0	A base menu group
acPartialMenuGroup	1	A partial menu group

AcMenuItemType

The following constants can be values for the Type property of the AcadPopupMenuItem object.

Constant	Value	Description
acMenuItem	0	Specifies a pop-up menu item
acMenuItemSeparator	1	Specifies a separator in a pop-up menu
acMenuItemSubMenu	2	Specifies a pop-up submenu

AcMLineJustification

The following constants can be values for the Justification property of an AcadMLine object. To access this property, use the IAcadMLine2 interface.

Constant	Value
acBottom	2
acTop	0
acZero	1

AcOleQuality

The following constants can be values for the OleQuality property of the AcadPreferencesOutput object.

Constant	Value	Description
acOQLineArt	0	Specifies line art quality (e.g., for an embedded spreadsheet)
acOQText	1	Specifies text quality (e.g., for an embedded Word document)
acOQGraphics	2	Specifies graphics quality (e.g., for an embedded pie chart)
acOQPhoto	3	Specifies photograph quality
acOQHighPhoto	4	High-quality photograph

AcPatternType

The following constants can be values for the PatternType property of the AcadHatch object, and they can also be supplied as the value for the second parameter of the AddHatch method of the AcadBlock, AcadModelSpace, and AcadPaperSpace objects and the SetPattern method of the AcadHatch object.

Constant	Value	Description	SetPattern method
acHatchPatternTypeUserDefined	0	Defines a pattern of lines using the current linetype	
acHatchPatternTypePreDefined	1	Selects the pattern name from those defined in the acad.pat file	
acHatchPatternTypeCustomDefined	2	Selects the pattern name from a .pat file other than the acad.pat file	

AcPlotPaperUnits

The following constants can be values for the PaperUnits property of the AcadLayout and AcadPlotConfiguration objects.

Constant	Value	Description
acInches	0	Units specified in inches
acMillimeters	1	Units specified in millimeters
acPixels	2	Units specified in pixels

AcPlotPolicy

The following constants can be values for the PlotPolicy property of the AcadPreferencesOutput object.

Constant	Value	Description
acPolicyNamed	0	No association is made between color and plot-style name. Plot-style name for each object is set to the default defined in the DefaultPlotStyleForObjects property.
acPolicyLegacy	1	An object's plot-style name is associated with its color per the naming convention ACI x , where x is the color number of the object according to the AutoCAD color index.

AcPlotRotation

The following constants can be values for the PlotRotation property of the AcadLayout and AcadPlotConfiguration objects.

Constant	Value	Description
ac0degrees	0	These values are read counterclockwise relative to the X-axis.
ac90degrees	1	
ac180degrees	2	
ac270degrees	3	

AcPlotScale

The following constants can be values for the StandardScale property of the AcadLayout and AcadPlotConfiguration objects.

Constant	Value	Description
acScaleToFit	0	Scale to fit
ac1_128in_1ft	1	1/128":1'
ac1_64in_1ft	2	1/64":1'
ac1_32in_1ft	3	1/32":1'
ac1_16in_1ft	4	1/16":1'
ac3_32in_1ft	5	3/32":1'
ac1_8in_1ft	6	1/8":1'
ac3_16in_1ft	7	3/16":1'
ac1_4in_1ft	8	1/4":1'
ac3_8in_1ft	9	3/8":1'
ac1_2in_1ft	10	1/2":1'
ac3_4in_1ft	11	3/4":1'
ac1in_1ft	12	1":1'
ac3in_1ft	13	3":1'
ac6in_1ft	14	6":1'
ac1ft_1ft	15	1':1'
ac1_1	16	1:1
ac1_2	17	1:2
ac1_4	18	1:4
ac1_8	19	1:8
ac1_10	20	1:10
ac1_16	21	1:16
ac1_20	22	1:20
ac1_30	23	1:30
ac1_40	24	1:40
ac1_50	25	1:50
ac1_100	26	1:100
ac2_1	27	2:1
ac4_1	28	4:1
ac8_1	29	8:1
ac10_1	30	10:1
ac100_1	31	100:1

AcPlotType

The following constants can be values for the `PlotType` property of the `AcadLayout` and `AcadPlotConfiguration` objects.

Constant	Value	Description
<code>acDisplay</code>	0	Prints everything that is in the current display.
<code>acExtents</code>	1	Prints everything that falls within the extents of the currently selected space.
<code>acLimits</code>	2	Prints everything that is in the limits of the current space.
<code>acView</code>	3	Prints the view named by the <code>ViewToPlot</code> property.
<code>acWindow</code>	4	Prints everything in the window specified by the <code>SetWindowToPlot</code> method.
<code>acLayout</code>	5	Prints everything that falls within the margins of the specified paper size with the origin being calculated from the 0,0 coordinate location in the Layout tab. This option isn't available when printing from model space.

AcPolylineType

The following constants can be values for the `Type` property of the `AcadPolyline` object.

Constant	Value	Description
<code>acSimplePoly</code>	0	Specifies a simple polyline
<code>acFitCurvePoly</code>	1	Specifies a fit curve polyline
<code>acQuadSplinePoly</code>	2	Specifies a quadratic B-spline polyline
<code>acCubicSplinePoly</code>	3	Specifies a cubic B-spline polyline

AcPolymeshType

The following constants can be values for the `Type` property of the `AcadPolygonMesh` object.

Constant	Value	Description
<code>acSimpleMesh</code>	0	Specifies a simple mesh with no surface fitting or smoothing
<code>acQuadSurfaceMesh</code>	5	Specifies a quadratic B-spline surface fit
<code>acCubicSurfaceMesh</code>	6	Specifies a cubic B-spline surface fit
<code>acBezierSurfaceMesh</code>	8	Specifies a Bezier surface fit

AcPreviewMode

The following constants can be values for the `DisplayPlotPreview` property of the `AcadPlot` object.

Constant	Value	Description
<code>acPartialPreview</code>	0	Shows an accurate representation of the effective plot area relative to the paper size. The final location of the affected area on the paper depends on the plotter.
<code>acFullPreview</code>	1	Displays the drawing on the screen as it will appear when plotted on paper. Because this requires a regeneration of the drawing, it's slower than a partial preview. It's faster than the normal plot regeneration because AutoCAD performs no vector sorting or optimization.

AcPrinterSpoolAlert

The following constants can be values for the `PrinterSpoolAlert` property of the `AcadPreferencesOutput` object.

Constant	Value	Description
<code>acPrinterAlwaysAlert</code>	0	Always alert and log errors.
<code>acPrinterAlertOnce</code>	1	Alert only the first error, and log all errors.
<code>acPrinterNeverAlertLogOnce</code>	2	Never alert, but log all errors.
<code>acPrinterNeverAlert</code>	3	Never alert, and don't log any errors.

AcProxyImage

The following constants can be values for the `ProxyImage` property of the `AcadPreferencesOpenSave` object.

Constant	Value	Description
<code>acProxyNotShow</code>	0	Don't show the proxy image.
<code>acProxyShow</code>	1	Show the proxy image.
<code>acProxyBoundingBox</code>	2	Show the proxy image in a bounding box.

AcRegenType

The following constants can be supplied as the value for the parameter of the `Regen` method of the `AcadDocument` object.

Constant	Value	Description
<code>acActiveViewport</code>	0	Regenerates drawing in active viewport only
<code>acAllViewports</code>	1	Regenerates drawing in all viewports

AcRotationAngle

The following constants can be values for the `TextRotation` parameter of the `SetTextRotation` property of an `AcadTable` object. Also, they can be return values of the `GetTextRotation` method of an `AcadTable` object.

Constant	Value
<code>acDegrees000</code>	0
<code>acDegrees090</code>	1
<code>acDegrees180</code>	2
<code>acDegrees270</code>	3
<code>acDegreesUnknown</code>	-1

AcRowType

The following constants can be values according to the following list.

Method	Parameter	Object(s)
<code>GetAlignment</code>	<code>RowType</code>	<code>AcadTable</code> , <code>AcadTableStyle</code>
<code>GetBackgroundColor</code>	<code>RowType</code>	<code>AcadTable</code> , <code>AcadTableStyle</code>
<code>GetBackgroundColorNone</code>	<code>RowType</code>	<code>AcadTable</code> , <code>AcadTableStyle</code>
<code>GetColor</code>	<code>RowType</code>	<code>AcadTableStyle</code>
<code>GetGridColor</code>	<code>RowType</code>	<code>AcadTable</code> , <code>AcadTableStyle</code>
<code>GetGridLineWeight</code>	<code>RowType</code>	<code>AcadTable</code> , <code>AcadTableStyle</code>
<code>GetGridVisibility</code>	<code>RowType</code>	<code>AcadTable</code> , <code>AcadTableStyle</code>
<code>GetTextHeight</code>	<code>RowType</code>	<code>AcadTable</code> , <code>AcadTableStyle</code>
<code>GetTextStyle</code>	<code>RowType</code>	<code>AcadTable</code> , <code>AcadTableStyle</code>
<code>GetContentColor</code>	<code>RowType</code>	<code>AcadTable</code>
<code>SetGridColor</code>	<code>RowTypes</code>	<code>AcadTable</code> , <code>AcadTableStyle</code>
<code>SetGridLineWeight</code>	<code>RowTypes</code>	<code>AcadTable</code> , <code>AcadTableStyle</code>
<code>SetGridVisibility</code>	<code>RowTypes</code>	<code>AcadTable</code> , <code>AcadTableStyle</code>

Also, they can be return values of the `GetRowType` method of an `AcadTable` object.

Constant	Value
<code>AcUnknown</code>	0
<code>acDataRow</code>	1
<code>acTitleRow</code>	2
<code>acHeaderRow</code>	4

AcSaveAsType

The following constants can be values for the SaveAsType property of the AcadPreferencesOpenSave object.

Constant	Value	Description
acUnknown	-1	This constant is read-only. The drawing type is unknown or invalid.
acR12_dxf	1	AutoCAD Release12/LT2 DXF (*.dxf).
acR13_dwg	4	AutoCAD Release13/LT95 DWG (*.dwg).
acR13_dxf	5	AutoCAD Release13/LT95 DXF (*.dxf).
acR14_dwg	8	AutoCAD Release14/LT97 DWG (*.dwg).
acR14_dxf	9	AutoCAD Release14/LT97 DXF (*.dxf).
acR15_dwg	12	AutoCAD 2000 DWG (*.dwg).
ac2000_dwg	12	AutoCAD 2000 DWG (*.dwg).
acR15_dxf	13	AutoCAD 2000 DXF (*.dxf).
ac2000_dxf	13	AutoCAD 2000 DXF (*.dxf).
acR15_Template	14	AutoCAD 2000 Drawing Template File (*.dwt).
ac2000_Template	14	AutoCAD 2000 Drawing Template File (*.dwt).
acR18	24	AutoCAD 2004 DWG (*.dwg)
ac2004_dwg	24	AutoCAD 2004 DWG (*.dwg)
acNative	24	A synonym for the current drawing release format. If you want your application to save the drawing in the format of whatever version of AutoCAD the application is running on, then use the acNative format.
acR18_dxf	25	AutoCAD 2004 DXF (*.dxf)
ac2004_dxf	25	AutoCAD 2004 DXF (*.dxf)
acR18_Template	26	AutoCAD 2004 Drawing Template File (*.dwt).
ac2004_Template	26	AutoCAD 2004 Drawing Template File (*.dwt).

Note The acR13_dwg, acR13_dxf, acR14_dwg, and acR14_dxf options aren't available in AutoCAD 2006.

AcSelect

The following constants can be supplied as the value for the first parameter in either of the Select or SelectByPolygon methods of the AcadSelectionSet object.

Constant	Value	Description
acSelectionSetWindow	0	Selects all entities completely inside a rectangular area whose corners are defined by Point1 and Point2 (Select method only).
acSelectionSetCrossing	1	Selects objects within and crossing a rectangular area whose corners are defined by Point1 and Point2 (Select method only).
acSelectionSetFence	2	Selects all entities crossing a selection fence. The fence is defined by coordinates in PointList. The fence mode is similar to CrossingPolygon mode except that AutoCAD doesn't close the last vector of the fence, and a fence can cross itself. The fence isn't affected by the PICKADD system variable (SelectByPolygon method only).
acSelectionSetPrevious	3	Selects the most recent selection set. This mode is ignored if you switch between paper and model space and attempt to use the selection set (Select method only).
acSelectionSetLast	4	Selects the most recently created visible entity (Select method only).
acSelectionSetAll	5	Selects all entities (Select method only).
acSelectionSetWindowPolygon	6	Selects objects within a polygon defined by PointList (SelectByPolygon method only).
acSelectionSetCrossingPolygon	7	Selects objects within and crossing an area defined by a polygon. Use PointList to define the coordinates of the polygon. AutoCAD will close the last vector of the polygon. A polygon definition can't cross itself (SelectByPolygon method only).

AcSelectType

The following constants can be values for the SelType parameter of the SelectSubRegion method of an AcadTable object.

Constant	Value	Description
acTableSelectCrossing	2	Cross-selection type
acTableSelectWindow	1	Window-selection type

AcShadePlot

The following constants can be values for the ShadePlot property of an Pviewport object.

Constant	Value	Description
acShadePlotAsDisplayed	0	Model space view plots the same way it is displayed.
acShadePlotWireframe	1	Model space view plots as wireframe regardless of display.
acShadePlotHidden	2	Model space view plots with hidden lines removed, regardless of display.
acShadePlotRendered	3	Model space view plots as rendered regardless of display.

AcTableDirection

The following constants can be values for the FlowDirection property of an AcadTable and AcadTableStyle object.

Constant	Value	Description
acTableBottomToTop	1	Table flow direction of bottom to top
acTableTopToBottom	0	Table flow direction of top to bottom

AcTableStyleOverrides

The following constants can be values for TableStyleOverrides property of an AcadTable object.

Constant	Value	Description
acTitleSuppressed	1	Indicates title-suppressed table style override for table object
acHeaderSuppressed	2	Indicates header-suppressed table style override for table object
acFlowDirection	3	Indicates flow direction table style override for table object
acHorzCellMargin	4	Indicates horizontal cell margin table style override for table object
acVertCellMargin	5	Indicates vertical cell margin table style override for table object
acTitleRowColor	6	Indicates title row color table style override for table object
acHeaderRowColor	7	Indicates header row color table style override for table object
acDataRowColor	8	Indicates data row color table style override for table object
acTitleRowFillNone	9	Indicates title row no fill table style override for table object
acHeaderRowFillNone	10	Indicates header row no fill table style override for table object
acDataRowFillNone	11	Indicates data row no fill table style override for table object
acTitleRowFillColor	12	Indicates title row fill color table style override for table object
acHeaderRowFillColor	13	Indicates header row fill color table style override for table object
acDataRowFillColor	14	Indicates data row fill color table style override for table object
acTitleRowAlignment	15	Indicates title row alignment table style override for table object
acHeaderRowAlignment	16	Indicates header row alignment table style override for table object

Continued

Constant	Value	Description
acDataRowAlignment	17	Indicates data row alignment table style override for table object
acTitleRowTextStyle	18	Indicates title row text style table style override for table object
acHeaderRowTextStyle	19	Indicates header row text style table style override for table object
acDataRowTextStyle	20	Indicates data row text style table style override for table object
acTitleRowTextHeight	21	Indicates title row text height table style override for table object
acHeaderRowTextHeight	22	Indicates header row text height table style override for table object
acDataRowTextHeight	23	Indicates data row text height table style override for table object
acTitleHorzTopColor	40	Indicates title row horizontal top gridline color table style override for table object
acTitleHorzInsideColor	41	Indicates title row horizontal inside gridline color table style override for table object
acTitleHorzBottomColor	42	Indicates title row horizontal bottom gridline color table style override for table object
acTitleVertLeftColor	43	Indicates title row vertical left gridline color table style override for table object
acTitleVertInsideColor	44	Indicates title row vertical inside gridline color table style override for table object
acTitleVertRightColor	45	Indicates title row vertical right gridline color table style override for table object
acHeaderHorzTopColor	46	Indicates header row horizontal top gridline color table style override for table object
acHeaderHorzInsideColor	47	Indicates header row horizontal inside gridline color table style override for table object
acHeaderHorzBottomColor	48	Indicates header row horizontal bottom gridline color table style override for table object
acHeaderVertLeftColor	49	Indicates header row vertical left gridline color table style override for table object
acHeaderVertInsideColor	50	Indicates header row vertical inside gridline color table style override for table object
acHeaderVertRightColor	51	Indicates header row vertical right gridline color table style override for table object
acDataHorzTopColor	52	Indicates data row horizontal top gridline color table style override for table object
acDataHorzInsideColor	53	Indicates data row horizontal inside gridline color table style override for table object
acDataHorzBottomColor	54	Indicates data row horizontal bottom gridline color table style override for table object
acDataVertLeftColor	55	Indicates data row vertical left gridline color table style override for table object

Constant	Value	Description
acDataVertInsideColor	56	Indicates data row vertical inside gridline color table style override for table object
acDataVertRightColor	57	Indicates data row vertical right gridline color table style override for table object
acTitleHorzTopLineWeight	70	Indicates title row horizontal top gridline lineweight table style override for table object
acTitleHorzInsideLineWeight	71	Indicates title row horizontal inside gridline lineweight table style override for table object
acTitleHorzBottomLineWeight	72	Indicates title row horizontal bottom gridline lineweight table style override for table object
acTitleVertLeftLineWeight	73	Indicates title row vertical left gridline lineweight table style override for table object
acTitleVertInsideLineWeight	74	Indicates title row vertical inside gridline lineweight table style override for table object
acTitleVertRightLineWeight	75	Indicates title row vertical right gridline lineweight table style override for table object
acHeaderHorzTopLineWeight	76	Indicates header row horizontal top gridline lineweight table style override for table object
acHeaderHorzInsideLineWeight	77	Indicates header row horizontal inside gridline lineweight table style override for table object
acHeaderHorzBottomLineWeight	78	Indicates header row horizontal bottom gridline lineweight table style override for table object
acHeaderVertLeftLineWeight	79	Indicates header row vertical left gridline lineweight table style override for table object
acHeaderVertInsideLineWeight	80	Indicates header row vertical inside gridline lineweight table style override for table object
acHeaderVertRightLineWeight	81	Indicates header row vertical right gridline lineweight table style override for table object
acDataHorzTopLineWeight	82	Indicates data row horizontal top gridline lineweight table style override for table object
acDataHorzInsideLineWeight	83	Indicates data row horizontal inside gridline lineweight table style override for table object
acDataHorzBottomLineWeight	84	Indicates data row horizontal bottom gridline lineweight table style override for table object
acDataVertLeftLineWeight	85	Indicates data row vertical left gridline lineweight table style override for table object
acDataVertInsideLineWeight	86	Indicates data row vertical inside gridline lineweight table style override for table object
acDataVertRightLineWeight	87	Indicates data row vertical right gridline lineweight table style override for table object
acTitleHorzTopVisibility	100	Indicates title row horizontal top gridline visibility table style override for table object
acTitleHorzInsideVisibility	101	Indicates title row horizontal inside gridline visibility table style override for table object

Continued

Constant	Value	Description
acTitleHorzBottomVisibility	102	Indicates title row horizontal bottom gridline visibility table style override for table object
acTitleVertLeftVisibility	103	Indicates title row vertical left gridline visibility table style override for table object
acTitleVertInsideVisibility	104	Indicates title row vertical inside gridline visibility table style override
acTitleVertRightVisibility	105	Indicates title row vertical right gridline visibility table style override for table object
acHeaderHorzTopVisibility	106	Indicates header row horizontal top gridline visibility table style override for table object
acHeaderHorzInsideVisibility	107	Indicates header row horizontal inside gridline visibility table style override for table object
acHeaderHorzBottomVisibility	108	Indicates header row horizontal bottom gridline visibility table style override for table object
acHeaderVertLeftVisibility	109	Indicates header row vertical left gridline visibility table style override for table object
acHeaderVertInsideVisibility	110	Indicates header row vertical inside gridline visibility table style override for table object
acHeaderVertRightVisibility	111	Indicates header row vertical right gridline visibility table style override for table object
acDataHorzTopVisibility	112	Indicates data row horizontal top gridline visibility table style override for table object
acDataHorzInsideVisibility	113	Indicates data row horizontal inside gridline visibility table style override for table object
acDataHorzBottomVisibility	114	Indicates data row horizontal bottom gridline visibility table style override for table object
acDataVertLeftVisibility	115	Indicates data row vertical left gridline visibility table style override for table object
acDataVertInsideVisibility	116	Indicates data row vertical inside gridline visibility table style override for table object
acDataVertRightVisibility	117	Indicates data row vertical right gridline visibility table style override for table object
acCellAlignment	130	Indicates alignment table style override for table cell
acCellBackgroundFillNone	131	Indicates background no fill table style override for table cell
acCellBackgroundColor	132	Indicates background color table style override for table cell
acCellContentColor	133	Indicates content color table style override for table cell
acCellTextStyle	134	Indicates text style table style override for table cell
acCellTextHeight	135	Indicates text height table style override for table cell
acCellTopGridColor	136	Indicates top-edge grid color table style override for table cell

Constant	Value	Description
acCellRightGridColor	137	Indicates right-edge grid color table style override for table cell
acCellBottomGridColor	138	Indicates bottom-edge grid color table style override for table cell
acCellLeftGridColor	139	Indicates left-edge grid color table style override for table cell
acCellTopGridLineWeight	140	Indicates top-edge grid lineweight table style override for table cell
acCellRightGridLineWeight	141	Indicates right-edge grid lineweight table style override for table cell
acCellBottomGridLineWeight	142	Indicates bottom-edge grid lineweight table style override for table cell
acCellLeftGridLineWeight	143	Indicates left-edge grid lineweight table style override for table cell
acCellTopVisibility	144	Indicates top-edge grid visibility table style override for table cell
acCellRightVisibility	145	Indicates right-edge grid visibility table style override for table cell
acCellBottomVisibility	146	Indicates bottom-edge grid visibility table style override for table cell
acCellLeftVisibility	147	Indicates left-edge grid visibility table style override for table cell

AcTextFontStyle

The following constants and their combinations can be values for the TextFontStyle property of the AcadPreferencesDisplay object.

Constant	Value	Description
acFontRegular	0	Specifies a normal font style
acFontItalic	1	Specifies italic style
acFontBold	2	Specifies bold style
acFontBoldItalic	3	Specifies both bold and italic styles

AcTextGenerationFlag

The following constants can be values for the TextGenerationFlag property of the AcadAttribute, AcadAttributeReference, and AcadText objects.

Constant	Value	Description
acTextFlagBackward	2	Text to be displayed backward
acTextFlagUpsideDown	4	Text to be displayed upside-down

AcToolbarDockStatus

The following constants can be values for the DockStatus property of the AcadToolbar object, or they can be supplied as the value for the parameter of the Dock method of the AcadToolbar object.

Constant	Value	Description
acToolbarDockTop	0	Toolbar docked at the top of the window
acToolbarDockBottom	1	Toolbar docked at the bottom of the window
acToolbarDockLeft	2	Toolbar docked at the left of the window
acToolbarDockRight	3	Toolbar docked at the right of the window
acToolbarFloating	4	Toolbar not docked at all

AcToolbarItemType

The following constants can be values for the Type property of the AcadToolbarItem object.

Constant	Value	Description
acToolbarButton	0	Specifies a generic button
acToolbarSeparator	1	Specifies a toolbar separator
acToolbarControl	2	Specifies a control button
acToolbarFlyout	3	Specifies a flyout button

AcUnits

The following constants can be supplied as the value for the second parameter of the DistanceToReal and RealToString methods of the AcadUtility object.

Constant	Value	Description
acDefaultUnits	-1	Displays measurements in default units
acScientific	1	Displays measurements in scientific notation
acDecimal	2	Displays measurements in decimal notation
acEngineering	3	Displays measurements in feet and decimal inches
acArchitectural	4	Displays measurements in feet, inches, and fractional inches
acFractional	5	Displays measurements in mixed-number (integer and fractional) notation

AcVerticalAlignment

The following constants can be values for the `VerticalAlignment` property of the `AcadAttribute`, `AcadAttributeReference`, and `AcadText` objects.

Constant	Value	Description
<code>acVerticalAlignmentBaseline</code>	0	Align on the baseline. Note that the baseline isn't the same as the bottom. The baseline is defined at a level partway between the bottom position and the middle position.
<code>acVerticalAlignmentBottom</code>	1	Align at the bottom.
<code>acVerticalAlignmentMiddle</code>	2	Align at the middle. This is midway between the bottom and the top positions.
<code>acVerticalAlignmentTop</code>	3	Align at the top.

AcViewportScale

The following constants can be values for the `StandardScale` property of the `AcadPViewport` object.

While these constants are still accepted as parameters for the `StandardScale` property of the `AcadPViewport` object (and are listed in AutoCAD's VBA Help file), they are not the constants shown in the VBA Object Browser. Also, they are not shown in the `AutoListMembers` tool. Those constants are named `acVp1` through `acVp100`, and their value is 1 greater than the numeral of their name. For values 1 through 29, the description scale corresponds to the scale listed.

Constant	Value	Description
<code>acVpScaleToFit</code>	0	Scale to fit
<code>acVpCustomScale</code>	1	Custom scale
<code>acVp1_1</code>	2	1 : 1
<code>acVp1_2</code>	3	1 : 2
<code>acVp1_4</code>	4	1 : 4
<code>acVp1_8</code>	5	1 : 8
<code>acVp1_10</code>	6	1 : 10
<code>acVp1_16</code>	7	1 : 16
<code>acVp1_20</code>	8	1 : 20
<code>acVp1_30</code>	9	1 : 30
<code>acVp1_40</code>	10	1 : 40
<code>acVp1_50</code>	11	1 : 50
<code>acVp1_100</code>	12	1 : 100
<code>acVp2_1</code>	13	2 : 1
<code>acVp4_1</code>	14	4 : 1
<code>acVp8_1</code>	15	8 : 1

Continued

Constant	Value	Description
acVp10_1	16	10 : 1
acVp100_1	17	100 : 1
acVp1_128in_1ft	18	$\frac{1}{128}'' : 1'$
acVp1_64in_1ft	19	$\frac{1}{64}'' : 1'$
acVp1_32in_1ft	20	$\frac{1}{32}'' : 1'$
acVp1_16in_1ft	21	$\frac{1}{16}'' : 1'$
acVp1_32in_1ft	22	$\frac{3}{32}'' : 1'$
acVp1_8in_1ft	23	$\frac{1}{8}'' : 1'$
acVp1_16in_1ft	24	$\frac{3}{16}'' : 1'$
acVp1_4in_1ft	25	$\frac{1}{4}'' : 1'$
acVp1_8in_1ft	26	$\frac{3}{8}'' : 1'$
acVp1_2in_1ft	27	$\frac{1}{2}'' : 1'$
acVp1_4in_1ft	28	$\frac{3}{4}'' : 1'$
acVp1in_1ft	29	1'' : 1'
acVp29	30	1 $\frac{1}{2}'' : 1'$
acVp3in_1ft	30	3'' : 1'
acVp6in_1ft	31	6'' : 1'
acVp1ft_1ft	32	1' : 1'

AcViewportSplitType

The following constants can be supplied as the value for the parameter of the Split method of the AcadViewport object.

Constant	Value	Description
acViewport2Horizontal	0	Splits the viewport horizontally into two equal sections.
acViewport2Vertical	1	Splits the viewport vertically into two equal sections.
acViewport3Left	2	Splits the viewport into two vertical halves. The left half is split horizontally into two equal sections.
acViewport3Right	3	Splits the viewport into two vertical halves. The right half is split horizontally into two equal sections.
acViewport3Horizontal	4	Splits the viewport horizontally into three equal sections.
acViewport3Vertical	5	Splits the viewport vertically into three equal sections.
acViewport3Above	6	Splits the viewport into two horizontal halves. The top half will be a single viewport. The bottom half is split vertically into two equal sections.
acViewport3Below	7	Splits the viewport into two horizontal halves. The bottom half will be a single viewport. The top half is split vertically into two equal sections.
acViewport4	8	Splits the viewport horizontally and vertically into four equal sections.

AcWindowState

The following constants can be values for the WindowState property of the AcadApplication and AcadDocument objects.

Constant	Value	Description
acNorm	1	Neither maximized nor minimized
acMin	2	Minimized window
acMax	3	Maximized window

AcXRefDemandLoad

The following constants can be values for the XrefDemandLoad property of the AcadPreference-sOpenSave object.

Constant	Value	Description
acDemandLoadDisabled	0	Turns off demand loading. The entire drawing is loaded.
acDemandLoadEnabled	1	Turns on demand loading and improves AutoCAD performance. Other users can't edit the file while it's being referenced.
acDemandLoadEnabledWithCopy	2	Turns on demand loading but uses a copy of the referenced drawing. Other users can edit the original drawing.

AcZoomScaleType

The following constants can be supplied as the value for the second parameter of the ZoomScale method of the AcadApplication object.

Constant	Value	Description
acZoomScaledAbsolute	0	The scale factor is applied relative to the drawing limits.
acZoomScaledRelative	1	The scale factor is applied relative to the current view.
acZoomScaledRelativePSpace	2	The scale factor is applied relative to paper space units.



System Variables

AutoCAD provides several system variables that let the user control interface elements such as object snap mode, grids, and limits. This appendix explains all of AutoCAD's system variables. It also demonstrates how to use the `GetVariable` and `SetVariable` methods to set and check a system variable's value. Each successive release of AutoCAD had introduced new system variables, modified existing ones, and removed others. Therefore, before you use a system variable, research it to determine whether your code needs to check the AutoCAD version to use the right system variable.

You can also use certain properties of AutoCAD objects to access some of these variables. This appendix lists those properties.

The GetVariable Method

Use the Document object's `GetVariable` method to retrieve a system variable's value. This method has the following syntax:

```
CurrentValue = DocumentObject.GetVariable(VariableName)
```

This method returns the value of the system variable you specify in the `VariableName` parameter. This parameter is a String.

This example retrieves and displays the `ANGDIR` system variable's current value.

This variable's value determines whether AutoCAD measures angles counterclockwise or clockwise with respect to the current OCS.

```
Public Sub GetANGDIR()  
Dim varAngleDirection As Variant  
  
varAngleDirection = ThisDrawing.GetVariable("ANGDIR")  
If varAngleDirection = 0 Then  
    MsgBox "Angles are measured counterclockwise"  
Else  
    MsgBox "Angles are measured clockwise"  
End If  
End Sub
```

The SetVariable Method

Use the Document object's SetVariable method to set a system variable. It has the following syntax:

```
DocumentObject.SetVariable VariableName, VariableValue
```

Table C-1 explains this method's parameters.

Table C-1. *The SetVariable Method's Parameters*

Name	Data Type	Description
VariableName	String	The name of the system variable
VariableValue	Variant	The system variable's new value

This example lets the user set the ANGDIR system variable's value:

```
Public Sub SetANGDIR()  
Dim strResponse As String  
  
On Error Resume Next  
    strResponse = InputBox("Please specify the direction that angles should " & _  
        "be measured (Clockwise or Counterclockwise )", , "Counterclockwise")  
    If UCase(strResponse) = "COUNTERCLOCKWISE" Then  
        ThisDrawing.SetVariable "ANGDIR", 0  
    ElseIf UCase(strResponse) = "CLOCKWISE" Then  
        ThisDrawing.SetVariable "ANGDIR", 1  
    ElseIf strResponse = "" Then  
        MsgBox "You cancelled, the system variable setting has not been changed"  
    Else  
        MsgBox "The value you typed has not been recognized - " & _  
            "the system variable setting has not been changed"  
    End If  
End Sub
```

A

Table C-2 explains the system variables that start with A.

Table C-2. *The A System Variables*

Name	Description
ACADLSPASDOC	Controls whether the acad.lsp file is loaded into every open drawing or just the first one in the session. This variable's value is also stored in the LoadAcadLspInAllDocuments property of the AcadPreferencesSelection object.
ACADPREFIX	Stores the directory path specified by the ACAD environment variable, including path separators where necessary.

Name	Description
ACADVER	Stores the AutoCAD version number. AutoCAD 2000 = 15.0, AutoCAD 2000i = 15.05, AutoCAD 2002 = 15.06, AutoCAD 2004 = 16.0, AutoCAD 2005 = 16.1, and AutoCAD 2006 = 16.2. This variable's value is also stored in the Version property of the AcadApplication object.
ACISOUTVER	Controls the ACIS version of .sat files created using the ACISOUT command.
ADCSTATE	Determines whether DesignCenter is active.
AFLAGS	Sets attribute flags for ATTDEF bit code. This variable's value is also stored in the Mode property of the AcadAttribute object.
ANGBASE	Sets the base angle 0 with respect to the current UCS.
ANGDIR	Sets the positive angle direction from angle 0 with respect to the current UCS.
APBOX	Turns the AutoSnap aperture box on or off. This variable's value is also stored in the AutoSnapAperture property of the AcadPreferencesDrafting object.
APERTURE	Sets object snap target height in pixels. This variable's value is also stored in the AutoSnapApertureSize property of the AcadPreferencesDrafting object.
AREA	Stores the last area computed by AREA, LIST, or DBLIST.
ASSISTSTATE	Indicates whether the Info palette that displays Quick Help is active.
ATTDIA	Controls whether -INSERT uses a dialog box when entering attribute values.
ATTMODE	Controls the display of attributes.
ATTREQ	Determines whether INSERT uses default attribute settings during the insertion of blocks.
AUDITCTL	Controls whether AUDIT creates an audit report (.adt) file.
AUNITS	Sets angle units.
AUPREC	Sets the number of decimal places for angular units.
AUTOSNAP	Controls the AutoSnap marker, tooltip, and magnet. This variable's value is also stored in the AutoSnapMagnet, AutoSnapMarker, AutoSnapToolTip, and AutoTrackToolTip properties of the AcadPreferencesDrafting object.

B

Table C-3 explains the system variables that start with *B*.

Table C-3. *The B System Variables*

Name	Description
BACKGROUNDPLOT	Controls whether background plotting is turned on or off for plotting and publishing. By default, background plotting is turned off for plotting and on for publishing.
BACKZ	Stores the current viewport's back clipping plane offset from the target plane.
BACTIONCOLOR	Sets the text color of actions in the Block Editor. Valid values include BYLAYER, BYBLOCK, and an integer from 1 to 255. Valid values for True Colors are a string of integers each from 1 to 255 separated by commas and preceded by RGB.

Continued

Table C-3. *Continued*

Name	Description
BDEPENDENCYHIGHLIGHT	Controls whether dependent objects are dependency highlighted when a parameter, action, or grip is selected in the Block Editor.
BGRIPOBJCOLOR	Sets the color of grips in the Block Editor. Valid values include BYLAYER, BYBLOCK, and an integer from 1 to 255. Valid values for True Colors are a string of integers each from 1 to 255 separated by commas and preceded by RGB.
BGRIPOBJSIZE	Sets the display size of custom grips in the Block Editor relative to the screen display. Valid values include an integer from 1 to 255. Use the REGEN command to update the display size of custom grips in the Block Editor.
BINDTYPE	Controls how Xref names are handled when binding them or when editing them in place.
BLIPMODE	Controls whether marker blips are visible.
BLOCKEDITLOCK	Disallows opening of the Block Editor and editing of dynamic block definitions. When BLOCKEDITLOCK is set to 1, double-clicking a dynamic block in a drawing opens the Reference Edit dialog box. If the block contains attributes, double-clicking the block reference opens the Enhanced Attribute Editor.
BLOCKEDITOR	Indicates whether the Block Editor is open.
BPARAMETERCOLOR	Sets the color of parameters in the Block Editor. Valid values include BYLAYER, BYBLOCK, and an integer from 1 to 255. Valid values for True Colors are a string of integers each from 1 to 255 separated by commas and preceded by RGB.
BPARAMETERFONT	Sets the font used for parameters and actions in the Block Editor. You can specify either a TrueType font or an SHX font (for example, Verdana or Verdana.ttf). You must add the .shx extension to specify an AutoCAD SHX font. When specifying an Asian Big Font, use the following naming convention: an SHX file followed by a comma (,), followed by the Big Font file name (for example, Simplex.shx,Bigfont.shx).
BPARAMETERSIZE	Sets the size of parameter text and features in the Block Editor relative to the screen display. Valid values include an integer from 1 to 255.
BTMARKDISPLAY	Controls whether value set markers are displayed for dynamic block references.
BVMODE	Controls how objects that are made invisible for the current visibility state are displayed in the Block Editor.

C

Table C-4 explains the system variables that start with C.

Table C-4. *The C System Variables*

Name	Description
CALCINPUT	Controls whether mathematical expressions and global constants are evaluated in text and numeric entry boxes of windows and dialog boxes.
CDATE	Sets the calendar's date and time.
CECOLOR	Sets the color of new objects.

Name	Description
CELTSCALE	Sets the linetype scaling factor for the current object.
CELTYPE	Sets the linetype of new objects.
CELWEIGHT	Sets the lineweight of new objects.
CENTERMT	Controls how grips stretch multiline text that is centered horizontally. CENTERMT does not apply to stretching multiline text by using the ruler in the In-Place Text Editor.
CHAMFERA	Sets the first chamfer distance.
CHAMFERB	Sets the second chamfer distance.
CHAMFERC	Sets the chamfer length.
CHAMFERD	Sets the chamfer angle.
CHAMMODE	Sets the input method by which chamfers are created.
CIRCLERAD	Sets the default radius for a circle.
CLAYER	Sets the current layer.
CLEANSCREENSTATE	Stores a value that indicates whether AutoCAD is in Clean Screen mode.
CLISTATE	Stores a value that indicates whether the command window is hidden or displayed.
CMDACTIVE	Stores the bit code that indicates whether an ordinary command, a transparent command, a script, or a dialog box is active.
CMDDIA	Controls whether the command line or dialog boxes are used in certain commands.
CMDECHO	Controls whether prompts and input are echoed during the AutoLISP Command function.
CMDINPUTHISTORYMAX	Sets the maximum number of previous input values that are stored for a prompt in a command. Display of the history of user input is controlled by the INPUTHISTORYMODE system variable.
CMDNAMES	Displays the names of active and transparent commands.
CMLJUST	Sets the justification of a multiline.
CMLSCALE	Sets the width of a multiline.
CMLSTYLE	Sets the style of a multiline.
COMPASS	Controls whether the 3-D compass is switched on or off for the current viewport.
COORDS	Controls when coordinates are updated on the status line.
CPLOTSTYLE	Sets the current plot style for new objects.
CPROFILE	Stores the current profile's name.
CROSSINGAREACOLOR	Controls the color of the selection area during crossing selection. The valid range is 1 to 255. The SELECTIONAREA system variable must be on.
CTAB	Returns the name of the current Model tab or Layout tab in the drawing.
CTABLESTYLE	Sets the name of the current table style.
CURSORSIZE	Determines the size of the crosshairs as a percentage of the screen size. This variable's value is also stored in the CursorSize property of the AcadPreferencesDisplay object.
CVPORT	Sets the current viewport's identification number.

D

Table C-5 explains the system variables that start with *D*.

Table C-5. *The D System Variables*

Name	Description
DATE	Stores the current date and time.
DBCSTATE	Stores the state of the dbConnect Manager, active or not active.
DBMOD	Indicates the status of drawing modifications using bit code.
DCTCUST	Displays the current custom dictionary's path and file name.
DCTMAIN	Displays the current main dictionary's file name.
DEFLPLSTYLE	Specifies the default plot style for new layers. This variable's value is also stored in the DefaultPlotStyleForLayer property of the AcadPreferencesOutput object.
DEFPLSTYLE	Specifies the default plot style for new objects. This variable's value is also stored in the DefaultPlotStyleForObjects property of the AcadPreferencesOutput object.
DELOBJ	Controls whether any objects used to create other objects are retained or deleted from the drawing database.
DEMANDLOAD	Specifies the conditions that cause AutoCAD to demand load a third-party application, if a drawing contains objects created in that application. This variable's value is also stored in the DemandLoadARXApp property of the AcadPreferencesOpenSave object.
DIASSTAT	Stores the exit method of the most recently used dialog box.
DIMADEC	Controls the number of places of precision displayed for angular dimensions.
DIMALT	Controls the display of alternate units in dimensions.
DIMALTD	Controls the number of decimal places in alternate units.
DIMALTF	Controls the multiplier for alternate units.
DIMALTRND	Determines the rounding of alternate units.
DIMALTTD	Sets the number of decimal places for tolerance values in a dimension's alternate units.
DIMALTTZ	Toggles suppression of zeros in tolerance values.
DIMALTU	Sets the units format for alternate units of all dimension style family members except angular.
DIMALTZ	Controls the suppression of zeros for alternate unit dimension values.
DIMAPOST	Specifies a text prefix and/or suffix to the alternate dimension measurement for all types of dimensions except angular.
DIMASO	Controls the associativity of dimension objects (replaced by DIMASSOC in AutoCAD 2002 and later versions).
DIMASSOC	Controls the associativity of dimension objects.
DIMASZ	Controls the size of dimension line and leader line arrowheads.
DIMATFIT	Determines how dimension text and arrows are arranged when space is not sufficient to place both within the extension lines.
DIMAUNIT	Sets the units format for angular dimensions.
DIMAZIN	Suppresses zeros for angular dimensions.

Name	Description
DIMBLK	Sets the arrowhead block displayed at the ends of dimension lines or leader lines.
DIMBLK1	Sets the arrowhead for the first end of the dimension line when DIMSAH is on.
DIMBLK2	Sets the arrowhead for the second end of the dimension line when DIMSAH is on.
DIMCEN	Controls drawing of circle or arc center marks and centerlines by DIMCENTER, DIMDIAMETER, and DIMRADIUS.
DIMCLRD	Assigns colors to dimension lines, arrowheads, and dimension leader lines.
DIMCLRE	Assigns colors to dimension extension lines.
DIMCLRT	Assigns colors to dimension text.
DIMDEC	Sets the number of decimal places displayed for a dimension's primary units.
DIMDLE	Sets the distance the dimension line extends beyond the extension line when oblique strokes are drawn instead of arrowheads.
DIMDLI	Controls the spacing of dimension lines in baseline dimensions.
DIMDSEP	Specifies a single character decimal separator to use when creating dimensions whose unit format is decimal.
DIMEXE	Specifies how far to extend the extension line beyond the dimension line.
DIMEXO	Specifies how far extension lines are offset from origin points.
DIMFIT	Obsolete. Replaced by DIMATFIT and DIMTMOVE. Retained for backward compatibility only.
DIMFRAC	Sets the fraction format when DIMLUNIT is set to 4 or 5.
DIMGAP	Sets the distance around dimension text when the dimension line breaks to accommodate dimension text.
DIMJUST	Controls the horizontal positioning of dimension text.
DIMLDRBLK	Specifies the arrow type for leaders.
DIMLFAC	Sets a scale factor for linear dimension measurements.
DIMLIM	Generates dimension limits as the default text.
DIMLUNIT	Sets units for all dimension types except angular.
DIMLWD	Assigns lineweight to dimension lines.
DIMLWE	Assigns lineweight to extension lines.
DIMPOST	Specifies a text prefix and/or suffix to the dimension measurement.
DIMRND	Rounds all dimensioning distances to the specified value.
DIMSAH	Controls the display of dimension line arrowhead blocks.
DIMSCALE	Sets the overall scale factor applied to dimensioning variables that specify sizes, distances, or offsets.
DIMSD1	Controls suppression of the first dimension line.
DIMSD2	Controls suppression of the second dimension line.
DIMSE1	Suppresses display of the first extension line.
DIMSE2	Suppresses display of the second extension line.
DIMSHO	Controls redefinition of dimension objects while dragging.
DIMSOXD	Suppresses drawing of dimension lines outside the extension lines.

Continued

Table C-5. *Continued*

Name	Description
DIMSTYLE	Shows the current dimension style.
DIMTAD	Controls the vertical position of text in relation to the dimension line.
DIMTDEC	Sets the number of decimal places to display in tolerance values for a dimension's primary units.
DIMTFAC	Sets a scale factor used to calculate the height of text for dimension fractions and tolerances.
DIMTIH	Controls the position of dimension text inside the extension lines for all dimension types except ordinate.
DIMTIX	Draws text between extension lines.
DIMTM	When DIMTOL or DIMLIM is on, sets the minimum tolerance limit for dimension text.
DIMTMOVE	Sets dimension text movement rules.
DIMTOFL	Controls whether a dimension line is drawn between the extension lines even if text is placed outside.
DIMTOH	Controls the position of dimension text outside the extension lines.
DIMTOL	Appends tolerances to dimension text.
DIMTOLJ	Sets the vertical justification for tolerance values relative to the nominal dimension text.
DIMTP	When DIMTOL or DIMLIM is on, sets the maximum tolerance limit for dimension text.
DIMTSZ	Specifies the size of oblique strokes drawn instead of arrowheads for linear, radial, and diameter dimensioning.
DIMTVP	Controls the vertical position of dimension text above or below the dimension line.
DIMTXSTY	Specifies the text style of the dimension.
DIMTXT	Specifies the height of dimension text, unless the current text style has a fixed height.
DIMTZIN	Controls the suppression of zeros in tolerance values.
DIMUNIT	Obsolete. Replaced by DIMLUNIT and DIMFRAC. Retained for backward compatibility only.
DIMUPT	Controls options for user-positioned text.
DIMZIN	Controls the suppression of zeroes in the primary unit value.
DISPSILH	Controls display of silhouette curves of solid objects in Wireframe mode. This variable's value is also stored in the DisplaySilhouette property of the AcadDatabasePreferences object.
DISTANCE	Stores the distance computed by DIST.
DONUTID	Sets the default for the inside diameter of a torus (donut).
DONUTOD	Sets the default for the outside diameter of a torus (donut).
DRAGMODE	Controls the display of objects being dragged.
DRAGP1	Sets the sampling rate for regen-drag input.
DRAGP2	Sets the sampling rate for fast-drag input.
DRAWORDERCTL	Controls the display order of overlapping objects. Use this setting to improve the speed of editing operations in large drawings. The commands that are affected by inheritance are BREAK, FILLET, HATCH, HATCHEDIT, EXPLODE, TRIM, JOIN, PEDIT, and OFFSET.

Name	Description
DRSTATE	Determines whether the Drawing Recovery window is active.
DTEXTED	Specifies the user interface for editing single-line text.
DWGCHECK	Determines whether a drawing was last edited by a product other than AutoCAD.
DWGCODEPAGE	Stores the same value as SYSCODEPAGE (retained for compatibility reasons).
DWGNAME	Stores the drawing name as entered by the user.
DWGPREFIX	Stores the drive or directory prefix for the drawing.
DWGTITLED	Indicates whether the current drawing has been named.
DYNDIGRIP	Controls which dynamic dimensions are displayed during grip stretch editing. The DYNDIVIS system variable must be set to 2, which displays all dynamic dimensions. The DYNMODE system variable turns Dynamic Input features on and off.
DYNDIVIS	Controls how many dynamic dimensions are displayed during grip stretch editing. DYNDIGRIP controls which dynamic dimensions are displayed during grip stretch editing. The DYNMODE system variable turns Dynamic Input features on and off.
DYNMODE	Turns Dynamic Input features on and off. When all features are on, the context governs what is displayed. When DYNMODE is set to a negative value, the Dynamic Input features are not visible, but the setting is stored. Click the Dyn button in the status bar to set DYNMODE to the corresponding positive value. When DYNMODE is set to anything but 0, you can turn off all features temporarily by holding down the temporary override key, F12. Settings are on the Dynamic Input tab in the Drafting Settings dialog box.
DYNPICOORDS	Controls whether pointer input uses relative or absolute format for coordinates. The DYNMODE system variable turns Dynamic Input features on and off.
DYNPIFORMAT	Controls whether pointer input uses polar or Cartesian format for coordinates. This setting applies only to a second or next point. The DYNMODE system variable turns Dynamic Input features on and off.
DYNPIVIS	Controls when pointer input is displayed. The DYNMODE system variable turns Dynamic Input features on and off.
DYNPROMPT	Controls display of prompts in Dynamic Input tooltips. The DYNMODE system variable turns Dynamic Input features on and off.
DYNTOOLTIPS	Controls which tooltips are affected by tooltip appearance settings.

E

Table C-6 explains the system variables that start with *E*.

Table C-6. *The E System Variables*

Name	Description
EDGEMODE	Controls how TRIM and EXTEND determine cutting and boundary edges.
ELEVATION	Stores the current elevation relative to the current UCS for the current viewport.
ENTERPRISEMENU	Stores the CUI file name (if defined), including the path for the file name.
EXPERT	Controls whether certain prompts are issued.
EXPLMODE	Controls whether EXPLODE supports nonuniformly scaled (NUS) blocks.

Continued

Table C-6. *Continued*

Name	Description
EXTMAX	Stores the upper-right point of the drawing extents.
EXTMIN	Stores the lower-left point of the drawing extents.
EXTNAMES	Sets the parameters for named object names (such as linetypes and layers) stored in symbol tables. This variable's value is also stored in the AllowLongSymbolNames property of the AcadDataBasePreferences object.

F

Table C-7 explains the system variables that start with *F*.

Table C-7. *The F System Variables*

Name	Description
FACETRATIO	Controls the aspect ratio of faceting for cylindrical and conic ACIS solids.
FACETRES	Adjusts the smoothness of shaded and rendered objects and objects with hidden lines removed. This variable's value is also stored in the RenderSmoothness property of the AcadDatabasePreferences object.
FIELDDISPLAY	Controls whether fields are displayed with a gray background. The background is not plotted.
FIELDEVAL	Controls how fields are updated. The setting is stored as a bit code using the sum of the values.
FILEDIA	Suppresses display of file dialog boxes.
FILLETRAD	Stores the current fillet radius.
FILLMODE	Specifies whether multilines, traces, solids, hatches (including solid-fill hatches), and wide polylines are filled in. This variable's value is also stored in the SolidFill property of the AcadDatabasePreferences object.
FONTALT	Stores the alternate font that will be used when the specified font file can't be located.
FONTMAP	Specifies the font mapping file that will be used.
FRONTZ	Stores the front clipping plane offset from the target plane for the current viewport.
FULLOPEN	Indicates whether the current drawing is partially open or fully open.
FULLPLOTPATH	Controls whether the full path of the drawing file is sent to the plot spooler.

G

Table C-8 explains the system variables that start with *G*.

Table C-8. *The G System Variables*

Name	Description
GFANG	Specifies the angle for a gradient fill.
GFCLR1	Specifies the color for a one-color gradient fill or the first color for a two-color gradient fill.
GFCLR2	Specifies the second color for a two-color gradient fill.
GFCLRLUM	Makes the color a tint (mixed with white) or a shade (mixed with black) in a one-color gradient fill.
GFCLRSTATE	Specifies whether a gradient fill uses one color or two colors.
GFNAME	Specifies the pattern of a gradient fill.
GFSHIFT	Specifies whether the pattern in a gradient fill is centered or is shifted up and to the left.
GRIDMODE	Specifies whether the grid is turned on or off. This variable's value is also stored in the <code>GridOn</code> property of the <code>AcadPViewport</code> and <code>AcadViewport</code> objects.
GRIDUNIT	Specifies the grid spacing for the current viewport.
GRIPBLOCK	Controls the assignment of grips in blocks. This variable's value is also stored in the <code>DisplayGripsWithinBlocks</code> property of the <code>AcadPreferencesSelection</code> object.
GRIPCOLOR	Controls the color of nonselected grips, which are drawn as box outlines. This variable's value is also stored in the <code>GripColorUnselected</code> property of the <code>AcadPreferencesSelection</code> object.
GRIPDYNCOLOR	Controls the color of custom grips for dynamic blocks. The valid range is 1 to 255.
GRIPHOT	Controls the color of selected grips, which are drawn as filled boxes. This variable's value is also stored in the <code>GripColorSelected</code> property of the <code>AcadPreferencesSelection</code> object.
GRIPHOVER	Controls the fill color of a grip when the cursor pauses over the grip.
GRIPOBJLIMIT	Determines the maximum number of objects on which grips will be displayed.
GRIPS	Controls the use of selection set grips for the Stretch, Move, Rotate, Scale, and Mirror grip modes. This variable's value is also stored in the <code>DisplayGrips</code> property of the <code>AcadPreferencesSelection</code> object.
GRIPSIZE	Sets the size of the grip box in pixels. This variable's value is also stored in the <code>GripSize</code> property of the <code>AcadPreferencesSelection</code> object.
GRIPTIPS	Controls the display of grip tips when the cursor hovers over grips on custom objects that support grip tips.

H

Table C-9 explains the system variables that start with *H*.

Table C-9. *The H System Variables*

Name	Description
HALOGAP	Specifies the distance to shorten a haloed line.
HANDLES	Reports whether applications can access object handles.
HIDEPRECISION	Controls the accuracy of hides and shades.
HIDETEXT	Specifies whether text objects created by the TEXT, DTEXT, or MTEXT command are processed during a HIDE command.
HIGHLIGHT	Controls object highlighting. This variable's value does not affect objects selected with grips.
HPANG	Specifies the angle of the hatch pattern. This variable's value is also stored in the PatternAngle property of the AcadHatch object.
HPASSOC	Controls whether hatch patterns and gradient fills are associative.
HPBOUND	Controls the object type created by BHATCH and BOUNDARY.
HPDOUBLE	Specifies hatch pattern doubling for user-defined patterns. This variable's value is also stored in the PatternDouble property of the AcadHatch object.
HPDRAWORDER	Controls the draw order of hatches and fills. Stores the Draw Order setting from the Hatch and Fill dialog box.
HPGAPTOL	Treats a set of objects that almost enclose an area as a closed hatch boundary. The default value, 0, specifies that the objects enclose the area, with no gaps. Enter a value, in drawing units, from 0 to 5000 to set the maximum size of gaps that can be ignored when the objects serve as a hatch boundary.
HPINHERIT	Controls the hatch origin of the resulting hatch when using Inherit Properties in HATCH and HATCHEDIT.
HPNAME	Sets the default name for the hatch pattern. This variable's value is also stored in the PatternName property of the AcadHatch object.
HPOBJWARNING	Sets the number of hatch boundary objects that can be selected before displaying a warning message.
HPORIGIN	Sets the hatch origin point for new hatch objects relative to the current user coordinate system.
HPORIGINMODE	Controls how HATCH determines the default hatch origin point.
HPSCALE HPINHERIT	Specifies the scale factor for the hatch pattern. This variable's value is also stored in the PatternScale property of the AcadHatch object.
HPSEPARATE	Controls whether HATCH creates a single hatch object or separate hatch objects when operating on several closed boundaries.
HPSPACE	Specifies the hatch pattern's line spacing for user-defined simple patterns. This variable's value is also stored in the PatternSpace property of the AcadHatch object.
HYPERLINKBASE	Specifies the path used for all relative hyperlinks in the drawing.

Table C-10 explains the system variables that start with *I*.

Table C-10. *The I System Variables*

Name	Description
IMAGEHLT	Specifies whether the entire raster image or only the image frame is highlighted. This variable's value is also stored in the ImageFrameHighlight property of the AcadPreferencesDisplay object.
INDEXCTL	Controls whether layer and spatial indexes are created and saved in drawing files.
INETLOCATION	Stores the Internet location used by BROWSER and the Browse the Web dialog box. This variable's value is also stored in the DefaultInternetURL property of the AcadPreferencesFiles object.
INPUTHISTORYMODE	Controls the content and location of the display of a history of user input. The setting is stored as a bit code using the sum of the values.
INSBASE	Stores the insertion base point set by BASE.
INSNAME	Sets the default block name for INSERT.
INSUNITS	Specifies a value for the drawing units, when a block is dragged from the AutoCAD DesignCenter.
INSUNITSDEFSOURCE	Sets the source content's units value. This variable's value is also stored in the ADCInsertUnitsDefaultSource property of the AcadPreferencesUser object.
INSUNITSDEFTARGET	Sets the target drawing's units value. This variable's value is also stored in the ADCInsertUnitsDefaultTarget property of the AcadPreferencesUser object.
INTELLIGENTUPDATE	Controls the graphics refresh rate. The default value is 20 frames per second. If you encounter problems related to graphics generation or timing, turn off the variable by setting it to 0. INTELLIGENTUPDATE works by suppressing the graphics update until the timer expires. Subsequent updates reset the timer. The performance improvement significantly affects updates for scripts and AutoLISP graphics. Those using regular AutoCAD commands will not see a noticeable difference in performance.
INTERSECTIONCOLOR	Specifies the color of intersection polylines.
INTERSECTIONDISPLAY	Specifies the display of intersection polylines.
ISAVEBAK	Increases the speed of incremental saves for large drawings. This variable's value is also stored in the CreateBackup property of the AcadPreferencesSaveOpen object.
ISAVEPERCENT	Determines the amount of wasted space tolerated in a drawing file. This variable's value is also stored in the IncrementalSavePercent property of the AcadPreferencesOpenSave object.
ISOLINES	Specifies the number of isolines per surface on objects. This variable's value is also stored in the ContourLinesPerSurface property of the AcadDatabasePreferences object.

L

Table C-11 explains the system variables that start with *L*.

Table C-11. *The L System Variables*

Name	Description
LASTANGLE	Stores the end angle of the most recently drawn arc.
LASTPOINT	Stores the last point drawn.
LASTPROMPT	Stores the last string echoed to the command line.
LAYERFILTERALERT	Deletes excessive layer filters to improve performance. When a drawing has 100 or more layer filters, and the number of layer filters exceeds the number of layers, LAYERFILTERALERT provides a method for deleting layer filters to improve performance.
LAYOUTREGENCTL	Specifies how the display list is updated in the Model tab and Layout tabs.
LENSLENGTH	Stores the length of the lens (in millimeters) used in perspective viewing for the current viewport. This variable's value is also stored in the <code>LensLength</code> property of the <code>AcadPViewport</code> object.
LIMCHECK	Controls the creation of objects outside the drawing limits.
LIMMAX	Stores the upper-right drawing limits for the current space. This variable's value is also stored in the <code>Limits</code> property of the <code>AcadDocument</code> object.
LIMMIN	Stores the lower-left drawing limits for the current space. This variable's value is also stored in the <code>Limits</code> property of the <code>AcadDocument</code> object.
LISPINIT	In SDI mode, this specifies whether AutoLISP-defined functions and variables are preserved when a new drawing is opened.
LOCALE	Displays the ISO language code of the current AutoCAD version.
LOCALROOTPREFIX	Stores the full path to the root folder where local customizable files were installed.
LOCKUI	Locks the position and size of toolbars and windows such as DesignCenter and Properties palette. Locked toolbars and windows can still be opened and closed and items can be added and deleted. To unlock them temporarily, hold down Ctrl. A lock icon in the status bar tray indicates whether toolbars and windows are locked. Right-click the icon to display locking options. The setting is stored as a bit code using the sum of the values.
LOGFILEMODE	Specifies whether the contents of the text window are written to a log file. This variable's value is also stored in the <code>LogFileOn</code> property of the <code>AcadPreferencesOpenSave</code> object.
LOGFILENAME	Specifies the path and name of the log file.
LOGFILEPATH	Specifies the path for the log files for all drawings in a session. This variable's value is also stored in the <code>LogFilePath</code> property of the <code>PreferencesFiles</code> object.
LOGINNAME	Displays the user's login name when AutoCAD is loaded.
LTSCALE	Sets the scale factor for the global linetype.
LUNITS	Sets linear units.
LUPREC	Sets the number of decimal places displayed for linear units.
LWDEFAULT	Sets the lineweight's default value.
LWDISPLAY	Controls whether the lineweight is displayed in the Model or Layout tab.

Name	Description
LWSCALE	Controls whether the lineweight is scaled with the rest of the geometry when a layout is printed. This variable's value is also stored in the ScaleLineweights property of the AcadLayout and AcadPlotConfiguration objects.
LWUNITS	Controls whether lineweight units are displayed in inches or millimeters.

M

Table C-12 explains the system variables that start with *M*.

Table C-12. *The M System Variables*

Name	Description
MAXACTVP	Sets the maximum number of viewports that can be active at one time. This variable's value is also stored in the MaxActiveViewports property of the AcadDatabasePreferences object.
MAXSORT	Sets the maximum number of symbol names or block names that can be sorted.
MBUTTONPAN	Controls the behavior of the third button or wheel on the pointing device.
MEASUREINIT	Sets the initial drawing units as English or metric.
MEASUREMENT	Sets drawing units as English or metric for the current drawing only.
MENUCTL	Controls the page switching of the screen menu.
MENUECHO	Sets menu echo and prompt control bits.
MENUNAME	Stores the menu file name, including the path.
MIRRTEXT	Controls how MIRROR reflects text.
MODEMACRO	Displays a text string on the status line.
MSOLESCALE	Controls the size of an OLE object with text that is pasted into model space. MSOLESCALE controls only the initial size. If the scale factor value is changed, existing OLE objects in the drawing are not affected. Positive number scales by value. Zero (0) scales by DIMSCALE value.
MTEXTED	Sets the primary and secondary text editors to use for AcadMText objects.
MTEXTFIXED	Controls the appearance of the Multiline Text Editor.
MTJIGSTRING	Sets the content of the sample text displayed at the cursor location when the MTEXT command is started.
MYDOCUMENTSPREFIX	Stores the full path to the My Documents folder for the user currently logged on.

N

Table C-13 explains the system variables that start with *N*.

Table C-13. *The N System Variables*

Name	Description
NOMUTT	Controls whether message display (muttering) is suppressed in AutoCAD.

O

Table C-14 explains the system variables that start with *O*.

Table C-14. *The O System Variables*

Name	Description
OBSCUREDCOLOR	Specifies the color of obscured lines.
OBSCUREDLTTYPE	Specifies the linetype of obscured lines.
OFFSETDIST	Sets the default offset distance.
OFFSETGAPTYPE	Controls how to fill in gaps resulting from offsetting individual polyline segments.
OLEFRAME	Controls whether a frame is displayed and plotted on all OLE objects in the drawing. The frame on an OLE object must be displayed in order for grips to be visible.
OLEHIDE	Controls the display of OLE objects in AutoCAD.
OLEQUALITY	Controls the default quality level for embedded OLE objects. This variable's value is also stored in the OLEQuality property of the AcadPreferencesOutput object.
OLESTARTUP	Controls whether to load an embedded OLE object's source application when plotting. This variable's value is also stored in the OLELaunch property of the AcadDatabasePreferences object.
ORTHOMODE	Constrains cursor movement to perpendicular axes.
OSMODE	Sets running object snap modes using bit codes. Specifies whether the line-weight is scaled with the rest of the geometry when a layout is printed.
OSNAPCOORD	Controls whether coordinates entered on the command line override running object snaps. This variable's value is also stored in the KeyboardPriority property of the AcadPreferencesUser object.
OSNAPHATCH	Controls whether object snaps ignore hatch objects. The default setting 0 improves performance.
OSNAPZ	Controls whether object snaps are automatically projected onto a plane parallel to the XY plane of the current UCS at the current elevation.

P

Table C-15 explains the system variables that start with *P*.

Table C-15. *The P System Variables*

Name	Description
_PKSER	Stores the AutoCAD license serial number (for example, 400-12345678).
PALETTEOPAQUE	Controls whether windows can be made transparent.
PAPERUPDATE	Controls the display of a warning dialog when you try to print a layout with a paper size other than that specified for the plotter configuration file.
PDMODE	Controls how to display point objects. See Chapter 8 for more details.
PDSIZE	Sets the display size for point objects. See Chapter 8 for more details.
PEDITACCEPT	Suppresses display of the Object Selected Is Not a Polyline prompt in PEDIT.
PELLIPSE	Controls the ellipse type created with ELLIPSE.
PERIMETER	Stores the last perimeter value computed by AREA, LIST, or DBLIST.
PFACEVMAX	Sets the maximum number of vertices per face.
PICKADD	Controls whether subsequent selections replace the current selection set or add to it. This variable's value is also stored in the PickAdd property of the AcadPreferencesSelection object.
PICKAUTO	Controls automatic windowing at the Select Objects prompt. This variable's value is also stored in the PickAuto property of the AcadPreferencesSelection object.
PICKBOX	Sets object selection target height. This variable's value is also stored in the PickBoxSize property of the AcadPreferencesSelection object.
PICKDRAG	Controls the method of drawing a selection window. This variable's value is also stored in the PickDrag property of the AcadPreferencesSelection object.
PICKFIRST	Controls whether you select objects before or after you issue a command. This variable's value is also stored in the PickFirst property of the AcadPreferencesSelection object.
PICKSTYLE	Controls the use of group selection and associative hatch selection.
PLATFORM	Indicates which platform (operating system) of AutoCAD is being used.
PLINEGEN	Sets how linetype patterns generate around the vertices of a 2-D polyline.
PLINETYPE	Specifies whether optimized 2-D polylines are used.
PLINEWID	Stores the default width of the polyline.
PLOTID	Obsolete in AutoCAD 2000 and later versions; retained for backward compatibility.
PLOTLEGACY	Controls whether legacy plot scripts can be run. This variable's value is also stored in the PlotLegacy property of the AcadPreferencesOutput object.
PLOTOFFSET	Controls whether the plot offset is relative to the printable area or to the edge of the paper.
PLOTROTMODE	Controls the orientation of plots.
PLOTTER	Obsolete in AutoCAD 2000 and later versions; retained for backward compatibility.

Continued

Table C-15. *Continued*

Name	Description
PLQUIET	Controls the display of optional dialog boxes and nonfatal errors for batch plotting and scripts.
POLARADDANG	Contains user-defined polar angles.
POLARANG	Sets the polar angle increment.
POLARDIST	Sets the snap increment when the SNAPSTYL system variable is set to 1 (polar snap).
POLARMODE	Controls the settings for polar and object snap tracking.
POLYSIDES	Sets the default number of sides for POLYGON.
POPUPS	Displays the status of the currently configured display driver.
PREVIEWEFFECT	Specifies the visual effect used for previewing selection of objects.
PREVIEWFILTER	Excludes specified object types from selection previewing. The setting is stored as a bit code using the sum of the values.
PRODUCT	Returns the product name (for example, AutoCAD).
PROGRAM	Returns the program executable name (for example, acad).
PROJECTNAME	Assigns a project name to the current drawing.
PROJMODE	Sets the current projection mode for trimming or extending.
PROXYGRAPHICS	Specifies whether to save images of proxy objects.
PROXYNOTICE	Displays a dialog box when a drawing is opened that contains custom objects created by an application that is not present on the system. This variable's value is also stored in the ShowProxyDialogBox property of the AcadPreferencesOpenSave object.
PROXYSHOW	Controls the display of proxy objects in a drawing.
PROXYWEBSEARCH	Specifies how AutoCAD checks for Object Enablers.
PSLTSCALE	Controls linetype scaling for the paper space.
PSPOOLALERT	Specifies whether to alert the user when output needs to be spooled through a system printer because of a conflict with an I/O port. Not used in AutoCAD 2004 and later. This variable's value is also stored in the PrinterSpoolAlertproperty of the AcadPreferencesOutput object.
PSPROLOG	Assigns a name for a prolog section to be read from the acad.psf file when you are using PSOUT. This variable's value is also stored in the PostScriptPrologFile property of the PreferencesFiles object.
PSQUALITY	Controls the rendering quality of PostScript images.
PSTYLEMODE	Indicates whether the current drawing is in a Color Dependent or Named Plot Style mode.
PSTYLEPOLICY	Controls whether an object's color property is associated with its plot style. This variable's value is also stored in the PlotPolicy property of the AcadPreferencesOutput object.
PSVPSCALE	Sets the view scale factor for all newly created viewports.
PUCSBASE	Stores the name of the UCS that defines the origin and orientation of orthographic UCS settings in paper space.

Q

Table C-16 explains the system variables that start with *Q*.

Table C-16. *The Q System Variables*

Name	Description
QAFLAGS	An undocumented, but useful variable. Bit coded: 1 causes ^C in menu macros to cancel grips, 2 causes screen listings to scroll past without pausing, 4 prevents alert dialogs from appearing.
QCSTATE	Determines whether the QuickCalc calculator is active.
QTEXTMODE	Controls how text is displayed. This variable's value is also stored in the TextFrameDisplay property of the DatabasePreferences object.

R

Table C-17 explains the system variables that start with *R*.

Table C-17. *The R System Variables*

Name	Description
RASTERDPI	Controls paper size and plot scaling when changing from dimensional to dimensionless output devices, or vice versa. Converts millimeters or inches to pixels, or vice versa. Accepts an integer from 100 to 32,767 as a valid value.
RASTERPREVIEW	Controls whether to save .bmp preview images. This variable's value is also stored in the SavePreviewThumbnail property of the AcadPreferencesOpenSave object.
RECOVERYMODE	Controls whether drawing recovery information is recorded after a system failure.
REFEDITNAME	Indicates whether a drawing is in a reference-editing state and stores the reference file's name.
REGENMODE	Controls automatic drawing regeneration.
RE-INIT	Reinitializes the digitizer, digitizer port, and acad.pgp file.
REMEMBERFOLDERS	Controls the default path for the Look In or Save In option in standard file selection dialog boxes.
REPORTERROR	Controls whether an error report can be sent to Autodesk if AutoCAD closes unexpectedly.
ROAMABLEROOTPREFIX	Stores the full path to the root folder where roamable customizable files were installed.
RTDISPLAY	Controls the display of raster images during real-time ZOOM or PAN. This variable's value is also stored in the ShowRasterImage property of the AcadPreferencesDisplay object.

S

Table C-18 explains the system variables that start with S.

Table C-18. *The S System Variables*

Name	Description
SAVEFILE	Stores the current autosave file name.
SAVEFILEPATH	Specifies the directory path for all automatic save files for the session. This variable's value is also stored in the <code>AutoSavePath</code> property of the <code>AcadPreferencesFiles</code> object.
SAVENAME	Stores the file name and directory path of a drawing after it is saved.
SAVETIME	Sets the automatic save interval in minutes. This variable's value is also stored in the <code>AutoSaveInterval</code> property of the <code>AcadPreferencesOpenSave</code> object.
SCREENBOXES	Stores the number of boxes in the drawing window's screen menu area.
SCREENMODE	Stores a bit code indicating the display's graphics and text state.
SCREENSIZE	Stores current viewport size in pixels.
SDI	Controls whether AutoCAD runs in single document interface (SDI) mode or in multiple-document interface (MDI) mode. This variable's value is also stored in the <code>SingleDocumentMode</code> property of the <code>AcadPreferencesSelection</code> object.
SELECTIONAREA	Controls the display of effects for selection areas. Selection areas are created by the <code>Window</code> , <code>Crossing</code> , <code>WPolygon</code> , and <code>CPolygon</code> options of <code>SELECT</code> .
SELECTIONAREAOPACITY	Controls the transparency of the selection area during window and crossing selection. The valid range is 0 to 100. The lower the setting, the more transparent the area. A value of 100 makes the area opaque. The <code>SELECTIONAREA</code> system variable must be on.
SELECTIONPREVIEW	Controls the display of selection previewing. Objects are highlighted when the pickbox cursor rolls over them. This selection previewing indicates that the object would be selected if you clicked. The setting is stored as a bit code using the sum of the values.
SHADEGE	Controls the shading of edges in rendering.
SHADEDIF	Sets the ratio of diffuse reflective light to ambient light.
SHORTCUTMENU	Controls whether Default, Edit, and Command mode shortcut menus are available in the drawing area. This variable's value is also stored in the <code>SCMCommandMode</code> , <code>SCMDefaultMode</code> , <code>SCMEditMode</code> , and <code>ShortCutMenuDisplay</code> properties of the <code>AcadPreferencesUser</code> object.
SHOWLAYERUSAGE	Displays icons in the Layer Properties Manager to indicate whether layers are in use. Setting this system variable to Off improves performance in the Layer Properties Manager.
SHPNAME	Sets a default shape name.
SIGWARN	Controls whether a warning is presented when a file with an attached digital signature is opened.
SKETCHINC	Sets the record increment for <code>SKETCH</code> .
SKPOLY	Determines whether <code>SKETCH</code> generates lines or polylines.
SNAPANG	Sets snap and grid rotation angle for the current viewport. This variable's value is also stored in the <code>SnapRotationAngle</code> property of the <code>AcadPViewport</code> and <code>AcadViewport</code> objects.

Name	Description
SNAPBASE	Sets the snap and grid origin point for the current viewport relative to the current UCS. This variable's value is also stored in the SnapBasePoint property of the AcadPViewport and VAcadviewport objects.
SNAPISOPAIR	Controls the isometric plane for the current viewport.
SNAPMODE	Turns snap mode on and off. This variable's value is also stored in the SnapOn property of the AcadPViewport and AcadViewport objects.
SNAPSTYL	Sets the current viewport's snap style.
SNAPTYPE	Sets the current viewport's snap style.
SNAPUNIT	Sets the current viewport's snap spacing.
SOLIDCHECK	Turns solid validation on and off for the current session.
SORTENTS	Controls the OPTIONS command object sort order operations. This variable's value is also stored in the ObjectSortByPlotting, ObjectSortByRedraws, ObjectSortByPSOutput, ObjectSortByRegens, ObjectSortBySelection, and ObjectSortBySnap properties of the AcadDatabasePreferences object.
SPLFRAME	Controls the display of splines and spline-fit polylines.
SPLINESEGS	Sets the number of line segments to be generated for each spline-fit polyline. This variable's value is also stored in the SegmentPerPolyline property of the AcadDatabasePreferences object.
SPLINETYPE	Sets the type of curve generated by the Spline option of the PEDIT command.
SSFOUND	Displays the sheet set path and file name if a search for a sheet set is successful.
SSLOCATE	Locates and opens the sheet set associated with a drawing when the drawing is opened. SSMAUTOOPEN and SSLOCATE must both be set to 1 to open a sheet set automatically in the Sheet Set Manager.
SSMAUTOOPEN	Displays the Sheet Set Manager when a drawing associated with a sheet is opened. SSMAUTOOPEN and SSLOCATE must both be set to 1 to open a sheet set automatically in the Sheet Set Manager.
SSMPOLLTIME	Controls the time interval between automatic refreshes of the status data in a sheet set. The SSMPOLLTIME timer sets the time in seconds between automatic refreshes of the status data of sheets in a sheet set. Valid values are from 20 to 600. The SSMSHEETSTATUS system variable must be set to 2 for the timer to operate.
SSMSHEETSTATUS	Controls how the status data in a sheet set is refreshed. The status data for sheets in the current sheet set includes whether a sheet is locked and whether a sheet is missing (or found in an unexpected location). This status data can be updated automatically for all sheets. To refresh the sheet set manually, use the Refresh Sheet Status button on the Sheet List tab of the Sheet Set Manager.
SSMSTATE	Determines whether the Sheet Set Manager window is active.
STANDARDSVIOLATION	Specifies whether a user is notified of standards violations that exist in the current drawing.
STARTUP	Controls whether the Create New Drawing dialog box is displayed when starting a new drawing with the NEW and QNEW commands.
SURFTAB1	Sets the number of tabulations to be generated for RULESURF and TABSURF.

Continued

Table C-18. *Continued*

Name	Description
SURFTAB2	Sets the mesh density in the N direction for REVSURF and EDGESURF.
SURFTYPE	Controls the type of surface fitting to be performed by the Smooth option of the PEDIT command.
SURFU	Sets the surface density for PEDIT command's Smooth option in the M direction.
SURFV	Sets the surface density for PEDIT command's Smooth option in the N direction.
SYSCODEPAGE	Indicates the system code page specified in acad.xmlf.

T

Table C-19 explains the system variables that start with *T*.

Table C-19. *The T System Variables*

Name	Description
TABLEINDICATOR	Controls the display of row numbers and column letters when the In-Place Text Editor is open for editing a table cell.
TABMODE	Controls the use of the tablet.
TARGET	Stores the location of the target point for the current viewport.
TBCUSTOMIZE	Controls whether toolbars can be customized.
TDCREATE	Stores the local time and date the drawing was created.
TDINDWG	Stores the total editing time.
TDCREATE	Stores the universal time and date the drawing was created.
TDUPDATE	Stores the local time and date of the last update/save.
TDUSRTIMER	Stores the user-elapsed timer.
TDUUPDATE	Stores the universal time and date of the last update/save.
TEMPOVERRIDES	Turns temporary override keys on and off. A temporary override key is a key that you can hold down to temporarily turn on or turn off one of the drawing aids that are set in the Drafting Settings dialog box; for example, Ortho mode, object snaps, or Polar mode.
TEMPPREFIX	Contains the directory name for temporary files.
TEXTEVAL	Controls the method of evaluation for text strings.
TEXTFILL	Controls the filling of TrueType fonts while plotting, exporting with the PSOUT command and rendering.
TEXTQLTY	Sets the resolution of text outlines for TrueType fonts while plotting, exporting with the PSOUT command, and rendering.
TEXTSIZE	Sets the default height for new text objects drawn with the current text style.
TEXTSTYLE	Sets the name of the current text style.
THICKNESS	Sets the current 3-D solid thickness.

Name	Description
TILEMODE	Makes the Model tab or the Last Layout tab current. This variable's value is also stored in the <code>ActiveSpace</code> property of the <code>AcadDocument</code> object.
TOOLTIPMERGE	Combines drafting tooltips into a single tooltip. The appearance of the merged tooltip is controlled by the settings in the <code>Tooltip Appearance</code> dialog box.
TOOLTIPS	Controls the display of tooltips.
TRACEWID	Sets the default trace width.
TRACKPATH	Controls the display of polar and object snap tracking alignment paths. This variable's value is also stored in the <code>FullScreenTrackingVector</code> and <code>PolarTrackingVector</code> properties of the <code>AcadPreferencesDrafting</code> object.
TRAYICONS	Controls whether a tray is displayed on the status bar.
TRAYNOTIFY	Controls whether service notifications are displayed in the status bar tray.
TRAYTIMEOUT	Controls the length of time (in seconds) that service notifications are displayed.
TREEDEPTH	Specifies the number of times the tree-structured spatial index can divide into branches.
TREEMAX	Limits memory consumption during drawing regeneration by limiting the number of nodes in the spatial index (oct-tree).
TRIMMODE	Controls whether AutoCAD trims selected edges for chamfers and fillets.
TSPACEFAC	Controls the multiline text line spacing distance, measured as a factor of text height.
TSPACETYPE	Controls the type of line spacing used in multiline text.
TSTACKALIGN	Controls the vertical alignment of stacked text.
TSTACKSIZE	Controls the height of stacked text fraction, as a percentage relative to the height of the selected text.

U

Table C-20 explains the system variables that start with *U*.

Table C-20. *The U System Variables*

Name	Description
UCSAXISANG	Stores the default angle when rotating the UCS around one of its axes using the X, Y, or Z options of the UCS command.
UCSBASE	Stores the name of the UCS that defines the origin and orientation of orthographic UCS settings.
UCSFOLLOW	Generates a plan view whenever you change from one UCS to another.
UCSICON	Displays the UCS icon for the current viewport.
UCSNAME	Stores the name of the current coordinate system for the current viewport in the current space.
UCSORG	Stores the origin point of the coordinate system for the current viewport in the current space. This variable's value is also stored in the <code>UCSIconAtOrigin</code> property of the <code>PViewport</code> and <code>Viewport</code> objects.

Continued

Table C-20. *Continued*

Name	Description
UCSORTHO	Determines whether the related orthographic UCS setting is restored automatically when an orthographic view is restored.
UCSVIEW	Determines whether the current UCS is saved with a named view.
UCSVP	Determines whether the UCS in active viewports remains fixed or changes to reflect the currently active viewport.
UCSXDIR	Stores the X direction of the current UCS for the current viewport in the current space. This variable's value is also stored in the XVector property of the AcadUCS object.
UCSYDIR	Stores the Y direction of the current UCS for the current viewport in the current space. This variable's value is also stored in the YVector property of the AcadUCS object.
UNDOCTL	Stores a bit code indicating the state of the Auto and Control options of the UNDO command.
UNDOMARKS	Stores the number of marks that have been placed in the UNDO control stream by the Mark option.
UNITMODE	Controls the display format for units.
UPDATETHUMBNAIL	Controls updating of the thumbnail previews in the Sheet Set Manager. The setting is stored as a bit code using the sum of the values.
USERI1-5	Stores and retrieves integer values. There are five variables of this type: USERI1, USERI2, ..., USERI5.
USERR1-5	Stores and retrieves real numbers. There are five variables of this type: USERR1, USERR2, ..., USERR5.
USERS1-5	Stores and retrieves strings. There are five variables of this type: USERS1, USERS2, ..., USERS5.

V

Table C-21 explains the system variables that start with *V*.

Table C-21. *The V System Variables*

Name	Description
_VERNUM	Stores the internal build number of AutoCAD (such as V.0.86).
VIEWCTR	Stores the center of view in the current viewport.
VIEWDIR	Stores the viewing direction in the current viewport.
VIEWMODE	Controls the current viewport's view mode using a bit code.
VIEWSIZE	Stores the view's height in the current viewport.
VIEWTWIST	Stores the view's twist angle for the current viewport.
VISRETAIN	Controls the visibility, color, linetype, lineweight, and plot styles (if PSTYLEPOLICY is set to 0) of Xref-dependent layers and specifies whether nested Xref path changes are saved. This variable's value is also stored in the XrefLayerVisibility property of the AcadDatabasePreferences object.

Name	Description
VPMAXIMIZEDSTATE	Stores a value that indicates whether the viewport is maximized. The maximized viewport state is canceled if you start the PLOT command.
VSMAX	Stores the upper-right corner of the current viewport's virtual screen.
VSMIN	Stores the lower-left corner of the current viewport's virtual screen.
VTDURATION	Sets the duration of a smooth view transition, in milliseconds. The valid range is 0 to 5000.
VTENABLE	Controls when smooth view transitions are used. Smooth view transitions can be on or off for panning and zooming, for changes of view angle, or for scripts. The valid range is 0 to 7.
VTFPS	Sets the minimum speed of a smooth view transition, in frames per second. When a smooth view transition cannot maintain this speed, an instant transition is used. The valid range is 1.0 to 30.0.

W

Table C-22 explains the system variables that start with *W*.

Table C-22. *The W System Variables*

Name	Description
WHIPARC	Controls whether the display of circles and arcs is smooth.
WHIPTHREAD	Controls whether to use an additional processor (known as multithreaded processing) to improve the speed of operations such as ZOOM and PAN that redraw or regenerate the drawing.
WINDOWAREACOLOR	Controls the color of the transparent selection area during window selection. The valid range is 1 to 255. SELECTIONAREA must be on.
WMFBKGND	Controls the background of the output Windows metafile resulting from the WMFOUT command, and also the metafile format of objects placed on the clipboard or dragged and dropped into other applications.
WMFFOREGND	Controls the assignment of the foreground color of AutoCAD objects in other applications.
WORLDUCS	Indicates whether the UCS is the same as the WCS.
WORLDVIEW	Determines whether input to the 3DORBIT, DVIEW, and VPOINT commands is relative to the WCS (default), the current UCS, or the UCS specified by the UCSBASE system variable.
WRITESTAT	Indicates whether a drawing file is read-only, for developers who need to determine write status through AutoLISP.
WSCURRENT	Returns the current work space name in the command line interface and sets a work space to current.

X

Table C-23 explains the system variables that start with X.

Table C-23. *The X System Variables*

Name	Description
XCLIPFRAME	Controls the visibility of Xref clipping boundaries.
XEDIT	Specifies whether the current drawing can be edited in place when another drawing references it. This variable's value is also stored in the XRefEdit property of the AcadDatabasePreferences object.
XFADECTL	Controls the fading intensity for references being edited in place. This variable's value is also stored in the XrefFadeIntensity property of the AcadPreferencesDisplay object.
XLOADCTL	Turns Xref demand loading on and off and controls whether it opens the original drawing or a copy of the drawing. This variable's value is also stored in the XrefDemandLoad property of the AcadPreferencesOpenSave object.
XLOADPATH	Creates a path for storing temporary copies of demand-loaded Xref files.
XREFCTL	Controls whether AutoCAD writes external reference log (.xlg) files.
XREFNOTIFY	Controls the notification for changed or missing Xrefs.
XREFTYPE	Controls the default reference type when attaching or overlaying an external reference.

Z

Table C-24 explains the system variables that start with Z.

Table C-24. *The Z System Variables*

Name	Description
ZOOMFACTOR	Controls the incremental change in zoom with each IntelliMouse wheel action, whether forward or backward.

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